

Planar Kinetics of a Rigid Body: Impulse and Momentum

CHAPTER OBJECTIVES

- To develop formulations for the linear and angular momentum of a body.
- To apply the principles of linear and angular impulse and momentum to solve rigid-body planar kinetic problems that involve force, velocity, and time.
- To discuss application of the conservation of momentum.
- To analyze the mechanics of eccentric impact.



Video Solutions are available for selected questions in this chapter.

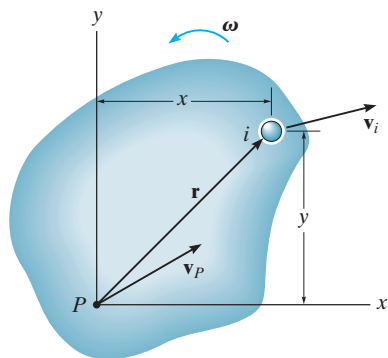
19.1 Linear and Angular Momentum

In this chapter we will use the principles of linear and angular impulse and momentum to solve problems involving force, velocity, and time as related to the planar motion of a rigid body. Before doing this, we will first formalize the methods for obtaining a body's linear and angular momentum, assuming the body is symmetric with respect to an inertial x - y reference plane.

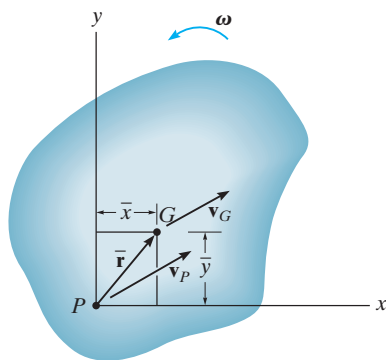
Linear Momentum. The linear momentum of a rigid body is determined by summing vectorially the linear momenta of all the particles of the body, i.e., $\mathbf{L} = \sum m_i \mathbf{v}_i$. Since $\sum m_i \mathbf{v}_i = m \mathbf{v}_G$ (see Sec. 15.2) we can also write

$$\mathbf{L} = m \mathbf{v}_G \quad (19-1)$$

This equation states that the body's linear momentum is a vector quantity having a *magnitude* $m v_G$, which is commonly measured in units of $\text{kg} \cdot \text{m/s}$ and a *direction* defined by $\mathbf{v}_G$ the velocity of the body's mass center.



(a)



(b)

Fig. 19-1

Angular Momentum. Consider the body in Fig. 19-1a, which is subjected to general plane motion. At the instant shown, the arbitrary point P has a known velocity $\mathbf{v}_P$, and the body has an angular velocity ω . Therefore the velocity of the i th particle of the body is

$$\mathbf{v}_i = \mathbf{v}_P + \mathbf{v}_{i/P} = \mathbf{v}_P + \omega \times \mathbf{r}$$

The angular momentum of this particle about point P is equal to the “moment” of the particle’s linear momentum about P , Fig. 19-1a. Thus,

$$(\mathbf{H}_P)_i = \mathbf{r} \times m_i \mathbf{v}_i$$

Expressing $\mathbf{v}_i$ in terms of $\mathbf{v}_P$ and using Cartesian vectors, we have

$$(\mathbf{H}_P)_i \mathbf{k} = m_i (\mathbf{x}\mathbf{i} + \mathbf{y}\mathbf{j}) \times [(v_P)_x \mathbf{i} + (v_P)_y \mathbf{j} + \omega \mathbf{k} \times (\mathbf{x}\mathbf{i} + \mathbf{y}\mathbf{j})]$$

$$(\mathbf{H}_P)_i = -m_i y (v_P)_x + m_i x (v_P)_y + m_i \omega r^2$$

Letting $m_i \rightarrow dm$ and integrating over the entire mass m of the body, we obtain

$$H_P = -\left(\int_m y dm\right)(v_P)_x + \left(\int_m x dm\right)(v_P)_y + \left(\int_m r^2 dm\right)\omega$$

Here H_P represents the angular momentum of the body about an axis (the z axis) perpendicular to the plane of motion that passes through point P . Since $\bar{y}m = \int y dm$ and $\bar{x}m = \int x dm$, the integrals for the first and second terms on the right are used to locate the body’s center of mass G with respect to P , Fig. 19-1b. Also, the last integral represents the body’s moment of inertia about point P . Thus,

$$H_P = -\bar{y}m(v_P)_x + \bar{x}m(v_P)_y + I_P \omega \quad (19-2)$$

This equation reduces to a simpler form if P coincides with the mass center G for the body,* in which case $\bar{x} = \bar{y} = 0$. Hence,

$$H_G = I_G \omega \quad (19-3)$$

*It also reduces to the same simple form, $H_P = I_P \omega$, if point P is a *fixed point* (see Eq. 19-9) or the velocity of P is directed along the line PG .

Here the angular momentum of the body about G is equal to the product of the moment of inertia of the body about an axis passing through G and the body's angular velocity. Realize that $\mathbf{H}_G$ is a vector quantity having a magnitude $I_G\omega$, which is commonly measured in units of $\text{kg} \cdot \text{m}^2/\text{s}$ and a direction defined by ω , which is always perpendicular to the plane of motion.

Equation 19-2 can also be rewritten in terms of the x and y components of the velocity of the body's mass center, $(\mathbf{v}_G)_x$ and $(\mathbf{v}_G)_y$, and the body's moment of inertia I_G . Since G is located at coordinates $(\bar{x}, \bar{y})$, then by the parallel-axis theorem, $I_P = I_G + m(\bar{x}^2 + \bar{y}^2)$. Substituting into Eq. 19-2 and rearranging terms, we have

$$H_P = \bar{y}m[-(v_P)_x + \bar{y}\omega] + \bar{x}m[(v_P)_y + \bar{x}\omega] + I_G\omega \quad (19-4)$$

From the kinematic diagram of Fig. 19-1*b*, $\mathbf{v}_G$ can be expressed in terms of $\mathbf{v}_P$ as

$$\begin{aligned} \mathbf{v}_G &= \mathbf{v}_P + \omega \times \bar{\mathbf{r}} \\ (v_G)_x \mathbf{i} + (v_G)_y \mathbf{j} &= (v_P)_x \mathbf{i} + (v_P)_y \mathbf{j} + \omega \mathbf{k} \times (\bar{x} \mathbf{i} + \bar{y} \mathbf{j}) \end{aligned}$$

Carrying out the cross product and equating the respective $\mathbf{i}$ and $\mathbf{j}$ components yields the two scalar equations

$$(v_G)_x = (v_P)_x - \bar{y}\omega$$

$$(v_G)_y = (v_P)_y + \bar{x}\omega$$

Substituting these results into Eq. 19-4 yields

$$(\zeta +)H_P = -\bar{y}m(v_G)_x + \bar{x}m(v_G)_y + I_G\omega \quad (19-5)$$

As shown in Fig. 19-1*c*, this result indicates that when the angular momentum of the body is computed about point P , it is equivalent to the moment of the linear momentum $m\mathbf{v}_G$, or its components $m(\mathbf{v}_G)_x$ and $m(\mathbf{v}_G)_y$, about P plus the angular momentum $I_G\omega$. Using these results, we will now consider three types of motion.

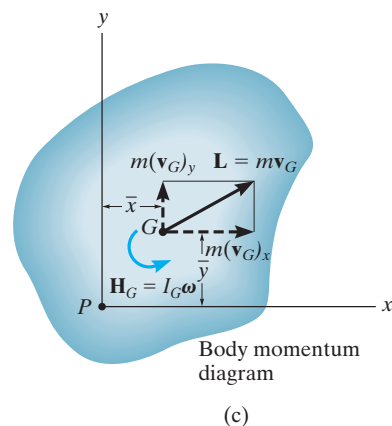
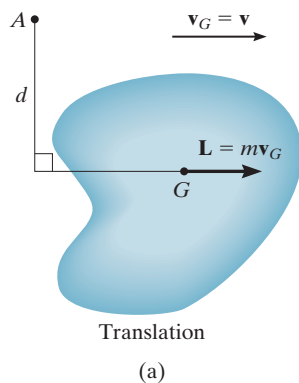


Fig. 19-1



Translation. When a rigid body is subjected to either rectilinear or curvilinear *translation*, Fig. 19-2a, then $\omega = 0$ and its mass center has a velocity of $\mathbf{v}_G = \mathbf{v}$. Hence, the linear momentum, and the angular momentum about G , become

$$\begin{aligned} L &= mv_G \\ H_G &= 0 \end{aligned} \quad (19-6)$$

If the angular momentum is computed about some other point A , the “moment” of the linear momentum $\mathbf{L}$ must be found about the point. Since d is the “moment arm” as shown in Fig. 19-2a, then in accordance with Eq. 19-5, $H_A = (d)(mv_G) \curvearrowleft$.

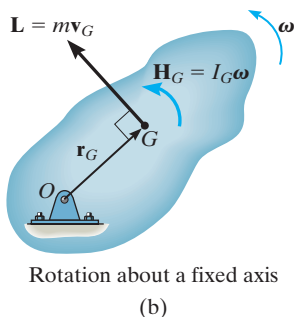


Fig. 19-2

Rotation About a Fixed Axis. When a rigid body is *rotating about a fixed axis*, Fig. 19-2b, the linear momentum, and the angular momentum about G , are

$$\begin{aligned} L &= mv_G \\ H_G &= I_G \omega \end{aligned} \quad (19-7)$$

It is sometimes convenient to compute the angular momentum about point O . Noting that $\mathbf{L}$ (or $\mathbf{v}_G$) is always *perpendicular to* $\mathbf{r}_G$, we have

$$(\curvearrowleft +) H_O = I_G \omega + r_G(mv_G) \quad (19-8)$$

Since $v_G = r_G \omega$, this equation can be written as $H_O = (I_G + mr_G^2) \omega$. Using the parallel-axis theorem,*

$$H_O = I_O \omega \quad (19-9)$$

For the calculation, then, either Eq. 19-8 or 19-9 can be used.

*The similarity between this derivation and that of Eq. 17-16 ($\Sigma M_O = I_O \alpha$) and Eq. 18-5 ($T = \frac{1}{2} I_O \omega^2$) should be noted. Also note that the same result can be obtained from Eq. 19-2 by selecting point P at O , realizing that $(v_O)_x = (v_O)_y = 0$.

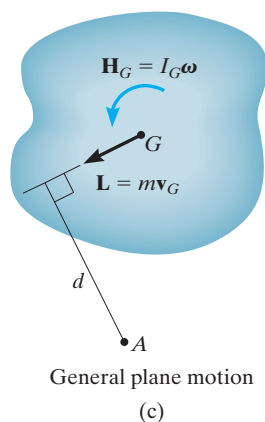


Fig. 19-2

General Plane Motion. When a rigid body is subjected to general plane motion, Fig. 19-2c, the linear momentum, and the angular momentum about G , become

$$\begin{aligned} L &= mv_G \\ H_G &= I_G \omega \end{aligned} \quad (19-10)$$

If the angular momentum is computed about point A , Fig. 19-2c, it is necessary to include the moment of $\mathbf{L}$ and $\mathbf{H}_G$ about this point. In this case,

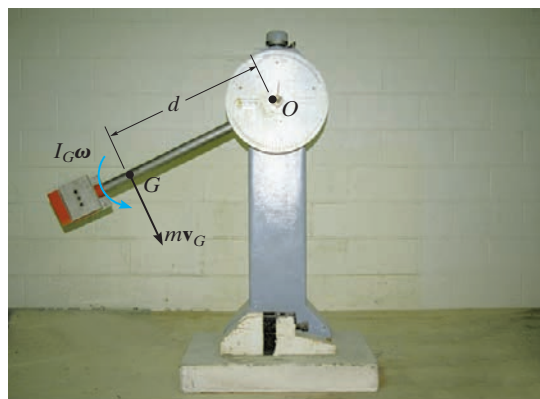
$$(\zeta +) \quad H_A = I_G \omega + (d)(mv_G)$$

Here d is the moment arm, as shown in the figure.

As a special case, if point A is the instantaneous center of zero velocity then, like Eq. 19-9, we can write the above equation in simplified form as

$$H_{IC} = I_{IC} \omega \quad (19-11)$$

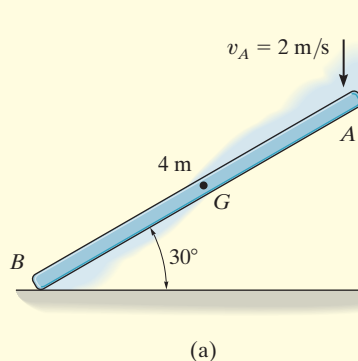
where I_{IC} is the moment of inertia of the body about the IC . (See Prob. 19-2.)



As the pendulum swings downward, its angular momentum about point O can be determined by computing the moment of $I_G \omega$ and $m\mathbf{v}_G$ about O . This is $H_O = I_G \omega + (mv_G)d$. Since $v_G = \omega d$, then $H_O = I_G \omega + m(\omega d)d = (I_G + md^2)\omega = I_O \omega$.

EXAMPLE 19.1

At a given instant the 5-kg slender bar has the motion shown in Fig. 19–3a. Determine its angular momentum about point G and about the IC at this instant.



SOLUTION

Bar. The bar undergoes *general plane motion*. The IC is established in Fig. 19–3b, so that

$$\omega = \frac{2 \text{ m/s}}{4 \text{ m} \cos 30^\circ} = 0.5774 \text{ rad/s}$$

$$v_G = (0.5774 \text{ rad/s})(2 \text{ m}) = 1.155 \text{ m/s}$$

Thus,

$$(\zeta +) H_G = I_G \omega = \left[\frac{1}{12} (5 \text{ kg})(4 \text{ m})^2 \right] (0.5774 \text{ rad/s}) = 3.85 \text{ kg} \cdot \text{m}^2/\text{s} \quad \text{Ans.}$$

Adding $I_G \omega$ and the moment of mv_G about the IC yields

$$\begin{aligned} (\zeta +) H_{IC} &= I_G \omega + d(mv_G) \\ &= \left[\frac{1}{12} (5 \text{ kg})(4 \text{ m})^2 \right] (0.5774 \text{ rad/s}) + (2 \text{ m})(5 \text{ kg})(1.155 \text{ m/s}) \\ &= 15.4 \text{ kg} \cdot \text{m}^2/\text{s} \quad \text{Ans.} \end{aligned}$$

We can also use

$$\begin{aligned} (\zeta +) H_{IC} &= I_{IC} \omega \\ &= \left[\frac{1}{12} (5 \text{ kg})(4 \text{ m})^2 + (5 \text{ kg})(2 \text{ m})^2 \right] (0.5774 \text{ rad/s}) \\ &= 15.4 \text{ kg} \cdot \text{m}^2/\text{s} \quad \text{Ans.} \end{aligned}$$

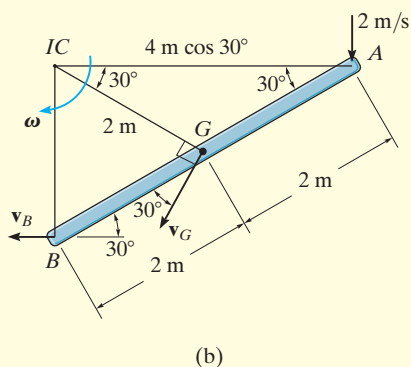


Fig. 19–3

19.2 Principle of Impulse and Momentum

Like the case for particle motion, the principle of impulse and momentum for a rigid body can be developed by *combining* the equation of motion with kinematics. The resulting equation will yield a *direct solution to problems involving force, velocity, and time*.

Principle of Linear Impulse and Momentum. The equation of translational motion for a rigid body can be written as $\Sigma \mathbf{F} = m\mathbf{a}_G = m(d\mathbf{v}_G/dt)$. Since the mass of the body is constant,

$$\Sigma \mathbf{F} = \frac{d}{dt}(m\mathbf{v}_G)$$

Multiplying both sides by dt and integrating from $t = t_1$, $\mathbf{v}_G = (\mathbf{v}_G)_1$ to $t = t_2$, $\mathbf{v}_G = (\mathbf{v}_G)_2$ yields

$$\Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m(\mathbf{v}_G)_2 - m(\mathbf{v}_G)_1$$

This equation is referred to as the *principle of linear impulse and momentum*. It states that the sum of all the impulses created by the external force system which acts on the body during the time interval t_1 to t_2 is equal to the change in the linear momentum of the body during this time interval, Fig. 19-4.

Principle of Angular Impulse and Momentum. If the body has *general plane motion* then $\Sigma M_G = I_G\alpha = I_G(d\omega/dt)$. Since the moment of inertia is constant,

$$\Sigma M_G = \frac{d}{dt}(I_G\omega)$$

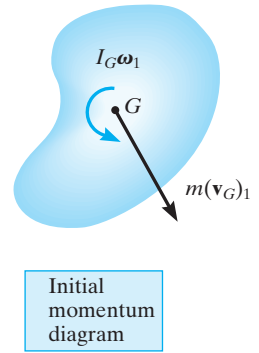
Multiplying both sides by dt and integrating from $t = t_1$, $\omega = \omega_1$ to $t = t_2$, $\omega = \omega_2$ gives

$$\Sigma \int_{t_1}^{t_2} M_G dt = I_G\omega_2 - I_G\omega_1 \quad (19-12)$$

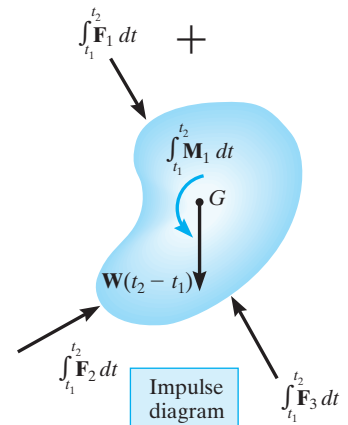
In a similar manner, for *rotation about a fixed axis* passing through point O , Eq. 17-16 ($\Sigma M_O = I_O\alpha$) when integrated becomes

$$\Sigma \int_{t_1}^{t_2} M_O dt = I_O\omega_2 - I_O\omega_1 \quad (19-13)$$

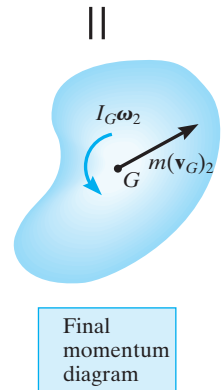
Equations 19-12 and 19-13 are referred to as the *principle of angular impulse and momentum*. Both equations state that the sum of the angular impulses acting on the body during the time interval t_1 to t_2 is equal to the change in the body's angular momentum during this time interval.



(a)

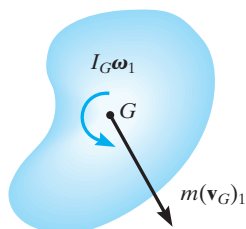


(b)



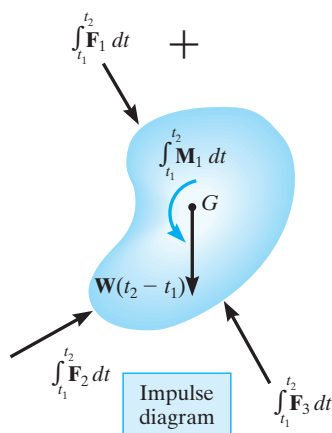
(c)

Fig. 19-4



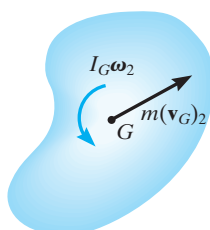
Initial
momentum
diagram

(a)



(b)

||



Final
momentum
diagram

(c)

Fig. 19-4 (repeated)

To summarize these concepts, if motion occurs in the x - y plane, the following *three scalar equations* can be written to describe the *planar motion* of the body.

$$\begin{aligned} m(v_{Gx})_1 + \Sigma \int_{t_1}^{t_2} F_x dt &= m(v_{Gx})_2 \\ m(v_{Gy})_1 + \Sigma \int_{t_1}^{t_2} F_y dt &= m(v_{Gy})_2 \\ I_G \omega_1 + \Sigma \int_{t_1}^{t_2} M_G dt &= I_G \omega_2 \end{aligned} \quad (19-14)$$

The terms in these equations can be shown graphically by drawing a set of impulse and momentum diagrams for the body, Fig. 19-4. Note that the linear momentum $m\mathbf{v}_G$ is applied at the body's mass center, Figs. 19-4a and 19-4c; whereas the angular momentum $I_G \omega$ is a free vector, and therefore, like a couple moment, it can be applied at any point on the body. When the impulse diagram is constructed, Fig. 19-4b, the forces $\mathbf{F}$ and moment $\mathbf{M}$ vary with time, and are indicated by the integrals. However, if $\mathbf{F}$ and $\mathbf{M}$ are *constant* integration of the impulses yields $\mathbf{F}(t_2 - t_1)$ and $\mathbf{M}(t_2 - t_1)$, respectively. Such is the case for the body's weight $\mathbf{W}$, Fig. 19-4b.

Equations 19-14 can also be applied to an entire system of connected bodies rather than to each body separately. This eliminates the need to include interaction impulses which occur at the connections since they are *internal* to the system. The resultant equations may be written in symbolic form as

$$\begin{aligned} \left(\Sigma \text{ syst. linear momentum} \right)_{x1} + \left(\Sigma \text{ syst. linear impulse} \right)_{x(1-2)} &= \left(\Sigma \text{ syst. linear momentum} \right)_{x2} \\ \left(\Sigma \text{ syst. linear momentum} \right)_{y1} + \left(\Sigma \text{ syst. linear impulse} \right)_{y(1-2)} &= \left(\Sigma \text{ syst. linear momentum} \right)_{y2} \\ \left(\Sigma \text{ syst. angular momentum} \right)_{O1} + \left(\Sigma \text{ syst. angular impulse} \right)_{O(1-2)} &= \left(\Sigma \text{ syst. angular momentum} \right)_{O2} \end{aligned}$$

(19-15)

As indicated by the third equation, the system's angular momentum and angular impulse must be computed with respect to the *same reference point* O for all the bodies of the system.

Procedure For Analysis

Impulse and momentum principles are used to solve kinetic problems that involve *velocity*, *force*, and *time* since these terms are involved in the formulation.

Free-Body Diagram.

- Establish the x, y, z inertial frame of reference and draw the free-body diagram in order to account for all the forces and couple moments that produce impulses on the body.
- The direction and sense of the initial and final velocity of the body's mass center, $\mathbf{v}_G$, and the body's angular velocity $\boldsymbol{\omega}$ should be established. If any of these motions is unknown, assume that the sense of its components is in the direction of the positive inertial coordinates.
- Compute the moment of inertia I_G or I_O .
- As an alternative procedure, draw the impulse and momentum diagrams for the body or system of bodies. Each of these diagrams represents an outlined shape of the body which graphically accounts for the data required for each of the three terms in Eqs. 19–14 or 19–15, Fig. 19–4. These diagrams are particularly helpful in order to visualize the “moment” terms used in the principle of angular impulse and momentum, if application is about the IC or another point other than the body's mass center G or a fixed point O .

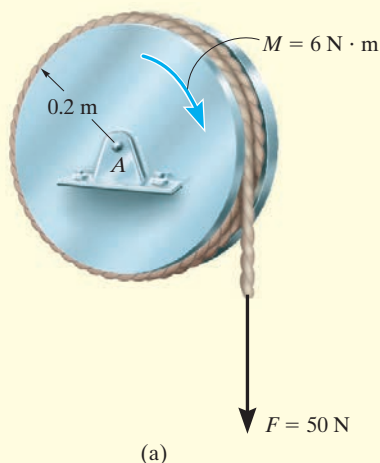
Principle of Impulse and Momentum.

- Apply the three scalar equations of impulse and momentum.
- The angular momentum of a rigid body rotating about a fixed axis is the moment of $m\mathbf{v}_G$ plus $I_G\boldsymbol{\omega}$ about the axis. This is equal to $H_O = I_O\boldsymbol{\omega}$, where I_O is the moment of inertia of the body about the axis.
- All the forces acting on the body's free-body diagram will create an impulse; however, some of these forces will do no work.
- Forces that are functions of time must be integrated to obtain the impulse.
- The principle of angular impulse and momentum is often used to eliminate unknown impulsive forces that are parallel or pass through a common axis, since the moment of these forces is zero about this axis.

Kinematics.

- If more than three equations are needed for a complete solution, it may be possible to relate the velocity of the body's mass center to the body's angular velocity using *kinematics*. If the motion appears to be complicated, kinematic (velocity) diagrams may be helpful in obtaining the necessary relation.

EXAMPLE 19.2



The 10-kg disk shown in Fig. 19-5a is acted upon by a constant couple moment of $6 \text{ N} \cdot \text{m}$ and a force of 50 N which is applied to a cord wrapped around its periphery. Determine the angular velocity of the disk two seconds after starting from rest. Also, what are the force components of reaction at the pin?

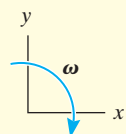
SOLUTION

Since angular velocity, force, and time are involved in the problems, we will apply the principles of impulse and momentum to the solution.

Free-Body Diagram. Fig. 19-5b. The disk's mass center does not move; however, the loading causes the disk to rotate clockwise.

The moment of inertia of the disk about its fixed axis of rotation is

$$I_A = \frac{1}{2}mr^2 = \frac{1}{2}(10 \text{ kg})(0.2 \text{ m})^2 = 0.2 \text{ kg} \cdot \text{m}^2$$



Principle of Impulse and Momentum.

($\pm$)

$$m(v_{Ax})_1 + \sum \int_{t_1}^{t_2} F_x dt = m(v_{Ax})_2$$

$$0 + A_x(2 \text{ s}) = 0$$

($+\uparrow$)

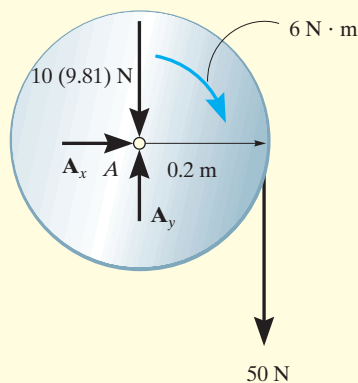
$$m(v_{Ay})_1 + \sum \int_{t_1}^{t_2} F_y dt = m(v_{Ay})_2$$

$$0 + A_y(2 \text{ s}) - [10(9.81) \text{ N}](2 \text{ s}) - 50 \text{ N}(2 \text{ s}) = 0$$

($\zeta +$)

$$I_A\omega_1 + \sum \int_{t_1}^{t_2} M_A dt = I_A\omega_2$$

$$0 + (6 \text{ N} \cdot \text{m})(2 \text{ s}) + [50 \text{ N}(2 \text{ s})](0.2 \text{ m}) = 0.2\omega_2$$



(b)

Fig. 19-5

Solving these equations yields

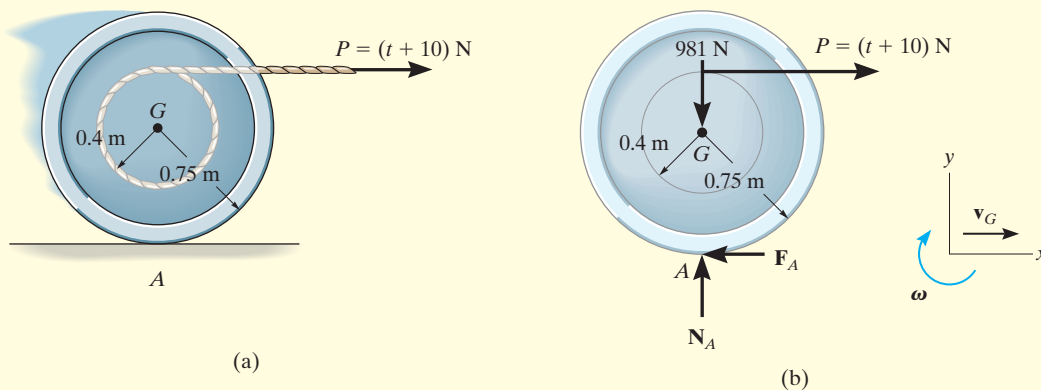
$$A_x = 0 \quad \text{Ans.}$$

$$A_y = 148 \text{ N} \quad \text{Ans.}$$

$$\omega_2 = 160 \text{ rad/s} \quad \text{Ans.}$$

EXAMPLE 19.3

The 100-kg spool shown in Fig. 19–6*a* has a radius of gyration $k_G = 0.35$ m. A cable is wrapped around the central hub of the spool, and a horizontal force having a variable magnitude of $P = (t + 10)$ N is applied, where t is in seconds. If the spool is initially at rest, determine its angular velocity in 5 s. Assume that the spool rolls without slipping at A .

**Fig. 19–6****SOLUTION**

Free-Body Diagram. From the free-body diagram, Fig. 19–6*b*, the *variable* force $\mathbf{P}$ will cause the friction force $\mathbf{F}_A$ to be variable, and thus the impulses created by both $\mathbf{P}$ and $\mathbf{F}_A$ must be determined by integration. Force $\mathbf{P}$ causes the mass center to have a velocity $\mathbf{v}_G$ to the right, and so the spool has a clockwise angular velocity ω .

Principle of Impulse and Momentum. A direct solution for ω can be obtained by applying the principle of angular impulse and momentum about point A , the *IC*, in order to eliminate the unknown friction impulse.

$$\begin{aligned}
 (\zeta +) \quad I_A \omega_1 + \Sigma \int M_A dt &= I_A \omega_2 \\
 0 + \left[\int_0^{5\text{ s}} (t + 10) \text{ N } dt \right] (0.75 \text{ m} + 0.4 \text{ m}) &= [100 \text{ kg } (0.35 \text{ m})^2 + (100 \text{ kg})(0.75 \text{ m})^2] \omega_2 \\
 62.5(1.15) &= 68.5 \omega_2 \\
 \omega_2 &= 1.05 \text{ rad/s } \curvearrowright
 \end{aligned}$$

Ans.

NOTE: Try solving this problem by applying the principle of impulse and momentum about G and using the principle of linear impulse and momentum in the x direction.

EXAMPLE 19.4

The cylinder B , shown in Fig. 19–7a has a mass of 6 kg. It is attached to a cord which is wrapped around the periphery of a 20-kg disk that has a moment of inertia $I_A = 0.40 \text{ kg} \cdot \text{m}^2$. If the cylinder is initially moving downward with a speed of 2 m/s, determine its speed in 3 s. Neglect the mass of the cord in the calculation.

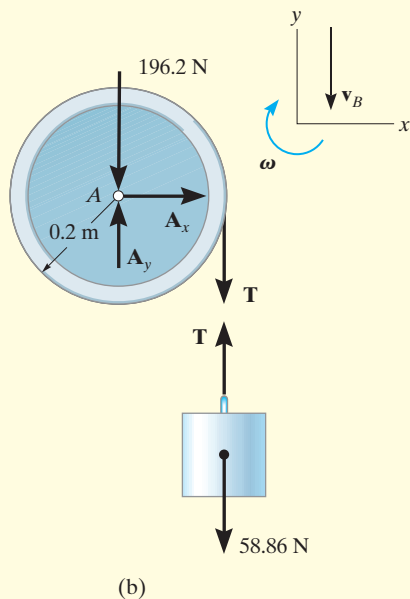
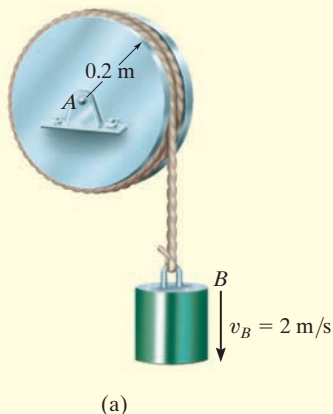


Fig. 19–7

SOLUTION I

Free-Body Diagram. The free-body diagrams of the cylinder and disk are shown in Fig. 19–7b. All the forces are *constant* since the weight of the cylinder causes the motion. The downward motion of the cylinder, v_B , causes ω of the disk to be clockwise.

Principle of Impulse and Momentum. We can eliminate A_x and A_y from the analysis by applying the principle of angular impulse and momentum about point A. Hence

Disk

$$(\zeta +) \quad I_A \omega_1 + \sum \int M_A dt = I_A \omega_2$$

$$0.40 \text{ kg} \cdot \text{m}^2 (\omega_1) + T(3 \text{ s})(0.2 \text{ m}) = (0.40 \text{ kg} \cdot \text{m}^2) \omega_2$$

Cylinder

$$(+ \uparrow) \quad m_B (v_B)_1 + \sum \int F_y dt = m_B (v_B)_2$$

$$-6 \text{ kg}(2 \text{ m/s}) + T(3 \text{ s}) - 58.86 \text{ N}(3 \text{ s}) = -6 \text{ kg}(v_B)_2$$

Kinematics. Since $\omega = v_B/r$, then $\omega_1 = (2 \text{ m/s})/(0.2 \text{ m}) = 10 \text{ rad/s}$ and $\omega_2 = (v_B)_2/0.2 \text{ m} = 5(v_B)_2$. Substituting and solving the equations simultaneously for $(v_B)_2$ yields

$$(v_B)_2 = 13.0 \text{ m/s} \downarrow$$

Ans.

SOLUTION II

Impulse and Momentum Diagrams. We can obtain $(v_B)_2$ directly by considering the *system* consisting of the cylinder, the cord, and the disk. The impulse and momentum diagrams have been drawn to clarify application of the principle of angular impulse and momentum about point *A*, Fig. 19–7c.

Principle of Angular Impulse and Momentum. Realizing that $\omega_1 = 10 \text{ rad/s}$ and $\omega_2 = 5(v_B)_2$, we have

$$(\zeta +) \left(\sum \text{syst. angular momentum} \right)_{A1} + \left(\sum \text{syst. angular impulse} \right)_{A(1-2)} = \left(\sum \text{syst. angular momentum} \right)_{A2}$$

$$(6 \text{ kg})(2 \text{ m/s})(0.2 \text{ m}) + (0.40 \text{ kg} \cdot \text{m}^2)(10 \text{ rad/s}) + (58.86 \text{ N})(3 \text{ s})(0.2 \text{ m})$$

$$= (6 \text{ kg})(v_B)_2(0.2 \text{ m}) + (0.40 \text{ kg} \cdot \text{m}^2)[5(v_B)_2]$$

$$(v_B)_2 = 13.0 \text{ m/s} \downarrow \quad \text{Ans.}$$

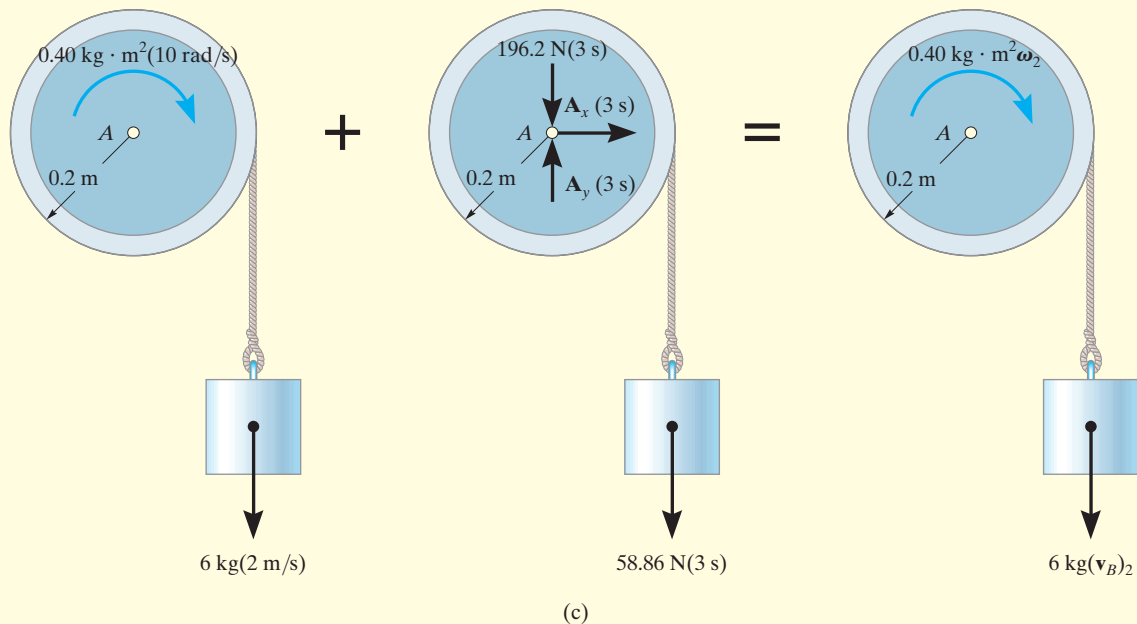
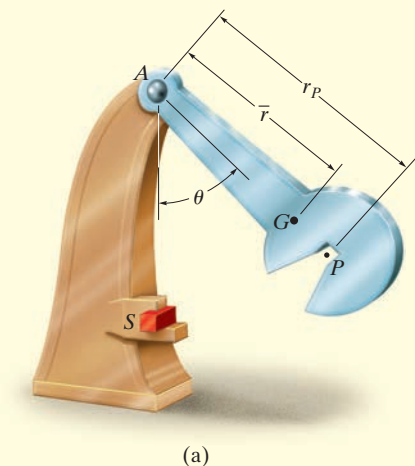
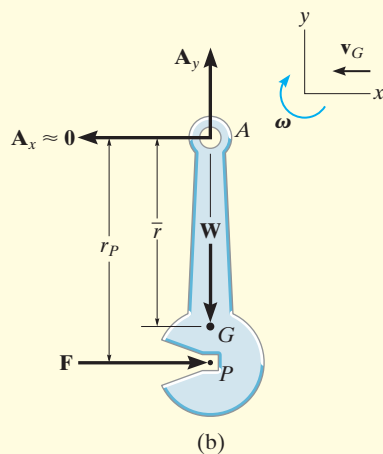


Fig. 19–7 (cont.)

EXAMPLE 19.5



(a)



(b)

Fig. 19-8

The Charpy impact test is used in materials testing to determine the energy absorption characteristics of a material during impact. The test is performed using the pendulum shown in Fig. 19-8a, which has a mass m , mass center at G , and a radius of gyration k_G about G . Determine the distance r_P from the pin at A to the point P where the impact with the specimen S should occur so that the horizontal force at the pin A is essentially zero during the impact. For the calculation, assume the specimen absorbs all the pendulum's kinetic energy gained during the time it falls and thereby stops the pendulum from swinging when $\theta = 0^\circ$.

SOLUTION

Free-Body Diagram. As shown on the free-body diagram, Fig. 19-8b, the conditions of the problem require the horizontal force at A to be zero. Just before impact, the pendulum has a clockwise angular velocity ω_1 , and the mass center of the pendulum is moving to the left at $(v_G)_1 = \bar{r}\omega_1$.

Principle of Impulse and Momentum. We will apply the principle of angular impulse and momentum about point A . Thus,

$$\begin{aligned}
 I_A \omega_1 + \Sigma \int M_A dt &= I_A \omega_2 \\
 (\zeta +) \quad I_A \omega_1 - \left(\int F dt \right) r_P &= 0 \\
 m(v_G)_1 + \Sigma \int F dt &= m(v_G)_2 \\
 (\pm) \quad -m(\bar{r}\omega_1) + \int F dt &= 0
 \end{aligned}$$

Eliminating the impulse $\int F dt$ and substituting $I_A = mk_G^2 + m\bar{r}^2$ yields

$$[mk_G^2 + m\bar{r}^2]\omega_1 - m(\bar{r}\omega_1)r_P = 0$$

Factoring out $m\omega_1$ and solving for r_P , we obtain

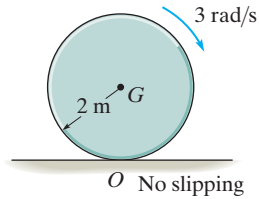
$$r_P = \bar{r} + \frac{k_G^2}{\bar{r}} \quad \text{Ans.}$$

NOTE: Point P , so defined, is called the *center of percussion*. By placing the striking point at P , the force developed at the pin will be minimized. Many sports rackets, clubs, etc. are designed so that collision with the object being struck occurs at the center of percussion. As a consequence, no “sting” or little sensation occurs in the hand of the player. (Also see Probs. 17-66 and 19-1.)

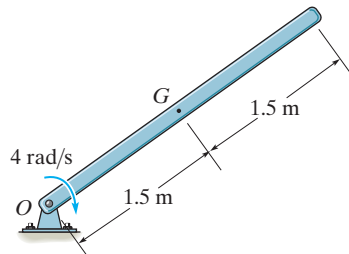
PRELIMINARY PROBLEMS

P19-1. Determine the angular momentum of the 100-kg disk or rod about point G and about point O .

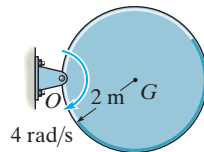
a)



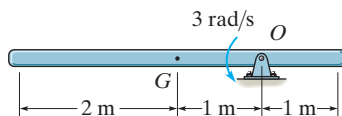
b)



c)



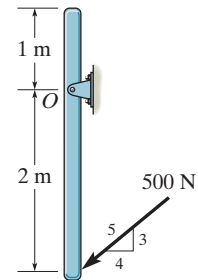
d)



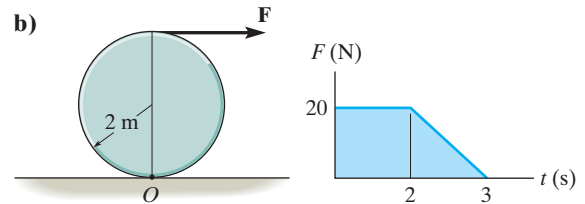
Prob. P19-1

P19-2. Determine the angular impulse about point O for $t = 3$ s.

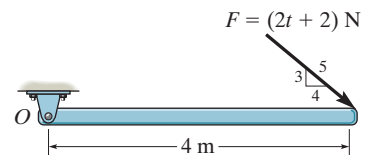
a)



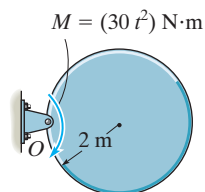
b)



c)



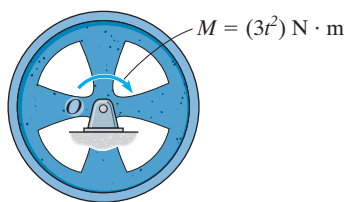
d)



Prob. P19-2

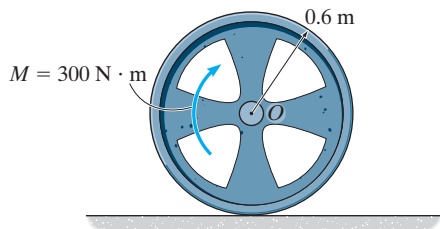
FUNDAMENTAL PROBLEMS

F19-1. The 60-kg wheel has a radius of gyration about its center O of $k_O = 300$ mm. If it is subjected to a couple moment of $M = (3t^2) \text{ N} \cdot \text{m}$, where t is in seconds, determine the angular velocity of the wheel when $t = 4$ s, starting from rest.



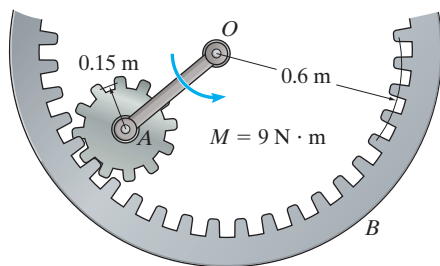
Prob. F19-1

F19-2. The 300-kg wheel has a radius of gyration about its mass center O of $k_O = 400$ mm. If the wheel is subjected to a couple moment of $M = 300 \text{ N} \cdot \text{m}$, determine its angular velocity 6 s after it starts from rest and no slipping occurs. Also, determine the friction force that the ground applies to the wheel.



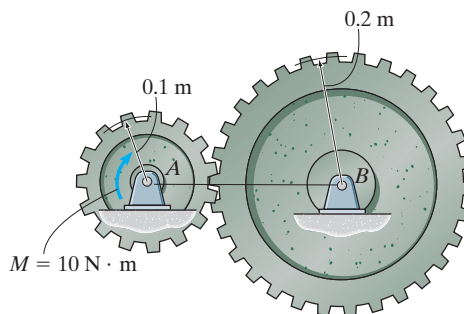
Prob. F19-2

F19-3. If rod OA of negligible mass is subjected to the couple moment $M = 9 \text{ N} \cdot \text{m}$, determine the angular velocity of the 10-kg inner gear $t = 5$ s after it starts from rest. The gear has a radius of gyration about its mass center of $k_A = 100$ mm, and it rolls on the fixed outer gear, B . Motion occurs in the horizontal plane.



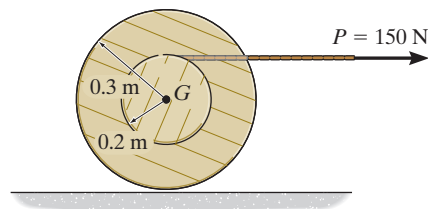
Prob. F19-3

F19-4. Gears A and B of mass 10 kg and 50 kg have radii of gyration about their respective mass centers of $k_A = 80$ mm and $k_B = 150$ mm. If gear A is subjected to the couple moment $M = 10 \text{ N} \cdot \text{m}$ when it is at rest, determine the angular velocity of gear B when $t = 5$ s.



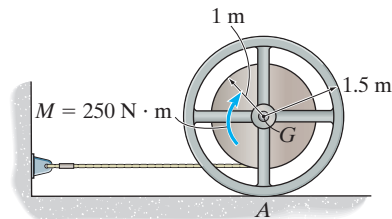
Prob. F19-4

F19-5. The 50-kg spool is subjected to a horizontal force of $P = 150 \text{ N}$. If the spool rolls without slipping, determine its angular velocity 3 s after it starts from rest. The radius of gyration of the spool about its center of mass is $k_G = 175$ mm.



Prob. F19-5

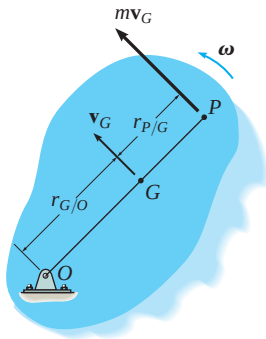
F19-6. The reel has a mass of 150 kg and a radius of gyration about its center of gravity of $k_G = 1.25$ m. If it is subjected to a torque of $M = 250 \text{ N} \cdot \text{m}$, and starts from rest when the torque is applied, determine its angular velocity in 3 seconds. The coefficient of kinetic friction between the reel and the horizontal plane is $\mu_k = 0.15$.



Prob. F19-6

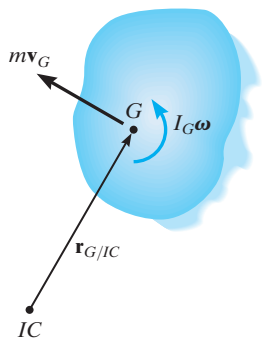
PROBLEMS

19-1. The rigid body (slab) has a mass m and rotates with an angular velocity ω about an axis passing through the fixed point O . Show that the momenta of all the particles composing the body can be represented by a single vector having a magnitude mv_G and acting through point P , called the *center of percussion*, which lies at a distance $r_{P/G} = k_G^2/r_{G/O}$ from the mass center G . Here k_G is the radius of gyration of the body, computed about an axis perpendicular to the plane of motion and passing through G .



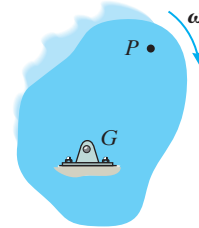
Prob. 19-1

19-2. At a given instant, the body has a linear momentum $\mathbf{L} = m\mathbf{v}_G$ and an angular momentum $\mathbf{H}_G = I_G\boldsymbol{\omega}$ computed about its mass center. Show that the angular momentum of the body computed about the instantaneous center of zero velocity IC can be expressed as $\mathbf{H}_{IC} = I_{IC}\boldsymbol{\omega}$, where I_{IC} represents the body's moment of inertia computed about the instantaneous axis of zero velocity. As shown, the IC is located at a distance $r_{G/IC}$ away from the mass center G .



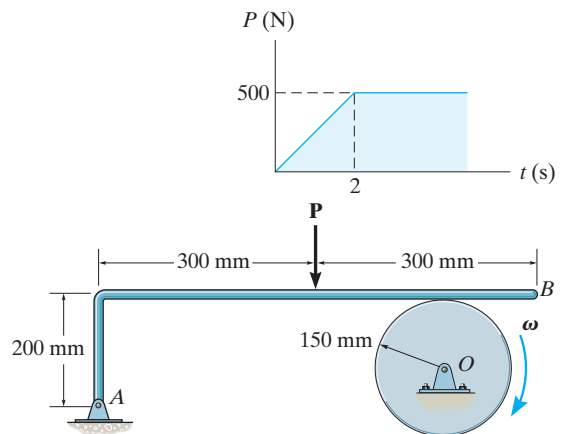
Prob. 19-2

19-3. Show that if a slab is rotating about a fixed axis perpendicular to the slab and passing through its mass center G , the angular momentum is the same when computed about any other point P .



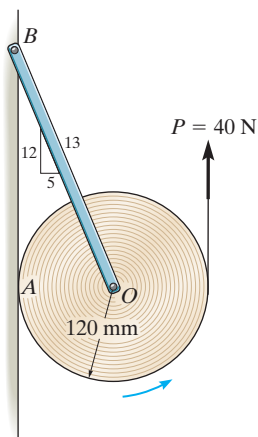
Prob. 19-3

***19-4.** The 40-kg disk is rotating at $\omega = 100$ rad/s when the force $\mathbf{P}$ is applied to the brake as indicated by the graph. If the coefficient of kinetic friction at B is $\mu_k = 0.3$, determine the time t needed to stay the disk from rotating. Neglect the thickness of the brake.



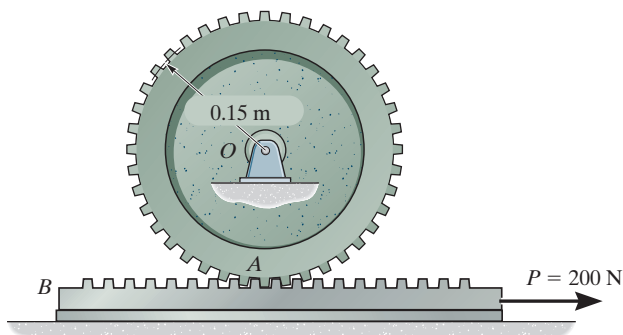
Prob. 19-4

19-5. The 40-kg roll of paper rests along the wall where the coefficient of kinetic friction is $\mu_k = 0.2$. If a vertical force of $P = 40$ N is applied to the paper, determine the angular velocity of the roll when $t = 6$ s starting from rest. Neglect the mass of the unraveled paper and take the radius of gyration of the spool about the axle O to be $k_O = 80$ mm.



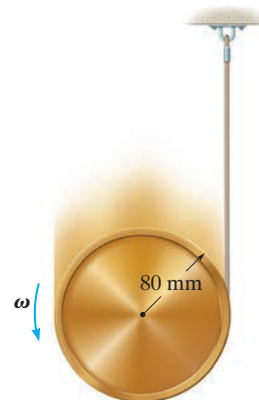
Prob. 19-5

19-6. The 30-kg gear A has a radius of gyration about its center of mass O of $k_O = 125$ mm. If the 20-kg gear rack B is subjected to a force of $P = 200$ N, determine the time required for the gear to obtain an angular velocity of 20 rad/s, starting from rest. The contact surface between the gear rack and the horizontal plane is smooth.



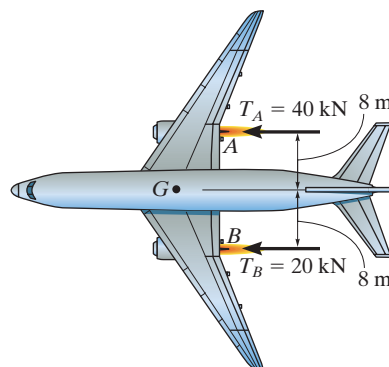
Prob. 19-6

19-7. A wire of negligible mass is wrapped around the outer surface of the 2-kg disk. If the disk is released from rest, determine its angular velocity in 3 s.



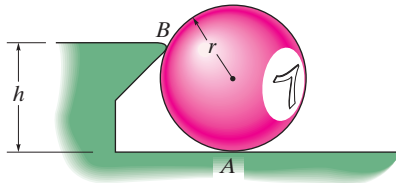
Prob. 19-7

***19-8.** The airplane is traveling in a straight line with a speed of 300 km/h, when the engines A and B produce a thrust of $T_A = 40$ kN and $T_B = 20$ kN, respectively. Determine the angular velocity of the airplane in $t = 5$ s. The plane has a mass of 200 Mg, its center of mass is located at G , and its radius of gyration about G is $k_G = 15$ m.



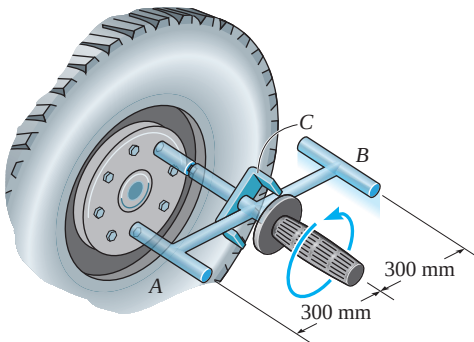
Prob. 19-8

19-9. Determine the height h of the bumper of the pool table, so that when the pool ball of mass m strikes it, no frictional force will be developed between the ball and the table at A . Assume the bumper exerts only a horizontal force on the ball.



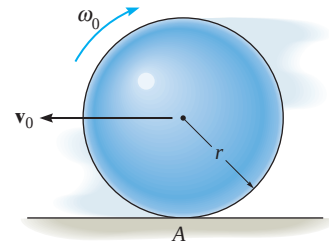
Prob. 19-9

19-10. The impact wrench consists of a slender 1-kg rod AB which is 580 mm long, and cylindrical end weights at A and B that each have a diameter of 20 mm and a mass of 1 kg. This assembly is free to turn about the handle and socket, which are attached to the lug nut on the wheel of a car. If the rod AB is given an angular velocity of 4 rad/s and it strikes the bracket C on the handle without rebounding, determine the angular impulse imparted to the lug nut.



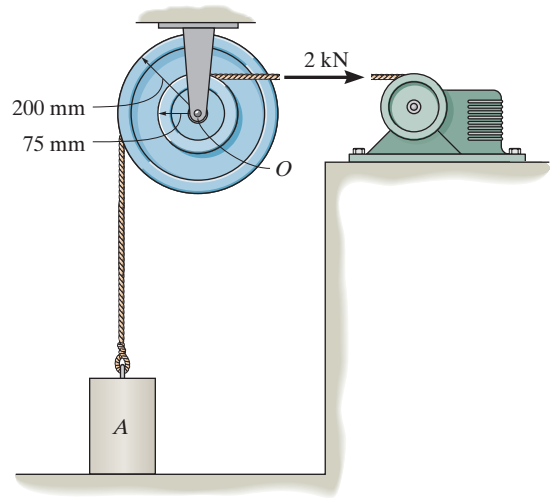
Prob. 19-10

19-11. If the ball has a weight W and radius r and is thrown onto a rough surface with a velocity v_0 parallel to the surface, determine the amount of backspin, ω_0 , it must be given so that it stops spinning at the same instant that its forward velocity is zero. It is not necessary to know the coefficient of friction at A for the calculation.



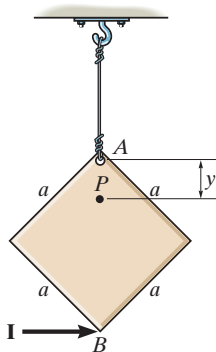
Prob. 19-11

***19-12.** The double pulley consists of two wheels which are attached to one another and turn at the same rate. The pulley has a mass of 15 kg and a radius of gyration of $k_O = 110$ mm. If the block at A has a mass of 40 kg, determine the speed of the block in 3 s after a constant force of 2 kN is applied to the rope wrapped around the inner hub of the pulley. The block is originally at rest.



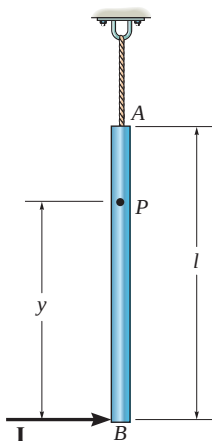
Prob. 19-12

19-13. The square plate has a mass m and is suspended at its corner A by a cord. If it receives a horizontal impulse $\mathbf{I}$ at corner B , determine the location y of the point P about which the plate appears to rotate during the impact.



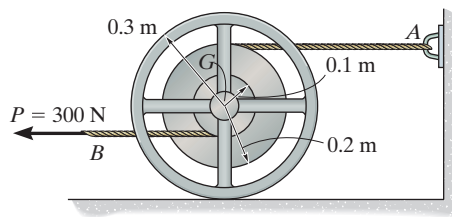
Prob. 19-13

19-14. The slender rod has a mass m and is suspended at its end A by a cord. If the rod receives a horizontal blow giving it an impulse $\mathbf{I}$ at its bottom B , determine the location y of the point P about which the rod appears to rotate during the impact.



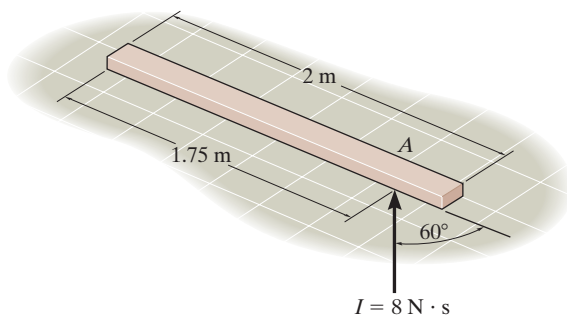
Prob. 19-14

19-15. The 100-kg reel has a radius of gyration about its center of mass G of $k_G = 200$ mm. If the cable B is subjected to a force of $P = 300$ N, determine the time required for the reel to obtain an angular velocity of 20 rad/s. The coefficient of kinetic friction between the reel and the plane is $\mu_k = 0.15$.



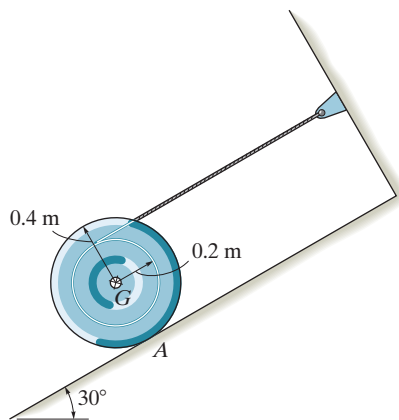
Prob. 19-15

***19-16.** The 4-kg slender rod rests on a smooth floor. If it is kicked so as to receive a horizontal impulse $I = 8 \text{ N} \cdot \text{s}$ at point A as shown, determine its angular velocity and the speed of its mass center.



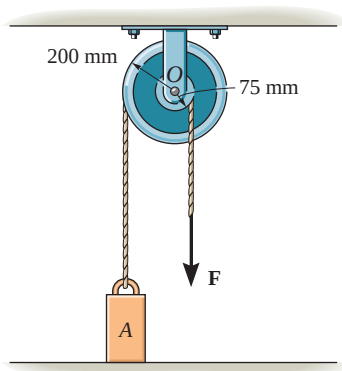
Prob. 19-16

19-17. The 100-kg spool is resting on the inclined surface for which the coefficient of kinetic friction is $\mu_k = 0.1$. Determine the angular velocity of the spool when $t = 4$ s after it is released from rest. The radius of gyration about the mass center is $k_G = 0.25$ m.



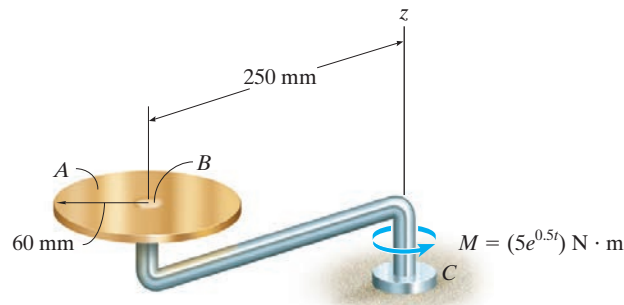
Prob. 19-17

19-18. The double pulley consists of two wheels which are attached to one another and turn at the same rate. The pulley has a mass of 15 kg and a radius of gyration $k_O = 110$ mm. If the block at A has a mass of 40 kg, determine the speed of the block in 3 s after a constant force $F = 2$ kN is applied to the rope wrapped around the inner hub of the pulley. The block is originally at rest. Neglect the mass of the rope.



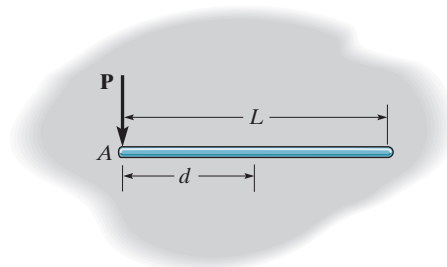
Prob. 19-18

19-19. A 4-kg disk A is mounted on arm BC , which has a negligible mass. If a torque of $M = (5e^{0.5t})$ N·m, where t is in seconds, is applied to the arm at C , determine the angular velocity of BC in 2 s starting from rest. Solve the problem assuming that (a) the disk is set in a smooth bearing at B so that it moves with curvilinear translation, (b) the disk is fixed to the shaft BC , and (c) the disk is given an initial freely spinning angular velocity of $\omega_D = \{-80\mathbf{k}\}$ rad/s prior to application of the torque.



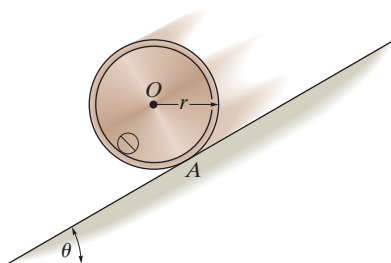
Prob. 19-19

***19-20.** The rod of length L and mass m lies on a smooth horizontal surface and is subjected to a force $\mathbf{P}$ at its end A as shown. Determine the location d of the point about which the rod begins to turn, i.e., the point that has zero velocity.



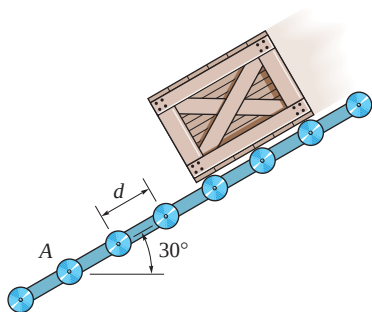
Prob. 19-20

19–21. The drum has a mass of 70 kg, a radius of 300 mm, and radius of gyration $k_O = 125$ mm. If the coefficients of static and kinetic friction at A are $\mu_s = 0.4$ and $\mu_k = 0.3$, respectively, determine the drum's angular velocity 2 s after it is released from rest. Take $\theta = 30^\circ$.



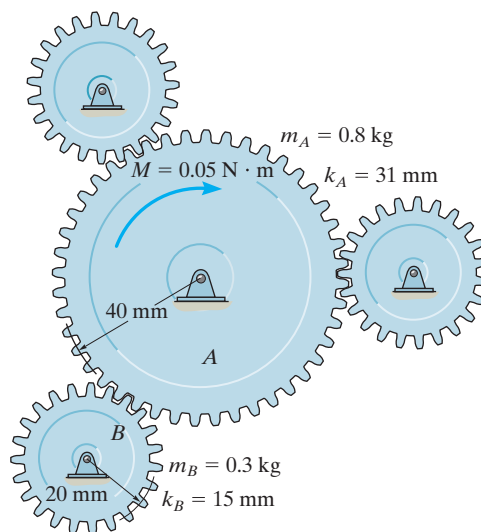
Prob. 19–21

19–22. The crate has a mass m_c . Determine the constant speed v_0 it acquires as it moves down the conveyor. The rollers each have a radius of r , mass m , and are spaced d apart. Note that friction causes each roller to rotate when the crate comes in contact with it.



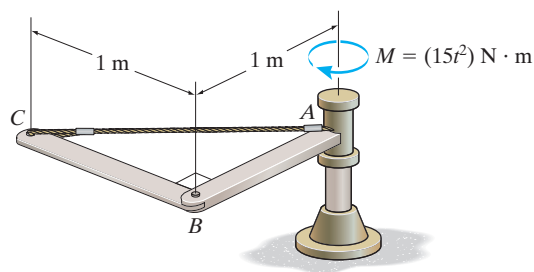
Prob. 19–22

19–23. A motor transmits a torque of $M = 0.05 \text{ N} \cdot \text{m}$ to the center of gear A . Determine the angular velocity of each of the three (equal) smaller gears in 2 s starting from rest. The smaller gears (B) are pinned at their centers, and the masses and centroidal radii of gyration of the gears are given in the figure.



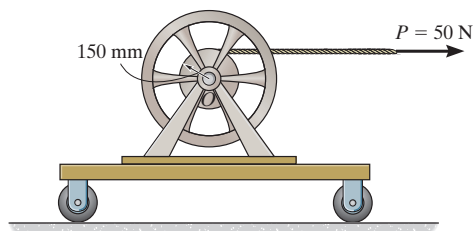
Prob. 19–23

***19–24.** If the shaft is subjected to a torque of $M = (15t^2) \text{ N} \cdot \text{m}$, where t is in seconds, determine the angular velocity of the assembly when $t = 3$ s, starting from rest. Rods AB and BC each have a mass of 9 kg.



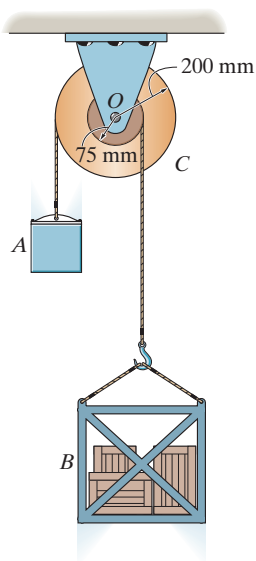
Prob. 19–24

19–25. The 30-kg reel is mounted on the 20-kg cart. If the cable wrapped around the inner hub of the reel is subjected to a force of $P = 50$ N, determine the velocity of the cart and the angular velocity of the reel when $t = 4$ s. The radius of gyration of the reel about its center of mass O is $k_O = 250$ mm. Neglect the size of the small wheels.



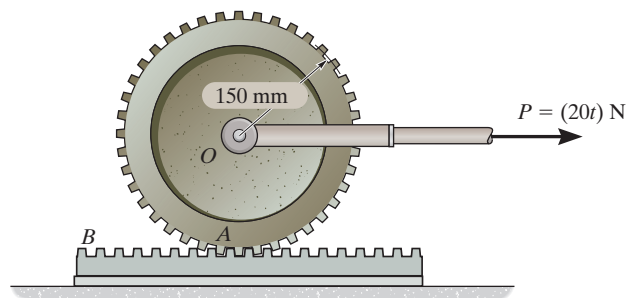
Prob. 19–25

19–26. The double pulley consists of two wheels which are attached to one another and turn at the same rate. The pulley has a mass of 15 kg and a radius of gyration of $k_O = 110$ mm. If the block at A has a mass of 40 kg and the container at B has a mass of 85 kg, including its contents, determine the speed of the container when $t = 3$ s after it is released from rest.



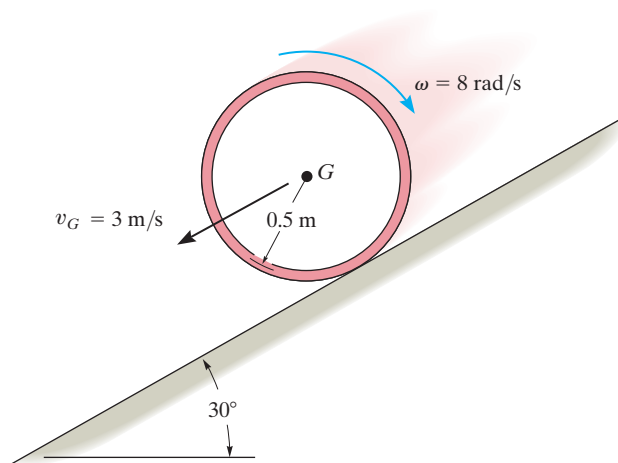
Prob. 19–26

19–27. The 30-kg gear is subjected to a force of $P = (20t)$ N, where t is in seconds. Determine the angular velocity of the gear at $t = 4$ s, starting from rest. Gear rack B is fixed to the horizontal plane, and the gear's radius of gyration about its mass center O is $k_O = 125$ mm.



Prob. 19–27

***19–28.** The hoop (thin ring) has a mass of 5 kg and is released down the inclined plane such that it has a backspin $\omega = 8$ rad/s and its center has a velocity $v_G = 3$ m/s as shown. If the coefficient of kinetic friction between the hoop and the plane is $\mu_k = 0.6$, determine how long the hoop rolls before it stops slipping.



Prob. 19–28

19.3 Conservation of Momentum

Conservation of Linear Momentum. If the sum of all the *linear impulses* acting on a system of connected rigid bodies is *zero* in a specific direction, then the linear momentum of the system is constant, or conserved in this direction, that is,

$$\left(\sum \text{syst. linear momentum} \right)_1 = \left(\sum \text{syst. linear momentum} \right)_2 \quad (19-16)$$

This equation is referred to as the *conservation of linear momentum*.

Without introducing appreciable errors in the calculations, it may be possible to apply Eq. 19-16 in a specified direction for which the linear impulses are small or *nonimpulsive*. Specifically, nonimpulsive forces occur when small forces act over very short periods of time. Typical examples include the force of a slightly deformed spring, the initial contact force with soft ground, and in some cases the weight of the body.

Conservation of Angular Momentum. The angular momentum of a system of connected rigid bodies is conserved about the system's center of mass G , or a fixed point O , when the sum of all the angular impulses about these points is zero or appreciably small (nonimpulsive). The third of Eqs. 19-15 then becomes

$$\left(\sum \text{syst. angular momentum} \right)_{O1} = \left(\sum \text{syst. angular momentum} \right)_{O2} \quad (19-17)$$

This equation is referred to as the *conservation of angular momentum*. In the case of a single rigid body, Eq. 19-17 applied to point G becomes $(I_G \omega)_1 = (I_G \omega)_2$. For example, consider a swimmer who executes a somersault after jumping off a diving board. By tucking his arms and legs in close to his chest, he *decreases* his body's moment of inertia and thus *increases* his angular velocity ($I_G \omega$ must be constant). If he straightens out just before entering the water, his body's moment of inertia is *increased*, and so his angular velocity *decreases*. Since the weight of his body creates a linear impulse during the time of motion, this example also illustrates how the angular momentum of a body can be conserved and yet the linear momentum is *not*. Such cases occur whenever the external forces creating the linear impulse pass through either the center of mass of the body or a fixed axis of rotation.

Procedure for Analysis

The conservation of linear or angular momentum should be applied using the following procedure.

Free-Body Diagram.

- Establish the x, y inertial frame of reference and draw the free-body diagram for the body or system of bodies during the time of impact. From this diagram classify each of the applied forces as being either “impulsive” or “nonimpulsive.”
- By inspection of the free-body diagram, the *conservation of linear momentum* applies in a given direction when *no* external impulsive forces act on the body or system in that direction; whereas the *conservation of angular momentum* applies about a fixed point O or at the mass center G of a body or system of bodies when all the external impulsive forces acting on the body or system create zero moment (or zero angular impulse) about O or G .
- As an alternative procedure, draw the impulse and momentum diagrams for the body or system of bodies. These diagrams are particularly helpful in order to visualize the “moment” terms used in the conservation of angular momentum equation, when it has been decided that angular momenta are to be computed about a point other than the body’s mass center G .

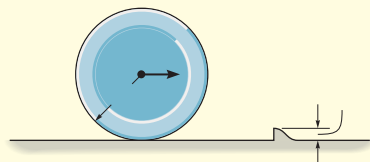
Conservation of Momentum.

- Apply the conservation of linear or angular momentum in the appropriate directions.

Kinematics.

- If the motion appears to be complicated, kinematic (velocity) diagrams may be helpful in obtaining the necessary kinematic relations.

EXAMPLE 19.6



The 10-kg wheel shown in Fig. 19-9a has a moment of inertia $I_G = 0.156 \text{ kg} \cdot \text{m}^2$. Assuming that the wheel does not slip or rebound, determine the minimum velocity v_G it must have to just roll over the obstruction at A.

SOLUTION

Impulse and Momentum Diagrams. Since no slipping or rebounding occurs, the wheel essentially *pivots* about point A during contact. This condition is shown in Fig. 19-9b, which indicates, respectively, the momentum of the wheel *just before impact*, the impulses given to the wheel *during impact*, and the momentum of the wheel *just after impact*. Only two impulses (forces) act on the wheel. By comparison, the force at A is much greater than that of the weight, and since the time of impact is very short, the weight can be considered nonimpulsive. The impulsive force $\mathbf{F}$ at A has both an unknown magnitude and an unknown direction θ . To eliminate this force from the analysis, note that angular momentum about A is essentially *conserved* since $(98.1 \Delta t)d \approx 0$.

Conservation of Angular Momentum. With reference to Fig. 19-9b,

($\zeta +$)

$$(H_A)_1 = (H_A)_2$$

$$r'm(v_G)_1 + I_G\omega_1 = rm(v_G)_2 + I_G\omega_2$$

$$(0.2 \text{ m} - 0.03 \text{ m})(10 \text{ kg})(v_G)_1 + (0.156 \text{ kg} \cdot \text{m}^2)(\omega_1) =$$

$$(0.2 \text{ m})(10 \text{ kg})(v_G)_2 + (0.156 \text{ kg} \cdot \text{m}^2)(\omega_2)$$

Kinematics. Since no slipping occurs, in general $\omega = v_G/r = v_G/0.2 \text{ m} = 5v_G$. Substituting this into the above equation and simplifying yields

$$(v_G)_2 = 0.8921(v_G)_1 \quad (1)$$

Conservation of Energy.* In order to roll over the obstruction, the wheel must pass position 3 shown in Fig. 19-9c. Hence, if $(v_G)_2$ [or $(v_G)_1$] is to be a minimum, it is necessary that the kinetic energy of the wheel at position 2 be equal to the potential energy at position 3. Placing the datum through the center of gravity, as shown in the figure, and applying the conservation of energy equation, we have

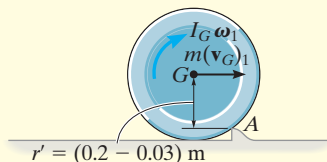
$$\{T_2\} + \{V_2\} = \{T_3\} + \{V_3\}$$

$$\left\{ \frac{1}{2}(10 \text{ kg})(v_G)_2^2 + \frac{1}{2}(0.156 \text{ kg} \cdot \text{m}^2)\omega_2^2 \right\} + \{0\} = \{0\} + \{(98.1 \text{ N})(0.03 \text{ m})\}$$

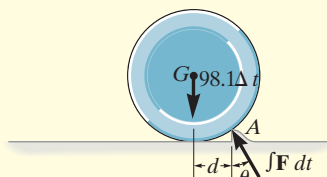
Substituting $\omega_2 = 5(v_G)_2$ and Eq. 1 into this equation, and solving,

$$(v_G)_1 = 0.729 \text{ m/s} \rightarrow \text{Ans.}$$

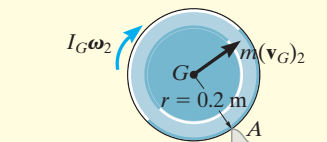
*This principle *does not apply during impact*, since energy is *lost* during the collision. However, just after impact, as in Fig. 19-9c, it can be used.



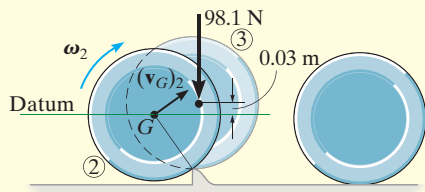
+



|



(b)



(c)

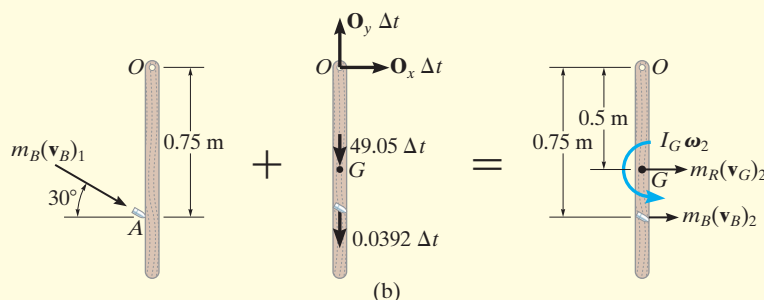
Fig. 19-9

EXAMPLE 19.7

The 5-kg slender rod shown in Fig. 19–10a is pinned at O and is initially at rest. If a 4-g bullet is fired into the rod with a velocity of 400 m/s, as shown in the figure, determine the angular velocity of the rod just after the bullet becomes embedded in it.

SOLUTION

Impulse and Momentum Diagrams. The impulse which the bullet exerts on the rod can be eliminated from the analysis, and the angular velocity of the rod just after impact can be determined by considering the bullet and rod as a single system. To clarify the principles involved, the impulse and momentum diagrams are shown in Fig. 19–10b. The momentum diagrams are drawn *just before and just after impact*. During impact, the bullet and rod exert equal but *opposite internal impulses* at A . As shown on the impulse diagram, the impulses that are external to the system are due to the reactions at O and the weights of the bullet and rod. Since the time of impact, Δt , is very short, the rod moves only a slight amount, and so the “moments” of the weight impulses about point O are essentially zero. Therefore angular momentum is conserved about this point.



Conservation of Angular Momentum. From Fig. 19–10b, we have

$$\begin{aligned}
 (\zeta +) \quad \Sigma(H_O)_1 &= \Sigma(H_O)_2 \\
 m_B(v_B)_1 \cos 30^\circ(0.75 \text{ m}) &= m_B(v_B)_2(0.75 \text{ m}) + m_R(v_G)_2(0.5 \text{ m}) + I_G\omega_2 \\
 (0.004 \text{ kg})(400 \cos 30^\circ \text{ m/s})(0.75 \text{ m}) &= \\
 (0.004 \text{ kg})(v_B)_2(0.75 \text{ m}) + (5 \text{ kg})(v_G)_2(0.5 \text{ m}) + \left[\frac{1}{12}(5 \text{ kg})(1 \text{ m})^2\right]\omega_2 \quad (1)
 \end{aligned}$$

or

$$1.039 = 0.003(v_B)_2 + 2.50(v_G)_2 + 0.4167\omega_2$$

Kinematics. Since the rod is pinned at O , from Fig. 19–9c we have

$$(v_G)_2 = (0.5 \text{ m})\omega_2 \quad (v_B)_2 = (0.75 \text{ m})\omega_2$$

Substituting into Eq. 1 and solving yields

$$\omega_2 = 0.623 \text{ rad/s} \quad \curvearrowright$$

Ans.

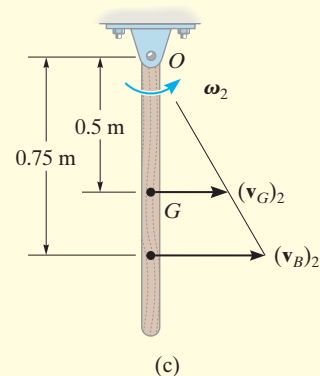
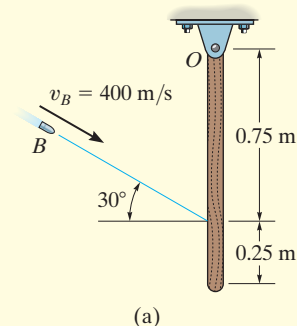


Fig. 19–10

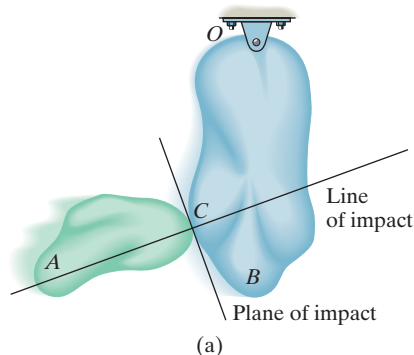


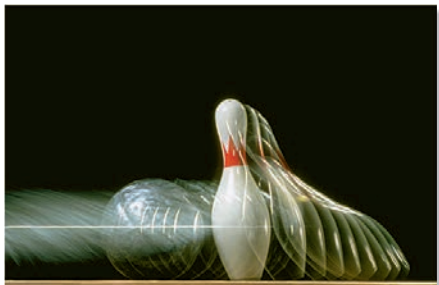
Fig. 19–11

*19.4 Eccentric Impact

The concepts involving central and oblique impact of particles were presented in Sec. 15.4. We will now expand this treatment and discuss the eccentric impact of two bodies. *Eccentric impact* occurs when the line connecting the *mass centers* of the two bodies *does not* coincide with the line of impact.* This type of impact often occurs when one or both of the bodies are constrained to rotate about a fixed axis. Consider, for example, the collision at C between the two bodies A and B, shown in Fig. 19–11a. It is assumed that just before collision B is rotating counterclockwise with an angular velocity $(\omega_B)_1$, and the velocity of the contact point C located on A is $(\mathbf{u}_A)_1$. Kinematic diagrams for both bodies just before collision are shown in Fig. 19–11b. Provided the bodies are smooth, the *impulsive forces* they exert on each other *are directed along the line of impact*. Hence, the component of velocity of point C on body B, which is directed along the line of impact, is $(v_B)_1 = (\omega_B)_1 r$, Fig. 19–11b. Likewise, on body A the component of velocity $(\mathbf{u}_A)_1$ along the line of impact is $(v_A)_1$. In order for a collision to occur, $(v_A)_1 > (v_B)_1$.

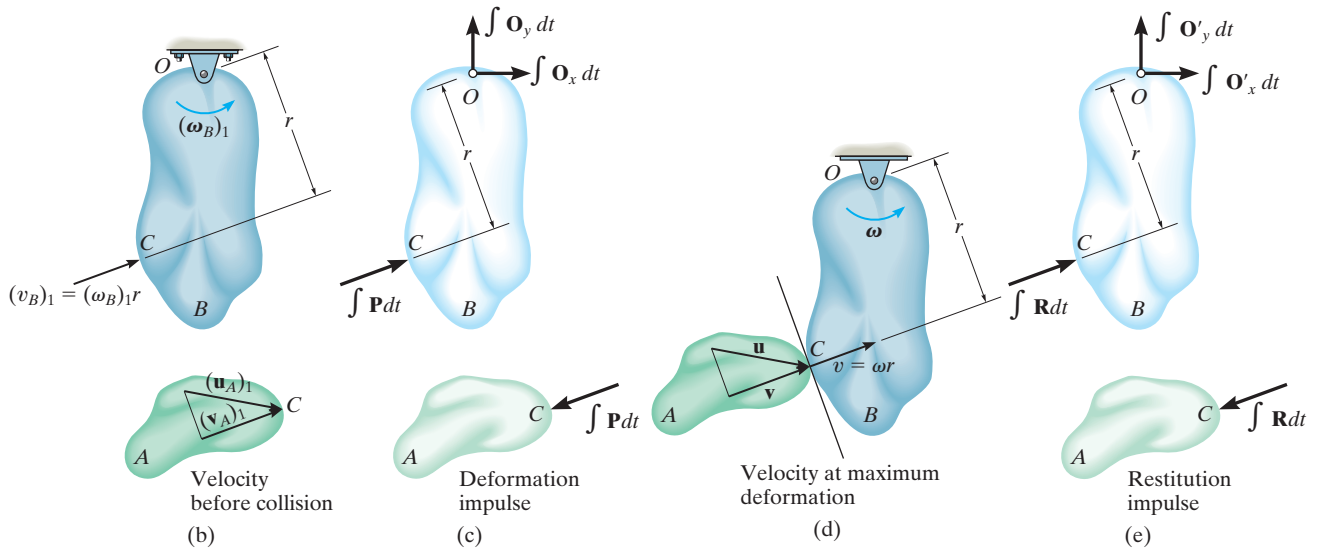
During the impact an equal but opposite impulsive force $\mathbf{P}$ is exerted between the bodies which *deforms* their shapes at the point of contact. The resulting impulse is shown on the impulse diagrams for both bodies, Fig. 19–11c. Note that the impulsive force at point C on the rotating body creates impulsive pin reactions at O. On these diagrams it is assumed that the impact creates forces which are much larger than the nonimpulsive weights of the bodies, which are not shown. When the deformation at point C is a maximum, C on both the bodies moves with a common velocity $\mathbf{v}$ along the line of impact, Fig. 19–11d. A period of *restitution* then occurs in which the bodies tend to regain their original shapes. The restitution phase creates an equal but opposite impulsive force $\mathbf{R}$ acting between the bodies as shown on the impulse diagram, Fig. 19–11e. After restitution the bodies move apart such that point C on body B has a velocity $(\mathbf{v}_B)_2$ and point C on body A has a velocity $(\mathbf{u}_A)_2$, Fig. 19–11f, where $(v_B)_2 > (v_A)_2$.

In general, a problem involving the impact of two bodies requires determining the *two unknowns* $(v_A)_2$ and $(v_B)_2$, assuming $(v_A)_1$ and $(v_B)_1$ are known (or can be determined using kinematics, energy methods, the equations of motion, etc.). To solve such problems, two equations must be written. The *first equation* generally involves application of the *conservation of angular momentum to the two bodies*. In the case of both bodies A and B, we can state that angular momentum is conserved about point O since the impulses at C are internal to the system and the impulses at O create zero moment (or zero angular impulse) about O. The *second equation* can be obtained using the definition of the *coefficient of restitution*, e , which is a ratio of the restitution impulse to the deformation impulse.



Here is an example of eccentric impact occurring between this bowling ball and pin.

*When these lines coincide, central impact occurs and the problem can be analyzed as discussed in Sec. 15.4.



It is important to realize, however, that *this analysis has only a very limited application in engineering, because values of e for this case have been found to be highly sensitive to the material, geometry, and the velocity of each of the colliding bodies.* To establish a useful form of the coefficient of restitution equation we must first apply the principle of angular impulse and momentum about point O to bodies B and A separately. Combining the results, we then obtain the necessary equation. Proceeding in this manner, the principle of impulse and momentum applied to body B from the time just before the collision to the instant of maximum deformation, Figs. 19-11b, 19-11c, and 19-11d, becomes

$$(\zeta +) \quad I_O(\omega_B)_1 + r \int P \, dt = I_O\omega \quad (19-18)$$

Here I_O is the moment of inertia of body B about point O . Similarly, applying the principle of angular impulse and momentum from the instant of maximum deformation to the time just after the impact, Figs. 19-11d, 19-11e, and 19-11f, yields

$$(\zeta +) \quad I_O\omega + r \int R \, dt = I_O(\omega_B)_2 \quad (19-19)$$

Solving Eqs. 19-18 and 19-19 for $\int P \, dt$ and $\int R \, dt$, respectively, and formulating e , we have

$$e = \frac{\int R \, dt}{\int P \, dt} = \frac{r(\omega_B)_2 - r\omega}{r\omega - r(\omega_B)_1} = \frac{(v_B)_2 - v}{v - (v_B)_1}$$

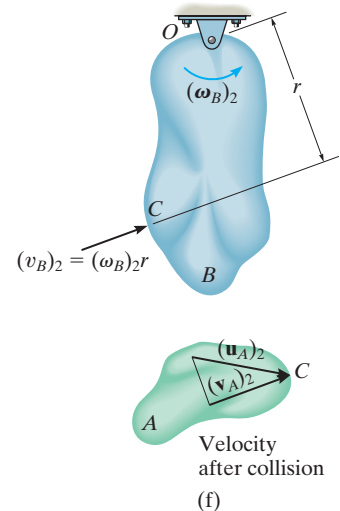


Fig. 19-11 (cont.)

In the same manner, we can write an equation which relates the magnitudes of velocity $(v_A)_1$ and $(v_A)_2$ of body A . The result is

$$e = \frac{v - (v_A)_2}{(v_A)_1 - v}$$

Combining the above two equations by eliminating the common velocity v yields the desired result, i.e.,

$$(+\nearrow) \quad e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1} \quad (19-20)$$

This equation is identical to Eq. 15-11, which was derived for the central impact between two particles. It states that the coefficient of restitution is equal to the ratio of the relative velocity of *separation* of the points of contact (C) *just after impact* to the relative velocity at which the points *approach* one another *just before impact*. In deriving this equation, we assumed that the points of contact for both bodies move up and to the right *both* before and after impact. If motion of any one of the contacting points occurs down and to the left, the velocity of this point should be considered a negative quantity in Eq. 19-20.



During impact the columns of many highway signs are intended to break out of their supports and easily collapse at their joints. This is shown by the slotted connections at their base and the breaks at the column's midsection.

EXAMPLE 19.8

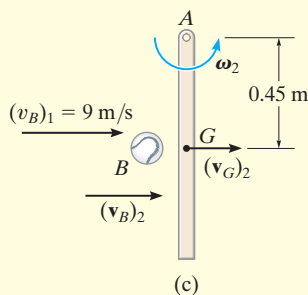
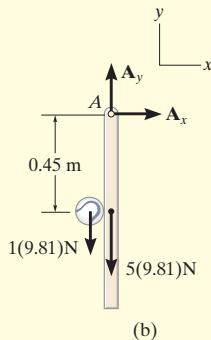
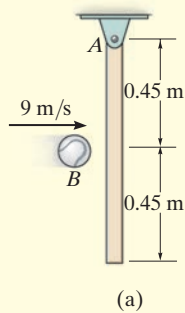


Fig. 19-12

The 5-kg slender rod is suspended from the pin at A , Fig. 19-12a. If a 1-kg ball B is thrown at the rod and strikes its center with a velocity of 9 m/s, determine the angular velocity of the rod just after impact. The coefficient of restitution is $e = 0.4$.

SOLUTION

Conservation of Angular Momentum. Consider the ball and rod as a system, Fig. 19-12b. Angular momentum is conserved about point A since the impulsive force between the rod and ball is *internal*. Also, the *weights* of the ball and rod are *nonimpulsive*. Noting the directions of the velocities of the ball and rod just after impact as shown on the kinematic diagram, Fig. 19-12c, we require

$$(\zeta +)$$

$$(H_A)_1 = (H_A)_2$$

$$m_B(v_B)_1(0.45 \text{ m}) = m_B(v_B)_2(0.45 \text{ m}) + m_R(v_G)_2(0.45 \text{ m}) + I_G\omega_2$$

$$(1 \text{ kg})(9 \text{ m/s})(0.45 \text{ m}) = (1 \text{ kg})(v_B)_2(0.45 \text{ m}) +$$

$$(5 \text{ kg})(v_G)_2(0.45 \text{ m}) + \left[\frac{1}{12}(5 \text{ kg})(0.9 \text{ m})^2 \right]\omega_2$$

Since $(v_G)_2 = 0.45\omega_2$ then

$$4.05 = 0.45(v_B)_2 + 1.35\omega_2 \quad (1)$$

Coefficient of Restitution. With reference to Fig. 19-12c, we have

$$(\pm)$$

$$e = \frac{(v_G)_2 - (v_B)_2}{(v_B)_1 - (v_G)_1} \quad 0.4 = \frac{(0.45 \text{ m})\omega_2 - (v_B)_2}{9 \text{ m/s} - 0}$$

$$3.6 = 0.45\omega_2 - (v_B)_2 \quad (2)$$

Solving Eqs. 1 and 2, yields

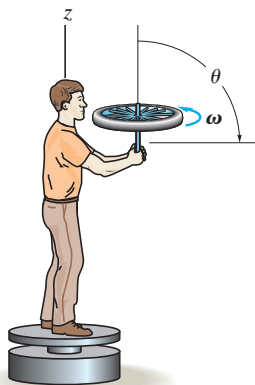
$$(v_B)_2 = -1.957 \text{ m/s} = 1.957 \text{ m/s} \leftarrow$$

$$\omega_2 = 3.65 \text{ rad/s} \curvearrowright$$

Ans.

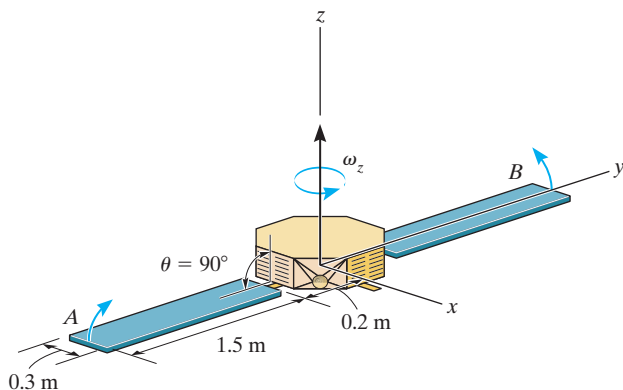
PROBLEMS

19–29. A man has a moment of inertia I_z about the z axis. He is originally at rest and standing on a small platform which can turn freely. If he is handed a wheel which is rotating at ω and has a moment of inertia I about its spinning axis, determine his angular velocity if (a) he holds the wheel upright as shown, (b) turns the wheel out, $\theta = 90^\circ$, and (c) turns the wheel downward, $\theta = 180^\circ$. Neglect the effect of holding the wheel a distance d away from the z axis.



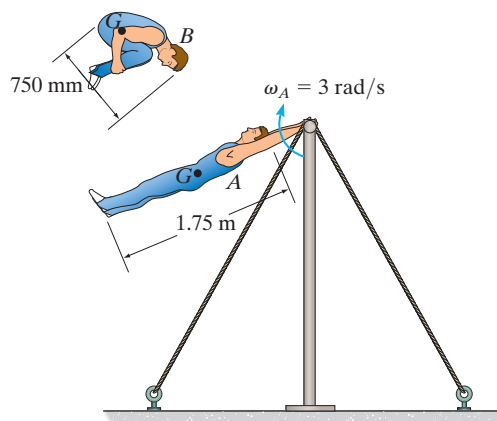
Prob. 19–29

19–30. The satellite has a mass of 200 kg and a radius of gyration about z axis of $k_z = 0.1$ m, excluding the two solar panels A and B . Each solar panel has a mass of 15 kg and can be approximated as a thin plate. If the satellite is originally spinning about the z axis at a constant rate $\omega_z = 0.5$ rad/s when $\theta = 90^\circ$, determine the rate of spin if both panels are raised and reach the upward position, $\theta = 0^\circ$, at the same instant.



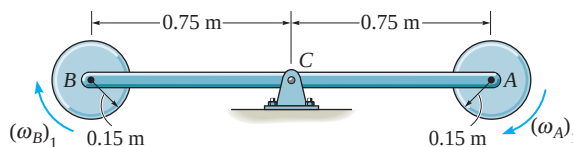
Prob. 19–30

19–31. The 75-kg gymnast lets go of the horizontal bar in a fully stretched position A , rotating with an angular velocity of $\omega_A = 3$ rad/s. Estimate his angular velocity when he assumes a tucked position B . Assume the gymnast at positions A and B as a uniform slender rod and a uniform circular disk, respectively.



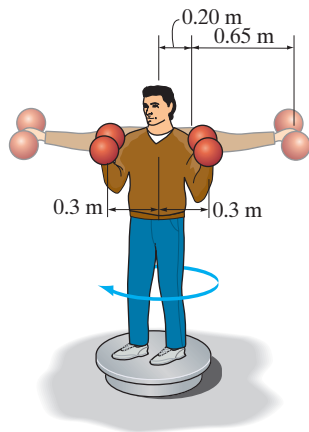
Prob. 19–31

***19–32.** The 2-kg rod ACB supports the two 4-kg disks at its ends. If both disks are given a clockwise angular velocity $(\omega_A)_1 = (\omega_B)_1 = 5$ rad/s while the rod is held stationary and then released, determine the angular velocity of the rod after both disks have stopped spinning relative to the rod due to frictional resistance at the pins A and B . Motion is in the horizontal plane. Neglect friction at pin C .



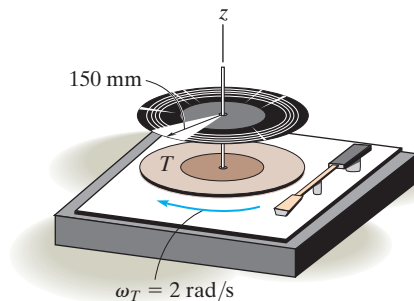
Prob. 19–32

19–33. The 80-kg man is holding two dumbbells while standing on a turntable of negligible mass, which turns freely about a vertical axis. When his arms are fully extended, the turntable is rotating with an angular velocity of 0.5 rev/s. Determine the angular velocity of the man when he retracts his arms to the position shown. When his arms are fully extended, approximate each arm as a uniform 6-kg rod having a length of 650 mm, and his body as a 68-kg solid cylinder of 400-mm diameter. With his arms in the retracted position, assume the man is an 80-kg solid cylinder of 450-mm diameter. Each dumbbell consists of two 5-kg spheres of negligible size.



Prob. 19–33

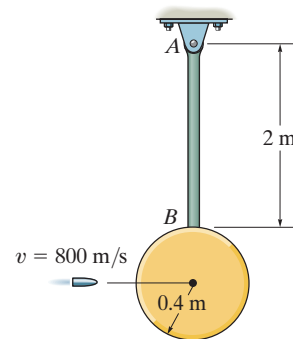
19–34. The turntable T of a record player has a mass of 0.75 kg and a radius of gyration $k_z = 125$ mm. It is *turning freely* at $\omega_T = 2$ rad/s when a 50-g record (thin disk) falls on it. Determine the final angular velocity of the turntable just after the record stops slipping on the turntable.



Prob. 19–34

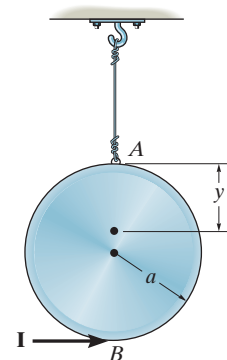
19–35. The 10-g bullet having a velocity of 800 m/s is fired into the edge of the 5-kg disk as shown. Determine the angular velocity of the disk just after the bullet becomes embedded into its edge. Also, calculate the angle θ the disk will swing when it stops. The disk is originally at rest. Neglect the mass of the rod AB .

***19–36.** The 10-g bullet having a velocity of 800 m/s is fired into the edge of the 5-kg disk as shown. Determine the angular velocity of the disk just after the bullet becomes embedded into its edge. Also, calculate the angle θ the disk will swing when it stops. The disk is originally at rest. The rod AB has a mass of 3 kg.



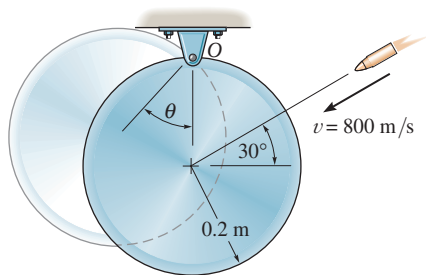
Probs. 19–35/36

19–37. The circular disk has a mass m and is suspended at A by the wire. If it receives a horizontal impulse $\mathbf{I}$ at its edge B , determine the location y of the point P about which the disk appears to rotate during the impact.



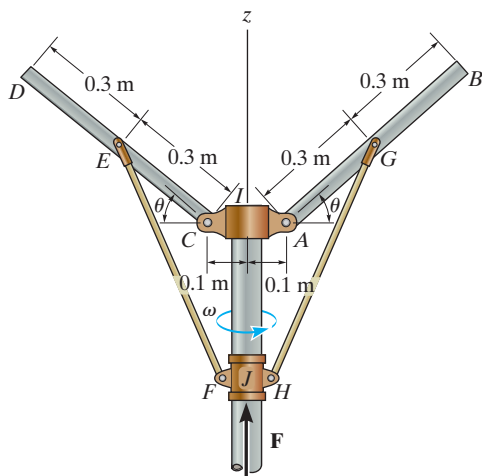
Prob. 19–37

19-38. A 7-g bullet having a velocity of 800 m/s is fired into the edge of the 5-kg disk as shown. Determine the angular velocity of the disk just after the bullet becomes embedded in it. Also, calculate how far θ the disk will swing until it stops. The disk is originally at rest.



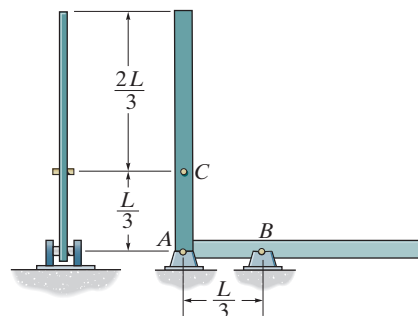
Prob. 19-38

19-39. The vertical shaft is rotating with an angular velocity of 3 rad/s when $\theta = 0^\circ$. If a force $\mathbf{F}$ is applied to the collar so that $\theta = 90^\circ$, determine the angular velocity of the shaft. Also, find the work done by force $\mathbf{F}$. Neglect the mass of rods GH and EF and the collars I and J . The rods AB and CD each have a mass of 10 kg.



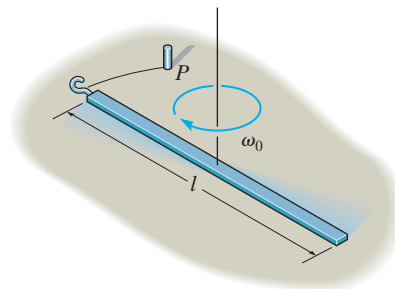
Prob. 19-39

***19-40.** The slender bar of mass m pivots at support A when it is released from rest in the vertical position. When it falls and rotates 90° , pin C will strike support B , and pin at A will leave its support. Determine the angular velocity of the bar immediately after the impact. Assume the pin at B will not rebound.



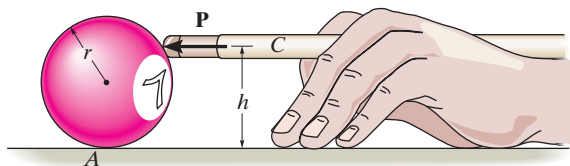
Prob. 19-40

19-41. A thin rod of mass m has an angular velocity ω_0 while rotating on a smooth surface. Determine its new angular velocity just after its end strikes and hooks onto the peg and the rod starts to rotate about P without rebounding. Solve the problem (a) using the parameters given, (b) setting $m = 2$ kg, $\omega_0 = 4$ rad/s, $l = 1.5$ m.



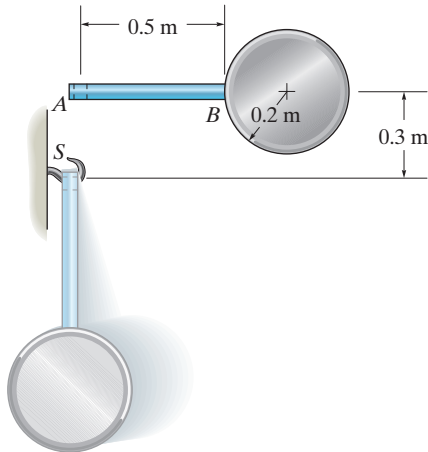
Prob. 19-41

19-42. Determine the height h at which a billiard ball of mass m must be struck so that no frictional force develops between it and the table at A . Assume that the cue C only exerts a horizontal force $\mathbf{P}$ on the ball.



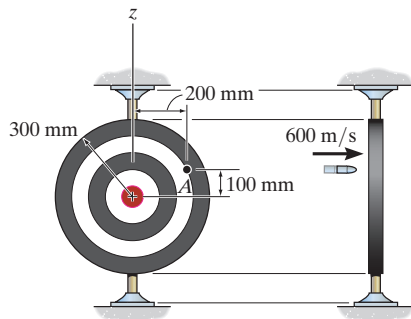
Prob. 19-42

19–43. The pendulum consists of a slender 2-kg rod AB and 5-kg disk. It is released from rest without rotating. When it falls 0.3 m, the end A strikes the hook S , which provides a permanent connection. Determine the angular velocity of the pendulum after it has rotated 90° . Treat the pendulum's weight during impact as a nonimpulsive force.



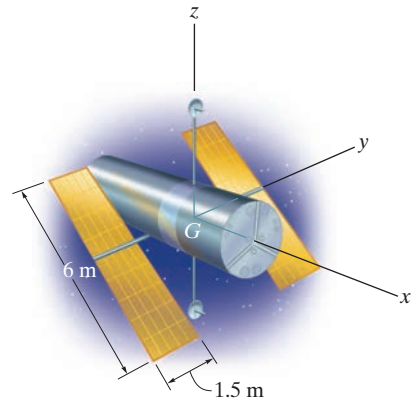
Prob. 19–43

***19–44.** The target is a thin 5-kg circular disk that can rotate freely about the z axis. A 25-g bullet, traveling at 600 m/s, strikes the target at A and becomes embedded in it. Determine the angular velocity of the target after the impact. Initially, it is at rest.



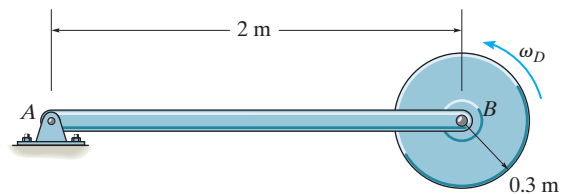
Prob. 19–44

19–45. The Hubble Space Telescope is powered by two solar panels as shown. The body of the telescope has a mass 11 Mg and radii of gyration $k_x = 1.64$ m and $k_y = 3.85$ m, whereas the solar panels can be considered as thin plates, each having a mass of 54 kg. Due to an internal drive, the panels are given an angular velocity of $\{0.6\mathbf{j}\}$ rad/s, measured relative to the telescope. Determine the angular velocity of the telescope due to the rotation of the panels. Prior to rotating the panels, the telescope was originally traveling at $\mathbf{v}_G = \{-400\mathbf{i} + 250\mathbf{j} + 175\mathbf{k}\}$ m/s. Neglect its orbital rotation.



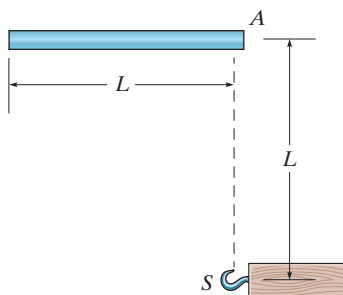
Prob. 19–45

19–46. The 12-kg rod AB is pinned to the 40-kg disk. If the disk is given an angular velocity $\omega_D = 100$ rad/s while the rod is held stationary, and the assembly is then released, determine the angular velocity of the rod after the disk has stopped spinning relative to the rod due to frictional resistance at the bearing B . Motion is in the *horizontal plane*. Neglect friction at the pin A .



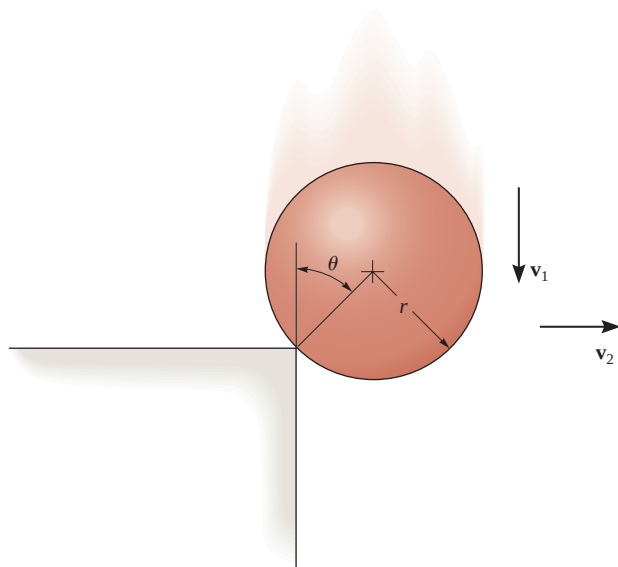
Prob. 19–46

19-47. The rod of mass m and length L is released from rest without rotating. When it falls a distance L , the end A strikes the hook S , which provides a permanent connection. Determine the angular velocity ω of the rod after it has rotated 90° . Treat the rod's weight during impact as a nonimpulsive force.



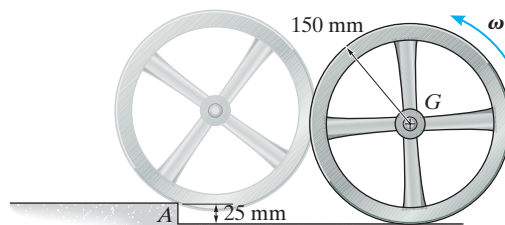
Prob. 19-47

***19-48.** The solid ball of mass m is dropped with a velocity $\mathbf{v}_1$ onto the edge of the rough step. If it rebounds horizontally off the step with a velocity $\mathbf{v}_2$, determine the angle θ at which contact occurs. Assume no slipping when the ball strikes the step. The coefficient of restitution is e .



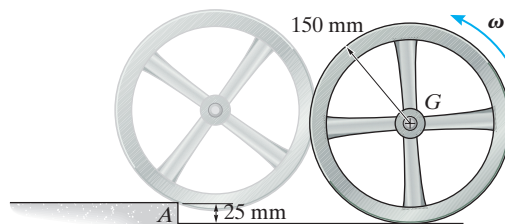
Prob. 19-48

19-49. The wheel has a mass of 50 kg and a radius of gyration of 125 mm about its center of mass G . Determine the minimum value of the angular velocity ω_1 of the wheel, so that it strikes the step at A without rebounding and then rolls over it without slipping.



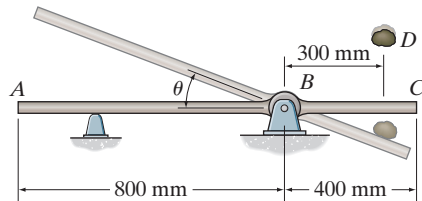
Prob. 19-49

19-50. The wheel has a mass of 50 kg and a radius of gyration of 125 mm about its center of mass G . If it rolls without slipping with an angular velocity of $\omega_1 = 5 \text{ rad/s}$ before it strikes the step at A , determine its angular velocity after it rolls over the step. The wheel does not lose contact with the step when it strikes it.



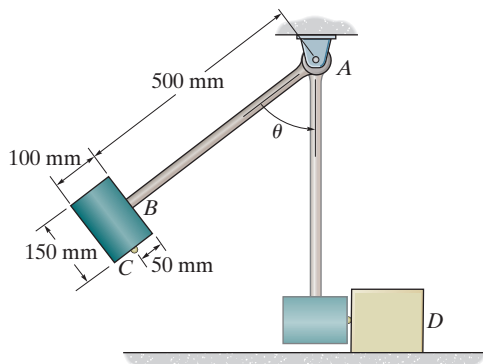
Prob. 19-50

19-51. A 2-kg mass of putty D strikes the uniform 10-kg plank ABC with a velocity of 10 m/s. If the putty remains attached to the plank, determine the maximum angle θ of swing before the plank momentarily stops. Neglect the size of the putty.



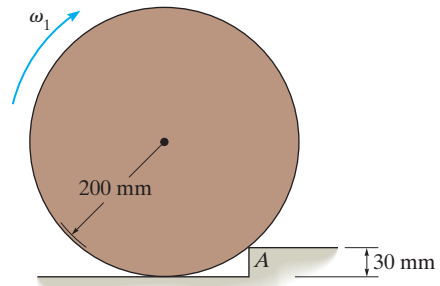
Prob. 19-51

***19-52.** The hammer consists of a 10-kg solid cylinder C and 6-kg uniform slender rod AB . If the hammer is released from rest when $\theta = 90^\circ$ and strikes the 30-kg block D when $\theta = 0^\circ$, determine the velocity of block D and the angular velocity of the hammer immediately after the impact. The coefficient of restitution between the hammer and the block is $e = 0.6$.



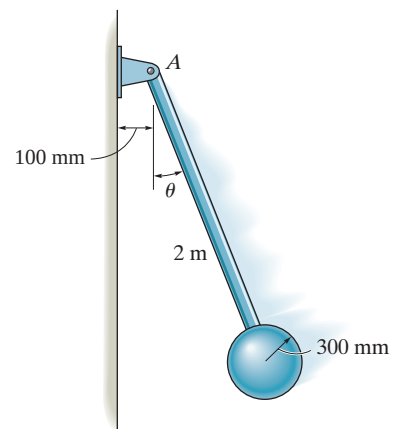
Prob. 19-52

19-53. The 20-kg disk strikes the step without rebounding. Determine the largest angular velocity ω_1 the disk can have and not lose contact with the step, A .



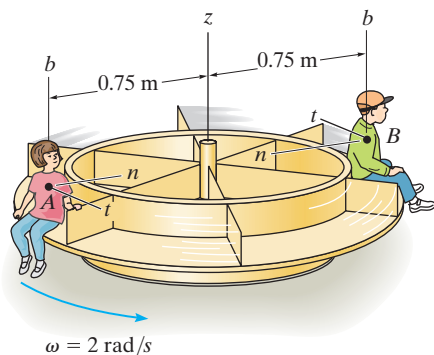
Prob. 19-53

19-54. The pendulum consists of a 15-kg solid ball and 6-kg rod. If it is released from rest when $\theta_1 = 90^\circ$, determine the angle θ_2 after the ball strikes the wall, rebounds, and the pendulum swings up to the point of momentary rest. Take $e = 0.6$.



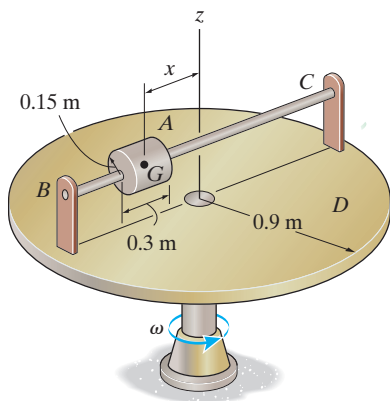
Prob. 19-54

19-55. Two children *A* and *B*, each having a mass of 30 kg, sit at the edge of the merry-go-round which is rotating at $\omega = 2 \text{ rad/s}$. Excluding the children, the merry-go-round has a mass of 180 kg and a radius of gyration $k_z = 0.6 \text{ m}$. Determine the angular velocity of the merry-go-round if *A* jumps off horizontally in the $-n$ direction with a speed of 2 m/s, measured with respect to the merry-go-round. What is the merry-go-round's angular velocity if *B* then jumps off horizontally in the $+t$ direction with a speed of 2 m/s, measured with respect to the merry-go-round? Neglect friction and the size of each child.



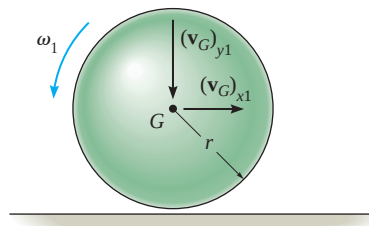
Prob. 19-55

***19-56.** The 20-kg cylinder *A* is free to slide along rod *BC*. When the cylinder is at $x = 0$, the 50-kg circular disk *D* is rotating with an angular velocity of 5 rad/s. If the cylinder is given a slight push, determine the angular velocity of the disk when the cylinder strikes *B* at $x = 600 \text{ mm}$. Neglect the mass of the brackets and the smooth rod.



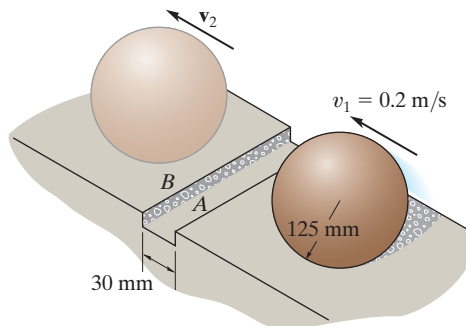
Prob. 19-56

19-57. A solid ball with a mass m is thrown on the ground such that at the instant of contact it has an angular velocity ω_1 and velocity components $(v_G)_{x1}$ and $(v_G)_{y1}$ as shown. If the ground is rough so no slipping occurs, determine the components of the velocity of its mass center just after impact. The coefficient of restitution is e .



Prob. 19-57

19-58. A ball having a mass of 8 kg and initial speed of $v_1 = 0.2 \text{ m/s}$ rolls over a 30-mm-long depression. Assuming that the ball rolls off the edges of contact first *A*, then *B*, without slipping, determine its final velocity v_2 when it reaches the other side.



Prob. 19-58

CONCEPTUAL PROBLEMS

C19-1. The soil compactor moves forward at constant velocity by supplying power to the rear wheels. Use appropriate numerical data for the wheel, roller, and body and calculate the angular momentum of this system about point A at the ground, point B on the rear axle, and point G , the center of gravity for the system.



Prob. C19-1

C19-2. The swing bridge opens and closes by turning 90° using a motor located under the center of the deck at A that applies a torque $\mathbf{M}$ to the bridge. If the bridge was supported at its end B , would the same torque open the bridge at the same time, or would it open slower or faster? Explain your answer using numerical values and an impulse and momentum analysis. Also, what are the benefits of making the bridge have the variable depth as shown?



Prob. C19-2

C19-3. Why is it necessary to have the tail blade B on the helicopter that spins perpendicular to the spin of the main blade A ? Explain your answer using numerical values and an impulse and momentum analysis.



Prob. C19-3

C19-4. The amusement park ride consists of two gondolas A and B , and counterweights C and D that swing in opposite directions. Using realistic dimensions and mass, calculate the angular momentum of this system for any angular position of the gondolas. Explain through analysis why it is a good idea to design this system to have counterweights with each gondola.



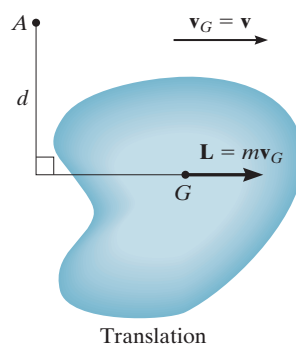
Prob. C19-4

CHAPTER REVIEW

Linear and Angular Momentum

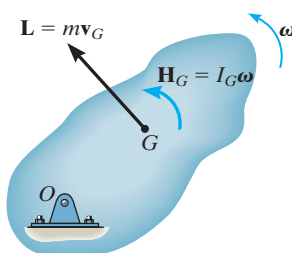
The linear and angular momentum of a rigid body can be referenced to its mass center G .

If the angular momentum is to be determined about an axis other than the one passing through the mass center, then the angular momentum is determined by summing vector $\mathbf{H}_G$ and the moment of vector $\mathbf{L}$ about this axis.



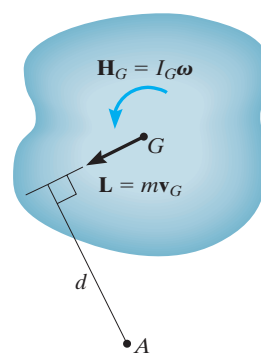
Translation

$$\begin{aligned} L &= mv_G \\ H_G &= 0 \\ H_A &= (mv_G)d \end{aligned}$$



Rotation about a fixed axis

$$\begin{aligned} L &= mv_G \\ H_G &= I_G \omega \\ H_O &= I_O \omega \end{aligned}$$



General plane motion

$$\begin{aligned} L &= mv_G \\ H_G &= I_G \omega \\ H_A &= I_G \omega + (mv_G)d \end{aligned}$$

Principle of Impulse and Momentum

The principles of linear and angular impulse and momentum are used to solve problems that involve force, velocity, and time. Before applying these equations, it is important to establish the x , y , z inertial coordinate system. The free-body diagram for the body should also be drawn in order to account for all of the forces and couple moments that produce impulses on the body.

$$m(v_{Gx})_1 + \sum \int_{t_1}^{t_2} F_x dt = m(v_{Gx})_2$$

$$m(v_{Gy})_1 + \sum \int_{t_1}^{t_2} F_y dt = m(v_{Gy})_2$$

$$I_G \omega_1 + \sum \int_{t_1}^{t_2} M_G dt = I_G \omega_2$$

Conservation of Momentum

Provided the sum of the linear impulses acting on a system of connected rigid bodies is zero in a particular direction, then the linear momentum for the system is conserved in this direction. Conservation of angular momentum occurs if the impulses pass through an axis or are parallel to it. Momentum is also conserved if the external forces are small and thereby create nonimpulsive forces on the system. A free-body diagram should accompany any application in order to classify the forces as impulsive or nonimpulsive and to determine an axis about which the angular momentum may be conserved.

$$\left(\sum_{\text{momentum}}^{\text{syst. linear}} \right)_1 = \left(\sum_{\text{momentum}}^{\text{syst. linear}} \right)_2$$

$$\left(\sum_{\text{momentum}}^{\text{syst. angular}} \right)_{O1} = \left(\sum_{\text{momentum}}^{\text{syst. angular}} \right)_{O2}$$

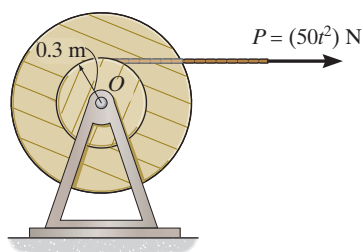
Eccentric Impact

If the line of impact does not coincide with the line connecting the mass centers of two colliding bodies, then eccentric impact will occur. If the motion of the bodies just after the impact is to be determined, then it is necessary to consider a conservation of momentum equation for the system and use the coefficient of restitution equation.

$$e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$$

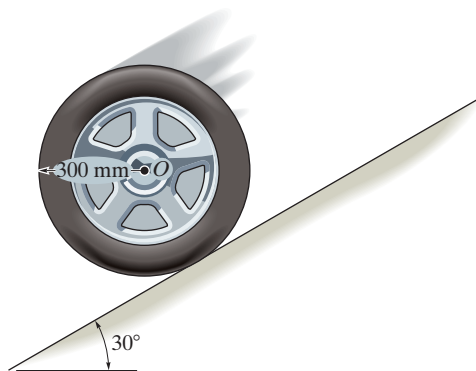
REVIEW PROBLEMS

R19-1. The cable is subjected to a force of $P = (50t^2)$ N, where t is in seconds. Determine the angular velocity of the spool 3 s after $\mathbf{P}$ is applied, starting from rest. The spool has a mass of 75 kg and a radius of gyration of 0.375 m about its center, O .



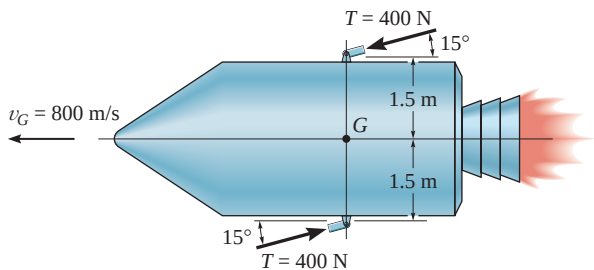
Prob. R19-1

R19-3. The tire has a mass of 9 kg and a radius of gyration $k_O = 225$ mm. If it is released from rest and rolls down the plane without slipping, determine the speed of its center O when $t = 3$ s.



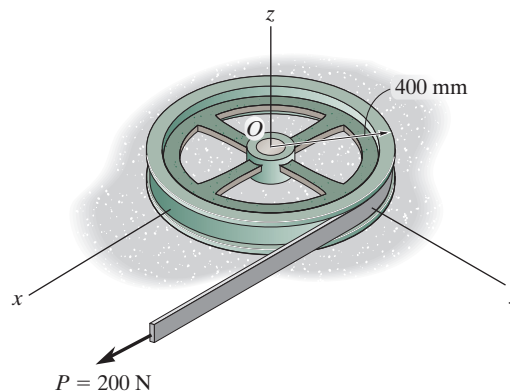
Prob. R19-3

R19-2. The space capsule has a mass of 1200 kg and a moment of inertia $I_G = 900 \text{ kg} \cdot \text{m}^2$ about an axis passing through G and directed perpendicular to the page. If it is traveling forward with a speed $v_G = 800$ m/s and executes a turn by means of two jets, which provide a constant thrust of 400 N for 0.3 s, determine the capsule's angular velocity just after the jets are turned off.



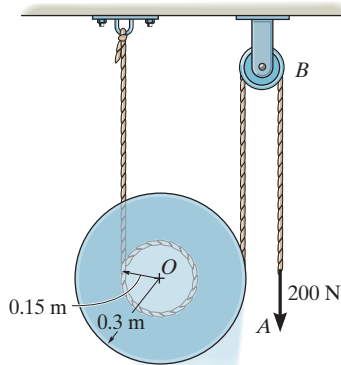
Prob. R19-2

R19-4. The wheel having a mass of 100 kg and a radius of gyration about the z axis of $k_z = 300$ mm, rests on the smooth horizontal plane. If the belt is subjected to a force of $P = 200$ N, determine the angular velocity of the wheel and the speed of its center of mass O , three seconds after the force is applied.



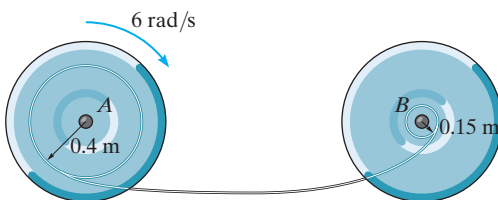
Prob. R19-4

R19-5. The spool has a mass of 15 kg and a radius of gyration $k_O = 0.2$ m. If a force of 200 N is applied to the cord at A , determine the angular velocity of the spool in $t = 3$ s starting from rest. Neglect the mass of the pulley and cord.



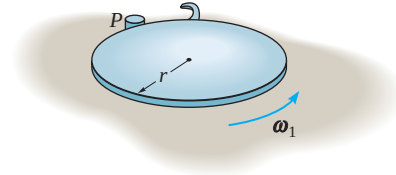
Prob. R19-5

R19-6. Spool B is at rest and spool A is rotating at 6 rad/s when the slack in the cord connecting them is taken up. If the cord does not stretch, determine the angular velocity of each spool immediately after the cord is jerked tight. The spools A and B have masses and radii of gyration $m_A = 15$ kg, $k_A = 0.25$ m, $m_B = 7.5$ kg, $k_B = 0.18$ m, respectively.



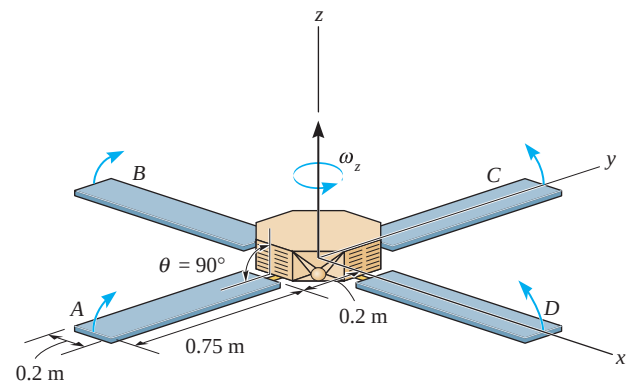
Prob. R19-6

R19-7. A thin disk of mass m has an angular velocity ω_1 while rotating on a smooth surface. Determine its new angular velocity just after the hook at its edge strikes the peg P and the disk starts to rotate about P without rebounding.



Prob. R19-7

R19-8. The space satellite has a mass of 125 kg and a moment of inertia $I_z = 0.940$ kg $\cdot$ m², excluding the four solar panels A , B , C , and D . Each solar panel has a mass of 20 kg and can be approximated as a thin plate. If the satellite is originally spinning about the z axis at a constant rate $\omega_z = 0.5$ rad/s when $\theta = 90^\circ$, determine the rate of spin if all the panels are raised and reach the upward position, $\theta = 0^\circ$, at the same instant.



Prob. R19-8

Chapter 20



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Design of industrial robots requires knowing the kinematics of their three-dimensional motions.

Three-Dimensional Kinematics of a Rigid Body

CHAPTER OBJECTIVES

- To analyze the kinematics of a body subjected to rotation about a fixed point and to general plane motion.
- To provide a relative-motion analysis of a rigid body using translating and rotating axes.



Video Solutions are available for selected questions in this chapter.

20.1 Rotation about a Fixed Point

When a rigid body rotates about a fixed point, the distance r from the point to a particle located on the body is the *same* for *any position* of the body. Thus, the path of motion for the particle lies on the *surface of a sphere* having a radius r and centered at the fixed point. Since motion along this path occurs only from a series of rotations made during a finite time interval, we will first develop a familiarity with some of the properties of rotational displacements.



The boom can rotate up and down, and because it is hinged at a point on the vertical axis about which it turns, it is subjected to rotation about a fixed point.

Euler's Theorem. Euler's theorem states that two “component” rotations about different axes passing through a point are equivalent to a single resultant rotation about an axis passing through the point. If more than two rotations are applied, they can be combined into pairs, and each pair can be further reduced and combined into one rotation.

Finite Rotations. If component rotations used in Euler's theorem are *finite*, it is important that the *order* in which they are applied be maintained. To show this, consider the two finite rotations $\theta_1 + \theta_2$ applied to the block in Fig. 20–1a. Each rotation has a magnitude of 90° and a direction defined by the right-hand rule, as indicated by the arrow. The final position of the block is shown at the right. When these two rotations are applied in the order $\theta_2 + \theta_1$, as shown in Fig. 20–1b, the final position of the block is *not* the same as it is in Fig. 20–1a. Because *finite rotations* do not obey the commutative law of addition ($\theta_1 + \theta_2 \neq \theta_2 + \theta_1$), they *cannot be classified as vectors*. If smaller, yet finite, rotations had been used to illustrate this point, e.g., 10° instead of 90° , the *final position* of the block after each combination of rotations would also be different; however, in this case, the difference is only a small amount.

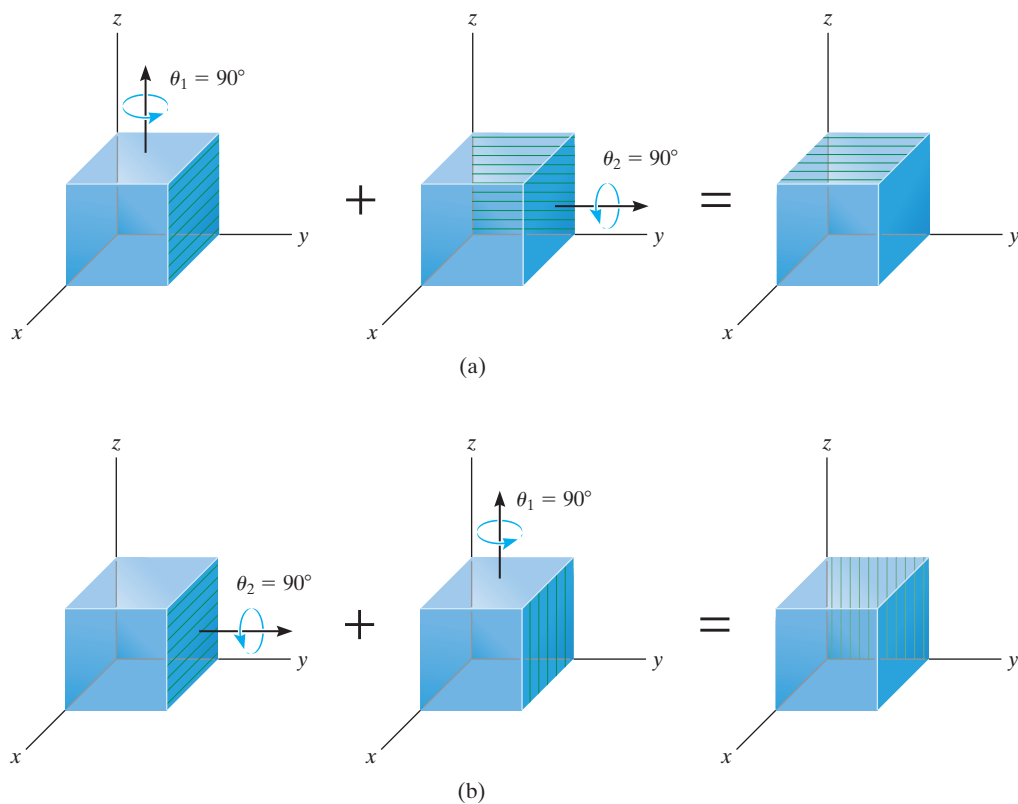


Fig. 20–1

Infinitesimal Rotations. When defining the angular motions of a body subjected to three-dimensional motion, only rotations which are *infinitesimally small* will be considered. Such rotations can be classified as *vectors*, since they can be added vectorially in any manner. To show this, for purposes of simplicity let us consider the rigid body itself to be a sphere which is allowed to rotate about its central fixed point O , Fig. 20–2a. If we impose two infinitesimal rotations $d\theta_1 + d\theta_2$ on the body, it is seen that point P moves along the path $d\theta_1 \times \mathbf{r} + d\theta_2 \times \mathbf{r}$ and ends up at P' . Had the two successive rotations occurred in the order $d\theta_2 + d\theta_1$, then the resultant displacements of P would have been $d\theta_2 \times \mathbf{r} + d\theta_1 \times \mathbf{r}$. Since the vector cross product obeys the distributive law, by comparison $(d\theta_1 + d\theta_2) \times \mathbf{r} = (d\theta_2 + d\theta_1) \times \mathbf{r}$. Here infinitesimal rotations $d\theta$ are vectors, since these quantities have both a magnitude and direction for which the order of (vector) addition is not important, i.e., $d\theta_1 + d\theta_2 = d\theta_2 + d\theta_1$. As a result, as shown in Fig. 20–2a, the two “component” rotations $d\theta_1$ and $d\theta_2$ are equivalent to a single resultant rotation $d\theta = d\theta_1 + d\theta_2$, a consequence of Euler’s theorem.

Angular Velocity. If the body is subjected to an angular rotation $d\theta$ about a fixed point, the angular velocity of the body is defined by the time derivative,

$$\omega = \dot{\theta} \quad (20-1)$$

The line specifying the direction of ω , which is collinear with $d\theta$, is referred to as the *instantaneous axis of rotation*, Fig. 20–2b. In general, this axis changes direction during each instant of time. Since $d\theta$ is a vector quantity, so too is ω , and it follows from vector addition that if the body is subjected to two component angular motions, $\omega_1 = \dot{\theta}_1$ and $\omega_2 = \dot{\theta}_2$, the resultant angular velocity is $\omega = \omega_1 + \omega_2$.

Angular Acceleration. The body’s angular acceleration is determined from the time derivative of its angular velocity, i.e.,

$$\alpha = \dot{\omega} \quad (20-2)$$

For motion about a fixed point, α must account for a change in *both* the magnitude and direction of ω , so that, in general, α is not directed along the instantaneous axis of rotation, Fig. 20–3.

As the direction of the instantaneous axis of rotation (or the line of action of ω) changes in space, the locus of the axis generates a fixed *space cone*, Fig. 20–4. If the change in the direction of this axis is viewed with respect to the rotating body, the locus of the axis generates a *body cone*.

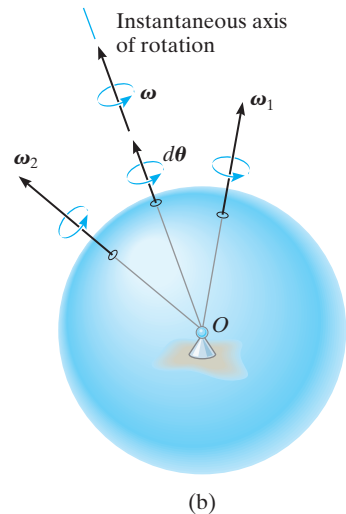
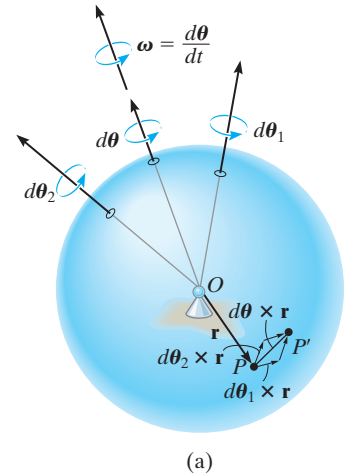


Fig. 20–2

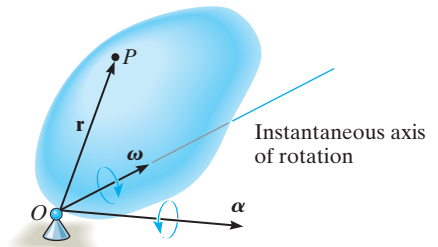


Fig. 20–3

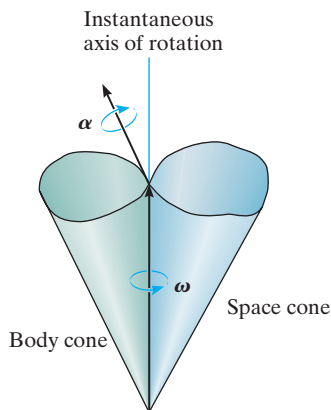
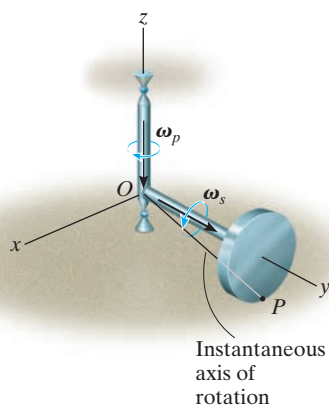
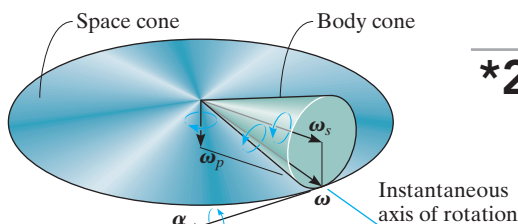


Fig. 20-4



(a)



(b)

Fig. 20-5

At any given instant, these cones meet along the instantaneous axis of rotation, and when the body is in motion, the body cone appears to roll either on the inside or the outside surface of the fixed space cone. Provided the paths defined by the open ends of the cones are described by the head of the ω vector, then α must act tangent to these paths at any given instant, since the time rate of change of ω is equal to α . Fig. 20-4.

To illustrate this concept, consider the disk in Fig. 20-5a that spins about the rod at ω_s , while the rod and disk precess about the vertical axis at ω_p . The resultant angular velocity of the disk is therefore $\omega = \omega_s + \omega_p$. Since both point O and the contact point P have zero velocity, then all points on a line between these points must have zero velocity. Thus, both ω and the instantaneous axis of rotation are along OP . Therefore, as the disk rotates, this axis appears to move along the surface of the fixed space cone shown in Fig. 20-5b. If the axis is observed from the rotating disk, the axis then appears to move on the surface of the body cone. At any instant, though, these two cones meet each other along the axis OP . If ω has a constant magnitude, then α indicates only the change in the direction of ω , which is tangent to the cones at the tip of ω as shown in Fig. 20-5b.

Velocity. Once ω is specified, the velocity of any point on a body rotating about a fixed point can be determined using the same methods as for a body rotating about a fixed axis. Hence, by the cross product,

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad (20-3)$$

Here $\mathbf{r}$ defines the position of the point measured from the fixed point O , Fig. 20-3.

Acceleration. If ω and α are known at a given instant, the acceleration of a point can be obtained from the time derivative of Eq. 20-3, which yields

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (20-4)$$

*20.2 The Time Derivative of a Vector Measured from Either a Fixed or Translating-Rotating System

In many types of problems involving the motion of a body about a fixed point, the angular velocity ω is specified in terms of its components. Then, if the angular acceleration α of such a body is to be determined, it is often easier to compute the time derivative of ω using a coordinate system that has a *rotation* defined by one or more of the components of ω . For example, in the case of the disk in Fig. 20-5a, where $\omega = \omega_s + \omega_p$, the x , y , z axes can be given an angular velocity of ω_p . For this reason, and for other uses later, an equation will now be derived, which relates the time derivative of any vector $\mathbf{A}$ defined from a translating-rotating reference to its time derivative defined from a fixed reference.

Consider the x, y, z axes of the moving frame of reference to be rotating with an angular velocity Ω , which is measured from the fixed X, Y, Z axes, Fig. 20-6a. In the following discussion, it will be convenient to express vector $\mathbf{A}$ in terms of its $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components, which define the directions of the moving axes. Hence,

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

In general, the time derivative of $\mathbf{A}$ must account for the change in both its magnitude and direction. However, if this derivative is taken *with respect to the moving frame of reference*, only the change in the magnitudes of the components of $\mathbf{A}$ must be accounted for, since the directions of the components do not change with respect to the moving reference. Hence,

$$(\dot{\mathbf{A}})_{xyz} = \dot{A}_x \mathbf{i} + \dot{A}_y \mathbf{j} + \dot{A}_z \mathbf{k} \quad (20-5)$$

When the time derivative of $\mathbf{A}$ is taken *with respect to the fixed frame of reference*, the *directions* of $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$ change only on account of the rotation Ω of the axes and not their translation. Hence, in general,

$$\dot{\mathbf{A}} = \dot{A}_x \mathbf{i} + \dot{A}_y \mathbf{j} + \dot{A}_z \mathbf{k} + A_x \dot{\mathbf{i}} + A_y \dot{\mathbf{j}} + A_z \dot{\mathbf{k}}$$

The time derivatives of the unit vectors will now be considered. For example, $\dot{\mathbf{i}} = d\mathbf{i}/dt$ represents only the change in the *direction* of $\mathbf{i}$ with respect to time, since $\mathbf{i}$ always has a magnitude of 1 unit. As shown in Fig. 20-6b, the change, $d\mathbf{i}$, is *tangent to the path* described by the arrowhead of $\mathbf{i}$ as $\mathbf{i}$ swings due to the rotation Ω . Accounting for both the magnitude and direction of $d\mathbf{i}$, we can therefore define $\dot{\mathbf{i}}$ using the cross product, $\dot{\mathbf{i}} = \Omega \times \mathbf{i}$. In general, then

$$\dot{\mathbf{i}} = \Omega \times \mathbf{i} \quad \dot{\mathbf{j}} = \Omega \times \mathbf{j} \quad \dot{\mathbf{k}} = \Omega \times \mathbf{k}$$

These formulations were also developed in Sec. 16.8, regarding planar motion of the axes. Substituting these results into the above equation and using Eq. 20-5 yields

$$\dot{\mathbf{A}} = (\dot{\mathbf{A}})_{xyz} + \Omega \times \mathbf{A} \quad (20-6)$$

This result is important, and will be used throughout Sec. 20.4 and Chapter 21. It states that the time derivative of *any vector* $\mathbf{A}$ as observed from the fixed X, Y, Z frame of reference is equal to the time rate of change of $\mathbf{A}$ as observed from the x, y, z translating-rotating frame of reference, Eq. 20-5, plus $\Omega \times \mathbf{A}$, the change of $\mathbf{A}$ caused by the rotation of the x, y, z frame. As a result, Eq. 20-6 should always be used whenever Ω produces a change in the direction of $\mathbf{A}$ as seen from the X, Y, Z reference. If this change does not occur, i.e., $\Omega = \mathbf{0}$, then $\dot{\mathbf{A}} = (\dot{\mathbf{A}})_{xyz}$, and so the time rate of change of $\mathbf{A}$ as observed from both coordinate systems will be the *same*.

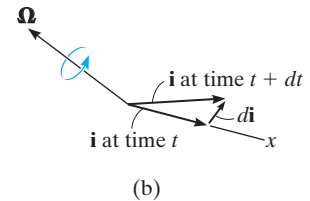
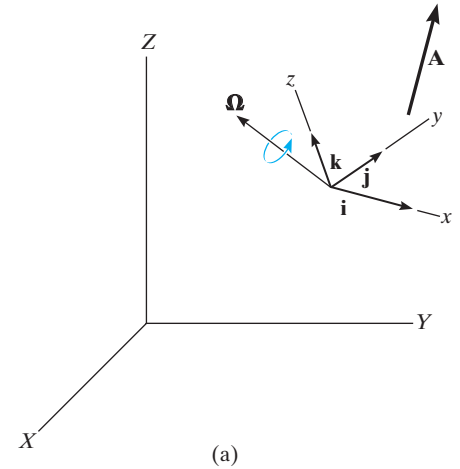


Fig. 20-6

EXAMPLE 20.1



Please refer to the Companion Website for the animation: *A Dish Experiencing 2 Perpendicular Rotations*

The disk shown in Fig. 20–7 spins about its axle with a constant angular velocity $\omega_s = 3 \text{ rad/s}$, while the horizontal platform on which the disk is mounted rotates about the vertical axis at a constant rate $\omega_p = 1 \text{ rad/s}$. Determine the angular acceleration of the disk and the velocity and acceleration of point A on the disk when it is in the position shown.

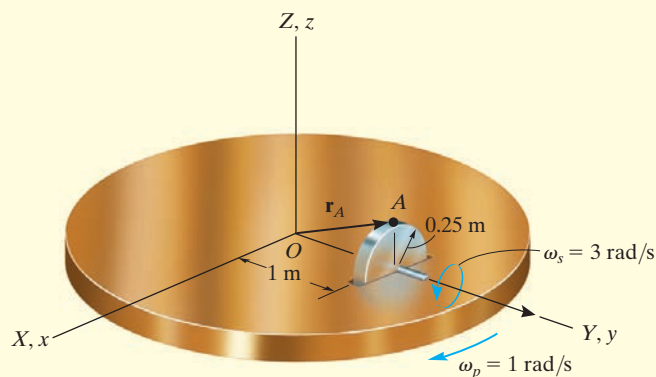


Fig. 20–7

SOLUTION

Point O represents a fixed point of rotation for the disk if one considers a hypothetical extension of the disk to this point. To determine the velocity and acceleration of point A , it is first necessary to determine the angular velocity $\boldsymbol{\omega}$ and angular acceleration $\boldsymbol{\alpha}$ of the disk, since these vectors are used in Eqs. 20–3 and 20–4.

Angular Velocity. The angular velocity, which is measured from X, Y, Z , is simply the vector addition of its two component motions. Thus,

$$\boldsymbol{\omega} = \boldsymbol{\omega}_s + \boldsymbol{\omega}_p = \{3\mathbf{j} - 1\mathbf{k}\} \text{ rad/s}$$

Angular Acceleration. Since the magnitude of ω is constant, only a change in its direction, as seen from the fixed reference, creates the angular acceleration α of the disk. One way to obtain α is to compute the time derivative of *each of the two components* of ω using Eq. 20-6. At the instant shown in Fig. 20-7, imagine the fixed X, Y, Z and a rotating x, y, z frame to be coincident. If the rotating x, y, z frame is chosen to have an angular velocity of $\Omega = \omega_p = \{-1\mathbf{k}\}$ rad/s, then ω_s will *always* be directed along the y (not Y) axis, and the time rate of change of ω_s *as seen from* x, y, z is *zero*; i.e., $(\dot{\omega}_s)_{xyz} = \mathbf{0}$ (the magnitude and direction of ω_s is constant). Thus,

$$\dot{\omega}_s = (\dot{\omega}_s)_{xyz} + \omega_p \times \omega_s = \mathbf{0} + (-1\mathbf{k}) \times (3\mathbf{j}) = \{3\mathbf{i}\} \text{ rad/s}^2$$

By the same choice of axes rotation, $\Omega = \omega_p$, or even with $\Omega = \mathbf{0}$, the time derivative $(\dot{\omega}_p)_{xyz} = \mathbf{0}$, since ω_p has a constant magnitude and direction with respect to x, y, z . Hence,

$$\dot{\omega}_p = (\dot{\omega}_p)_{xyz} + \omega_p \times \omega_p = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

The angular acceleration of the disk is therefore

$$\alpha = \dot{\omega} = \dot{\omega}_s + \dot{\omega}_p = \{3\mathbf{i}\} \text{ rad/s}^2 \quad \text{Ans.}$$

Velocity and Acceleration. Since ω and α have now been determined, the velocity and acceleration of point A can be found using Eqs. 20-3 and 20-4. Realizing that $\mathbf{r}_A = \{1\mathbf{j} + 0.25\mathbf{k}\}$ m, Fig. 20-7, we have

$$\mathbf{v}_A = \omega \times \mathbf{r}_A = (3\mathbf{j} - 1\mathbf{k}) \times (1\mathbf{j} + 0.25\mathbf{k}) = \{1.75\mathbf{i}\} \text{ m/s} \quad \text{Ans.}$$

$$\begin{aligned} \mathbf{a}_A &= \alpha \times \mathbf{r}_A + \omega \times (\omega \times \mathbf{r}_A) \\ &= (3\mathbf{i}) \times (1\mathbf{j} + 0.25\mathbf{k}) + (3\mathbf{j} - 1\mathbf{k}) \times [(3\mathbf{j} - 1\mathbf{k}) \times (1\mathbf{j} + 0.25\mathbf{k})] \\ &= \{-2.50\mathbf{j} - 2.25\mathbf{k}\} \text{ m/s}^2 \quad \text{Ans.} \end{aligned}$$



Please refer to the Companion Website for the animation: *A Dish Experiencing 2 Perpendicular Rotations*

EXAMPLE 20.2

At the instant $\theta = 60^\circ$, the gyrotop in Fig. 20–8 has three components of angular motion directed as shown and having magnitudes defined as:

Spin: $\omega_s = 10 \text{ rad/s}$, increasing at the rate of 6 rad/s^2

Nutation: $\omega_n = 3 \text{ rad/s}$, increasing at the rate of 2 rad/s^2

Precession: $\omega_p = 5 \text{ rad/s}$, increasing at the rate of 4 rad/s^2

Determine the angular velocity and angular acceleration of the top.

SOLUTION

Angular Velocity. The top rotates about the fixed point O . If the fixed and rotating frames are coincident at the instant shown, then the angular velocity can be expressed in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components, with reference to the x, y, z frame; i.e.,

$$\begin{aligned}\boldsymbol{\omega} &= -\omega_n \mathbf{i} + \omega_s \sin \theta \mathbf{j} + (\omega_p + \omega_s \cos \theta) \mathbf{k} \\ &= -3\mathbf{i} + 10 \sin 60^\circ \mathbf{j} + (5 + 10 \cos 60^\circ) \mathbf{k} \\ &= \{-3\mathbf{i} + 8.66\mathbf{j} + 10\mathbf{k}\} \text{ rad/s} \quad \text{Ans.}\end{aligned}$$

Angular Acceleration. As in the solution of Example 20.1, the angular acceleration $\boldsymbol{\alpha}$ will be determined by investigating separately the time rate of change of *each of the angular velocity components* as observed from the fixed X, Y, Z reference. We will choose an $\boldsymbol{\Omega}$ for the x, y, z reference so that the component of $\boldsymbol{\omega}$ being considered is viewed as having a *constant direction* when observed from x, y, z .

Careful examination of the motion of the top reveals that $\boldsymbol{\omega}_s$ has a *constant direction* relative to x, y, z if these axes rotate at $\boldsymbol{\Omega} = \boldsymbol{\omega}_n + \boldsymbol{\omega}_p$. Thus,

$$\begin{aligned}\dot{\boldsymbol{\omega}}_s &= (\dot{\boldsymbol{\omega}}_s)_{xyz} + (\boldsymbol{\omega}_n + \boldsymbol{\omega}_p) \times \boldsymbol{\omega}_s \\ &= (6 \sin 60^\circ \mathbf{j} + 6 \cos 60^\circ \mathbf{k}) + (-3\mathbf{i} + 5\mathbf{k}) \times (10 \sin 60^\circ \mathbf{j} + 10 \cos 60^\circ \mathbf{k}) \\ &= \{-43.30\mathbf{i} + 20.20\mathbf{j} - 22.98\mathbf{k}\} \text{ rad/s}^2\end{aligned}$$

Since $\boldsymbol{\omega}_n$ *always* lies in the fixed X – Y plane, this vector has a *constant direction* if the motion is viewed from axes x, y, z having a rotation of $\boldsymbol{\Omega} = \boldsymbol{\omega}_p$ (not $\boldsymbol{\Omega} = \boldsymbol{\omega}_s + \boldsymbol{\omega}_p$). Thus,

$$\dot{\boldsymbol{\omega}}_n = (\dot{\boldsymbol{\omega}}_n)_{xyz} + \boldsymbol{\omega}_p \times \boldsymbol{\omega}_n = -2\mathbf{i} + (5\mathbf{k}) \times (-3\mathbf{i}) = \{-2\mathbf{i} - 15\mathbf{j}\} \text{ rad/s}^2$$

Finally, the component $\boldsymbol{\omega}_p$ is *always directed* along the Z axis so that here it is not necessary to think of x, y, z as rotating, i.e., $\boldsymbol{\Omega} = \mathbf{0}$. Expressing the data in terms of the $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components, we therefore have

$$\dot{\boldsymbol{\omega}}_p = (\dot{\boldsymbol{\omega}}_p)_{xyz} + \mathbf{0} \times \boldsymbol{\omega}_p = \{4\mathbf{k}\} \text{ rad/s}^2$$

Thus, the angular acceleration of the top is

$$\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}_s + \dot{\boldsymbol{\omega}}_n + \dot{\boldsymbol{\omega}}_p = \{-45.3\mathbf{i} + 5.20\mathbf{j} - 19.0\mathbf{k}\} \text{ rad/s}^2 \quad \text{Ans.}$$

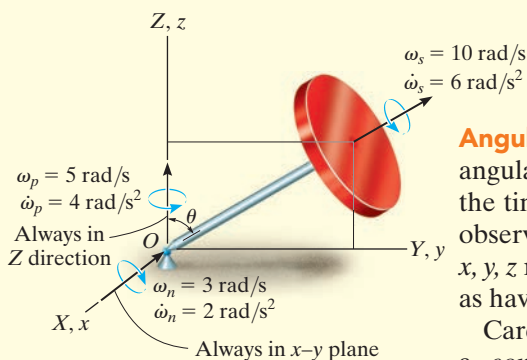


Fig. 20–8

20.3 General Motion

Shown in Fig. 20–9 is a rigid body subjected to general motion in three dimensions for which the angular velocity is $\boldsymbol{\omega}$ and the angular acceleration is $\boldsymbol{\alpha}$. If point A has a known motion of $\mathbf{v}_A$ and $\mathbf{a}_A$, the motion of any other point B can be determined by using a relative-motion analysis. In this section a *translating coordinate system* will be used to define the relative motion, and in the next section a reference that is both rotating and translating will be considered.

If the origin of the translating coordinate system x, y, z ($\boldsymbol{\Omega} = \mathbf{0}$) is located at the “base point” A , then, at the instant shown, the motion of the body can be regarded as the sum of an instantaneous translation of the body having a motion of $\mathbf{v}_A$, and $\mathbf{a}_A$, and a rotation of the body about an instantaneous axis passing through point A . Since the body is rigid, the motion of point B measured by an observer located at A is therefore the same as *the rotation of the body about a fixed point*. This relative motion occurs about the instantaneous axis of rotation and is defined by $\mathbf{v}_{B/A} = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$, Eq. 20–3, and $\mathbf{a}_{B/A} = \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A})$, Eq. 20–4. For translating axes, the relative motions are related to absolute motions by $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$ and $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$, Eqs. 16–15 and 16–17, so that the absolute velocity and acceleration of point B can be determined from the equations

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \quad (20-7)$$

and

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A}) \quad (20-8)$$

These two equations are essentially the same as those describing the general plane motion of a rigid body, Eqs. 16–16 and 16–18. However, difficulty in application arises for three-dimensional motion, because $\boldsymbol{\alpha}$ now measures the change in *both* the magnitude and direction of $\boldsymbol{\omega}$.

Although this may be the case, a direct solution for $\mathbf{v}_B$ and $\mathbf{a}_B$ can be obtained by noting that $\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A$, and so Eq. 20–7 becomes $\mathbf{v}_{B/A} = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$. The cross product indicates that $\mathbf{v}_{B/A}$ is *perpendicular* to $\mathbf{r}_{B/A}$, and so, as noted by Eq. C–14 of Appendix C, we require

$$\mathbf{r}_{B/A} \cdot \mathbf{v}_{B/A} = 0 \quad (20-9)$$

Taking the time derivative, we have

$$\mathbf{v}_{B/A} \cdot \mathbf{v}_{B/A} + \mathbf{r}_{B/A} \cdot \mathbf{a}_{B/A} = 0 \quad (20-10)$$

Solution II of the following example illustrates application of this idea.

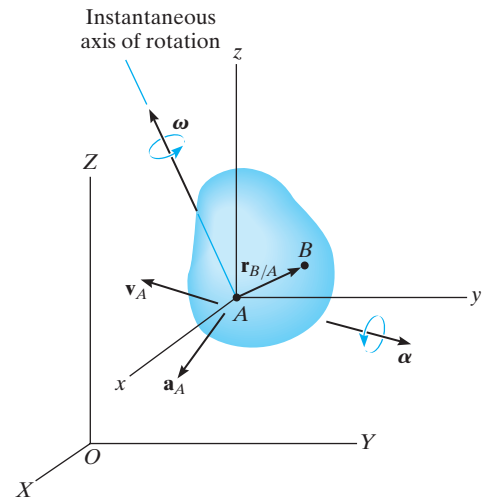
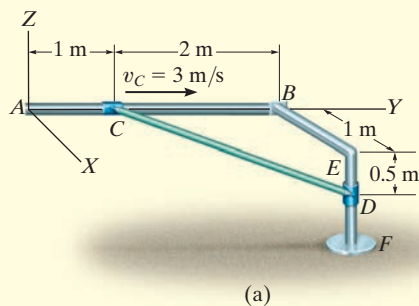
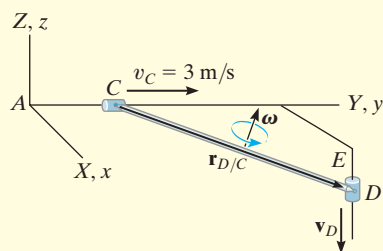


Fig. 20–9

EXAMPLE 20.3



(a)



(b)

Fig. 20-10

If the collar at C in Fig. 20-10a moves toward B with a speed of 3 m/s, determine the velocity of the collar at D and the angular velocity of the bar at the instant shown. The bar is connected to the collars at its end points by ball-and-socket joints.

SOLUTION I

Bar CD is subjected to general motion. Why? The velocity of point D on the bar can be related to the velocity of point C by the equation

$$\mathbf{v}_D = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_{D/C}$$

The fixed and translating frames of reference are assumed to coincide at the instant considered, Fig. 20-10b. We have

$$\begin{aligned} \mathbf{v}_D &= -v_D \mathbf{k} & \mathbf{v}_C &= \{3\mathbf{j}\} \text{ m/s} \\ \mathbf{r}_{D/C} &= \{1\mathbf{i} + 2\mathbf{j} - 0.5\mathbf{k}\} \text{ m} & \boldsymbol{\omega} &= \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k} \end{aligned}$$

Substituting into the above equation we get

$$-v_D \mathbf{k} = 3\mathbf{j} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \omega_x & \omega_y & \omega_z \\ 1 & 2 & -0.5 \end{vmatrix}$$

Expanding and equating the respective $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components yields

$$-0.5\omega_y - 2\omega_z = 0 \quad (1)$$

$$0.5\omega_x + 1\omega_z + 3 = 0 \quad (2)$$

$$2\omega_x - 1\omega_y + v_D = 0 \quad (3)$$

These equations contain four unknowns.* A fourth equation can be written if the direction of $\boldsymbol{\omega}$ is specified. In particular, any component of $\boldsymbol{\omega}$ acting along the bar's axis has no effect on moving the collars. This is because the bar is *free to rotate* about its axis. Therefore, if $\boldsymbol{\omega}$ is specified as acting *perpendicular* to the axis of the bar, then $\boldsymbol{\omega}$ must have a *unique magnitude* to satisfy the above equations. Hence,

$$\begin{aligned} \boldsymbol{\omega} \cdot \mathbf{r}_{D/C} &= (\omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}) \cdot (1\mathbf{i} + 2\mathbf{j} - 0.5\mathbf{k}) = 0 \\ 1\omega_x + 2\omega_y - 0.5\omega_z &= 0 \quad (4) \end{aligned}$$

*Although this is the case, the magnitude of v_D can be obtained. For example, solve Eqs. 1 and 2 for ω_y and ω_x in terms of ω_z and substitute this into Eq. 3. Then ω_z will cancel out, which will allow a solution for v_D .

Solving Eqs. 1 through 4 simultaneously yields

$$\begin{aligned}\omega_x &= -4.86 \text{ rad/s} & \omega_y &= 2.29 \text{ rad/s} & \omega_z &= -0.571 \text{ rad/s}, \\ v_D &= 12.0 \text{ m/s, so that} & \omega &= 5.40 \text{ rad/s}\end{aligned}$$

Ans.

SOLUTION II

Applying Eq. 20-9, $\mathbf{v}_{D/C} = \mathbf{v}_D - \mathbf{v}_C = -v_D \mathbf{k} - 3\mathbf{j}$, so that

$$\begin{aligned}\mathbf{r}_{D/C} \cdot \mathbf{v}_{D/C} &= (1\mathbf{i} + 2\mathbf{j} - 0.5\mathbf{k}) \cdot (-v_D \mathbf{k} - 3\mathbf{j}) = 0 \\ (1)(0) + (2)(-3) + (-0.5)(-v_D) &= 0 \\ v_D &= 12 \text{ m/s}\end{aligned}$$

Ans.

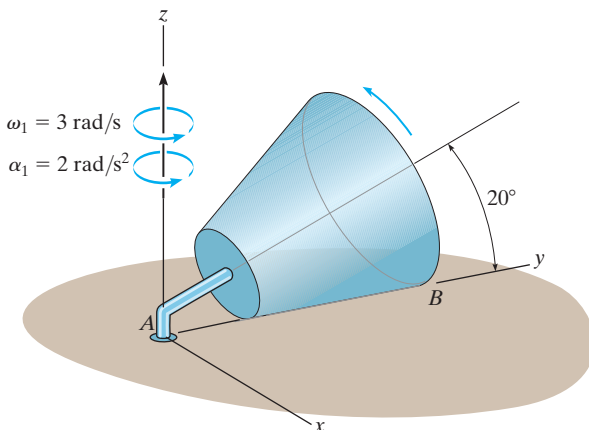
Since $\boldsymbol{\omega}$ is perpendicular to $\mathbf{r}_{D/C}$ then $\mathbf{v}_{D/C} = \boldsymbol{\omega} \times \mathbf{r}_{D/C}$ or

$$\begin{aligned}v_{D/C} &= \omega r_{D/C} \\ \sqrt{(-12)^2 + (-3)^2} &= \omega \sqrt{(1)^2 + (2)^2 + (-0.5)^2} \\ \omega &= 5.40 \text{ rad/s}\end{aligned}$$

Ans.

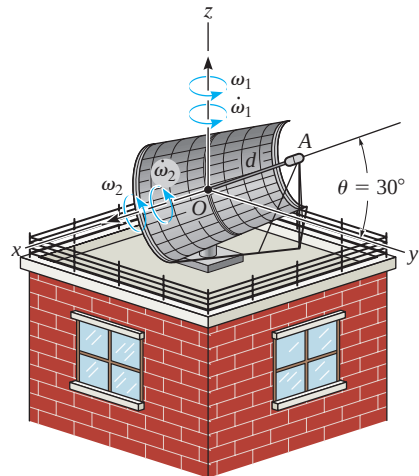
PROBLEMS

20-1. The conical spool rolls on the plane without slipping. If the axle has an angular velocity of $\omega_1 = 3 \text{ rad/s}$ and an angular acceleration of $\alpha_1 = 2 \text{ rad/s}^2$ at the instant shown, determine the angular velocity and angular acceleration of the spool at this instant.



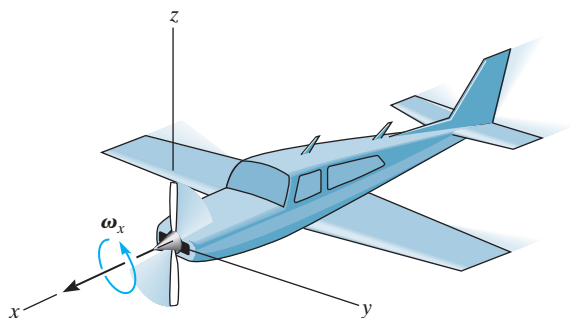
Prob. 20-1

20-2. At a given instant, the antenna has an angular motion $\omega_1 = 3 \text{ rad/s}$ and $\dot{\omega}_1 = 2 \text{ rad/s}^2$ about the z axis. At this same instant $\theta = 30^\circ$, the angular motion about the x axis is $\omega_2 = 1.5 \text{ rad/s}$, and $\dot{\omega}_2 = 4 \text{ rad/s}^2$. Determine the velocity and acceleration of the signal horn A at this instant. The distance from O to A is $d = 1 \text{ m}$.



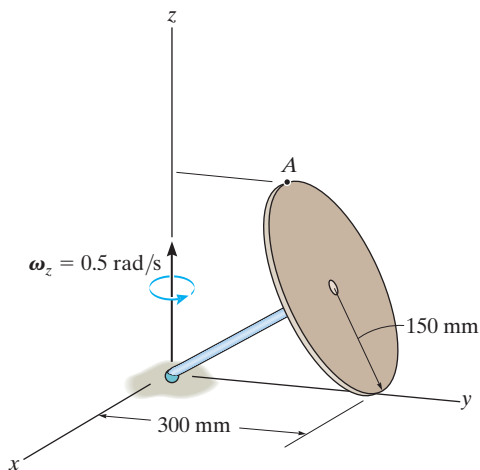
Prob. 20-2

20-3. The propeller of an airplane is rotating at a constant speed $\omega_x \mathbf{i}$, while the plane is undergoing a turn at a constant rate ω_t . Determine the angular acceleration of the propeller if (a) the turn is horizontal, i.e., $\omega_t \mathbf{k}$, and (b) the turn is vertical, downward, i.e., $\omega_t \mathbf{j}$.



Prob. 20-3

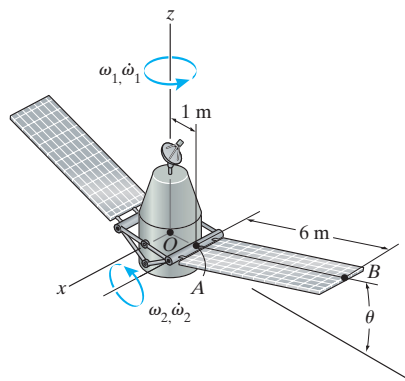
***20-4.** The disk rotates about the z axis at a constant rate $\omega_z = 0.5$ rad/s without slipping on the horizontal plane. Determine the velocity and the acceleration of point A on the disk.



Prob. 20-4

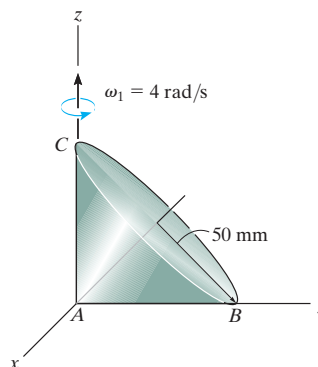
20-5. At the instant $\theta = 30^\circ$, the satellite's body is rotating with an angular velocity of $\omega_1 = 20$ rad/s, and it has an angular acceleration of $\dot{\omega}_1 = 5$ rad/s². Simultaneously, the solar panels rotate with a constant angular velocity of $\omega_2 = 5$ rad/s. Determine the velocity and acceleration of point B located at the end of one of the solar panels at this instant.

20-6. At the instant $\theta = 30^\circ$, the satellite's body is rotating with an angular velocity of $\omega_1 = 20$ rad/s, and it has an angular acceleration of $\dot{\omega}_1 = 5$ rad/s². At the same instant, the satellite travels in the x direction with a velocity of $\mathbf{v}_O = \{5000\mathbf{i}\}$ m/s, and it has an acceleration of $\mathbf{a}_O = \{500\mathbf{i}\}$ m/s². Simultaneously, the solar panels rotate with a constant angular speed of $\omega_2 = 5$ rad/s. Determine the velocity and acceleration of point B located at the end of one of the solar panels at this instant.



Probs. 20-5/6

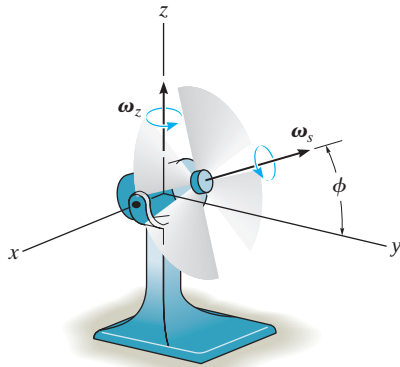
20-7. The right circular cone rotates about the z axis at a constant rate of $\omega_1 = 4$ rad/s without slipping on the horizontal plane. Determine the magnitudes of the velocity and acceleration of points B and C .



Prob. 20-7

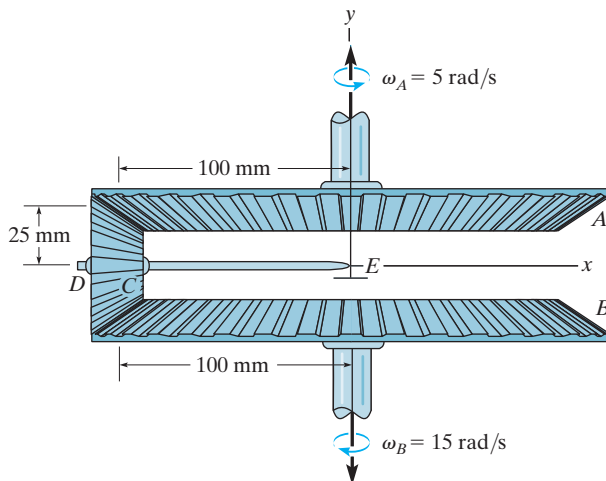
***20-8.** The electric fan is mounted on a swivel support such that the fan rotates about the z axis at a constant rate of $\omega_z = 1$ rad/s and the fan blade is spinning at a constant rate $\omega_s = 60$ rad/s. If $\phi = 45^\circ$ for the motion, determine the angular velocity and the angular acceleration of the blade.

20-9. The electric fan is mounted on a swivel support such that the fan rotates about the z axis at a constant rate of $\omega_z = 1$ rad/s and the fan blade is spinning at a constant rate $\omega_s = 60$ rad/s. If at the instant $\phi = 45^\circ$, $\dot{\phi} = 2$ rad/s for the motion, determine the angular velocity and the angular acceleration of the blade.



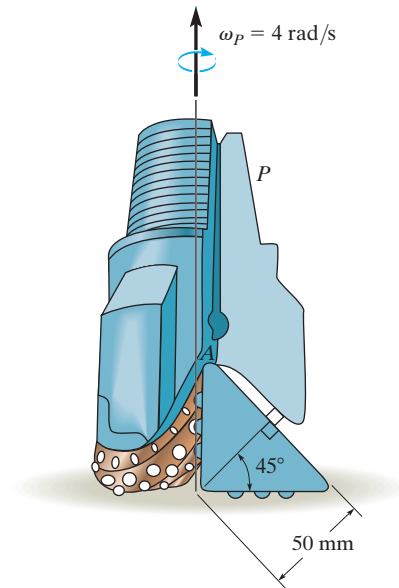
Probs. 20-8/9

20-10. If the plate gears A and B are rotating with the angular velocities shown, determine the angular velocity of gear C about the shaft DE . What is the angular velocity of DE about the y axis?



Prob. 20-10

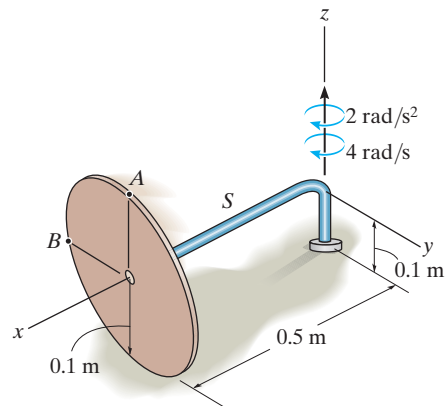
20-11. The drill pipe P turns at a constant angular rate $\omega_P = 4$ rad/s. Determine the angular velocity and angular acceleration of the conical rock bit, which rolls without slipping. Also, what are the velocity and acceleration of point A ?



Prob. 20-11

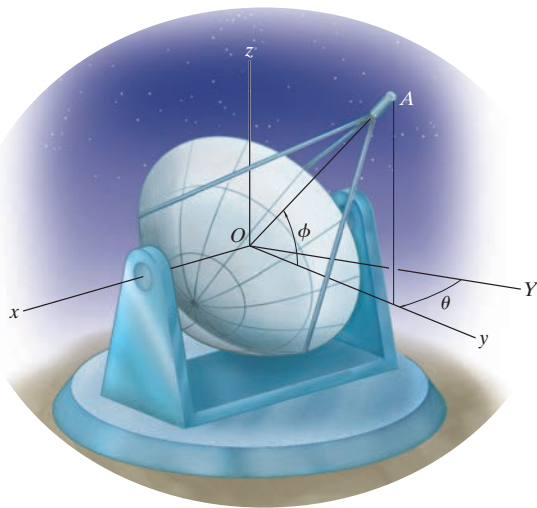
***20-12.** The disk rotates about the shaft S , while the shaft is turning about the z axis at a rate of $\omega_z = 4$ rad/s, which is increasing at 2 rad/s². Determine the velocity and acceleration of point A on the disk at the instant shown. No slipping occurs.

20-13. The disk rotates about the shaft S , while the shaft is turning about the z axis at a rate of $\omega_z = 4$ rad/s, which is increasing at 2 rad/s². Determine the velocity and acceleration of point B on the disk at the instant shown. No slipping occurs.



Probs. 20-12/13

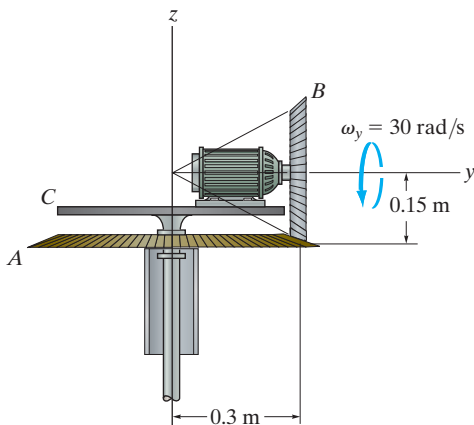
20-14. The antenna is following the motion of a jet plane. At the instant $\theta = 25^\circ$ and $\phi = 75^\circ$, the constant angular rates of change are $\dot{\theta} = 0.4$ rad/s and $\dot{\phi} = 0.6$ rad/s. Determine the velocity and acceleration of the signal horn A at this instant. The distance OA is 0.8 m.



Prob. 20-14

20-15. Gear B is driven by a motor mounted on turntable C . If gear A is held fixed, and the motor shaft rotates with a constant angular velocity of $\omega_y = 30$ rad/s, determine the angular velocity and angular acceleration of gear B .

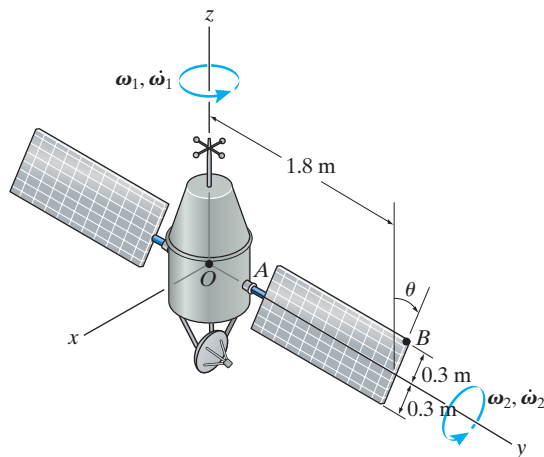
***20-16.** Gear B is driven by a motor mounted on turntable C . If gear A and the motor shaft rotate with constant angular speeds of $\omega_A = \{10\mathbf{k}\}$ rad/s and $\omega_y = \{30\mathbf{j}\}$ rad/s, respectively, determine the angular velocity and angular acceleration of gear B .



Probs. 20-15/16

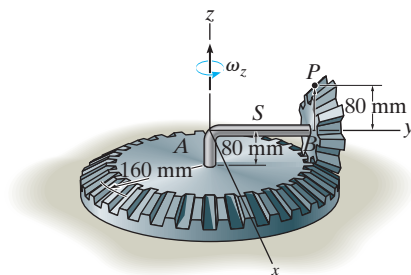
20-17. At the instant when $\theta = 90^\circ$, the satellite's body is rotating with an angular velocity of $\omega_1 = 15$ rad/s and angular acceleration of $\dot{\omega}_1 = 3$ rad/s². Simultaneously, the solar panels rotate with an angular velocity of $\omega_2 = 6$ rad/s and angular acceleration of $\dot{\omega}_2 = 1.5$ rad/s². Determine the velocity and acceleration of point B on the solar panel at this instant.

20-18. At the instant when $\theta = 90^\circ$, the satellite's body travels in the x direction with a velocity of $\mathbf{v}_O = \{500\mathbf{i}\}$ m/s and acceleration of $\mathbf{a}_O = \{500\mathbf{i}\}$ m/s². Simultaneously, the body also rotates with an angular velocity of $\omega_1 = 15$ rad/s and angular acceleration of $\dot{\omega}_1 = 3$ rad/s². At the same time, the solar panels rotate with an angular velocity of $\omega_2 = 6$ rad/s and angular acceleration of $\dot{\omega}_2 = 1.5$ rad/s². Determine the velocity and acceleration of point B on the solar panel.



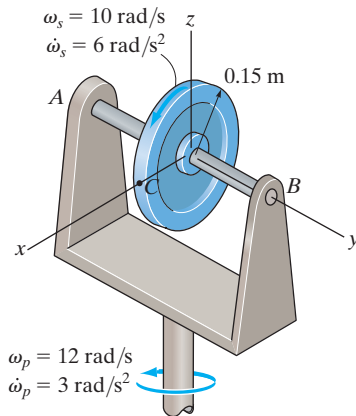
Probs. 20-17/18

20-19. Gear A is fixed while gear B is free to rotate on the shaft S . If the shaft is turning about the z axis at $\omega_z = 5$ rad/s, while increasing at 2 rad/s², determine the velocity and acceleration of point P at the instant shown. The face of gear B lies in a vertical plane.



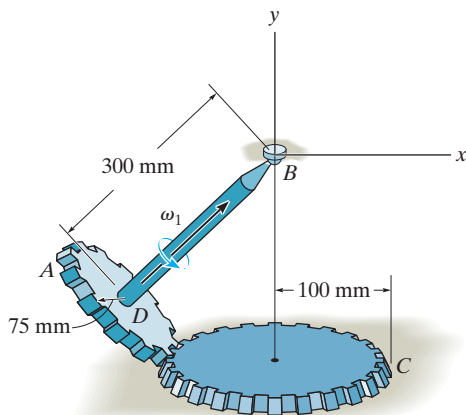
Prob. 20-19

***20–20.** The wheel is spinning about shaft AB with an angular velocity of $\omega_s = 10 \text{ rad/s}$, which is increasing at a constant rate of $\dot{\omega}_s = 6 \text{ rad/s}^2$, while the frame precesses about the z axis with an angular velocity of $\omega_p = 12 \text{ rad/s}$, which is increasing at a constant rate of $\dot{\omega}_p = 3 \text{ rad/s}^2$. Determine the velocity and acceleration of point C located on the rim of the wheel at this instant.



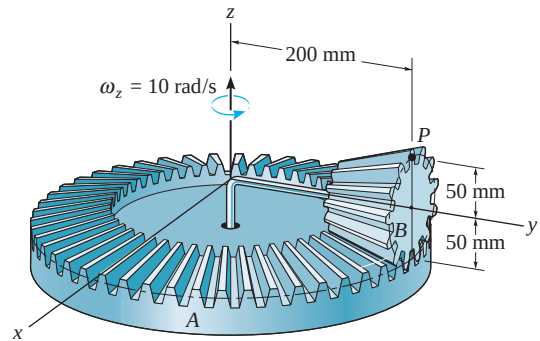
Prob. 20–20

20–21. Shaft BD is connected to a ball-and-socket joint at B , and a beveled gear A is attached to its other end. The gear is in mesh with a fixed gear C . If the shaft and gear A are spinning with a constant angular velocity $\omega_1 = 8 \text{ rad/s}$, determine the angular velocity and angular acceleration of gear A .



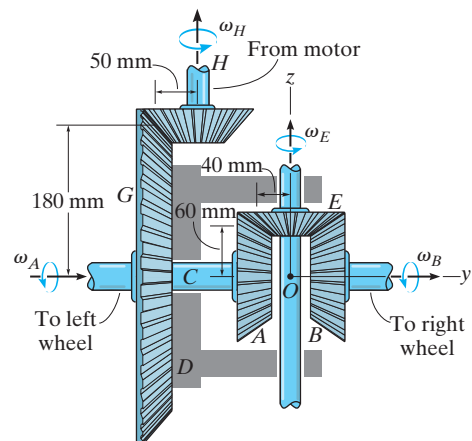
Prob. 20–21

20–22. Gear B is connected to the rotating shaft, while the plate gear A is fixed. If the shaft is turning at a constant rate of $\omega_z = 10 \text{ rad/s}$ about the z axis, determine the magnitudes of the angular velocity and the angular acceleration of gear B . Also, determine the magnitudes of the velocity and acceleration of point P .



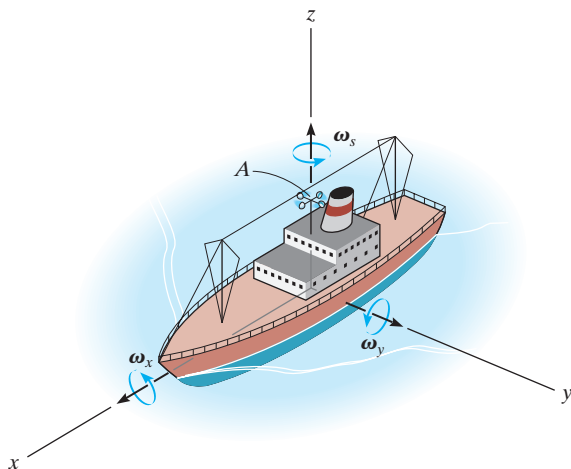
Prob. 20–22

20–23. The differential of an automobile allows the two rear wheels to rotate at different speeds when the automobile travels along a curve. For operation, the rear axles are attached to the wheels at one end and have beveled gears A and B on their other ends. The differential case D is placed over the left axle but can rotate about C independent of the axle. The case supports a pinion gear E on a shaft, which meshes with gears A and B . Finally, a ring gear G is fixed to the differential case so that the case rotates with the ring gear when the latter is driven by the drive pinion H . This gear, like the differential case, is free to rotate about the left wheel axle. If the drive pinion is turning at $\omega_H = 100 \text{ rad/s}$ and the pinion gear E is spinning about its shaft at $\omega_E = 30 \text{ rad/s}$, determine the angular velocity, ω_A and ω_B , of each axle.



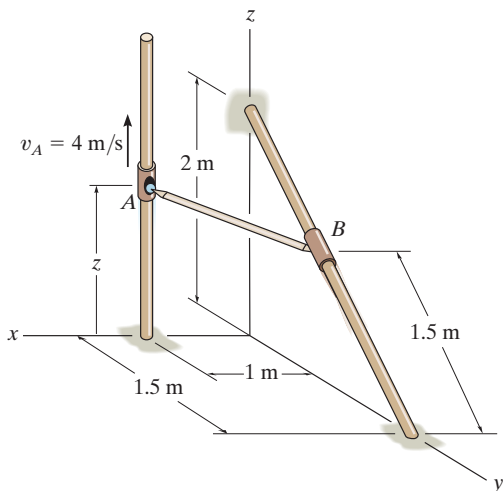
Prob. 20–23

***20–24.** The anemometer located on the ship at A spins about its own axis at a rate ω_s , while the ship rolls about the x axis at the rate ω_x and about the y axis at the rate ω_y . Determine the angular velocity and angular acceleration of the anemometer at the instant the ship is level as shown. Assume that the magnitudes of all components of angular velocity are constant and that the rolling motion caused by the sea is independent in the x and y directions.



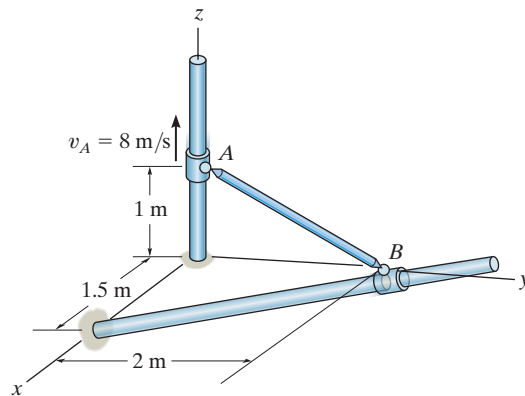
Prob. 20–24

20–25. Rod AB is attached to collars at its ends by ball-and-socket joints. If collar A has a speed $v_A = 4$ m/s, determine the speed of collar B at the instant $z = 2$ m. Assume the angular velocity of the rod is directed perpendicular to the rod.



Prob. 20–25

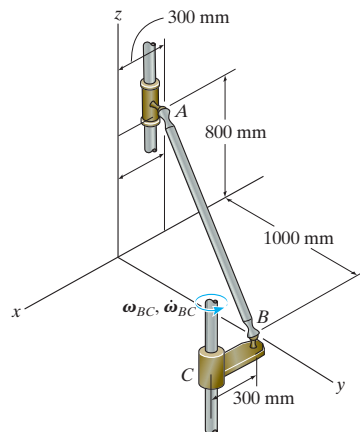
20–26. The rod is attached to smooth collars A and B at its ends using ball-and-socket joints. Determine the speed of B at the instant shown if A is moving at $v_A = 8$ m/s. Also, determine the angular velocity of the rod if it is directed perpendicular to the axis of the rod.



Probs. 20–26/27

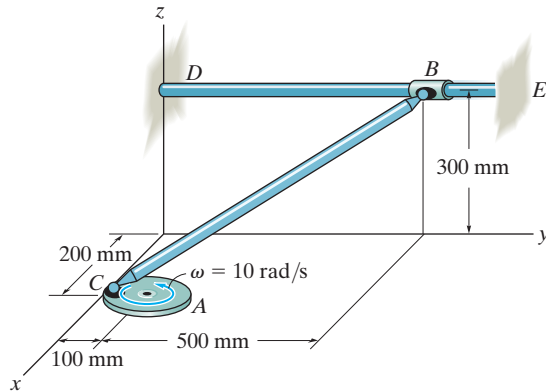
***20–28.** If crank BC rotates with a constant angular velocity of $\omega_{BC} = 6$ rad/s, determine the velocity of the collar at A . Assume the angular velocity of AB is perpendicular to the rod.

20–29. If crank BC is rotating with an angular velocity of $\omega_{BC} = 6$ rad/s and an angular acceleration of $\dot{\omega}_{BC} = 1.5$ rad/s², determine the acceleration of collar A at this instant. Assume the angular velocity and angular acceleration of AB are perpendicular to the rod.



Probs. 20–28/29

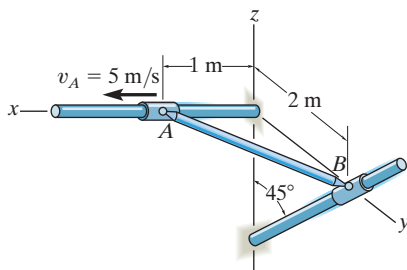
20–30. Disk A rotates at a constant angular velocity of 10 rad/s . If rod BC is joined to the disk and a collar by ball-and-socket joints, determine the velocity of collar B at the instant shown. Also, what is the rod's angular velocity ω_{BC} if it is directed perpendicular to the axis of the rod?



Prob. 20–30

20–31. Rod AB is attached to collars at its ends by using ball-and-socket joints. If collar A moves along the fixed rod at $v_A = 5 \text{ m/s}$, determine the angular velocity of the rod and the velocity of collar B at the instant shown. Assume that the rod's angular velocity is directed perpendicular to the axis of the rod.

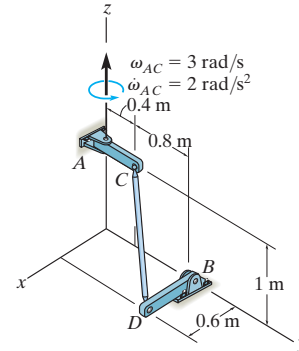
***20–32.** Rod AB is attached to collars at its ends by using ball-and-socket joints. If collar A moves along the fixed rod with a velocity of $v_A = 5 \text{ m/s}$ and has an acceleration $a_A = 2 \text{ m/s}^2$ at the instant shown, determine the angular acceleration of the rod and the acceleration of collar B at this instant. Assume that the rod's angular velocity and angular acceleration are directed perpendicular to the axis of the rod.



Probs. 20–31/32

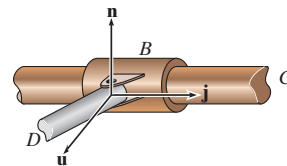
20–33. Rod CD is attached to the rotating arms using ball-and-socket joints. If AC has the motion shown, determine the angular velocity of link BD at the instant shown.

20–34. Rod CD is attached to the rotating arms using ball-and-socket joints. If AC has the motion shown, determine the angular acceleration of link BD at this instant.



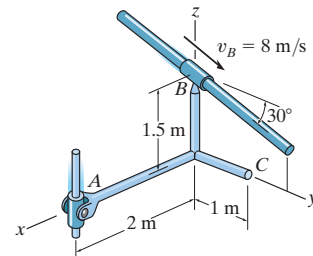
Probs. 20–33/34

20–35. Solve Prob. 20–30 if the connection at B consists of a pin as shown in the figure below, rather than a ball-and-socket joint. *Hint:* The constraint allows rotation of the rod both along the bar ($\mathbf{j}$ direction) and along the axis of the pin ($\mathbf{n}$ direction). Since there is no rotational component in the $\mathbf{u}$ direction, i.e., perpendicular to $\mathbf{n}$ and $\mathbf{j}$ where $\mathbf{u} = \mathbf{j} \times \mathbf{n}$, an additional equation for solution can be obtained from $\omega \cdot \mathbf{u} = 0$. The vector $\mathbf{n}$ is in the same direction as $\mathbf{r}_{B/C} \times \mathbf{r}_{D/C}$.



Prob. 20–35

***20–36.** Member ABC is pin connected at A and has a ball-and-socket joint at B . If the collar at B is moving along the inclined rod at $v_B = 8 \text{ m/s}$, determine the velocity of point C at the instant shown. *Hint:* See Prob. 20–35.



Prob. 20–36

*20.4 Relative-Motion Analysis Using Translating and Rotating Axes

The most general way to analyze the three-dimensional motion of a rigid body requires the use of x, y, z axes that both translate and rotate relative to a second frame X, Y, Z . This analysis also provides a means to determine the motions of two points A and B located on separate members of a mechanism, and the relative motion of one particle with respect to another when one or both particles are moving along *curved paths*.

As shown in Fig. 20–11, the locations of points A and B are specified relative to the X, Y, Z frame of reference by position vectors $\mathbf{r}_A$ and $\mathbf{r}_B$. The base point A represents the origin of the x, y, z coordinate system, which is translating and rotating with respect to X, Y, Z . At the instant considered, the velocity and acceleration of point A are $\mathbf{v}_A$ and $\mathbf{a}_A$, and the angular velocity and angular acceleration of the x, y, z axes are $\boldsymbol{\Omega}$ and $\dot{\boldsymbol{\Omega}} = d\boldsymbol{\Omega}/dt$. All these vectors are *measured* with respect to the X, Y, Z frame of reference, although they can be expressed in Cartesian component form along either set of axes.

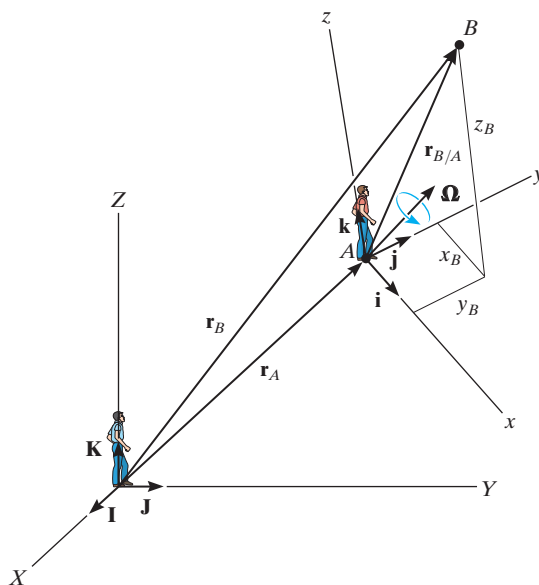


Fig. 20–11

Position. If the position of “ B with respect to A ” is specified by the *relative-position vector* $\mathbf{r}_{B/A}$, Fig. 20–11, then, by vector addition,

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} \quad (20-11)$$

where

$\mathbf{r}_B$ = position of B

$\mathbf{r}_A$ = position of the origin A

$\mathbf{r}_{B/A}$ = position of “ B with respect to A ”

Velocity. The velocity of point B measured from X, Y, Z can be determined by taking the time derivative of Eq. 20–11,

$$\dot{\mathbf{r}}_B = \dot{\mathbf{r}}_A + \dot{\mathbf{r}}_{B/A}$$

The first two terms represent $\mathbf{v}_B$ and $\mathbf{v}_A$. The last term must be evaluated by applying Eq. 20–6, since $\mathbf{r}_{B/A}$ is measured with respect to a rotating reference. Hence,

$$\dot{\mathbf{r}}_{B/A} = (\dot{\mathbf{r}}_{B/A})_{xyz} + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} = (\mathbf{v}_{B/A})_{xyz} + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} \quad (20-12)$$

Therefore,

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz} \quad (20-13)$$

where

$\mathbf{v}_B$ = velocity of B

$\mathbf{v}_A$ = velocity of the origin A of the x, y, z frame of reference

$(\mathbf{v}_{B/A})_{xyz}$ = velocity of “ B with respect to A ” as measured by an observer attached to the rotating x, y, z frame of reference

$\boldsymbol{\Omega}$ = angular velocity of the x, y, z frame of reference

$\mathbf{r}_{B/A}$ = position of “ B with respect to A ”

Acceleration. The acceleration of point B measured from X, Y, Z is determined by taking the time derivative of Eq. 20-13.

$$\dot{\mathbf{v}}_B = \dot{\mathbf{v}}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times \dot{\mathbf{r}}_{B/A} + \frac{d}{dt}(\mathbf{v}_{B/A})_{xyz}$$

The time derivatives defined in the first and second terms represent $\mathbf{a}_B$ and $\mathbf{a}_A$, respectively. The fourth term can be evaluated using Eq. 20-12, and the last term is evaluated by applying Eq. 20-6, which yields

$$\frac{d}{dt}(\mathbf{v}_{B/A})_{xyz} = (\dot{\mathbf{v}}_{B/A})_{xyz} + \boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} = (\mathbf{a}_{B/A})_{xyz} + \boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz}$$

Here $(\mathbf{a}_{B/A})_{xyz}$ is the acceleration of B with respect to A measured from x, y, z . Substituting this result and Eq. 20-12 into the above equation and simplifying, we have

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz} \quad (20-14)$$

where

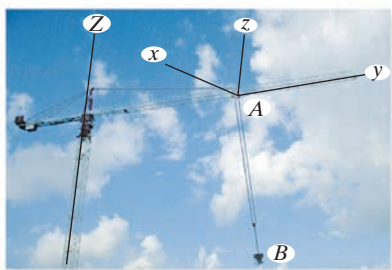
$\mathbf{a}_B$ = acceleration of B

$\mathbf{a}_A$ = acceleration of the origin A of the x, y, z frame of reference

$(\mathbf{a}_{B/A})_{xyz}, (\mathbf{v}_{B/A})_{xyz}$ = relative acceleration and relative velocity of “ B with respect to A ” as measured by an observer attached to the rotating x, y, z frame of reference

$\dot{\boldsymbol{\Omega}}, \boldsymbol{\Omega}$ = angular acceleration and angular velocity of the x, y, z frame of reference

$\mathbf{r}_{B/A}$ = position of “ B with respect to A ”



Complicated spatial motion of the concrete bucket B occurs due to the rotation of the boom about the Z axis, motion of the carriage A along the boom, and extension and swinging of the cable AB . A translating-rotating x, y, z coordinate system can be established on the carriage, and a relative-motion analysis can then be applied to study this motion.

Equations 20-13 and 20-14 are identical to those used in Sec. 16.8 for analyzing relative plane motion.* In that case, however, application is simplified since $\boldsymbol{\Omega}$ and $\dot{\boldsymbol{\Omega}}$ have a *constant direction* which is always perpendicular to the plane of motion. For three-dimensional motion, $\dot{\boldsymbol{\Omega}}$ must be computed by using Eq. 20-6, since $\dot{\boldsymbol{\Omega}}$ depends on the change in *both* the magnitude and direction of $\boldsymbol{\Omega}$.

*Refer to Sec. 16.8 for an interpretation of the terms.

Procedure for Analysis

Three-dimensional motion of particles or rigid bodies can be analyzed with Eqs. 20–13 and 20–14 by using the following procedure.

Coordinate Axes.

- Select the location and orientation of the X, Y, Z and x, y, z coordinate axes. Most often solutions can be easily obtained if at the instant considered:
 - (1) the origins are *coincident*
 - (2) the axes are collinear
 - (3) the axes are parallel
- If several components of angular velocity are involved in a problem, the calculations will be reduced if the x, y, z axes are selected such that only one component of angular velocity is observed with respect to this frame (Ω_{xyz}) and the frame rotates with Ω defined by the other components of angular velocity.

Kinematic Equations.

- After the origin of the moving reference, A , is defined and the moving point B is specified, Eqs. 20–13 and 20–14 should then be written in symbolic form as

$$\mathbf{v}_B = \mathbf{v}_A + \Omega \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz}$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\Omega} \times \mathbf{r}_{B/A} + \Omega \times (\Omega \times \mathbf{r}_{B/A}) + 2\Omega \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz}$$

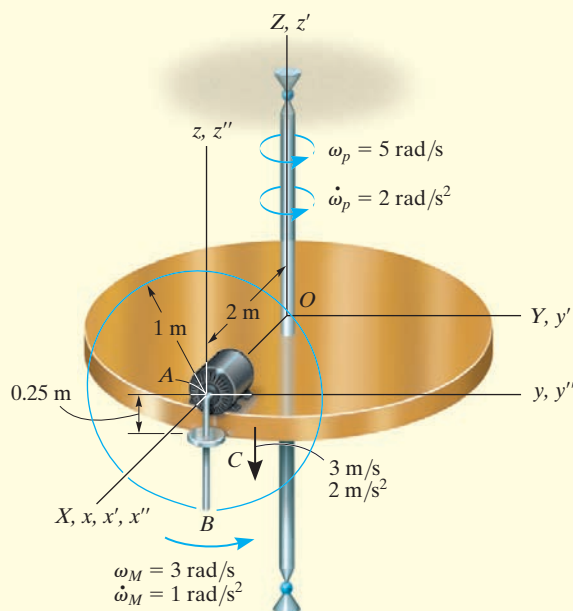
- If $\mathbf{r}_A$ and Ω appear to *change direction* when observed from the fixed X, Y, Z reference then use a set of primed reference axes, x', y', z' having a rotation $\Omega' = \Omega$. Equation 20–6 is then used to determine $\dot{\Omega}$ and the motion $\mathbf{v}_A$ and $\mathbf{a}_A$ of the origin of the moving x, y, z axes.
- If $\mathbf{r}_{B/A}$ and Ω_{xyz} appear to change direction as observed from x, y, z , then use a set of double-primed reference axes x'', y'', z'' having $\Omega'' = \Omega_{xyz}$ and apply Eq. 20–6 to determine $\dot{\Omega}_{xyz}$ and the relative motion $(\mathbf{v}_{B/A})_{xyz}$ and $(\mathbf{a}_{B/A})_{xyz}$.
- After the final forms of $\dot{\Omega}$, $\mathbf{v}_A$, $\mathbf{a}_A$, $\dot{\Omega}_{xyz}$, $(\mathbf{v}_{B/A})_{xyz}$, and $(\mathbf{a}_{B/A})_{xyz}$ are obtained, numerical problem data can be substituted and the kinematic terms evaluated. The components of all these vectors can be selected either along the X, Y, Z or along the x, y, z axes. The choice is arbitrary, provided a consistent set of unit vectors is used.

EXAMPLE 20.4

A motor and attached rod AB have the angular motions shown in Fig. 20–12. A collar C on the rod is located 0.25 m from A and is moving downward along the rod with a velocity of 3 m/s and an acceleration of 2 m/s². Determine the velocity and acceleration of C at this instant.

SOLUTION**Coordinate Axes.**

The origin of the fixed X, Y, Z reference is chosen at the center of the platform, and the origin of the moving x, y, z frame at point A , Fig. 20–12. Since the collar is subjected to two components of angular motion, ω_p and ω_M , it will be viewed as having an angular velocity of $\Omega_{xyz} = \omega_M$ in x, y, z . Therefore, the x, y, z axes will be attached to the platform so that $\Omega = \omega_p$.

**Fig. 20–12**

Kinematic Equations. Equations 20–13 and 20–14, applied to points C and A , become

$$\mathbf{v}_C = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{C/A} + (\mathbf{v}_{C/A})_{xyz}$$

$$\mathbf{a}_C = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/A})_{xyz} + (\mathbf{a}_{C/A})_{xyz}$$

Motion of A . Here $\mathbf{r}_A$ changes direction relative to X, Y, Z . To find the time derivatives of $\mathbf{r}_A$ we will use a set of x', y', z' axes coincident with the X, Y, Z axes that rotate at $\boldsymbol{\Omega}' = \boldsymbol{\omega}_p$. Thus,

$$\boldsymbol{\Omega} = \boldsymbol{\omega}_p = \{5\mathbf{k}\} \text{ rad/s } (\boldsymbol{\Omega} \text{ does not change direction relative to } X, Y, Z.)$$

$$\dot{\boldsymbol{\Omega}} = \dot{\boldsymbol{\omega}}_p = \{2\mathbf{k}\} \text{ rad/s}^2$$

$$\mathbf{r}_A = \{2\mathbf{i}\} \text{ m}$$

$$\mathbf{v}_A = \dot{\mathbf{r}}_A = (\dot{\mathbf{r}}_A)_{x'y'z'} + \boldsymbol{\omega}_p \times \mathbf{r}_A = \mathbf{0} + 5\mathbf{k} \times 2\mathbf{i} = \{10\mathbf{j}\} \text{ m/s}$$

$$\begin{aligned} \mathbf{a}_A = \ddot{\mathbf{r}}_A &= [(\ddot{\mathbf{r}}_A)_{x'y'z'} + \boldsymbol{\omega}_p \times (\dot{\mathbf{r}}_A)_{x'y'z'}] + \dot{\boldsymbol{\omega}}_p \times \mathbf{r}_A + \boldsymbol{\omega}_p \times \dot{\mathbf{r}}_A \\ &= [\mathbf{0} + \mathbf{0}] + 2\mathbf{k} \times 2\mathbf{i} + 5\mathbf{k} \times 10\mathbf{j} = \{-50\mathbf{i} + 4\mathbf{j}\} \text{ m/s}^2 \end{aligned}$$

Motion of C with Respect to A . Here $\mathbf{r}_{C/A}$ changes direction relative to x, y, z , and so to find its time derivatives use a set of x'', y'', z'' axes that rotate at $\boldsymbol{\Omega}'' = \boldsymbol{\Omega}_{xyz} = \boldsymbol{\omega}_M$. Thus,

$$\boldsymbol{\Omega}_{xyz} = \boldsymbol{\omega}_M = \{3\mathbf{i}\} \text{ rad/s } (\boldsymbol{\Omega}_{xyz} \text{ does not change direction relative to } x, y, z.)$$

$$\dot{\boldsymbol{\Omega}}_{xyz} = \dot{\boldsymbol{\omega}}_M = \{1\mathbf{i}\} \text{ rad/s}^2$$

$$\mathbf{r}_{C/A} = \{-0.25\mathbf{k}\} \text{ m}$$

$$\begin{aligned} (\mathbf{v}_{C/A})_{xyz} &= (\dot{\mathbf{r}}_{C/A})_{xyz} = (\dot{\mathbf{r}}_{C/A})_{x''y''z''} + \boldsymbol{\omega}_M \times \mathbf{r}_{C/A} \\ &= -3\mathbf{k} + [3\mathbf{i} \times (-0.25\mathbf{k})] = \{0.75\mathbf{j} - 3\mathbf{k}\} \text{ m/s} \end{aligned}$$

$$\begin{aligned} (\mathbf{a}_{C/A})_{xyz} &= (\ddot{\mathbf{r}}_{C/A})_{xyz} = [(\ddot{\mathbf{r}}_{C/A})_{x''y''z''} + \boldsymbol{\omega}_M \times (\dot{\mathbf{r}}_{C/A})_{x''y''z''}] + \dot{\boldsymbol{\omega}}_M \times \mathbf{r}_{C/A} + \boldsymbol{\omega}_M \times (\dot{\mathbf{r}}_{C/A})_{xyz} \\ &= [-2\mathbf{k} + 3\mathbf{i} \times (-3\mathbf{k})] + (1\mathbf{i}) \times (-0.25\mathbf{k}) + (3\mathbf{i}) \times (0.75\mathbf{j} - 3\mathbf{k}) \\ &= \{18.25\mathbf{j} + 0.25\mathbf{k}\} \text{ m/s}^2 \end{aligned}$$

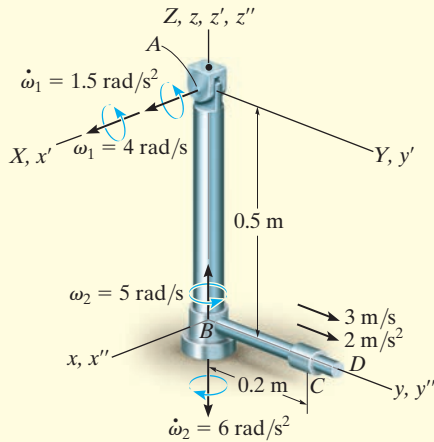
Motion of C .

$$\begin{aligned} \mathbf{v}_C &= \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{C/A} + (\mathbf{v}_{C/A})_{xyz} \\ &= 10\mathbf{j} + [5\mathbf{k} \times (-0.25\mathbf{k})] + (0.75\mathbf{j} - 3\mathbf{k}) \\ &= \{10.75\mathbf{j} - 3\mathbf{k}\} \text{ m/s} \end{aligned}$$

Ans.

$$\begin{aligned} \mathbf{a}_C &= \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/A})_{xyz} + (\mathbf{a}_{C/A})_{xyz} \\ &= (-50\mathbf{i} + 4\mathbf{j}) + [2\mathbf{k} \times (-0.25\mathbf{k})] + 5\mathbf{k} \times [5\mathbf{k} \times (-0.25\mathbf{k})] \\ &\quad + 2[5\mathbf{k} \times (0.75\mathbf{j} - 3\mathbf{k})] + (18.25\mathbf{j} + 0.25\mathbf{k}) \\ &= \{-57.5\mathbf{i} + 22.25\mathbf{j} + 0.25\mathbf{k}\} \text{ m/s}^2 \end{aligned}$$

Ans.

EXAMPLE 20.5**Fig. 20-13**

The pendulum shown in Fig. 20-13 consists of two rods; AB is pin supported at A and swings only in the Y - Z plane, whereas a bearing at B allows the attached rod BD to spin about rod AB . At a given instant, the rods have the angular motions shown. Also, a collar C , located 0.2 m from B , has a velocity of 3 m/s and an acceleration of 2 m/s² along the rod. Determine the velocity and acceleration of the collar at this instant.

SOLUTION I

Coordinate Axes. The origin of the fixed X, Y, Z frame will be placed at A . Motion of the collar is conveniently observed from B , so the origin of the x, y, z frame is located at this point. We will choose $\mathbf{\Omega} = \boldsymbol{\omega}_1$ and $\mathbf{\Omega}_{xyz} = \boldsymbol{\omega}_2$.

Kinematic Equations.

$$\mathbf{v}_C = \mathbf{v}_B + \mathbf{\Omega} \times \mathbf{r}_{C/B} + (\mathbf{v}_{C/B})_{xyz}$$

$$\mathbf{a}_C = \mathbf{a}_B + \dot{\mathbf{\Omega}} \times \mathbf{r}_{C/B} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_{C/B}) + 2\mathbf{\Omega} \times (\mathbf{v}_{C/B})_{xyz} + (\mathbf{a}_{C/B})_{xyz}$$

Motion of B . To find the time derivatives of $\mathbf{r}_B$ let the x', y', z' axes rotate with $\mathbf{\Omega}' = \boldsymbol{\omega}_1$. Then

$$\mathbf{\Omega}' = \boldsymbol{\omega}_1 = \{4\mathbf{i}\} \text{ rad/s} \quad \dot{\mathbf{\Omega}}' = \dot{\boldsymbol{\omega}}_1 = \{1.5\mathbf{i}\} \text{ rad/s}^2$$

$$\mathbf{r}_B = \{-0.5\mathbf{k}\} \text{ m}$$

$$\mathbf{v}_B = \dot{\mathbf{r}}_B = (\dot{\mathbf{r}}_B)_{x'y'z'} + \boldsymbol{\omega}_1 \times \mathbf{r}_B = \mathbf{0} + 4\mathbf{i} \times (-0.5\mathbf{k}) = \{2\mathbf{j}\} \text{ m/s}$$

$$\begin{aligned} \mathbf{a}_B = \ddot{\mathbf{r}}_B &= [(\ddot{\mathbf{r}}_B)_{x'y'z'} + \boldsymbol{\omega}_1 \times (\dot{\mathbf{r}}_B)_{x'y'z'}] + \dot{\boldsymbol{\omega}}_1 \times \mathbf{r}_B + \boldsymbol{\omega}_1 \times \dot{\mathbf{r}}_B \\ &= [\mathbf{0} + \mathbf{0}] + 1.5\mathbf{i} \times (-0.5\mathbf{k}) + 4\mathbf{i} \times 2\mathbf{j} = \{0.75\mathbf{j} + 8\mathbf{k}\} \text{ m/s}^2 \end{aligned}$$

Motion of C with Respect to B . To find the time derivatives of $\mathbf{r}_{C/B}$ relative to x, y, z , let the x'', y'', z'' axes rotate with $\mathbf{\Omega}_{xyz} = \boldsymbol{\omega}_2$. Then

$$\mathbf{\Omega}_{xyz} = \boldsymbol{\omega}_2 = \{5\mathbf{k}\} \text{ rad/s} \quad \dot{\mathbf{\Omega}}_{xyz} = \dot{\boldsymbol{\omega}}_2 = \{-6\mathbf{k}\} \text{ rad/s}^2$$

$$\mathbf{r}_{C/B} = \{0.2\mathbf{j}\} \text{ m}$$

$$(\mathbf{v}_{C/B})_{xyz} = (\dot{\mathbf{r}}_{C/B})_{xyz} = (\dot{\mathbf{r}}_{C/B})_{x''y''z''} + \boldsymbol{\omega}_2 \times \mathbf{r}_{C/B} = 3\mathbf{j} + 5\mathbf{k} \times 0.2\mathbf{j} = \{-1\mathbf{i} + 3\mathbf{j}\} \text{ m/s}$$

$$\begin{aligned} (\mathbf{a}_{C/B})_{xyz} &= (\ddot{\mathbf{r}}_{C/B})_{xyz} = [(\ddot{\mathbf{r}}_{C/B})_{x''y''z''} + \boldsymbol{\omega}_2 \times (\dot{\mathbf{r}}_{C/B})_{x''y''z''}] + \dot{\boldsymbol{\omega}}_2 \times \mathbf{r}_{C/B} + \boldsymbol{\omega}_2 \times (\dot{\mathbf{r}}_{C/B})_{xyz} \\ &= (2\mathbf{j} + 5\mathbf{k} \times 3\mathbf{j}) + (-6\mathbf{k} \times 0.2\mathbf{j}) + [5\mathbf{k} \times (-1\mathbf{i} + 3\mathbf{j})] \\ &= \{-28.8\mathbf{i} - 3\mathbf{j}\} \text{ m/s}^2 \end{aligned}$$

Motion of C .

$$\begin{aligned} \mathbf{v}_C &= \mathbf{v}_B + \mathbf{\Omega} \times \mathbf{r}_{C/B} + (\mathbf{v}_{C/B})_{xyz} = 2\mathbf{j} + 4\mathbf{i} \times 0.2\mathbf{j} + (-1\mathbf{i} + 3\mathbf{j}) \\ &= \{-1\mathbf{i} + 5\mathbf{j} + 0.8\mathbf{k}\} \text{ m/s} \end{aligned}$$

Ans.

$$\begin{aligned} \mathbf{a}_C &= \mathbf{a}_B + \dot{\mathbf{\Omega}} \times \mathbf{r}_{C/B} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_{C/B}) + 2\mathbf{\Omega} \times (\mathbf{v}_{C/B})_{xyz} + (\mathbf{a}_{C/B})_{xyz} \\ &= (0.75\mathbf{j} + 8\mathbf{k}) + (1.5\mathbf{i} \times 0.2\mathbf{j}) + [4\mathbf{i} \times (4\mathbf{i} \times 0.2\mathbf{j})] \\ &\quad + 2[4\mathbf{i} \times (-1\mathbf{i} + 3\mathbf{j})] + (-28.8\mathbf{i} - 3\mathbf{j}) \\ &= \{-28.8\mathbf{i} - 5.45\mathbf{j} + 32.3\mathbf{k}\} \text{ m/s}^2 \end{aligned}$$

Ans.

SOLUTION II

Coordinate Axes. Here we will let the x, y, z axes rotate at

$$\boldsymbol{\Omega} = \boldsymbol{\omega}_1 + \boldsymbol{\omega}_2 = \{4\mathbf{i} + 5\mathbf{k}\} \text{ rad/s}$$

Then $\boldsymbol{\Omega}_{xyz} = \mathbf{0}$.

Motion of B. From the constraints of the problem $\boldsymbol{\omega}_1$ does not change direction relative to X, Y, Z ; however, the direction of $\boldsymbol{\omega}_2$ is changed by $\boldsymbol{\omega}_1$. Thus, to obtain $\dot{\boldsymbol{\Omega}}$ consider x', y', z' axes coincident with the X, Y, Z axes at A , so that $\boldsymbol{\Omega}' = \boldsymbol{\omega}_1$. Then taking the derivative of the components of $\boldsymbol{\Omega}$,

$$\begin{aligned}\dot{\boldsymbol{\Omega}} &= \dot{\boldsymbol{\omega}}_1 + \dot{\boldsymbol{\omega}}_2 = [(\dot{\boldsymbol{\omega}}_1)_{x'y'z'} + \boldsymbol{\omega}_1 \times \boldsymbol{\omega}_1] + [(\dot{\boldsymbol{\omega}}_2)_{x'y'z'} + \boldsymbol{\omega}_1 \times \boldsymbol{\omega}_2] \\ &= [1.5\mathbf{i} + \mathbf{0}] + [-6\mathbf{k} + 4\mathbf{i} \times 5\mathbf{k}] = \{1.5\mathbf{i} - 20\mathbf{j} - 6\mathbf{k}\} \text{ rad/s}^2\end{aligned}$$

Also, $\boldsymbol{\omega}_1$ changes the direction of $\mathbf{r}_B$ so that the time derivatives of $\mathbf{r}_B$ can be found using the primed axes defined above. Hence,

$$\begin{aligned}\mathbf{v}_B &= \dot{\mathbf{r}}_B = (\dot{\mathbf{r}}_B)_{x'y'z'} + \boldsymbol{\omega}_1 \times \mathbf{r}_B \\ &= \mathbf{0} + 4\mathbf{i} \times (-0.5\mathbf{k}) = \{2\mathbf{j}\} \text{ m/s}\end{aligned}$$

$$\begin{aligned}\mathbf{a}_B &= \ddot{\mathbf{r}}_B = [(\ddot{\mathbf{r}}_B)_{x'y'z'} + \boldsymbol{\omega}_1 \times (\dot{\mathbf{r}}_B)_{x'y'z'}] + \dot{\boldsymbol{\omega}}_1 \times \mathbf{r}_B + \boldsymbol{\omega}_1 \times \dot{\mathbf{r}}_B \\ &= [\mathbf{0} + \mathbf{0}] + 1.5\mathbf{i} \times (-0.5\mathbf{k}) + 4\mathbf{i} \times 2\mathbf{j} = \{0.75\mathbf{j} + 8\mathbf{k}\} \text{ m/s}^2\end{aligned}$$

Motion of C with Respect to B.

$$\begin{aligned}\boldsymbol{\Omega}_{xyz} &= \mathbf{0} \\ \dot{\boldsymbol{\Omega}}_{xyz} &= \mathbf{0} \\ \mathbf{r}_{C/B} &= \{0.2\mathbf{j}\} \text{ m} \\ (\mathbf{v}_{C/B})_{xyz} &= \{3\mathbf{j}\} \text{ m/s} \\ (\mathbf{a}_{C/B})_{xyz} &= \{2\mathbf{j}\} \text{ m/s}^2\end{aligned}$$

Motion of C.

$$\begin{aligned}\mathbf{v}_C &= \mathbf{v}_B + \boldsymbol{\Omega} \times \mathbf{r}_{C/B} + (\mathbf{v}_{C/B})_{xyz} \\ &= 2\mathbf{j} + [(4\mathbf{i} + 5\mathbf{k}) \times (0.2\mathbf{j})] + 3\mathbf{j} \\ &= \{-1\mathbf{i} + 5\mathbf{j} + 0.8\mathbf{k}\} \text{ m/s} \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\mathbf{a}_C &= \mathbf{a}_B + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{C/B} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{C/B}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{C/B})_{xyz} + (\mathbf{a}_{C/B})_{xyz} \\ &= (0.75\mathbf{j} + 8\mathbf{k}) + [(1.5\mathbf{i} - 20\mathbf{j} - 6\mathbf{k}) \times (0.2\mathbf{j})] \\ &\quad + (4\mathbf{i} + 5\mathbf{k}) \times [(4\mathbf{i} + 5\mathbf{k}) \times 0.2\mathbf{j}] + 2[(4\mathbf{i} + 5\mathbf{k}) \times 3\mathbf{j}] + 2\mathbf{j} \\ &= \{-28.8\mathbf{i} - 5.45\mathbf{j} + 32.3\mathbf{k}\} \text{ m/s}^2 \quad \text{Ans.}\end{aligned}$$

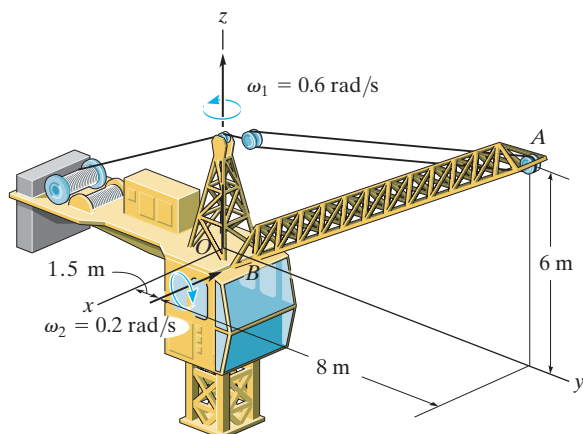
PROBLEMS

20–37. Solve Example 20.5 such that the x, y, z axes move with curvilinear translation, $\mathbf{\Omega} = \mathbf{0}$ in which case the collar appears to have both an angular velocity $\mathbf{\Omega}_{xyz} = \boldsymbol{\omega}_1 + \boldsymbol{\omega}_2$ and radial motion.

20–38. Solve Example 20.5 by fixing x, y, z axes to rod BD so that $\mathbf{\Omega} = \boldsymbol{\omega}_1 + \boldsymbol{\omega}_2$. In this case the collar appears only to move radially outward along BD ; hence $\mathbf{\Omega}_{xyz} = \mathbf{0}$.

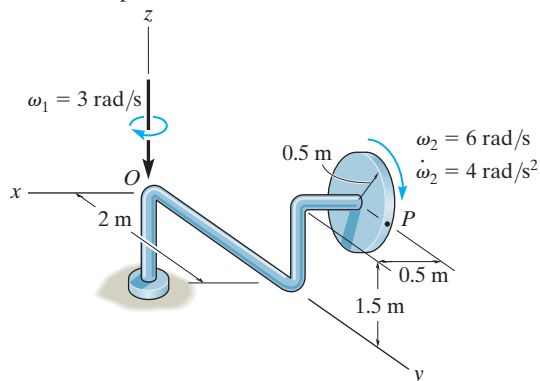
20–39. The crane rotates about the z axis with a constant rate $\omega_1 = 0.6$ rad/s, while the boom rotates downward with a constant rate $\omega_2 = 0.2$ rad/s. Determine the velocity and acceleration of point A located at the end of the boom at the instant shown.

***20–40.** The crane rotates about the z axis with a rate of $\omega_1 = 0.6$ rad/s, which is increasing at $\dot{\omega}_1 = 0.6$ rad/s². Also, the boom rotates downward at $\omega_2 = 0.2$ rad/s, which is increasing at $\dot{\omega}_2 = 0.3$ rad/s². Determine the velocity and acceleration of point A located at the end of the boom at the instant shown.



Probs. 20–39/40

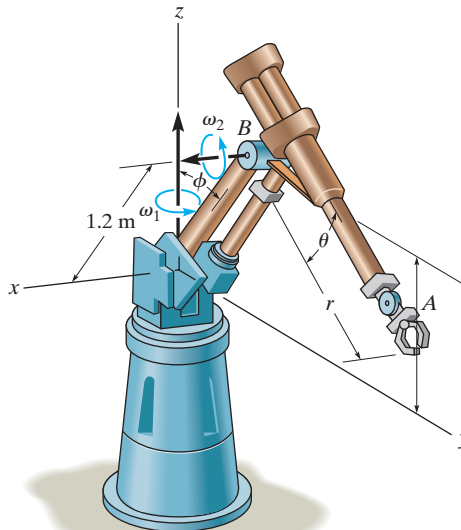
20–41. At the given instant, the rod is turning about the z axis with a constant angular velocity $\omega_1 = 3$ rad/s. At this same instant, the disk is spinning at $\omega_2 = 6$ rad/s when $\dot{\omega}_2 = 4$ rad/s², both measured *relative* to the rod. Determine the velocity and acceleration of point P on the disk at this instant.



Prob. 20–41

20–42. At the instant shown, the industrial manipulator is rotating about the z axis at $\omega_1 = 5$ rad/s, and about joint B at $\omega_2 = 2$ rad/s. Determine the velocity and acceleration of the grip A at this instant, when $\phi = 30^\circ$, $\theta = 45^\circ$, and $r = 1.6$ m.

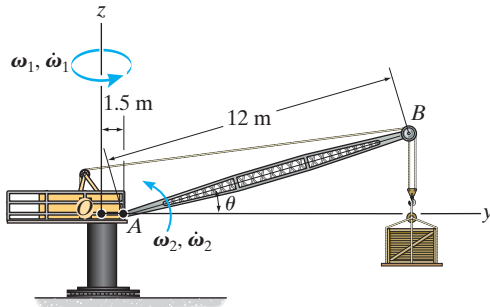
20–43. At the instant shown, the industrial manipulator is rotating about the z axis at $\omega_1 = 5$ rad/s, and $\dot{\omega}_1 = 2$ rad/s², and about joint B at $\omega_2 = 2$ rad/s and $\dot{\omega}_2 = 3$ rad/s². Determine the velocity and acceleration of the grip A at this instant, when $\phi = 30^\circ$, $\theta = 45^\circ$, and $r = 1.6$ m.



Probs. 20–42/43

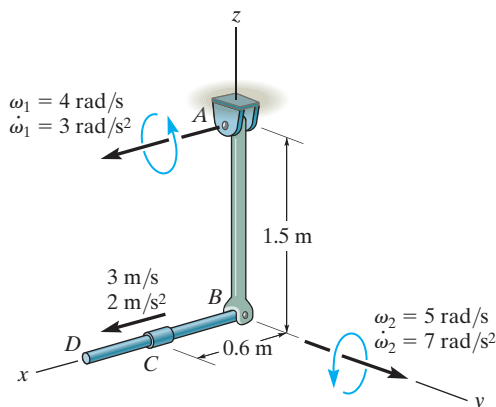
***20–44.** At the instant $\theta = 30^\circ$, the frame of the crane and the boom AB rotate with a constant angular velocity of $\omega_1 = 1.5 \text{ rad/s}$ and $\omega_2 = 0.5 \text{ rad/s}$, respectively. Determine the velocity and acceleration of point B at this instant.

20–45. At the instant $\theta = 30^\circ$, the frame of the crane is rotating with an angular velocity of $\omega_1 = 1.5 \text{ rad/s}$ and angular acceleration of $\dot{\omega}_1 = 0.5 \text{ rad/s}^2$, while the boom AB rotates with an angular velocity of $\omega_2 = 0.5 \text{ rad/s}$ and angular acceleration of $\dot{\omega}_2 = 0.25 \text{ rad/s}^2$. Determine the velocity and acceleration of point B at this instant.



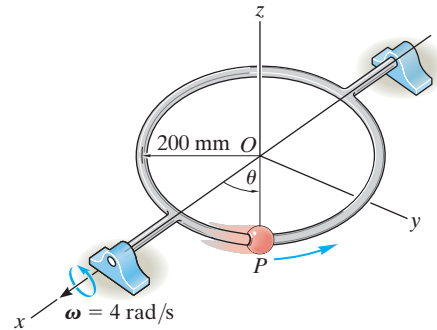
Probs. 20–44/45

20–46. At the instant shown, the arm AB is rotating about the fixed pin A with an angular velocity $\omega_1 = 4 \text{ rad/s}$ and angular acceleration $\dot{\omega}_1 = 3 \text{ rad/s}^2$. At this same instant, rod BD is rotating relative to rod AB with an angular velocity $\omega_2 = 5 \text{ rad/s}$, which is increasing at $\dot{\omega}_2 = 7 \text{ rad/s}^2$. Also, the collar C is moving along rod BD with a velocity of 3 m/s and an acceleration of 2 m/s^2 , both measured relative to the rod. Determine the velocity and acceleration of the collar at this instant.



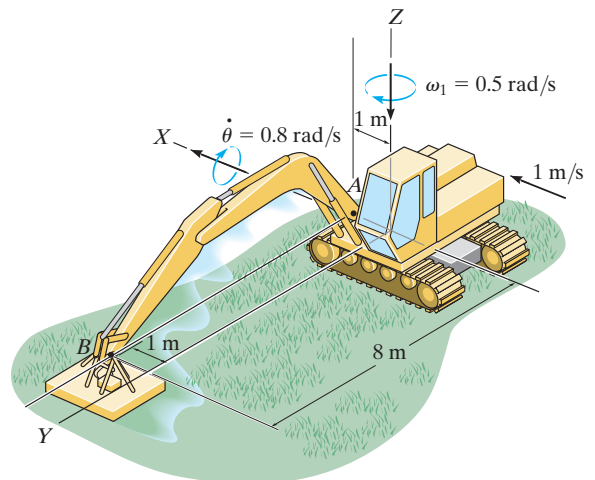
Prob. 20–46

20–47. The particle P slides around the circular hoop with a constant angular velocity of $\dot{\theta} = 6 \text{ rad/s}$, while the hoop rotates about the x axis at a constant rate of $\omega = 4 \text{ rad/s}$. If at the instant shown the hoop is in the x - y plane and the angle $\theta = 45^\circ$, determine the velocity and acceleration of the particle at this instant.



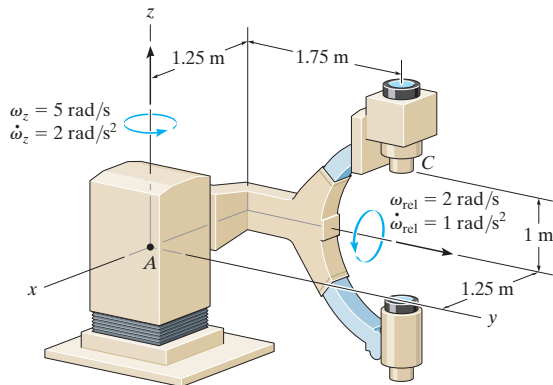
Prob. 20–47

***20–48.** At the instant shown, the frame of the brush cutter is traveling forward in the x direction with a constant velocity of 1 m/s , and the cab is rotating about the vertical axis with a constant angular velocity of $\omega_1 = 0.5 \text{ rad/s}$. At the same instant the boom AB has an angular velocity of $\dot{\theta} = 0.8 \text{ rad/s}$ in the direction shown. Determine the velocity and acceleration of point B at the connection to the mower at this instant.



Prob. 20–48

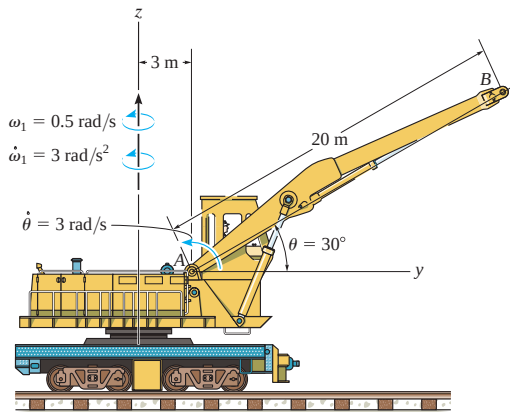
20-49. During the instant shown the frame of the X-ray camera is rotating about the vertical axis at $\omega_z = 5 \text{ rad/s}$ and $\dot{\omega}_z = 2 \text{ rad/s}^2$. Relative to the frame the arm is rotating at $\omega_{\text{rel}} = 2 \text{ rad/s}$ and $\dot{\omega}_{\text{rel}} = 1 \text{ rad/s}^2$. Determine the velocity and acceleration of the center of the camera C at this instant.



Prob. 20-49

20-50. The boom AB of the locomotive crane is rotating about the z axis with an angular velocity $\omega_1 = 0.5 \text{ rad/s}$, which is increasing at $\dot{\omega}_1 = 3 \text{ rad/s}^2$. At this same instant, $\theta = 30^\circ$ and the boom is rotating upward at a constant rate of $\dot{\theta} = 3 \text{ rad/s}$. Determine the velocity and acceleration of the tip B of the boom at this instant.

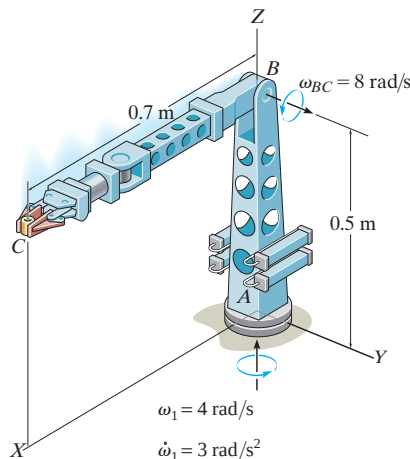
20-51. The locomotive crane is traveling to the right at 2 m/s and has an acceleration of 1.5 m/s^2 , while the boom is rotating about the z axis with an angular velocity $\omega_1 = 0.5 \text{ rad/s}$, which is increasing at $\dot{\omega}_1 = 3 \text{ rad/s}^2$. At this same instant, $\theta = 30^\circ$ and the boom is rotating upward at a constant rate $\dot{\theta} = 3 \text{ rad/s}$. Determine the velocity and acceleration of the tip B of the boom at this instant.



Probs. 20-50/51

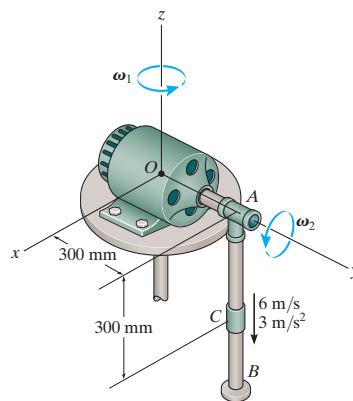
***20-52.** At the instant shown, the base of the robotic arm is turning about the z axis with an angular velocity of $\omega_1 = 4 \text{ rad/s}$, which is increasing at $\dot{\omega}_1 = 3 \text{ rad/s}^2$. Also, the boom segment BC is rotating at a constant rate of $\omega_{BC} = 8 \text{ rad/s}$. Determine the velocity and acceleration of the part C held in its grip at this instant.

20-53. At the instant shown, the base of the robotic arm is turning about the z axis with an angular velocity of $\omega_1 = 4 \text{ rad/s}$, which is increasing at $\dot{\omega}_1 = 3 \text{ rad/s}^2$. Also, the boom segment BC is rotating at a constant rate of $\omega_{BC} = 8 \text{ rad/s}$, which is increasing at $\dot{\omega}_{BC} = 2 \text{ rad/s}^2$. Determine the velocity and acceleration of the part C held in its grip at this instant.



Probs. 20-52/53

20-54. The motor rotates about the z axis with a constant angular velocity of $\omega_1 = 3 \text{ rad/s}$. Simultaneously, shaft OA rotates with a constant angular velocity of $\omega_2 = 6 \text{ rad/s}$. Also, collar C slides along rod AB with a velocity and acceleration of 6 m/s and 3 m/s^2 . Determine the velocity and acceleration of collar C at the instant shown.



Prob. 20-54

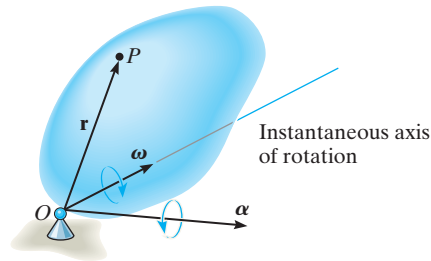
CHAPTER REVIEW

Rotation About a Fixed Point

When a body rotates about a fixed point O , then points on the body follow a path that lies on the surface of a sphere centered at O .

Since the angular acceleration is a time rate of change in the angular velocity, then it is necessary to account for both the magnitude and directional changes of $\boldsymbol{\omega}$ when finding its time derivative. To do this, the angular velocity is often specified in terms of its component motions, such that the direction of some of these components will remain constant relative to rotating x, y, z axes. If this is the case, then the time derivative relative to the fixed axis can be determined using $\dot{\mathbf{A}} = (\dot{\mathbf{A}})_{xyz} + \boldsymbol{\Omega} \times \mathbf{A}$.

Once $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ are known, the velocity and acceleration of any point P in the body can then be determined.



$$\mathbf{v}_P = \boldsymbol{\omega} \times \mathbf{r}$$

$$\mathbf{a}_P = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

General Motion

If the body undergoes general motion, then the motion of a point B on the body can be related to the motion of another point A using a relative motion analysis, with translating axes attached to A .

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A})$$

Relative Motion Analysis Using Translating and Rotating Axes

The motion of two points A and B on a body, a series of connected bodies, or each point located on two different paths, can be related using a relative motion analysis with rotating and translating axes at A .

When applying the equations, to find $\mathbf{v}_B$ and $\mathbf{a}_B$, it is important to account for both the magnitude and directional changes of $\mathbf{r}_A$, $\mathbf{r}_{B/A}$, $\boldsymbol{\Omega}$, and $\boldsymbol{\Omega}_{xyz}$ when taking their time derivatives to find $\mathbf{v}_A$, $\mathbf{a}_A$, $(\mathbf{v}_{B/A})_{xyz}$, $(\mathbf{a}_{B/A})_{xyz}$, $\dot{\boldsymbol{\Omega}}$, and $\dot{\boldsymbol{\Omega}}_{xyz}$. To do this properly, one must use Eq. 20–6.

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xyz}$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xyz} + (\mathbf{a}_{B/A})_{xyz}$$

Chapter 21



(© Derek Watt/Alamy)

The forces acting on each of these motorcycles can be determined using the equations of motion as discussed in this chapter.

Three-Dimensional Kinetics of a Rigid Body

CHAPTER OBJECTIVES

- To introduce the methods for finding the moments of inertia and products of inertia of a body about various axes.
- To show how to apply the principles of work and energy and linear and angular impulse and momentum to a rigid body having three-dimensional motion.
- To develop and apply the equations of motion in three dimensions.
- To study gyroscopic and torque-free motion.



Video Solutions are available for selected questions in this chapter.

*21.1 Moments and Products of Inertia

When studying the planar kinetics of a body, it was necessary to introduce the moment of inertia I_G , which was computed about an axis perpendicular to the plane of motion and passing through the body's mass center G . For the kinetic analysis of three-dimensional motion it will sometimes be necessary to calculate six inertial quantities. These terms, called the moments and products of inertia, describe in a particular way the distribution of mass for a body relative to a given coordinate system that has a specified orientation and point of origin.

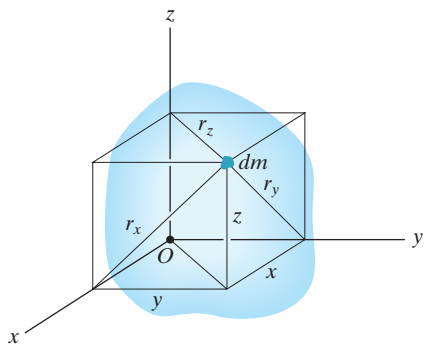


Fig. 21-1

Moment of Inertia. Consider the rigid body shown in Fig. 21-1. The *moment of inertia* for a differential element dm of the body about any one of the three coordinate axes is defined as the product of the mass of the element and the square of the shortest distance from the axis to the element. For example, as noted in the figure, $r_x = \sqrt{y^2 + z^2}$, so that the mass moment of inertia of the element about the x axis is

$$dI_{xx} = r_x^2 dm = (y^2 + z^2) dm$$

The moment of inertia I_{xx} for the body can be determined by integrating this expression over the entire mass of the body. Hence, for each of the axes, we can write

$$\begin{aligned} I_{xx} &= \int_m r_x^2 dm = \int_m (y^2 + z^2) dm \\ I_{yy} &= \int_m r_y^2 dm = \int_m (x^2 + z^2) dm \\ I_{zz} &= \int_m r_z^2 dm = \int_m (x^2 + y^2) dm \end{aligned} \quad (21-1)$$

Here it is seen that the moment of inertia is *always a positive quantity*, since it is the summation of the product of the mass dm , which is always positive, and the distances squared.

Product of Inertia. The *product of inertia* for a differential element dm with respect to a set of *two orthogonal planes* is defined as the product of the mass of the element and the perpendicular (or shortest) distances from the planes to the element. For example, this distance is x to the y - z plane and it is y to the x - z plane, Fig. 21-1. The product of inertia dI_{xy} for the element is therefore

$$dI_{xy} = xy dm$$

Note also that $dI_{yx} = dI_{xy}$. By integrating over the entire mass, the products of inertia of the body with respect to each combination of planes can be expressed as

$$\begin{aligned} I_{xy} &= I_{yx} = \int_m xy dm \\ I_{yz} &= I_{zy} = \int_m yz dm \\ I_{xz} &= I_{zx} = \int_m xz dm \end{aligned} \quad (21-2)$$

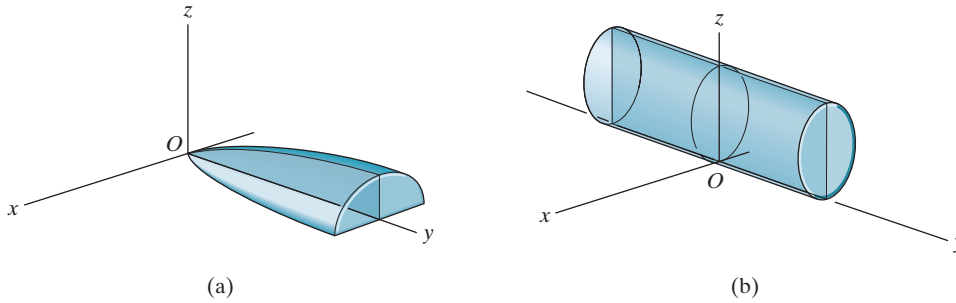


Fig. 21-2

Unlike the moment of inertia, which is always positive, the product of inertia may be positive, negative, or zero. The result depends on the algebraic signs of the two defining coordinates, which vary independently from one another. In particular, if either one or both of the orthogonal planes are *planes of symmetry* for the mass, the *product of inertia* with respect to these planes will be *zero*. In such cases, elements of mass occur in *pairs* located on each side of the plane of symmetry. On one side of the plane the product of inertia for the element will be positive, while on the other side the product of inertia of the corresponding element will be negative, the sum therefore yielding zero. Examples of this are shown in Fig. 21-2. In the first case, Fig. 21-2a, the y - z plane is a plane of symmetry, and hence $I_{xy} = I_{xz} = 0$. Calculation of I_{yz} will yield a *positive* result, since all elements of mass are located using only positive y and z coordinates. For the cylinder, with the coordinate axes located as shown in Fig. 21-2b, the x - z and y - z planes are both planes of symmetry. Thus, $I_{xy} = I_{yz} = I_{zx} = 0$.

Parallel-Axis and Parallel-Plane Theorems. The techniques of integration used to determine the moment of inertia of a body were described in Sec. 17.1. Also discussed were methods to determine the moment of inertia of a composite body, i.e., a body that is composed of simpler segments, as tabulated on the inside back cover. In both of these cases the *parallel-axis theorem* is often used for the calculations. This theorem, which was developed in Sec. 17.1, allows us to transfer the moment of inertia of a body from an axis passing through its mass center G to a parallel axis passing through some other point. If G has coordinates x_G, y_G, z_G defined with respect to the x, y, z axes, Fig. 21-3, then the parallel-axis equations used to calculate the moments of inertia about the x, y, z axes are

$$\begin{aligned} I_{xx} &= (I_{x'x'})_G + m(y_G^2 + z_G^2) \\ I_{yy} &= (I_{y'y'})_G + m(x_G^2 + z_G^2) \\ I_{zz} &= (I_{z'z'})_G + m(x_G^2 + y_G^2) \end{aligned}$$

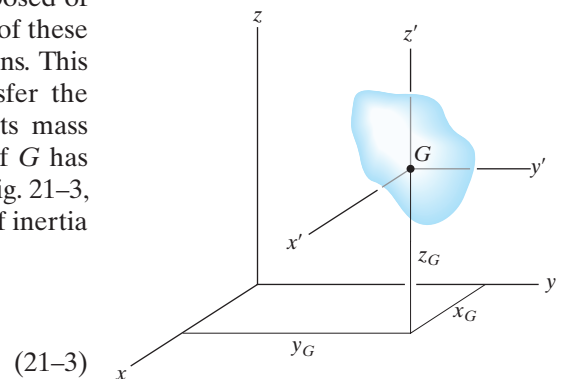


Fig. 21-3

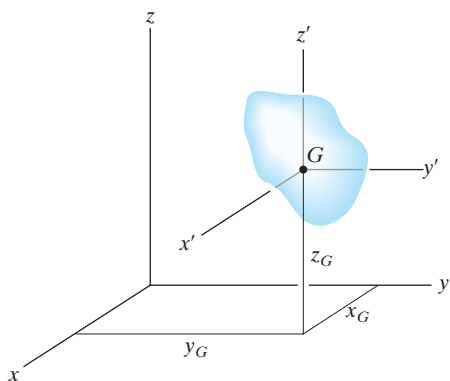


Fig. 21-3 (repeated)

The products of inertia of a composite body are computed in the same manner as the body's moments of inertia. Here, however, the *parallel-plane theorem* is important. This theorem is used to transfer the products of inertia of the body with respect to a set of three orthogonal planes passing through the body's mass center to a corresponding set of three parallel planes passing through some other point O . Defining the perpendicular distances between the planes as x_G , y_G , and z_G , Fig. 21-3, the parallel-plane equations can be written as

$$\begin{aligned} I_{xy} &= (I_{x'y'})_G + m x_G y_G \\ I_{yz} &= (I_{y'z'})_G + m y_G z_G \\ I_{zx} &= (I_{z'x'})_G + m z_G x_G \end{aligned} \quad (21-4)$$

The derivation of these formulas is similar to that given for the parallel-axis equation, Sec. 17.1.

Inertia Tensor. The inertial properties of a body are therefore completely characterized by nine terms, six of which are independent of one another. This set of terms is defined using Eqs. 21-1 and 21-2 and can be written as

$$\begin{pmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{pmatrix}$$

This array is called an *inertia tensor*.* It has a unique set of values for a body when it is determined for each location of the origin O and orientation of the coordinate axes.

In general, for point O we can specify a unique axes inclination for which the products of inertia for the body are zero when computed with respect to these axes. When this is done, the inertia tensor is said to be “diagonalized” and may be written in the simplified form

$$\begin{pmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{pmatrix}$$

Here $I_x = I_{xx}$, $I_y = I_{yy}$, and $I_z = I_{zz}$ are termed the *principal moments of inertia* for the body, which are computed with respect to the *principal axes of inertia*. Of these three principal moments of inertia, one will be a maximum and another a minimum of the body's moment of inertia.



The dynamics of the space shuttle while it orbits the earth can be predicted only if its moments and products of inertia are known relative to its mass center. (©Ablestock/Getty Images)

*The negative signs are here as a consequence of the development of angular momentum, Eqs. 21-10.

The mathematical determination of the directions of principal axes of inertia will not be discussed here (see Prob. 21–22). However, there are many cases in which the principal axes can be determined by inspection. From the previous discussion it was noted that if the coordinate axes are oriented such that *two* of the three orthogonal planes containing the axes are planes of *symmetry* for the body, then all the products of inertia for the body are zero with respect to these coordinate planes, and hence these coordinate axes are principal axes of inertia. For example, the x , y , z axes shown in Fig. 21–2*b* represent the principal axes of inertia for the cylinder at point O .

Moment of Inertia About an Arbitrary Axis. Consider the body shown in Fig. 21–4, where the nine elements of the inertia tensor have been determined with respect to the x , y , z axes having an origin at O . Here we wish to determine the moment of inertia of the body about the Oa axis, which has a direction defined by the unit vector $\mathbf{u}_a$. By definition $I_{Oa} = \int b^2 dm$, where b is the *perpendicular distance* from dm to Oa . If the position of dm is located using $\mathbf{r}$, then $b = r \sin \theta$, which represents the *magnitude* of the cross product $\mathbf{u}_a \times \mathbf{r}$. Hence, the moment of inertia can be expressed as

$$I_{Oa} = \int_m |(\mathbf{u}_a \times \mathbf{r})|^2 dm = \int_m (\mathbf{u}_a \times \mathbf{r}) \cdot (\mathbf{u}_a \times \mathbf{r}) dm$$

Provided $\mathbf{u}_a = u_x \mathbf{i} + u_y \mathbf{j} + u_z \mathbf{k}$ and $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$, then $\mathbf{u}_a \times \mathbf{r} = (u_y z - u_z y) \mathbf{i} + (u_z x - u_x z) \mathbf{j} + (u_x y - u_y x) \mathbf{k}$. After substituting and performing the dot-product operation, the moment of inertia is

$$\begin{aligned} I_{Oa} &= \int_m [(u_y z - u_z y)^2 + (u_z x - u_x z)^2 + (u_x y - u_y x)^2] dm \\ &= u_x^2 \int_m (y^2 + z^2) dm + u_y^2 \int_m (z^2 + x^2) dm + u_z^2 \int_m (x^2 + y^2) dm \\ &\quad - 2u_x u_y \int_m xy dm - 2u_y u_z \int_m yz dm - 2u_z u_x \int_m zx dm \end{aligned}$$

Recognizing the integrals to be the moments and products of inertia of the body, Eqs. 21–1 and 21–2, we have

$$I_{Oa} = I_{xx} u_x^2 + I_{yy} u_y^2 + I_{zz} u_z^2 - 2I_{xy} u_x u_y - 2I_{yz} u_y u_z - 2I_{zx} u_z u_x \quad (21-5)$$

Thus, if the inertia tensor is specified for the x , y , z axes, the moment of inertia of the body about the inclined Oa axis can be found. For the calculation, the direction cosines u_x , u_y , u_z of the axes must be determined. These terms specify the cosines of the coordinate direction angles α , β , γ made between the positive Oa axis and the positive x , y , z axes, respectively (see Appendix B).

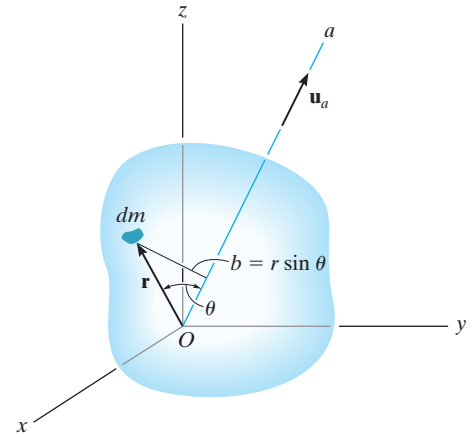
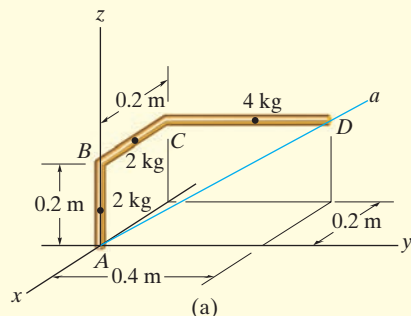


Fig. 21–4

EXAMPLE 21.1



Determine the moment of inertia of the bent rod shown in Fig. 21–5a about the Aa axis. The mass of each of the three segments is given in the figure.

SOLUTION

Before applying Eq. 21–5, it is first necessary to determine the moments and products of inertia of the rod with respect to the x , y , z axes. This is done using the formula for the moment of inertia of a slender rod, $I = \frac{1}{12}ml^2$, and the parallel-axis and parallel-plane theorems, Eqs. 21–3 and 21–4. Dividing the rod into three parts and locating the mass center of each segment, Fig. 21–5b, we have

$$I_{xx} = \left[\frac{1}{12}(2)(0.2)^2 + 2(0.1)^2 \right] + [0 + 2(0.2)^2] + \left[\frac{1}{12}(4)(0.4)^2 + 4((0.2)^2 + (0.2)^2) \right] = 0.480 \text{ kg} \cdot \text{m}^2$$

$$I_{yy} = \left[\frac{1}{12}(2)(0.2)^2 + 2(0.1)^2 \right] + \left[\frac{1}{12}(2)(0.2)^2 + 2((-0.1)^2 + (0.2)^2) \right] + [0 + 4((-0.2)^2 + (0.2)^2)] = 0.453 \text{ kg} \cdot \text{m}^2$$

$$I_{zz} = [0 + 0] + \left[\frac{1}{12}(2)(0.2)^2 + 2(-0.1)^2 \right] + \left[\frac{1}{12}(4)(0.4)^2 + 4((-0.2)^2 + (0.2)^2) \right] = 0.400 \text{ kg} \cdot \text{m}^2$$

$$I_{xy} = [0 + 0] + [0 + 0] + [0 + 4(-0.2)(0.2)] = -0.160 \text{ kg} \cdot \text{m}^2$$

$$I_{yz} = [0 + 0] + [0 + 0] + [0 + 4(0.2)(0.2)] = 0.160 \text{ kg} \cdot \text{m}^2$$

$$I_{zx} = [0 + 0] + [0 + 2(0.2)(-0.1)] + [0 + 4(0.2)(-0.2)] = -0.200 \text{ kg} \cdot \text{m}^2$$

The Aa axis is defined by the unit vector

$$\mathbf{u}_{Aa} = \frac{\mathbf{r}_D}{r_D} = \frac{-0.2\mathbf{i} + 0.4\mathbf{j} + 0.2\mathbf{k}}{\sqrt{(-0.2)^2 + (0.4)^2 + (0.2)^2}} = -0.408\mathbf{i} + 0.816\mathbf{j} + 0.408\mathbf{k}$$

Thus,

$$u_x = -0.408 \quad u_y = 0.816 \quad u_z = 0.408$$

Substituting these results into Eq. 21–5 yields

$$\begin{aligned} I_{Aa} &= I_{xx}u_x^2 + I_{yy}u_y^2 + I_{zz}u_z^2 - 2I_{xy}u_xu_y - 2I_{yz}u_yu_z - 2I_{zx}u_zu_x \\ &= 0.480(-0.408)^2 + (0.453)(0.816)^2 + 0.400(0.408)^2 \\ &\quad - 2(-0.160)(-0.408)(0.816) - 2(0.160)(0.816)(0.408) \\ &\quad - 2(-0.200)(0.408)(-0.408) \\ &= 0.169 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

Ans.

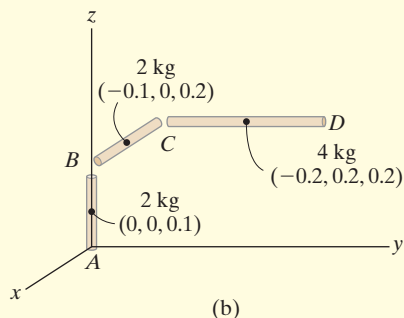
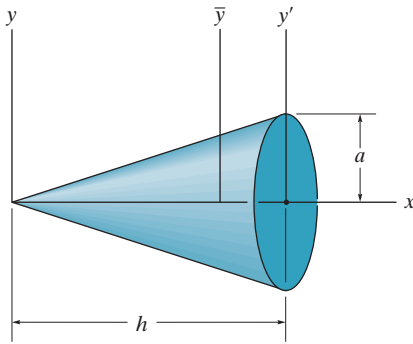


Fig. 21–5

PROBLEMS

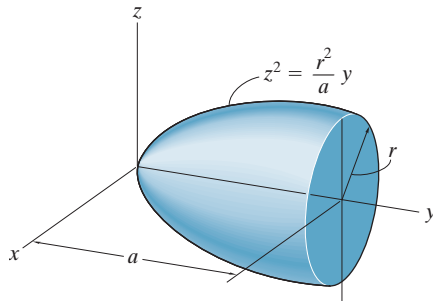
21-1. Show that the sum of the moments of inertia of a body, $I_{xx} + I_{yy} + I_{zz}$, is independent of the orientation of the x, y, z axes and thus depends only on the location of the origin.

21-2. Determine the moment of inertia of the cone with respect to a vertical $\bar{y}$ axis passing through the cone's center of mass. What is the moment of inertia about a parallel axis y' that passes through the diameter of the base of the cone? The cone has a mass m .



Prob. 21-2

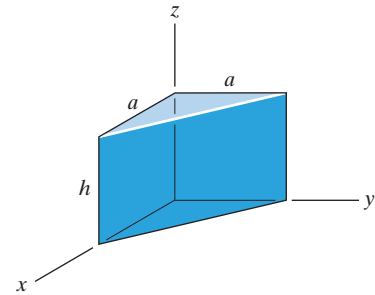
21-3. Determine the moments of inertia I_x and I_y of the paraboloid of revolution. The mass of the paraboloid is m .



Prob. 21-3

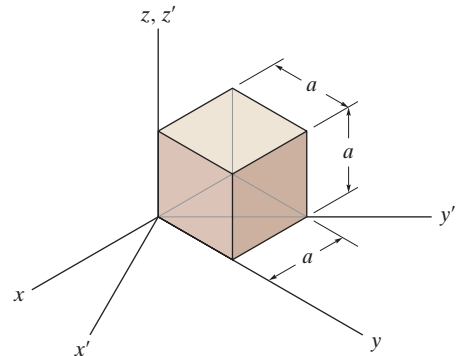
***21-4.** Determine by direct integration the product of inertia I_{yz} for the homogeneous prism. The density of the material is ρ . Express the result in terms of the total mass m of the prism.

21-5. Determine by direct integration the product of inertia I_{xy} for the homogeneous prism. The density of the material is ρ . Express the result in terms of the total mass m of the prism.



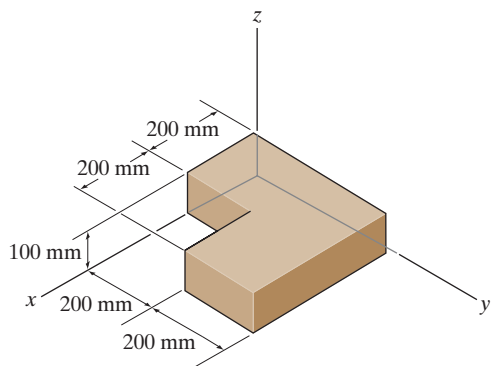
Probs. 21-4/5

21-6. Determine the elements of the inertia tensor for the cube with respect to the x, y, z coordinate system. The mass of the cube is m .



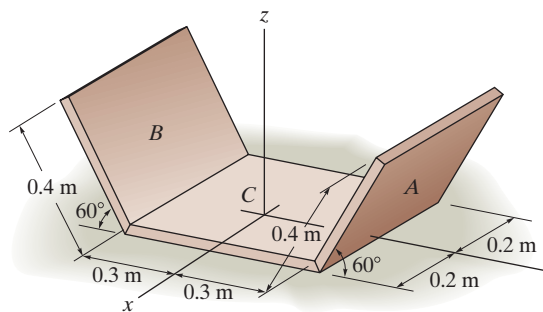
Prob. 21-6

21-7. Determine the products of inertia I_{xy} , I_{yz} , and I_{xz} of the homogeneous solid. The material has a density of 7.85 Mg/m^3 .



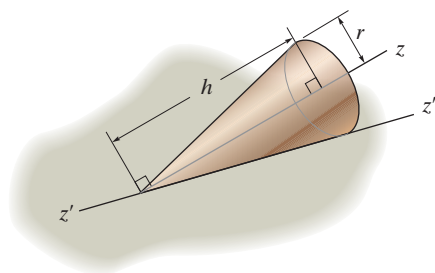
Prob. 21-7

***21-8.** The assembly consists of two square plates A and B which have a mass of 3 kg each and a rectangular plate C which has a mass of 4.5 kg . Determine the moments of inertia I_x , I_y and I_z .



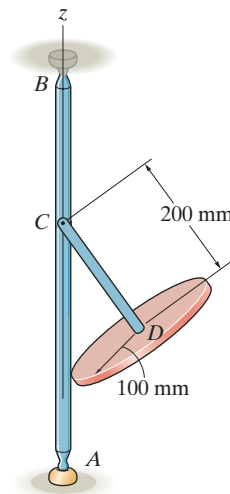
Prob. 21-8

21-9. Determine the moment of inertia of the cone about the z' axis. The mass of the cone is 15 kg , the height is $h = 1.5 \text{ m}$, and the radius is $r = 0.5 \text{ m}$.



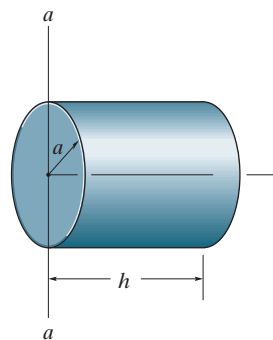
Prob. 21-9

21-10. Determine the moment of inertia about the z axis of the assembly which consists of the 1.5-kg rod CD and the 7-kg disk.



Prob. 21-10

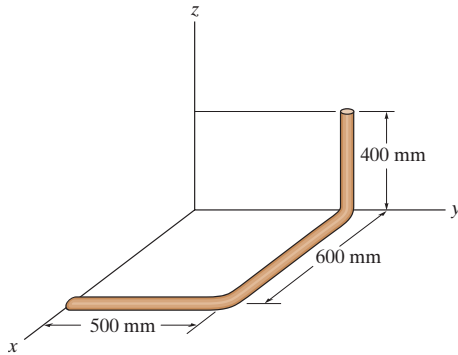
21-11. Determine the moment of inertia of the cylinder with respect to the $a-a$ axis of the cylinder. The cylinder has a mass m .



Prob. 21-11

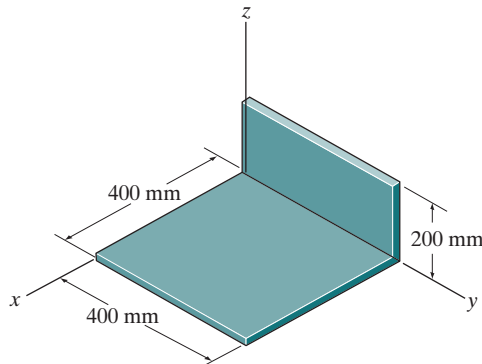
***21–12.** Determine the product of inertia I_{xy} for the bent rod. The rod has a mass per unit length of 2 kg/m .

21–13. Determine the moments of inertia I_{xx} , I_{yy} , I_{zz} for the bent rod. The rod has a mass per unit length of 2 kg/m .



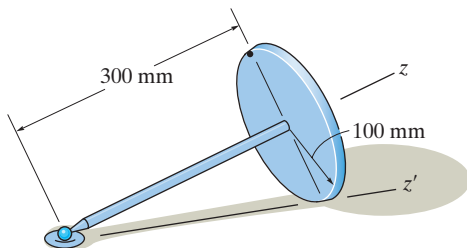
Probs. 21–12/13

21–14. Determine the products of inertia I_{xy} , I_{yz} , and I_{xz} , of the thin plate. The material has a mass per unit area of 50 kg/m^2 .



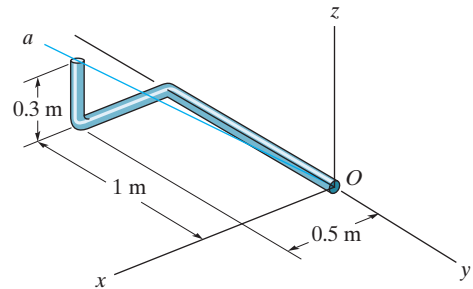
Prob. 21–14

21–15. Determine the moment of inertia of both the 1.5-kg rod and 4-kg disk about the z' axis.



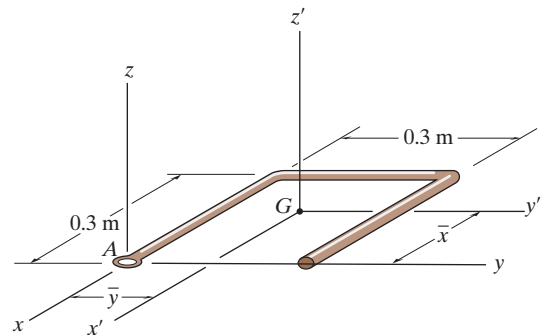
Prob. 21–15

***21–16.** The bent rod has a mass per unit length of 3 kg/m . Determine the moment of inertia of the rod about the O – a axis.



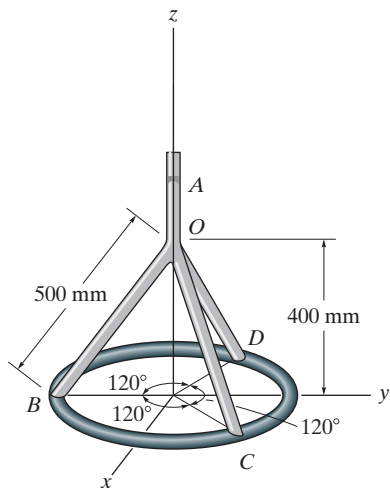
Prob. 21–16

21–17. The bent rod has a mass per unit length of 8.49 kg/m . Locate the center of gravity $G(\bar{x}, \bar{y})$ and determine the principal moments of inertia $I_{x'}$, $I_{y'}$, and $I_{z'}$ of the rod with respect to the x' , y' , z' axes.



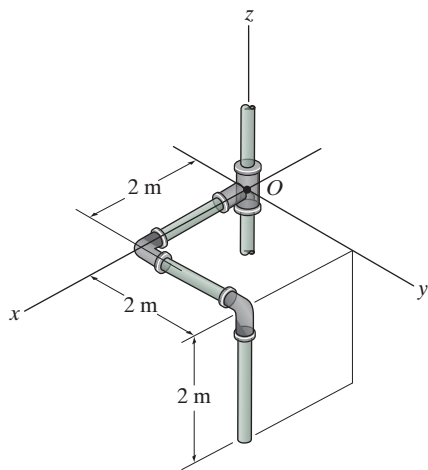
Prob. 21–17

21-18. Determine the moment of inertia of the rod-and-thin-ring assembly about the z axis. The rods and ring have a mass per unit length of 2 kg/m .



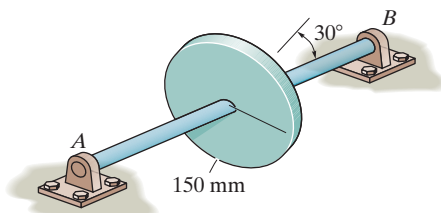
Prob. 21-18

21-19. The slender rod has a mass per unit length of 6 kg/m . Determine its moments and products of inertia with respect to the x , y , z axes.



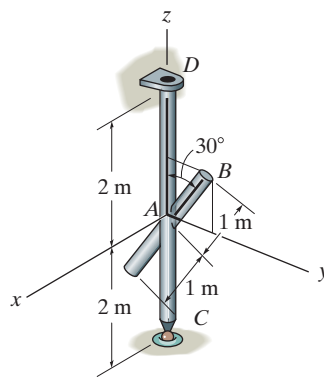
Prob. 21-19

***21-20.** Determine the moment of inertia of the disk about the axis of shaft AB . The disk has a mass of 15 kg .



Prob. 21-20

21-21. Determine the moments of inertia about the x , y , z axes of the rod assembly. The rods have a mass of 0.75 kg/m .



Prob. 21-21

21.2 Angular Momentum

In this section we will develop the necessary equations used to determine the angular momentum of a rigid body about an arbitrary point. These equations will provide a means for developing both the principle of impulse and momentum and the equations of rotational motion for a rigid body.

Consider the rigid body in Fig. 21–6, which has a mass m and center of mass at G . The X, Y, Z coordinate system represents an inertial frame of reference, and hence, its axes are fixed or translate with a constant velocity. The angular momentum as measured from this reference will be determined relative to the arbitrary point A . The position vectors $\mathbf{r}_A$ and $\boldsymbol{\rho}_A$ are drawn from the origin of coordinates to point A and from A to the i th particle of the body. If the particle's mass is m_i , the angular momentum about point A is

$$(\mathbf{H}_A)_i = \boldsymbol{\rho}_A \times m_i \mathbf{v}_i$$

where $\mathbf{v}_i$ represents the particle's velocity measured from the X, Y, Z coordinate system. If the body has an angular velocity $\boldsymbol{\omega}$ at the instant considered, $\mathbf{v}_i$ may be related to the velocity of A by applying Eq. 20–7, i.e.,

$$\mathbf{v}_i = \mathbf{v}_A + \boldsymbol{\omega} \times \boldsymbol{\rho}_A$$

Thus,

$$\begin{aligned} (\mathbf{H}_A)_i &= \boldsymbol{\rho}_A \times m_i(\mathbf{v}_A + \boldsymbol{\omega} \times \boldsymbol{\rho}_A) \\ &= (\boldsymbol{\rho}_A m_i) \times \mathbf{v}_A + \boldsymbol{\rho}_A \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) m_i \end{aligned}$$

Summing the moments of all the particles of the body requires an integration. Since $m_i \rightarrow dm$, we have

$$\mathbf{H}_A = \left(\int_m \boldsymbol{\rho}_A dm \right) \times \mathbf{v}_A + \int_m \boldsymbol{\rho}_A \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) dm \quad (21-6)$$

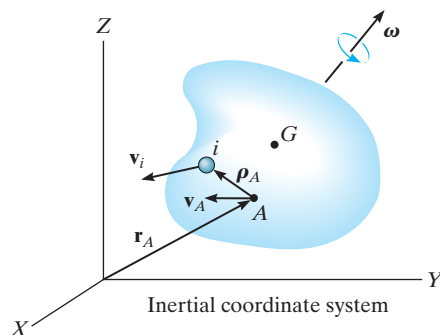


Fig. 21–6

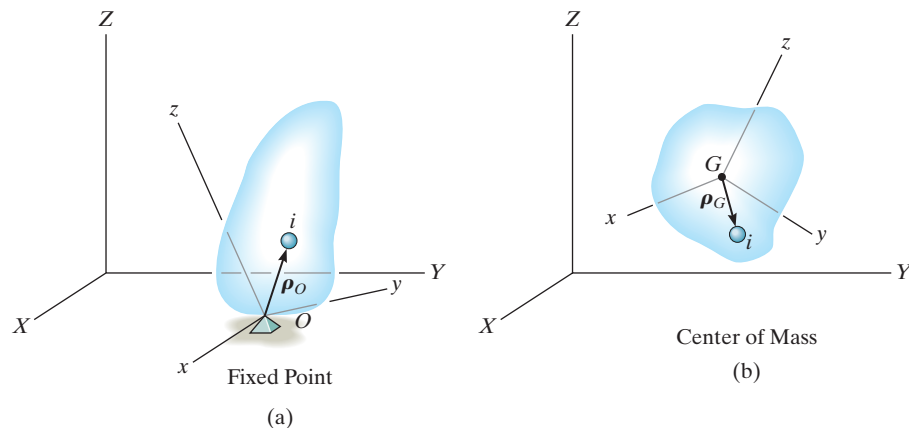


Fig. 21-7

Fixed Point O. If A becomes a *fixed point* O in the body, Fig. 21-7a, then $\mathbf{v}_A = \mathbf{0}$ and Eq. 21-6 reduces to

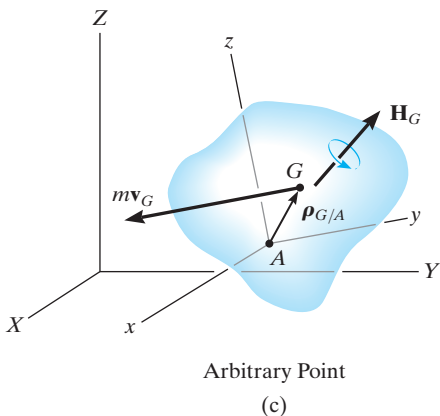
$$\mathbf{H}_O = \int_m \boldsymbol{\rho}_O \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_O) dm \quad (21-7)$$

Center of Mass G. If A is located at the *center of mass* G of the body, Fig. 21-7b, then $\int_m \boldsymbol{\rho}_A dm = \mathbf{0}$ and

$$\mathbf{H}_G = \int_m \boldsymbol{\rho}_G \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_G) dm \quad (21-8)$$

Arbitrary Point A. In general, A can be a point other than O or G , Fig. 21-7c, in which case Eq. 21-6 may nevertheless be simplified to the following form (see Prob. 21-23).

$$\mathbf{H}_A = \boldsymbol{\rho}_{G/A} \times m\mathbf{v}_G + \mathbf{H}_G \quad (21-9)$$



Here the angular momentum consists of two parts—the moment of the linear momentum $m\mathbf{v}_G$ of the body about point A added (vectorially) to the angular momentum $\mathbf{H}_G$. Equation 21-9 can also be used to determine the angular momentum of the body about a fixed point O . The results, of course, will be the same as those found using the more convenient Eq. 21-7.

Rectangular Components of H . To make practical use of Eqs. 21-7 through 21-9, the angular momentum must be expressed in terms of its scalar components. For this purpose, it is convenient to

choose a second set of x, y, z axes having an arbitrary orientation relative to the X, Y, Z axes, Fig. 21-7, and for a general formulation, note that Eqs. 21-7 and 21-8 are both of the form

$$\mathbf{H} = \int_m \boldsymbol{\rho} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}) dm$$

Expressing $\mathbf{H}$, $\boldsymbol{\rho}$, and $\boldsymbol{\omega}$ in terms of x, y, z components, we have

$$H_x \mathbf{i} + H_y \mathbf{j} + H_z \mathbf{k} = \int_m (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \times [(\omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}) \times (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})] dm$$

Expanding the cross products and combining terms yields

$$\begin{aligned} H_x \mathbf{i} + H_y \mathbf{j} + H_z \mathbf{k} = & \left[\omega_x \int_m (y^2 + z^2) dm - \omega_y \int_m xy dm - \omega_z \int_m xz dm \right] \mathbf{i} \\ & + \left[-\omega_x \int_m xy dm + \omega_y \int_m (x^2 + z^2) dm - \omega_z \int_m yz dm \right] \mathbf{j} \\ & + \left[-\omega_x \int_m zx dm - \omega_y \int_m yz dm + \omega_z \int_m (x^2 + y^2) dm \right] \mathbf{k} \end{aligned}$$

Equating the respective $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components and recognizing that the integrals represent the moments and products of inertia, we obtain

$$\begin{aligned} H_x &= I_{xx}\omega_x - I_{xy}\omega_y - I_{xz}\omega_z \\ H_y &= -I_{yx}\omega_x + I_{yy}\omega_y - I_{yz}\omega_z \\ H_z &= -I_{zx}\omega_x - I_{zy}\omega_y + I_{zz}\omega_z \end{aligned} \quad (21-10)$$

These equations can be simplified further if the x, y, z coordinate axes are oriented such that they become *principal axes of inertia* for the body at the point. When these axes are used, the products of inertia $I_{xy} = I_{yz} = I_{zx} = 0$, and if the principal moments of inertia about the x, y, z axes are represented as $I_x = I_{xx}$, $I_y = I_{yy}$, and $I_z = I_{zz}$, the three components of angular momentum become

$$H_x = I_x \omega_x \quad H_y = I_y \omega_y \quad H_z = I_z \omega_z \quad (21-11)$$



The motion of the astronaut is controlled by use of small directional jets attached to his or her space suit. The impulses these jets provide must be carefully specified in order to prevent tumbling and loss of orientation. (© NASA)

Principle of Impulse and Momentum. Now that the formulation of the angular momentum for a body has been developed, the *principle of impulse and momentum*, as discussed in Sec. 19.2, can be used to solve kinetic problems which involve *force, velocity, and time*. For this case, the following two vector equations are available:

$$m(\mathbf{v}_G)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{F} dt = m(\mathbf{v}_G)_2 \quad (21-12)$$

$$(\mathbf{H}_O)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{M}_O dt = (\mathbf{H}_O)_2 \quad (21-13)$$

In three dimensions each vector term can be represented by three scalar components, and therefore a total of *six scalar equations* can be written. Three equations relate the linear impulse and momentum in the x, y, z directions, and the other three equations relate the body's angular impulse and momentum about the x, y, z axes. Before applying Eqs. 21-12 and 21-13 to the solution of problems, the material in Secs. 19.2 and 19.3 should be reviewed.

21.3 Kinetic Energy

In order to apply the principle of work and energy to solve problems involving general rigid body motion, it is first necessary to formulate expressions for the kinetic energy of the body. To do this, consider the rigid body shown in Fig. 21-8, which has a mass m and center of mass at G . The kinetic energy of the i th particle of the body having a mass m_i and velocity $\mathbf{v}_i$, measured relative to the inertial X, Y, Z frame of reference, is

$$T_i = \frac{1}{2} m_i v_i^2 = \frac{1}{2} m_i (\mathbf{v}_i \cdot \mathbf{v}_i)$$

Provided the velocity of an arbitrary point A in the body is known, $\mathbf{v}_i$ can be related to $\mathbf{v}_A$ by the equation $\mathbf{v}_i = \mathbf{v}_A + \boldsymbol{\omega} \times \boldsymbol{\rho}_A$, where $\boldsymbol{\omega}$ is the angular velocity of the body, measured from the X, Y, Z coordinate system, and $\boldsymbol{\rho}_A$ is a position vector extending from A to i . Using this expression, the kinetic energy for the particle can be written as

$$\begin{aligned} T_i &= \frac{1}{2} m_i (\mathbf{v}_A + \boldsymbol{\omega} \times \boldsymbol{\rho}_A) \cdot (\mathbf{v}_A + \boldsymbol{\omega} \times \boldsymbol{\rho}_A) \\ &= \frac{1}{2} (\mathbf{v}_A \cdot \mathbf{v}_A) m_i + \mathbf{v}_A \cdot (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) m_i + \frac{1}{2} (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) \cdot (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) m_i \end{aligned}$$

The kinetic energy for the entire body is obtained by summing the kinetic energies of all the particles of the body. This requires an integration. Since $m_i \rightarrow dm$, we get

$$T = \frac{1}{2} m (\mathbf{v}_A \cdot \mathbf{v}_A) + \mathbf{v}_A \cdot \left(\boldsymbol{\omega} \times \int_m \boldsymbol{\rho}_A dm \right) + \frac{1}{2} \int_m (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) \cdot (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) dm$$

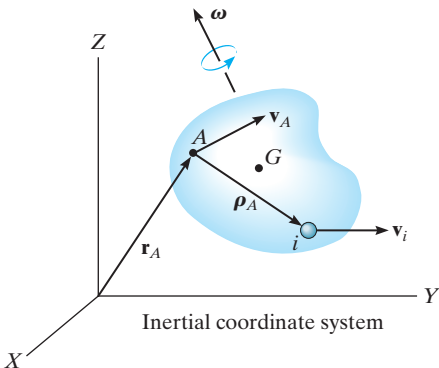


Fig. 21-8

The last term on the right can be rewritten using the vector identity $\mathbf{a} \times \mathbf{b} \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$, where $\mathbf{a} = \boldsymbol{\omega}$, $\mathbf{b} = \boldsymbol{\rho}_A$, and $\mathbf{c} = \boldsymbol{\omega} \times \boldsymbol{\rho}_A$. The final result is

$$T = \frac{1}{2}m(\mathbf{v}_A \cdot \mathbf{v}_A) + \mathbf{v}_A \cdot \left(\boldsymbol{\omega} \times \int_m \boldsymbol{\rho}_A dm \right) + \frac{1}{2}\boldsymbol{\omega} \cdot \int_m \boldsymbol{\rho}_A \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_A) dm \quad (21-14)$$

This equation is rarely used because of the computations involving the integrals. Simplification occurs, however, if the reference point A is either a fixed point or the center of mass.

Fixed Point O. If A is a *fixed point* O in the body, Fig. 21-7a, then $\mathbf{v}_A = \mathbf{0}$, and using Eq. 21-7, we can express Eq. 21-14 as

$$T = \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{H}_O$$

If the x, y, z axes represent the principal axes of inertia for the body, then $\boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$ and $\mathbf{H}_O = I_x \omega_x \mathbf{i} + I_y \omega_y \mathbf{j} + I_z \omega_z \mathbf{k}$. Substituting into the above equation and performing the dot-product operations yields

$$T = \frac{1}{2}I_x \omega_x^2 + \frac{1}{2}I_y \omega_y^2 + \frac{1}{2}I_z \omega_z^2 \quad (21-15)$$

Center of Mass G. If A is located at the *center of mass* G of the body, Fig. 21-7b, then $\int \boldsymbol{\rho}_A dm = \mathbf{0}$ and, using Eq. 21-8, we can write Eq. 21-14 as

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{H}_G$$

In a manner similar to that for a fixed point, the last term on the right side may be represented in scalar form, in which case

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_x \omega_x^2 + \frac{1}{2}I_y \omega_y^2 + \frac{1}{2}I_z \omega_z^2 \quad (21-16)$$

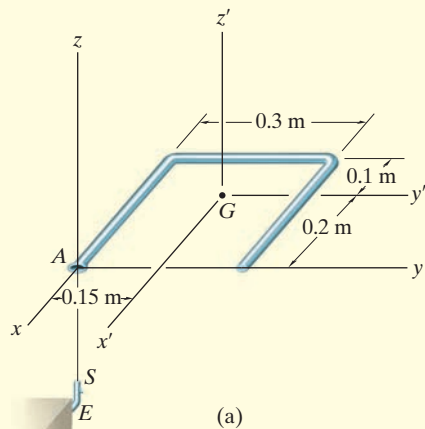
Here it is seen that the kinetic energy consists of two parts; namely, the translational kinetic energy of the mass center, $\frac{1}{2}mv_G^2$, and the body's rotational kinetic energy.

Principle of Work and Energy. Having formulated the kinetic energy for a body, the *principle of work and energy* can be applied to solve kinetics problems which involve *force, velocity, and displacement*. For this case only one scalar equation can be written for each body, namely,

$$T_1 + \Sigma U_{1-2} = T_2 \quad (21-17)$$

Before applying this equation, the material in Chapter 18 should be reviewed.

EXAMPLE 21.2



The rod in Fig. 21-9a has a mass per unit length of 8.49 kg/m. Determine its angular velocity just after the end A falls onto the hook at E. The hook provides a permanent connection for the rod due to the spring-lock mechanism S. Just before striking the hook the rod is falling downward with a speed $(v_G)_1 = 3$ m/s.

SOLUTION

The principle of impulse and momentum will be used since impact occurs.

Impulse and Momentum Diagrams. Fig. 21-9b. During the short time Δt , the impulsive force $\mathbf{F}$ acting at A changes the momentum of the rod. (The impulse created by the rod's weight $\mathbf{W}$ during this time is small compared to $\int \mathbf{F} dt$, so that it can be neglected, i.e., the weight is a nonimpulsive force.) Hence, the angular momentum of the rod is conserved about point A since the moment of $\int \mathbf{F} dt$ about A is zero.

Conservation of Angular Momentum. Equation 21-9 must be used to find the angular momentum of the rod, since A does not become a fixed point until after the impulsive interaction with the hook. Thus, with reference to Fig. 21-9b, $(\mathbf{H}_A)_1 = (\mathbf{H}_A)_2$, or

$$\mathbf{r}_{G/A} \times m(\mathbf{v}_G)_1 = \mathbf{r}_{G/A} \times m(\mathbf{v}_G)_2 + (\mathbf{H}_G)_2 \quad (1)$$

From Fig. 21-9a, $\mathbf{r}_{G/A} = \{-0.2\mathbf{i} + 0.15\mathbf{j}\}$ m. Furthermore, the primed axes are principal axes of inertia for the rod because $I_{x'y'} = I_{x'z'} = I_{y'z'} = 0$. Hence, from Eqs. 21-11, $(\mathbf{H}_G)_2 = I_{x'}\omega_x\mathbf{i} + I_{y'}\omega_y\mathbf{j} + I_{z'}\omega_z\mathbf{k}$. The principal moments of inertia are $I_{x'} = 0.1338$ kg $\cdot$ m², $I_{y'} = 0.0764$ kg $\cdot$ m², $I_{z'} = 0.2102$ kg $\cdot$ m² (see Prob. 21-17). Substituting into Eq. 1, we have

$$(-0.2\mathbf{i} + 0.15\mathbf{j}) \times [(8.49 \times 0.9)(-3\mathbf{k})] = (-0.2\mathbf{i} + 0.15\mathbf{j}) \times [(8.49 \times 0.9)(-v_G)_2\mathbf{k}] + 0.1338\omega_x\mathbf{i} + 0.0764\omega_y\mathbf{j} + 0.2102\omega_z\mathbf{k}$$

Expanding and equating the respective $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components yields

$$-3.4386 = -1.1462(v_G)_2 + 0.1338\omega_x \quad (2)$$

$$-4.5846 = -1.5282(v_G)_2 + 0.0764\omega_y \quad (3)$$

$$0 = 0.2102\omega_z \quad (4)$$

Kinematics. There are four unknowns in the above equations; however, another equation may be obtained by relating $\boldsymbol{\omega}$ to $(\mathbf{v}_G)_2$ using kinematics. Since $\omega_z = 0$ (Eq. 4) and after impact the rod rotates about the fixed point A, Eq. 20-3 can be applied, in which case $(\mathbf{v}_G)_2 = \boldsymbol{\omega} \times \mathbf{r}_{G/A}$, or

$$\begin{aligned} -(v_G)_2\mathbf{k} &= (\omega_x\mathbf{i} + \omega_y\mathbf{j}) \times (-0.2\mathbf{i} + 0.15\mathbf{j}) \\ -(v_G)_2 &= 0.15\omega_x + 0.2\omega_y \end{aligned} \quad (5)$$

Solving Eqs. 2, 3 and 5 simultaneously yields

$$(\mathbf{v}_G)_2 = \{-2.523\mathbf{k}\} \text{ m/s} \quad \boldsymbol{\omega} = \{-4.09\mathbf{i} - 9.55\mathbf{j}\} \text{ rad/s} \quad \text{Ans.}$$

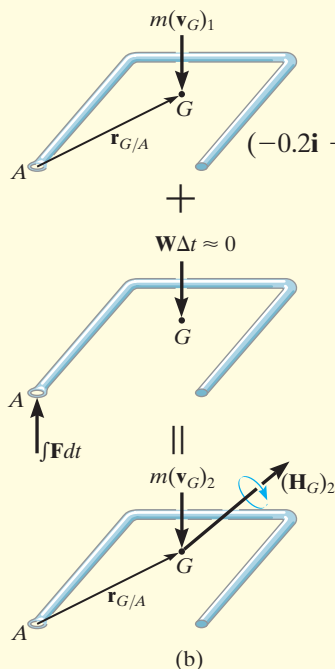


Fig. 21-9

EXAMPLE 21.3

A $5\text{-N} \cdot \text{m}$ torque is applied to the vertical shaft CD shown in Fig. 21–10a, which allows the 10-kg gear A to turn freely about CE . Assuming that gear A starts from rest, determine the angular velocity of CD after it has turned two revolutions. Neglect the mass of shaft CD and axle CE and assume that gear A can be approximated by a thin disk. Gear B is fixed.

SOLUTION

The principle of work and energy may be used for the solution. Why?

Work. If shaft CD , the axle CE , and gear A are considered as a system of connected bodies, only the applied torque $\mathbf{M}$ does work. For two revolutions of CD , this work is $\Sigma U_{1-2} = (5\text{ N} \cdot \text{m})(4\pi\text{ rad}) = 62.83\text{ J}$.

Kinetic Energy. Since the gear is initially at rest, its initial kinetic energy is zero. A kinematic diagram for the gear is shown in Fig. 21–10b. If the angular velocity of CD is taken as ω_{CD} , then the angular velocity of gear A is $\omega_A = \omega_{CD} + \omega_{CE}$. The gear may be imagined as a portion of a massless extended body which is rotating about the *fixed point* C . The instantaneous axis of rotation for this body is along line CH , because both points C and H on the body (gear) have zero velocity and must therefore lie on this axis. This requires that the components ω_{CD} and ω_{CE} be related by the equation $\omega_{CD}/0.1\text{ m} = \omega_{CE}/0.3\text{ m}$ or $\omega_{CE} = 3\omega_{CD}$. Thus,

$$\omega_A = -\omega_{CE}\mathbf{i} + \omega_{CD}\mathbf{k} = -3\omega_{CD}\mathbf{i} + \omega_{CD}\mathbf{k} \quad (1)$$

The x, y, z axes in Fig. 21–10a represent *principal axes of inertia* at C for the gear. Since point C is a fixed point of rotation, Eq. 21–15 may be applied to determine the kinetic energy, i.e.,

$$T = \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2 + \frac{1}{2}I_z\omega_z^2 \quad (2)$$

Using the parallel-axis theorem, the moments of inertia of the gear about point C are as follows:

$$I_x = \frac{1}{2}(10\text{ kg})(0.1\text{ m})^2 = 0.05\text{ kg} \cdot \text{m}^2$$

$$I_y = I_z = \frac{1}{4}(10\text{ kg})(0.1\text{ m})^2 + 10\text{ kg}(0.3\text{ m})^2 = 0.925\text{ kg} \cdot \text{m}^2$$

Since $\omega_x = -3\omega_{CD}$, $\omega_y = 0$, $\omega_z = \omega_{CD}$, Eq. 2 becomes

$$T_A = \frac{1}{2}(0.05)(-3\omega_{CD})^2 + 0 + \frac{1}{2}(0.925)(\omega_{CD})^2 = 0.6875\omega_{CD}^2$$

Principle of Work and Energy. Applying the principle of work and energy, we obtain

$$T_1 + \Sigma U_{1-2} = T_2$$

$$0 + 62.83 = 0.6875\omega_{CD}^2$$

$$\omega_{CD} = 9.56\text{ rad/s}$$

Ans.

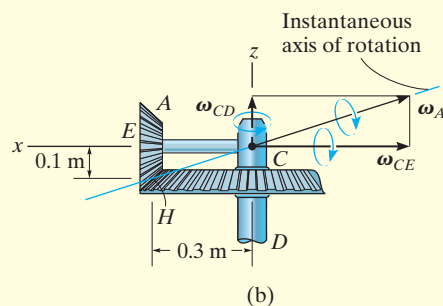
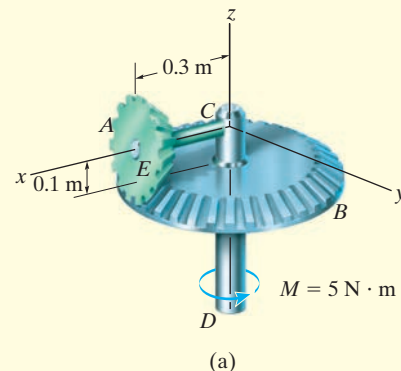


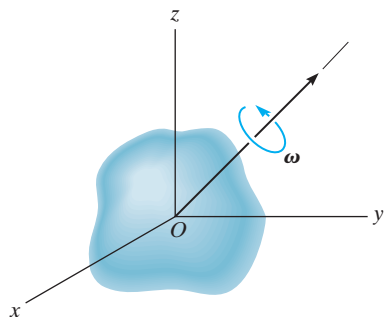
Fig. 21–10

PROBLEMS

21–22. If a body contains *no planes of symmetry*, the principal moments of inertia can be determined mathematically. To show how this is done, consider the rigid body which is spinning with an angular velocity $\boldsymbol{\omega}$, directed along one of its principal axes of inertia. If the principal moment of inertia about this axis is I , the angular momentum can be expressed as $\mathbf{H} = I\boldsymbol{\omega} = I\omega_x\mathbf{i} + I\omega_y\mathbf{j} + I\omega_z\mathbf{k}$. The components of $\mathbf{H}$ may also be expressed by Eqs. 21–10, where the inertia tensor is assumed to be known. Equate the $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ components of both expressions for $\mathbf{H}$ and consider ω_x , ω_y , and ω_z to be unknown. The solution of these three equations is obtained provided the determinant of the coefficients is zero. Show that this determinant, when expanded, yields the cubic equation

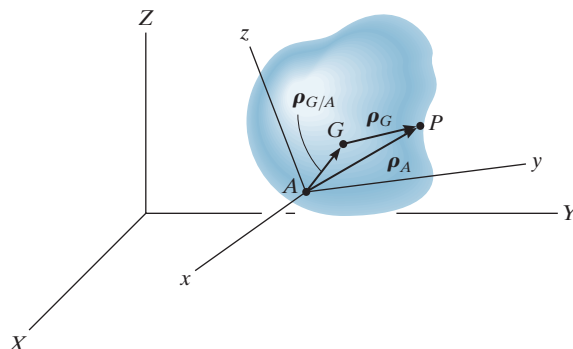
$$\begin{aligned} I^3 - (I_{xx} + I_{yy} + I_{zz})I^2 \\ + (I_{xx}I_{yy} + I_{yy}I_{zz} + I_{zz}I_{xx} - I_{xy}^2 - I_{yz}^2 - I_{zx}^2)I \\ - (I_{xx}I_{yy}I_{zz} - 2I_{xy}I_{yz}I_{zx} - I_{xx}I_{yz}^2 - I_{yy}I_{zx}^2 - I_{zz}I_{xy}^2) = 0 \end{aligned}$$

The three positive roots of I , obtained from the solution of this equation, represent the principal moments of inertia I_x , I_y , and I_z .



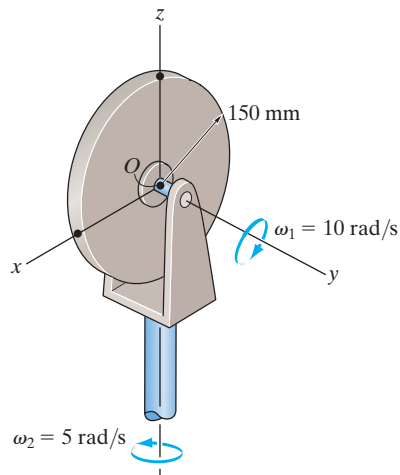
Prob. 21–22

21–23. Show that if the angular momentum of a body is determined with respect to an arbitrary point A , then $\mathbf{H}_A$ can be expressed by Eq. 21–9. This requires substituting $\boldsymbol{\rho}_A = \boldsymbol{\rho}_G + \boldsymbol{\rho}_{G/A}$ into Eq. 21–6 and expanding, noting that $\int \boldsymbol{\rho}_G dm = \mathbf{0}$ by definition of the mass center and $\mathbf{v}_G = \mathbf{v}_A + \boldsymbol{\omega} \times \boldsymbol{\rho}_{G/A}$.



Prob. 21–23

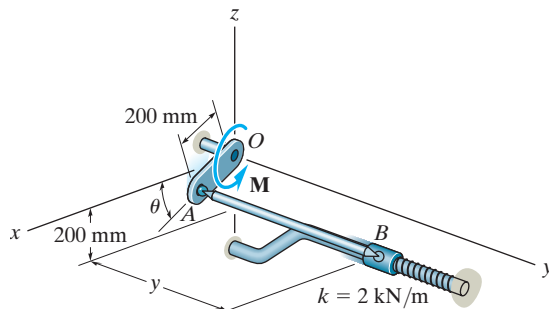
***21–24.** The 15-kg circular disk spins about its axle with a constant angular velocity of $\omega_1 = 10$ rad/s. Simultaneously, the yoke is rotating with a constant angular velocity of $\omega_2 = 5$ rad/s. Determine the angular momentum of the disk about its center of mass O , and its kinetic energy.



Prob. 21–24

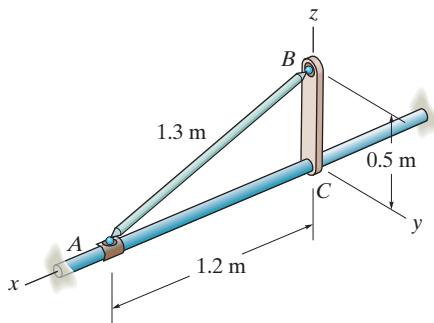
21–25. The assembly consists of a 4-kg rod AB which is connected to link OA and the collar at B by ball-and-socket joints. When $\theta = 0^\circ$, $y = 600$ mm, the system is at rest, the spring is unstretched, and a couple moment $M = 7 \text{ N} \cdot \text{m}$ is applied to the link at O . Determine the angular velocity of the link at the instant $\theta = 90^\circ$. Neglect the mass of the link.

21–26. The assembly consists of a 4-kg rod AB which is connected to link OA and the collar at B by ball-and-socket joints. When $\theta = 0^\circ$, $y = 600$ mm, the system is at rest, the spring is unstretched, and a couple moment $M = (4\theta + 2) \text{ N} \cdot \text{m}$, where θ is in radians, is applied to the link at O . Determine the angular velocity of the link at the instant $\theta = 90^\circ$. Neglect the mass of the link.



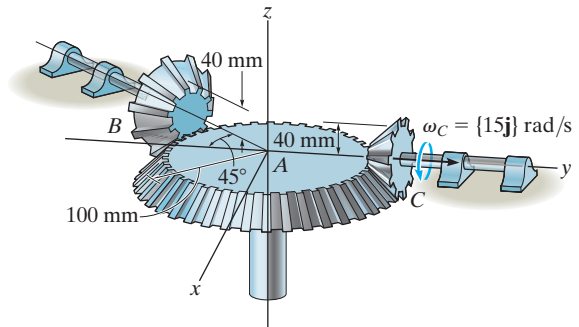
Probs. 21–25/26

21–27. The 4-kg rod AB is attached to the 1-kg collar at A and a 2-kg link BC using ball-and-socket joints. If the rod is released from rest in the position shown, determine the angular velocity of the link after it has rotated 180° .



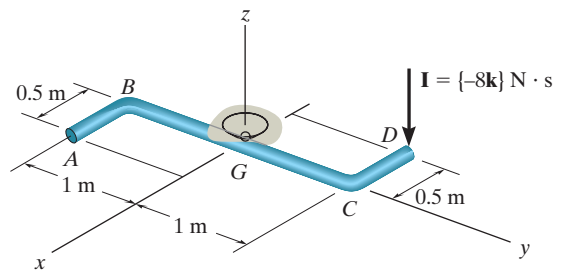
Prob. 21–27

***21–28.** The large gear has a mass of 5 kg and a radius of gyration of $k_z = 75$ mm. Gears B and C each have a mass of 200 g and a radius of gyration about the axis of their connecting shaft of 15 mm. If the gears are in mesh and C has an angular velocity of $\omega_c = \{15\mathbf{j}\} \text{ rad/s}$, determine the total angular momentum for the system of three gears about point A .



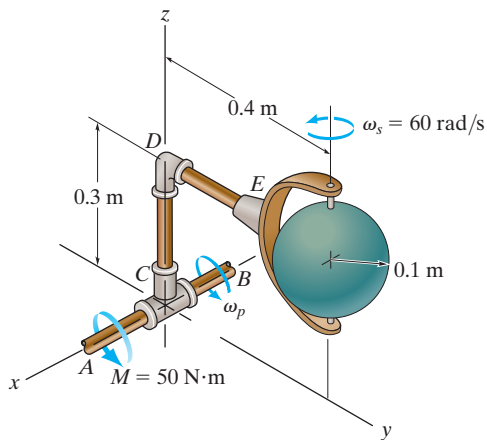
Prob. 21–28

21–29. The rod assembly is supported at G by a ball-and-socket joint. Each segment has a mass per unit length of 0.5 kg/m . If the assembly is originally at rest and an impulse of $\mathbf{I} = \{-8\mathbf{k}\} \text{ N} \cdot \text{s}$ is applied at D , determine the angular velocity of the assembly just after the impact.



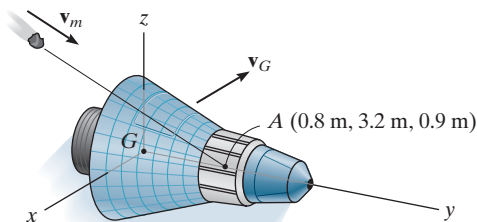
Prob. 21–29

21–30. The 20-kg sphere rotates about the axle with a constant angular velocity of $\omega_s = 60 \text{ rad/s}$. If shaft AB is subjected to a torque of $M = 50 \text{ N}\cdot\text{m}$, causing it to rotate, determine the value of ω_p after the shaft has turned 90° from the position shown. Initially, $\omega_p = 0$. Neglect the mass of arm CDE .



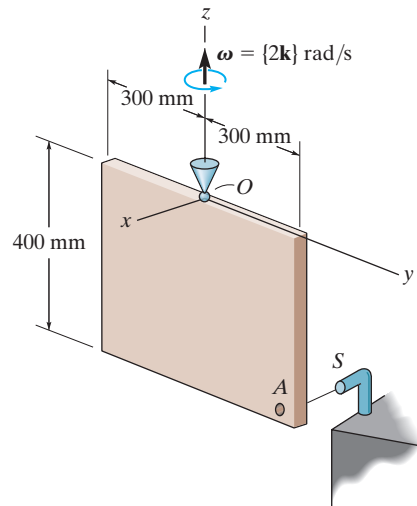
Prob. 21–30

21–31. The space capsule has a mass of 5 Mg and the radii of gyration are $k_x = k_z = 1.30 \text{ m}$ and $k_y = 0.45 \text{ m}$. If it travels with a velocity $\mathbf{v}_G = \{400\mathbf{j} + 200\mathbf{k}\} \text{ m/s}$, compute its angular velocity just after it is struck by a meteoroid having a mass of 0.80 kg and a velocity $\mathbf{v}_m = \{-300\mathbf{i} + 200\mathbf{j} - 150\mathbf{k}\} \text{ m/s}$. Assume that the meteoroid embeds itself into the capsule at point A and that the capsule initially has no angular velocity.



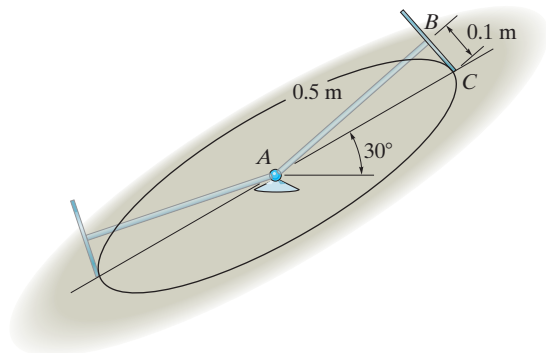
Prob. 21–31

***21–32.** The 5-kg thin plate is suspended at O using a ball-and-socket joint. It is rotating with a constant angular velocity $\boldsymbol{\omega} = \{2\mathbf{k}\} \text{ rad/s}$ when the corner A strikes the hook at S , which provides a permanent connection. Determine the angular velocity of the plate immediately after impact.



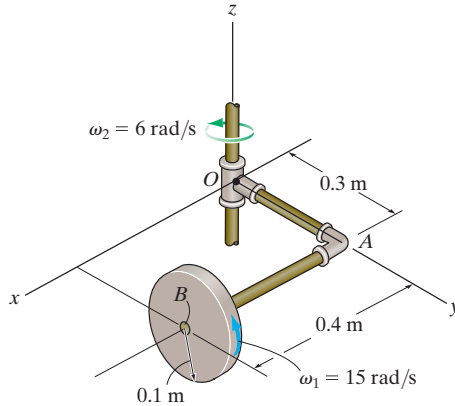
Prob. 21–32

21–33. The 2-kg thin disk is connected to the slender rod which is fixed to the ball-and-socket joint at A . If it is released from rest in the position shown, determine the spin of the disk about the rod when the disk reaches its lowest position. Neglect the mass of the rod. The disk rolls without slipping.



Prob. 21–33

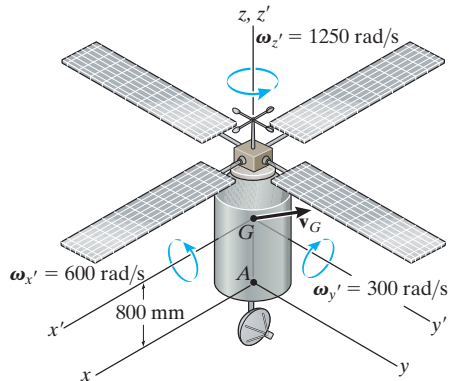
21-34. The 5-kg circular disk spins about AB with a constant angular velocity of $\omega_1 = 15 \text{ rad/s}$. Simultaneously, the shaft to which arm OAB is rigidly attached, rotates with a constant angular velocity of $\omega_2 = 6 \text{ rad/s}$. Determine the angular momentum of the disk about point O , and its kinetic energy.



Prob. 21-34

21-35. The 200-kg satellite has its center of mass at point G . Its radii of gyration about the z' , x' , y' axes are $k_{z'} = 300 \text{ mm}$, $k_{x'} = k_{y'} = 500 \text{ mm}$, respectively. At the instant shown, the satellite rotates about the x' , y' , and z' axes with the angular velocities shown, and its center of mass G has a velocity of $\mathbf{v}_G = \{-250\mathbf{i} + 200\mathbf{j} + 120\mathbf{k}\} \text{ m/s}$. Determine the angular momentum of the satellite about point A at this instant.

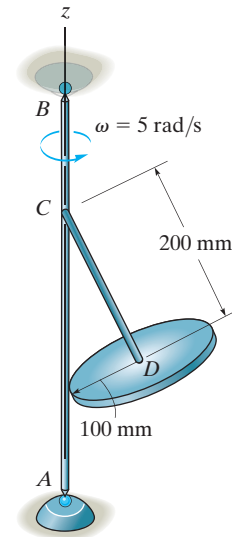
***21-36.** The 200-kg satellite has its center of mass at point G . Its radii of gyration about the z' , x' , y' axes are $k_{z'} = 300 \text{ mm}$, $k_{x'} = k_{y'} = 500 \text{ mm}$, respectively. At the instant shown, the satellite rotates about the x' , y' , and z' axes with the angular velocities shown, and its center of mass G has a velocity of $\mathbf{v}_G = \{-250\mathbf{i} + 200\mathbf{j} + 120\mathbf{k}\} \text{ m/s}$. Determine the kinetic energy of the satellite at this instant.



Probs. 21-35/36

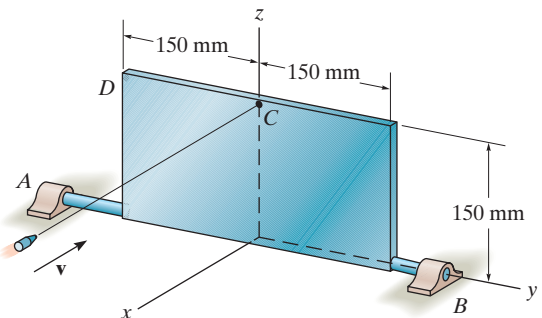
21-37. Determine the kinetic energy of the 7-kg disk and 1.5-kg rod when the assembly is rotating about the z axis at $\omega = 5 \text{ rad/s}$.

21-38. Determine the angular momentum $\mathbf{H}_z$ of the 7-kg disk and 1.5-kg rod when the assembly is rotating about the z axis at $\omega = 5 \text{ rad/s}$.



Probs. 21-37/38

21-39. The 15-kg rectangular plate is free to rotate about the y axis because of the bearing supports at A and B . When the plate is balanced in the vertical plane, a 3-g bullet is fired into it, perpendicular to its surface, with a velocity $\mathbf{v} = \{-2000\mathbf{i}\} \text{ m/s}$. Compute the angular velocity of the plate at the instant it has rotated 180° . If the bullet strikes corner D with the same velocity $\mathbf{v}$, instead of at C , does the angular velocity remain the same? Why or why not?



Prob. 21-39

*21.4 Equations of Motion

Having become familiar with the techniques used to describe both the inertial properties and the angular momentum of a body, we can now write the equations which describe the motion of the body in their most useful forms.

Equations of Translational Motion. The *translational motion* of a body is defined in terms of the acceleration of the body's mass center, which is measured from an inertial X, Y, Z reference. The equation of translational motion for the body can be written in vector form as

$$\Sigma \mathbf{F} = m\mathbf{a}_G \quad (21-18)$$

or by the three scalar equations

$$\begin{aligned} \Sigma F_x &= m(a_G)_x \\ \Sigma F_y &= m(a_G)_y \\ \Sigma F_z &= m(a_G)_z \end{aligned} \quad (21-19)$$

Here, $\Sigma \mathbf{F} = \Sigma F_x \mathbf{i} + \Sigma F_y \mathbf{j} + \Sigma F_z \mathbf{k}$ represents the sum of all the external forces acting on the body.

Equations of Rotational Motion. In Sec. 15.6, we developed Eq. 15-17, namely,

$$\Sigma \mathbf{M}_O = \dot{\mathbf{H}}_O \quad (21-20)$$

which states that the sum of the moments of all the external forces acting on a system of particles (contained in a rigid body) about a fixed point O is equal to the time rate of change of the total angular momentum of the body about point O . When moments of the external forces acting on the particles are summed about the system's *mass center* G , one again obtains the same simple form of Eq. 21-20, relating the moment summation $\Sigma \mathbf{M}_G$ to the angular momentum $\mathbf{H}_G$. To show this, consider the system of particles in Fig. 21-11, where X, Y, Z represents an inertial frame of reference and the x, y, z axes, with origin at G , *translate* with respect to this frame. In general, G is *accelerating*, so by definition the translating frame is *not* an inertial reference. The angular momentum of the i th particle with respect to this frame is, however,

$$(\mathbf{H}_i)_G = \mathbf{r}_{i/G} \times m_i \mathbf{v}_{i/G}$$

where $\mathbf{r}_{i/G}$ and $\mathbf{v}_{i/G}$ represent the position and velocity of the i th particle with respect to G . Taking the time derivative we have

$$(\dot{\mathbf{H}}_i)_G = \dot{\mathbf{r}}_{i/G} \times m_i \mathbf{v}_{i/G} + \mathbf{r}_{i/G} \times m_i \dot{\mathbf{v}}_{i/G}$$

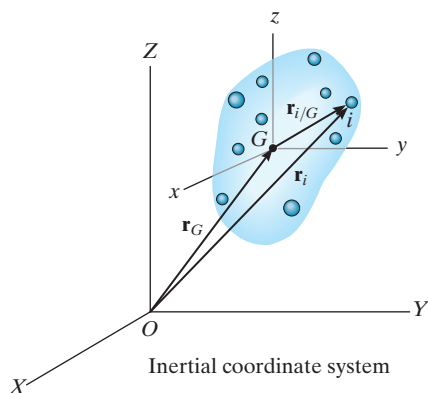


Fig. 21-11

By definition, $\mathbf{v}_{i/G} = \dot{\mathbf{r}}_{i/G}$. Thus, the first term on the right side is zero since the cross product of the same vectors is zero. Also, $\mathbf{a}_{i/G} = \dot{\mathbf{v}}_{i/G}$, so that

$$(\dot{\mathbf{H}}_i)_G = (\mathbf{r}_{i/G} \times m_i \mathbf{a}_{i/G})$$

Similar expressions can be written for the other particles of the body. When the results are summed, we get

$$\dot{\mathbf{H}}_G = \Sigma(\mathbf{r}_{i/G} \times m_i \mathbf{a}_{i/G})$$

Here $\dot{\mathbf{H}}_G$ is the time rate of change of the total angular momentum of the body computed about point G .

The relative acceleration for the i th particle is defined by the equation $\mathbf{a}_{i/G} = \mathbf{a}_i - \mathbf{a}_G$, where $\mathbf{a}_i$ and $\mathbf{a}_G$ represent, respectively, the accelerations of the i th particle and point G measured with respect to the *inertial frame of reference*. Substituting and expanding, using the distributive property of the vector cross product, yields

$$\dot{\mathbf{H}}_G = \Sigma(\mathbf{r}_{i/G} \times m_i \mathbf{a}_i) - (\Sigma m_i \mathbf{r}_{i/G}) \times \mathbf{a}_G$$

By definition of the mass center, the sum $(\Sigma m_i \mathbf{r}_{i/G}) = (\Sigma m_i) \bar{\mathbf{r}}$ is equal to zero, since the position vector $\bar{\mathbf{r}}$ relative to G is zero. Hence, the last term in the above equation is zero. Using the equation of motion, the product $m_i \mathbf{a}_i$ can be replaced by the resultant *external force* $\mathbf{F}_i$ acting on the i th particle. Denoting $\Sigma \mathbf{M}_G = \Sigma(\mathbf{r}_{i/G} \times \mathbf{F}_i)$, the final result can be written as

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (21-21)$$

The rotational equation of motion for the body will now be developed from either Eq. 21-20 or 21-21. In this regard, the scalar components of the angular momentum $\mathbf{H}_O$ or $\mathbf{H}_G$ are defined by Eqs. 21-10 or, if principal axes of inertia are used either at point O or G , by Eqs. 21-11. If these components are computed about x, y, z axes that are *rotating* with an angular velocity $\boldsymbol{\Omega}$ that is *different* from the body's angular velocity $\boldsymbol{\omega}$, then the time derivative $\dot{\mathbf{H}} = d\mathbf{H}/dt$, as used in Eqs. 21-20 and 21-21, must account for the rotation of the x, y, z axes as measured from the inertial X, Y, Z axes. This requires application of Eq. 20-6, in which case Eqs. 21-20 and 21-21 become

$$\begin{aligned} \Sigma \mathbf{M}_O &= (\dot{\mathbf{H}}_O)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}_O \\ \Sigma \mathbf{M}_G &= (\dot{\mathbf{H}}_G)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}_G \end{aligned} \quad (21-22)$$

Here $(\dot{\mathbf{H}})_{xyz}$ is the time rate of change of $\mathbf{H}$ measured from the x, y, z reference.

There are three ways in which one can define the motion of the x, y, z axes. Obviously, motion of this reference should be chosen so that it will yield the simplest set of moment equations for the solution of a particular problem.

x, y, z Axes Having Motion $\mathbf{\Omega} = \mathbf{0}$. If the body has general motion, the x, y, z axes can be chosen with origin at G , such that the axes only *translate* relative to the inertial X, Y, Z frame of reference. Doing this simplifies Eq. 21–22, since $\mathbf{\Omega} = \mathbf{0}$. However, the body may have a rotation $\boldsymbol{\omega}$ about these axes, and therefore the moments and products of inertia of the body would have to be expressed as *functions of time*. In most cases this would be a difficult task, so that such a choice of axes has restricted application.

x, y, z Axes Having Motion $\mathbf{\Omega} = \boldsymbol{\omega}$. The x, y, z axes can be chosen such that they are *fixed in and move with the body*. The moments and products of inertia of the body relative to these axes will then be *constant* during the motion. Since $\mathbf{\Omega} = \boldsymbol{\omega}$, Eqs. 21–22 become

$$\begin{aligned}\Sigma \mathbf{M}_O &= (\dot{\mathbf{H}}_O)_{xyz} + \boldsymbol{\omega} \times \mathbf{H}_O \\ \Sigma \mathbf{M}_G &= (\dot{\mathbf{H}}_G)_{xyz} + \boldsymbol{\omega} \times \mathbf{H}_G\end{aligned}\quad (21-23)$$

We can express each of these vector equations as three scalar equations using Eqs. 21–10. Neglecting the subscripts O and G yields

$$\begin{aligned}\Sigma M_x &= I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z - I_{xy}(\dot{\omega}_y - \omega_z\omega_x) \\ &\quad - I_{yz}(\omega_y^2 - \omega_z^2) - I_{zx}(\dot{\omega}_z + \omega_x\omega_y) \\ \Sigma M_y &= I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x - I_{yz}(\dot{\omega}_z - \omega_x\omega_y) \\ &\quad - I_{zx}(\omega_z^2 - \omega_x^2) - I_{xy}(\dot{\omega}_x + \omega_y\omega_z) \\ \Sigma M_z &= I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y - I_{zx}(\dot{\omega}_x - \omega_y\omega_z) \\ &\quad - I_{xy}(\omega_x^2 - \omega_y^2) - I_{yz}(\dot{\omega}_y + \omega_z\omega_x)\end{aligned}\quad (21-24)$$

If the x, y, z axes are chosen as *principal axes of inertia*, the products of inertia are zero, $I_{xx} = I_x$, etc., and the above equations become

$$\begin{aligned}\Sigma M_x &= I_x\dot{\omega}_x - (I_y - I_z)\omega_y\omega_z \\ \Sigma M_y &= I_y\dot{\omega}_y - (I_z - I_x)\omega_z\omega_x \\ \Sigma M_z &= I_z\dot{\omega}_z - (I_x - I_y)\omega_x\omega_y\end{aligned}\quad (21-25)$$

This set of equations is known historically as the *Euler equations of motion*, named after the Swiss mathematician Leonhard Euler, who first developed them. They apply *only* for moments summed about either point O or G .

When applying these equations it should be realized that $\dot{\omega}_x, \dot{\omega}_y, \dot{\omega}_z$ represent the time derivatives of the magnitudes of the x, y, z components of $\boldsymbol{\omega}$ as observed from x, y, z . To determine these components, it is first necessary to find $\omega_x, \omega_y, \omega_z$ when the x, y, z axes are oriented in a *general position* and *then* take the time derivative of the magnitude of these components, i.e., $(\dot{\omega})_{xyz}$. However, since the x, y, z axes are rotating at $\boldsymbol{\Omega} = \boldsymbol{\omega}$, then from Eq. 20–6, it should be noted that $\dot{\boldsymbol{\omega}} = (\dot{\boldsymbol{\omega}})_{xyz} + \boldsymbol{\omega} \times \boldsymbol{\omega}$. Since $\boldsymbol{\omega} \times \boldsymbol{\omega} = \mathbf{0}$, then $\dot{\boldsymbol{\omega}} = (\dot{\boldsymbol{\omega}})_{xyz}$. This important result indicates that the time derivative of $\boldsymbol{\omega}$ with respect to the fixed X, Y, Z axes, that is $\dot{\boldsymbol{\omega}}$, can also be used to obtain $(\dot{\boldsymbol{\omega}})_{xyz}$. Generally this is the easiest way to determine the result. See Example 21.5.

x, y, z Axes Having Motion $\boldsymbol{\Omega} \neq \boldsymbol{\omega}$. To simplify the calculations for the time derivative of $\boldsymbol{\omega}$, it is often convenient to choose the x, y, z axes having an angular velocity $\boldsymbol{\Omega}$ which is different from the angular velocity $\boldsymbol{\omega}$ of the body. This is particularly suitable for the analysis of spinning tops and gyroscopes which are *symmetrical* about their spinning axes.* When this is the case, the moments and products of inertia remain constant about the axis of spin.

Equations 21–22 are applicable for such a set of axes. Each of these two vector equations can be reduced to a set of three scalar equations which are derived in a manner similar to Eqs. 21–25,[†] i.e.,

$$\begin{aligned}\Sigma M_x &= I_x \dot{\omega}_x - I_y \Omega_z \omega_y + I_z \Omega_y \omega_z \\ \Sigma M_y &= I_y \dot{\omega}_y - I_z \Omega_x \omega_z + I_x \Omega_z \omega_x \\ \Sigma M_z &= I_z \dot{\omega}_z - I_x \Omega_y \omega_x + I_y \Omega_x \omega_y\end{aligned}\quad (21-26)$$

Here $\Omega_x, \Omega_y, \Omega_z$ represent the x, y, z components of $\boldsymbol{\Omega}$, measured from the inertial frame of reference, and $\dot{\omega}_x, \dot{\omega}_y, \dot{\omega}_z$ must be determined relative to the x, y, z axes that have the rotation $\boldsymbol{\Omega}$. See Example 21.6.

Any one of these sets of moment equations, Eqs. 21–24, 21–25, or 21–26, represents a series of three first-order nonlinear differential equations. These equations are “coupled,” since the angular-velocity components are present in all the terms. Success in determining the solution for a particular problem therefore depends upon what is unknown in these equations. Difficulty certainly arises when one attempts to solve for the unknown components of $\boldsymbol{\omega}$ when the external moments are functions of time. Further complications can arise if the moment equations are coupled to the three scalar equations of translational motion, Eqs. 21–19. This can happen because of the existence of kinematic constraints which relate the rotation of the body to the translation of its mass center, as in the case of a hoop which rolls

*A detailed discussion of such devices is given in Sec. 21.5.

[†]See Prob. 21–42.

without slipping. Problems that require the simultaneous solution of differential equations are generally solved using numerical methods with the aid of a computer. In many engineering problems, however, we are given information about the motion of the body and are required to determine the applied moments acting on the body. Most of these problems have direct solutions, so that there is no need to resort to computer techniques.

Procedure for Analysis

Problems involving the three-dimensional motion of a rigid body can be solved using the following procedure.

Free-Body Diagram.

- Draw a *free-body diagram* of the body at the instant considered and specify the x, y, z coordinate system. The origin of this reference must be located either at the body's mass center G , or at point O , considered fixed in an inertial reference frame and located either in the body or on a massless extension of the body.
- Unknown reactive force components can be shown having a positive sense of direction.
- Depending on the nature of the problem, decide what type of rotational motion $\boldsymbol{\Omega}$ the x, y, z coordinate system should have, i.e., $\boldsymbol{\Omega} = \mathbf{0}$, $\boldsymbol{\Omega} = \boldsymbol{\omega}$, or $\boldsymbol{\Omega} \neq \boldsymbol{\omega}$. When choosing, keep in mind that the moment equations are simplified when the axes move in such a manner that they represent principal axes of inertia for the body at all times.
- Compute the necessary moments and products of inertia for the body relative to the x, y, z axes.

Kinematics.

- Determine the x, y, z components of the body's angular velocity and find the time derivatives of $\boldsymbol{\omega}$.
- Note that if $\boldsymbol{\Omega} = \boldsymbol{\omega}$, then $\dot{\boldsymbol{\omega}} = (\dot{\boldsymbol{\omega}})_{xyz}$. Therefore we can either find the time derivative of $\boldsymbol{\omega}$ with respect to the X, Y, Z axes, $\dot{\boldsymbol{\omega}}$, and then determine its components $\dot{\omega}_x, \dot{\omega}_y, \dot{\omega}_z$, or we can find the components of $\boldsymbol{\omega}$ along the x, y, z axes, when the axes are oriented in a general position, and then take the time derivative of the magnitudes of these components, $(\dot{\boldsymbol{\omega}})_{xyz}$.

Equations of Motion.

- Apply either the two vector equations 21-18 and 21-22 or the six scalar component equations appropriate for the x, y, z coordinate axes chosen for the problem.

EXAMPLE 21.4

The gear shown in Fig. 21–12a has a mass of 10 kg and is mounted at an angle of 10° with the rotating shaft having negligible mass. If $I_z = 0.1 \text{ kg} \cdot \text{m}^2$, $I_x = I_y = 0.05 \text{ kg} \cdot \text{m}^2$, and the shaft is rotating with a constant angular velocity of $\omega = 30 \text{ rad/s}$, determine the components of reaction that the thrust bearing A and journal bearing B exert on the shaft at the instant shown.

SOLUTION

Free-Body Diagram. Fig. 21–12b. The origin of the x, y, z coordinate system is located at the gear's center of mass G , which is also a fixed point. The axes are fixed in and rotate with the gear so that these axes will then always represent the principal axes of inertia for the gear. Hence $\mathbf{\Omega} = \mathbf{\omega}$.

Kinematics. As shown in Fig. 21–12c, the angular velocity $\mathbf{\omega}$ of the gear is constant in magnitude and is always directed along the axis of the shaft AB . Since this vector is measured from the X, Y, Z inertial frame of reference, for any position of the x, y, z axes,

$$\omega_x = 0 \quad \omega_y = -30 \sin 10^\circ \quad \omega_z = 30 \cos 10^\circ$$

These components remain constant for any general orientation of the x, y, z axes, and so $\dot{\omega}_x = \dot{\omega}_y = \dot{\omega}_z = 0$. Also note that since $\mathbf{\Omega} = \mathbf{\omega}$, then $\dot{\mathbf{\omega}} = (\dot{\mathbf{\omega}})_{xyz}$. Therefore, we can find these time derivatives relative to the X, Y, Z axes. In this regard $\mathbf{\omega}$ has a constant magnitude and direction ($+Z$) since $\dot{\mathbf{\omega}} = \mathbf{0}$, and so $\dot{\omega}_x = \dot{\omega}_y = \dot{\omega}_z = 0$. Furthermore, since G is a fixed point, $(a_G)_x = (a_G)_y = (a_G)_z = 0$.

Equations of Motion. Applying Eqs. 21–25 ($\mathbf{\Omega} = \mathbf{\omega}$) yields

$$\begin{aligned} \Sigma M_x &= I_x \dot{\omega}_x - (I_y - I_z) \omega_y \omega_z \\ -(A_Y)(0.2) + (B_Y)(0.25) &= 0 - (0.05 - 0.1)(-30 \sin 10^\circ)(30 \cos 10^\circ) \\ -0.2A_Y + 0.25B_Y &= -7.70 \end{aligned} \quad (1)$$

$$\begin{aligned} \Sigma M_y &= I_y \dot{\omega}_y - (I_z - I_x) \omega_z \omega_x \\ A_X(0.2) \cos 10^\circ - B_X(0.25) \cos 10^\circ &= 0 - 0 \\ A_X &= 1.25B_X \end{aligned} \quad (2)$$

$$\begin{aligned} \Sigma M_z &= I_z \dot{\omega}_z - (I_x - I_y) \omega_x \omega_y \\ A_X(0.2) \sin 10^\circ - B_X(0.25) \sin 10^\circ &= 0 - 0 \\ A_X &= 1.25B_X \text{ (check)} \end{aligned}$$

Applying Eqs. 21–19, we have

$$\Sigma F_X = m(a_G)_X; \quad A_X + B_X = 0 \quad (3)$$

$$\Sigma F_Y = m(a_G)_Y; \quad A_Y + B_Y - 98.1 = 0 \quad (4)$$

$$\Sigma F_Z = m(a_G)_Z; \quad A_Z = 0 \quad \text{Ans.}$$

Solving Eqs. 1 through 4 simultaneously gives

$$A_X = B_X = 0 \quad A_Y = 71.6 \text{ N} \quad B_Y = 26.5 \text{ N} \quad \text{Ans.}$$

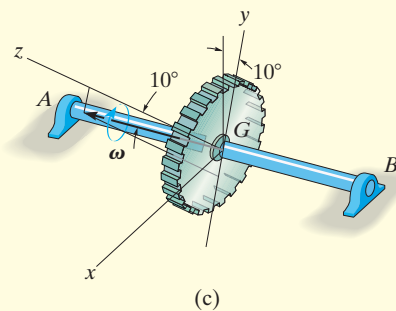
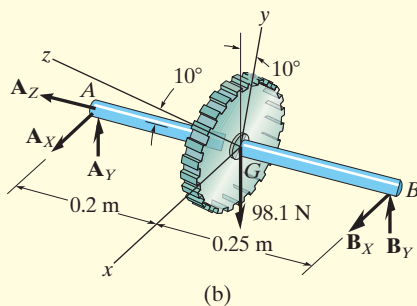
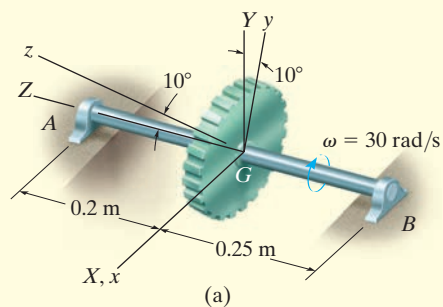
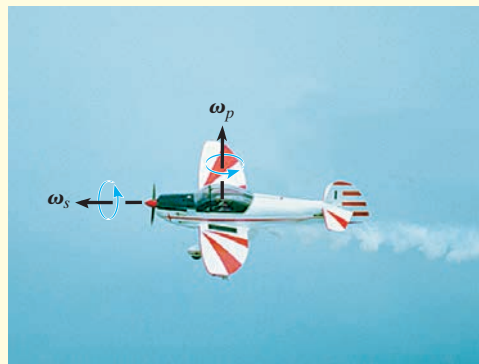


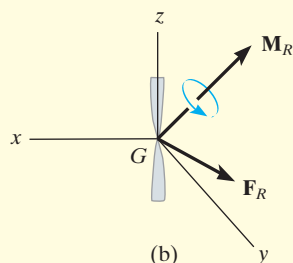
Fig. 21–12

EXAMPLE 21.5

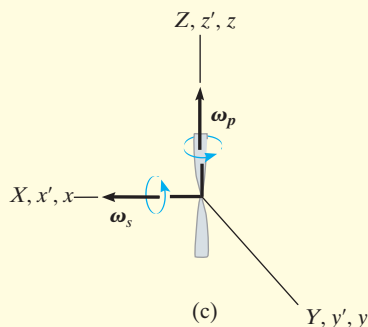
The airplane shown in Fig. 21–13a is in the process of making a steady *horizontal* turn at the rate of ω_p . During this motion, the propeller is spinning at the rate of ω_s . If the propeller has two blades, determine the moments which the propeller shaft exerts on the propeller at the instant the blades are in the vertical position. For simplicity, assume the blades to be a uniform slender bar having a moment of inertia I about an axis perpendicular to the blades passing through the center of the bar, and having zero moment of inertia about a longitudinal axis.



(a)



(b)



(c)

Fig. 21–13**SOLUTION**

Free-Body Diagram. Fig. 21–13b. The reactions of the connecting shaft on the propeller are indicated by the resultants $\mathbf{F}_R$ and $\mathbf{M}_R$. (The propeller's weight is assumed to be negligible.) The x, y, z axes will be taken fixed to the propeller, since these axes always represent the principal axes of inertia for the propeller. Thus, $\mathbf{\Omega} = \mathbf{\omega}$. The moments of inertia I_x and I_y are equal ($I_x = I_y = I$) and $I_z = 0$.

Kinematics. The angular velocity of the propeller observed from the X, Y, Z axes, coincident with the x, y, z axes, Fig. 21–13c, is $\mathbf{\omega} = \mathbf{\omega}_s + \mathbf{\omega}_p = \omega_s \mathbf{i} + \omega_p \mathbf{k}$, so that the x, y, z components of $\mathbf{\omega}$ are

$$\omega_x = \omega_s \quad \omega_y = 0 \quad \omega_z = \omega_p$$

Since $\mathbf{\Omega} = \mathbf{\omega}$, then $\dot{\mathbf{\omega}} = (\dot{\mathbf{\omega}})_{xyz}$. To find $\dot{\mathbf{\omega}}$, which is the time derivative with respect to the fixed X, Y, Z axes, we can use Eq. 20–6 since $\mathbf{\omega}$ changes direction relative to X, Y, Z . The time rate of change of each of these components $\dot{\mathbf{\omega}} = \dot{\mathbf{\omega}}_s + \dot{\mathbf{\omega}}_p$ relative to the X, Y, Z axes can be obtained by introducing a third coordinate system x', y', z' , which has an angular velocity $\mathbf{\Omega}' = \mathbf{\omega}_p$ and is coincident with the X, Y, Z axes at the instant shown. Thus

$$\begin{aligned}
 \dot{\omega} &= (\dot{\omega})_{x'y'z'} + \omega_p \times \omega \\
 &= (\dot{\omega}_s)_{x'y'z'} + (\dot{\omega}_p)_{x'y'z'} + \omega_p \times (\omega_s + \omega_p) \\
 &= \mathbf{0} + \mathbf{0} + \omega_p \times \omega_s + \omega_p \times \omega_p \\
 &= \mathbf{0} + \mathbf{0} + \omega_p \mathbf{k} \times \omega_s \mathbf{i} + \mathbf{0} = \omega_p \omega_s \mathbf{j}
 \end{aligned}$$

Since the X, Y, Z axes are coincident with the x, y, z axes at the instant shown, the components of $\dot{\omega}$ along x, y, z are therefore

$$\dot{\omega}_x = 0 \quad \dot{\omega}_y = \omega_p \omega_s \quad \dot{\omega}_z = 0$$

These same results can also be determined by direct calculation of $(\dot{\omega})_{xyz}$; however, this will involve a bit more work. To do this, it will be necessary to view the propeller (or the x, y, z axes) in some *general position* such as shown in Fig. 21–13d. Here the plane has turned through an angle ϕ (phi) and the propeller has turned through an angle ψ (psi) relative to the plane. Notice that ω_p is always directed along the fixed Z axis and ω_s follows the x axis. Thus the general components of ω are

$$\omega_x = \omega_s \quad \omega_y = \omega_p \sin \psi \quad \omega_z = \omega_p \cos \psi$$

Since ω_s and ω_p are constant, the time derivatives of these components become

$$\dot{\omega}_x = 0 \quad \dot{\omega}_y = \omega_p \cos \psi \dot{\psi} \quad \dot{\omega}_z = -\omega_p \sin \psi \dot{\psi}$$

But $\phi = \psi = 0^\circ$ and $\dot{\psi} = \omega_s$ at the instant considered. Thus,

$$\begin{aligned}
 \omega_x &= \omega_s & \omega_y &= 0 & \omega_z &= \omega_p \\
 \dot{\omega}_x &= 0 & \dot{\omega}_y &= \omega_p \omega_s & \dot{\omega}_z &= 0
 \end{aligned}$$

which are the same results as those obtained previously.

Equations of Motion. Using Eqs. 21–25, we have

$$\Sigma M_x = I_x \dot{\omega}_x - (I_y - I_z) \omega_y \omega_z = I(0) - (I - 0)(0) \omega_p$$

$$M_x = 0 \quad \text{Ans.}$$

$$\Sigma M_y = I_y \dot{\omega}_y - (I_z - I_x) \omega_z \omega_x = I(\omega_p \omega_s) - (0 - I) \omega_p \omega_s$$

$$M_y = 2I \omega_p \omega_s \quad \text{Ans.}$$

$$\Sigma M_z = I_z \dot{\omega}_z - (I_x - I_y) \omega_x \omega_y = 0(0) - (I - I) \omega_s(0)$$

$$M_z = 0 \quad \text{Ans.}$$

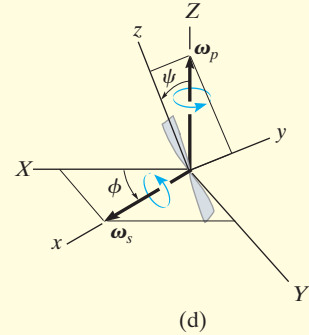
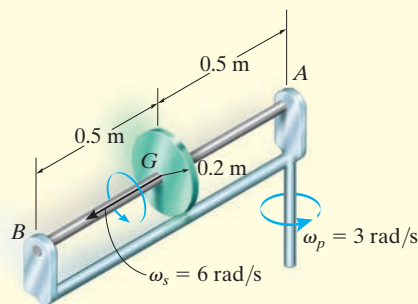
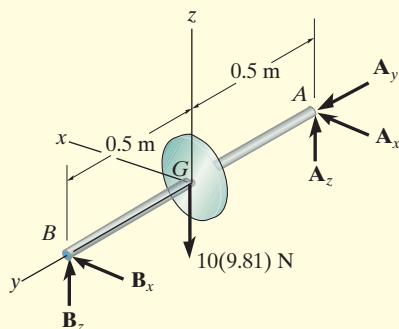


Fig. 21–13

EXAMPLE 21.6



(a)



(b)

Fig. 21-14

The 10-kg flywheel (or thin disk) shown in Fig. 21-14a rotates (spins) about the shaft at a constant angular velocity of $\omega_s = 6 \text{ rad/s}$. At the same time, the shaft rotates (precessing) about the bearing at A with an angular velocity of $\omega_p = 3 \text{ rad/s}$. If A is a thrust bearing and B is a journal bearing, determine the components of force reaction at each of these supports due to the motion.

SOLUTION I

Free-Body Diagram. Fig. 21-14b. The origin of the x, y, z coordinate system is located at the center of mass G of the flywheel. Here we will let these coordinates have an angular velocity of $\boldsymbol{\Omega} = \boldsymbol{\omega}_p = \{3\mathbf{k}\} \text{ rad/s}$. Although the wheel spins relative to these axes, the moments of inertia remain constant,* i.e.,

$$I_x = I_z = \frac{1}{4}(10 \text{ kg})(0.2 \text{ m})^2 = 0.1 \text{ kg} \cdot \text{m}^2$$

$$I_y = \frac{1}{2}(10 \text{ kg})(0.2 \text{ m})^2 = 0.2 \text{ kg} \cdot \text{m}^2$$

Kinematics. From the coincident inertial X, Y, Z frame of reference, Fig. 21-14c, the flywheel has an angular velocity of $\boldsymbol{\omega} = \{6\mathbf{j} + 3\mathbf{k}\} \text{ rad/s}$, so that

$$\omega_x = 0 \quad \omega_y = 6 \text{ rad/s} \quad \omega_z = 3 \text{ rad/s}$$

The time derivative of $\boldsymbol{\omega}$ must be determined relative to the x, y, z axes. In this case both $\boldsymbol{\omega}_p$ and $\boldsymbol{\omega}_s$ do not change their magnitude or direction, and so

$$\dot{\omega}_x = 0 \quad \dot{\omega}_y = 0 \quad \dot{\omega}_z = 0$$

Equations of Motion. Applying Eqs. 21-26 ($\boldsymbol{\Omega} \neq \boldsymbol{\omega}$) yields

$$\Sigma M_x = I_x \dot{\omega}_x - I_y \Omega_z \omega_y + I_z \Omega_y \omega_z$$

$$-A_z(0.5) + B_z(0.5) = 0 - (0.2)(3)(6) + 0 = -3.6$$

$$\Sigma M_y = I_y \dot{\omega}_y - I_z \Omega_x \omega_z + I_x \Omega_z \omega_x$$

$$0 = 0 - 0 + 0$$

$$\Sigma M_z = I_z \dot{\omega}_z - I_x \Omega_y \omega_x + I_y \Omega_x \omega_y$$

$$A_x(0.5) - B_x(0.5) = 0 - 0 + 0$$

*This would not be true for the propeller in Example 21.5.

Applying Eqs. 21–19, we have

$$\Sigma F_X = m(a_G)_X; \quad A_x + B_x = 0$$

$$\Sigma F_Y = m(a_G)_Y; \quad A_y = -10(0.5)(3)^2$$

$$\Sigma F_Z = m(a_G)_Z; \quad A_z + B_z - 10(9.81) = 0$$

Solving these equations, we obtain

$$A_x = 0 \quad A_y = -45.0 \text{ N} \quad A_z = 52.6 \text{ N} \quad \text{Ans.}$$

$$B_x = 0 \quad B_z = 45.4 \text{ N} \quad \text{Ans.}$$

NOTE: If the precession ω_p had not occurred, the z component of force at A and B would be equal to 49.05 N. In this case, however, the difference in these components is caused by the “gyroscopic moment” created whenever a spinning body precesses about another axis. We will study this effect in detail in the next section.

SOLUTION II

This example can also be solved using Euler’s equations of motion, Eqs. 21–25. In this case $\boldsymbol{\Omega} = \boldsymbol{\omega} = \{6\mathbf{j} + 3\mathbf{k}\}$ rad/s, and the time derivative $(\dot{\boldsymbol{\omega}})_{xyz}$ can be conveniently obtained with reference to the fixed X, Y, Z axes since $\dot{\boldsymbol{\omega}} = (\dot{\boldsymbol{\omega}})_{xyz}$. This calculation can be performed by choosing x', y', z' axes to have an angular velocity of $\boldsymbol{\Omega}' = \boldsymbol{\omega}_p$, Fig. 21–14c, so that

$$\dot{\boldsymbol{\omega}} = (\dot{\boldsymbol{\omega}})_{x'y'z'} + \boldsymbol{\omega}_p \times \boldsymbol{\omega} = \mathbf{0} + 3\mathbf{k} \times (6\mathbf{j} + 3\mathbf{k}) = \{-18\mathbf{i}\} \text{ rad/s}^2$$

$$\dot{\omega}_x = -18 \text{ rad/s}^2 \quad \dot{\omega}_y = 0 \quad \dot{\omega}_z = 0$$

The moment equations then become

$$\Sigma M_x = I_x \dot{\omega}_x - (I_y - I_z) \omega_y \omega_z$$

$$-A_z(0.5) + B_z(0.5) = 0.1(-18) - (0.2 - 0.1)(6)(3) = -3.6$$

$$\Sigma M_y = I_y \dot{\omega}_y - (I_z - I_x) \omega_z \omega_x$$

$$0 = 0 - 0$$

$$\Sigma M_z = I_z \dot{\omega}_z - (I_x - I_y) \omega_x \omega_y$$

$$A_x(0.5) - B_x(0.5) = 0 - 0$$

The solution then proceeds as before.

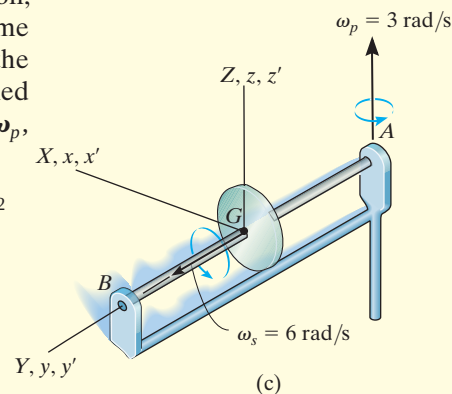


Fig. 21–14

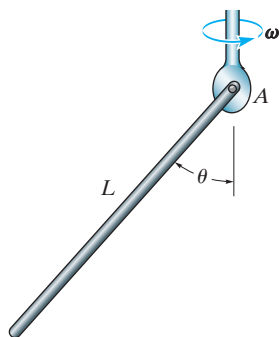
PROBLEMS

***21–40.** Derive the scalar form of the rotational equation of motion about the x axis if $\mathbf{\Omega} \neq \mathbf{\omega}$ and the moments and products of inertia of the body are *not constant* with respect to time.

21–41. Derive the scalar form of the rotational equation of motion about the x axis if $\mathbf{\Omega} \neq \mathbf{\omega}$ and the moments and products of inertia of the body are *constant* with respect to time.

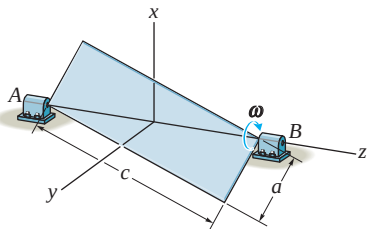
21–42. Derive the Euler equations of motion for $\mathbf{\Omega} \neq \mathbf{\omega}$, i.e., Eqs. 21–26.

21–43. The conical pendulum consists of a bar of mass m and length L that is supported by the pin at its end A . If the pin is subjected to a rotation ω , determine the angle θ that the bar makes with the vertical as it rotates.



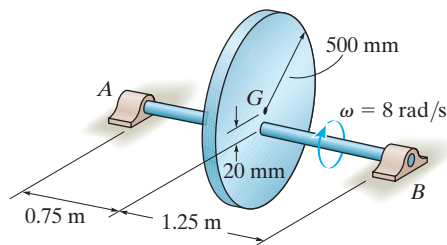
Prob. 21–43

***21–44.** The uniform plate has a mass of $m = 2$ kg and is given a rotation of $\omega = 4$ rad/s about its bearings at A and B . If $a = 0.2$ m and $c = 0.3$ m, determine the vertical reactions at the instant shown. Use the x , y , z axes shown and note that $I_{zx} = -\left(\frac{mac}{12}\right)\left(\frac{c^2 - a^2}{c^2 + a^2}\right)$.



Prob. 21–44

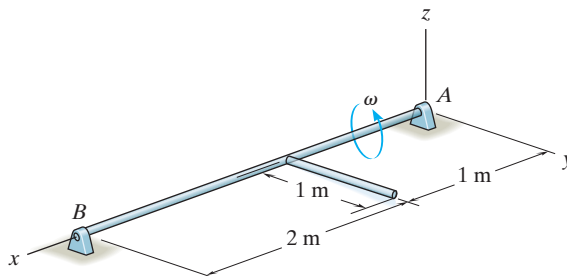
21–45. The 40-kg flywheel (disk) is mounted 20 mm off its true center at G . If the shaft is rotating at a constant speed $\omega = 8$ rad/s, determine the maximum reactions exerted on the journal bearings at A and B .



Prob. 21–45

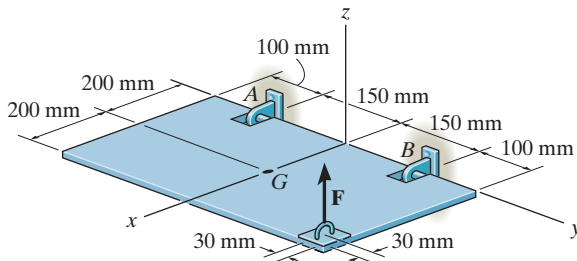
21–46. The assembly is supported by journal bearings at A and B , which develop only y and z force reactions on the shaft. If the shaft is rotating in the direction shown at $\mathbf{\omega} = \{2\mathbf{i}\}$ rad/s, determine the reactions at the bearings when the assembly is in the position shown. Also, what is the shaft's angular acceleration? The mass per unit length of each rod is 5 kg/m.

21–47. The assembly is supported by journal bearings at A and B , which develop only y and z force reactions on the shaft. If the shaft A is subjected to a couple moment $\mathbf{M} = \{40\mathbf{i}\}$ N·m, and at the instant shown the shaft has an angular velocity of $\mathbf{\omega} = \{2\mathbf{i}\}$ rad/s, determine the reactions at the bearings of the assembly at this instant. Also, what is the shaft's angular acceleration? The mass per unit length of each rod is 5 kg/m.



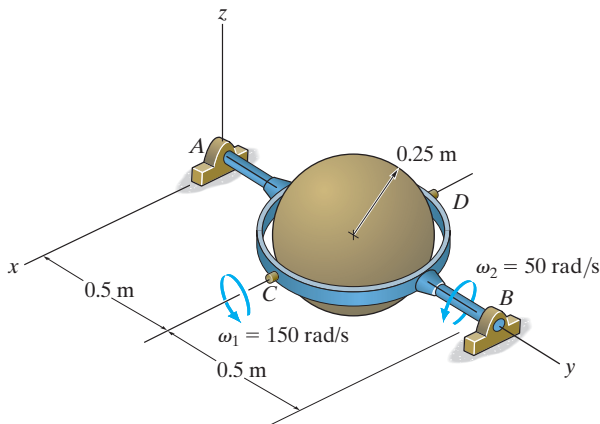
Probs. 21–46/47

***21–48.** The uniform hatch door, having a mass of 15 kg and a mass center at G , is supported in the horizontal plane by bearings at A and B . If a vertical force $F = 300$ N is applied to the door as shown, determine the components of reaction at the bearings and the angular acceleration of the door. The bearing at A will resist a component of force in the y direction, whereas the bearing at B will not. For the calculation, assume the door to be a thin plate and neglect the size of each bearing. The door is originally at rest.



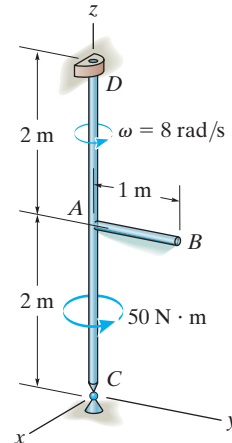
Prob. 21–48

21–49. The 20-kg sphere is rotating with a constant angular speed of $\omega_1 = 150$ rad/s about axle CD , which is mounted on the circular ring. The ring rotates about shaft AB with a constant angular speed of $\omega_2 = 50$ rad/s. If shaft AB is supported by a thrust bearing at A and a journal bearing at B , determine the X, Y, Z components of reaction at these bearings at the instant shown. Neglect the mass of the ring and shaft.



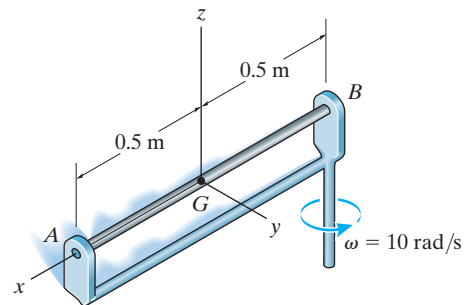
Prob. 21–49

21–50. The rod assembly is supported by a ball-and-socket joint at C and a journal bearing at D , which develops only x and y force reactions. The rods have a mass per unit length of 0.75 kg/m. Determine the angular acceleration of the rods and the components of reaction at the supports at the instant $\omega = 8$ rad/s as shown.



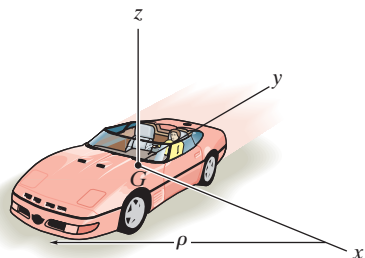
Prob. 21–50

21–51. The 5-kg rod AB is supported by a rotating arm. The support at A is a journal bearing, which develops reactions normal to the rod. The support at B is a thrust bearing, which develops reactions both normal to the rod and along the axis of the rod. Neglecting friction, determine the x, y, z components of reaction at these supports when the frame rotates with a constant angular velocity of $\omega = 10$ rad/s.



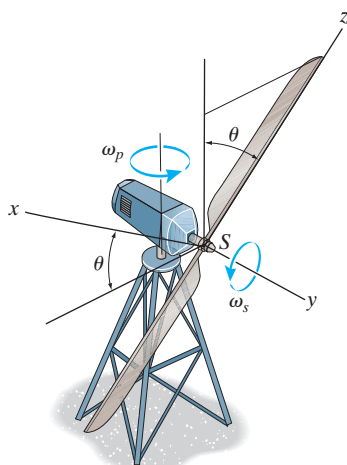
Prob. 21–51

***21-52.** The car travels around the curved road of radius ρ such that its mass center has a constant speed v_G . Write the equations of rotational motion with respect to the x , y , z axes. Assume that the car's six moments and products of inertia with respect to these axes are known.



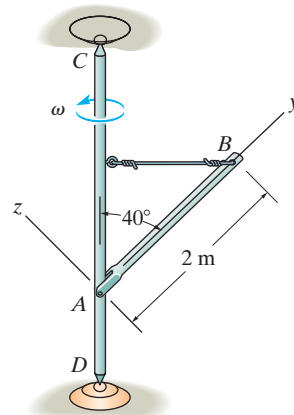
Prob. 21-52

21-53. The blades of a wind turbine spin about the shaft S with a constant angular speed of ω_s , while the frame precesses about the vertical axis with a constant angular speed of ω_p . Determine the x , y , and z components of moment that the shaft exerts on the blades as a function of θ . Consider each blade as a slender rod of mass m and length l .



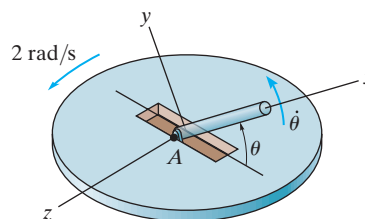
Prob. 21-53

21-54. The 4-kg slender rod AB is pinned at A and held at B by a cord. The axle CD is supported at its ends by ball-and-socket joints and is rotating with a constant angular velocity of 2 rad/s. Determine the tension developed in the cord and the magnitude of force developed at the pin A .



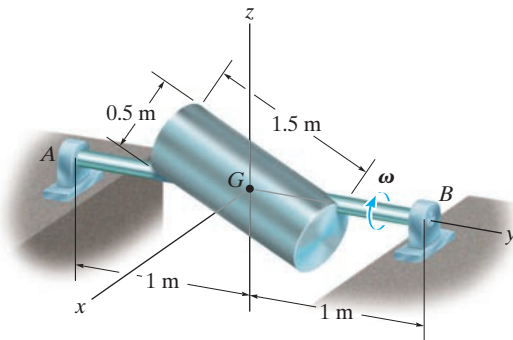
Prob. 21-54

21-55. The *thin rod* has a mass of 0.8 kg and a total length of 150 mm. It is rotating about its midpoint at a constant rate $\dot{\theta} = 6$ rad/s, while the table to which its axle A is fastened is rotating at 2 rad/s. Determine the x , y , z moment components which the axle exerts on the rod when the rod is in any position θ .



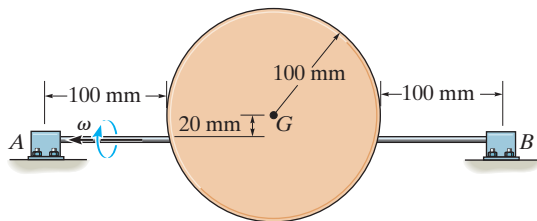
Prob. 21-55

***21–56.** The cylinder has mass of 30 kg and is mounted on an axle that is supported by bearings at A and B . If the axle is subjected to a couple moment $\mathbf{M} = \{-30\mathbf{j}\} \text{ N} \cdot \text{m}$, and at the instant shown has an angular velocity $\boldsymbol{\omega} = \{-40\mathbf{j}\} \text{ rad/s}$, determine the vertical components of force acting at the bearings at this instant.



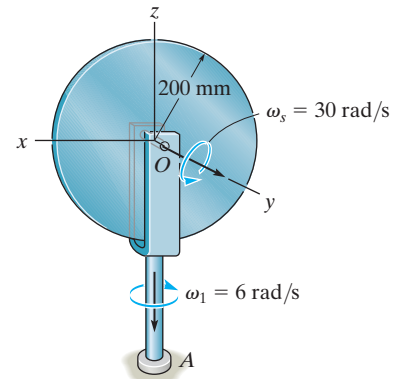
Prob. 21–56

21–57. The 5-kg circular disk is mounted off center on a shaft which is supported by bearings at A and B . If the shaft is rotating at a constant rate of $\omega = 10 \text{ rad/s}$, determine the vertical reactions at the bearings when the disk is in the position shown.



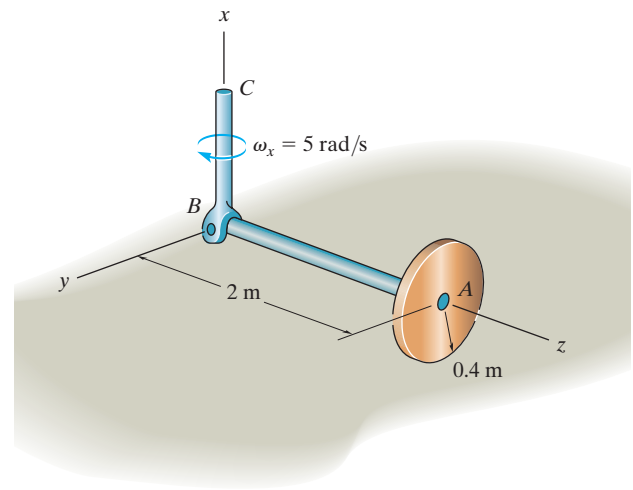
Prob. 21–57

21–58. The 20-kg disk is spinning on its axle at $\omega_s = 30 \text{ rad/s}$, while the forked rod is turning at $\omega_1 = 6 \text{ rad/s}$. Determine the x and z moment components the axle exerts on the disk during the motion.



Prob. 21–58

21–59. The 10-kg disk turns around the shaft AB , while the shaft rotates about BC at a constant rate of $\omega_x = 5 \text{ rad/s}$. If the disk does not slip, determine the normal and frictional force it exerts on the ground. Neglect the mass of shaft AB .



Prob. 21–59

*21.5 Gyroscopic Motion

In this section we will develop the equations defining the motion of a body (top) which is symmetrical with respect to an axis and rotating about a fixed point. These equations also apply to the motion of a particularly interesting device, the gyroscope.

The body's motion will be analyzed using *Euler angles* ϕ , θ , ψ (phi, theta, psi). To illustrate how they define the position of a body, consider the top shown in Fig. 21–15a. To define its final position, Fig. 21–15d, a second set of x, y, z axes is fixed in the top. Starting with the X, Y, Z and x, y, z axes in coincidence, Fig. 21–15a, the final position of the top can be determined using the following three steps:

1. Rotate the top about the Z (or z) axis through an angle ϕ ($0 \leq \phi < 2\pi$), Fig. 21–15b.
2. Rotate the top about the x axis through an angle θ ($0 \leq \theta \leq \pi$), Fig. 21–15c.
3. Rotate the top about the z axis through an angle ψ ($0 \leq \psi < 2\pi$) to obtain the final position, Fig. 21–15d.

The sequence of these three angles, ϕ , θ , then ψ , must be maintained, since finite rotations are *not* vectors (see Fig. 20–1). Although this is the case, the differential rotations $d\phi$, $d\theta$, and $d\psi$ are vectors, and thus the angular velocity $\boldsymbol{\omega}$ of the top can be expressed in terms of the time derivatives of the Euler angles. The angular-velocity components $\dot{\phi}$, $\dot{\theta}$, and $\dot{\psi}$ are known as the *precession*, *nutation*, and *spin*, respectively.

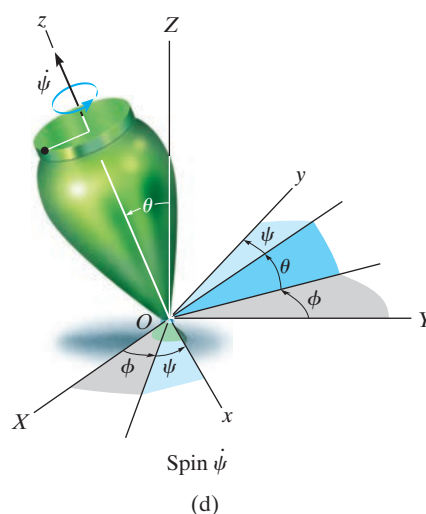
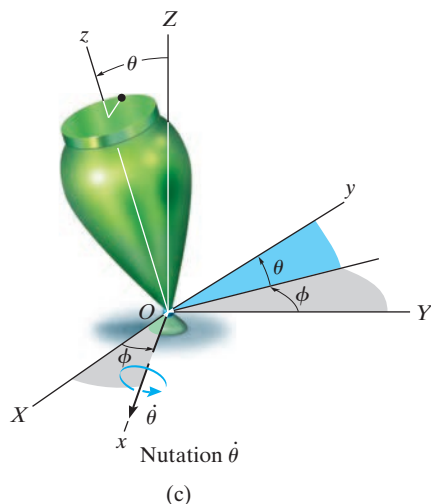
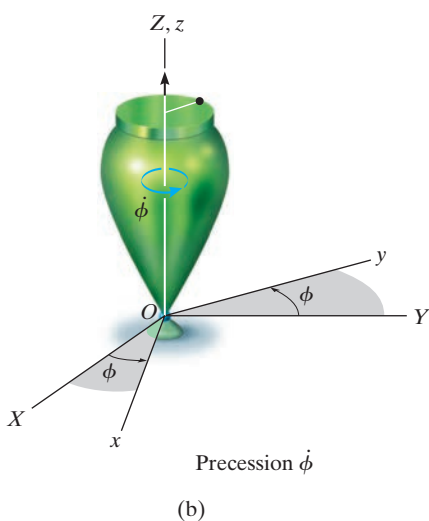
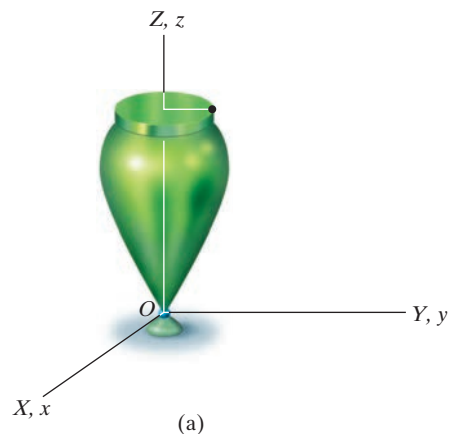


Fig. 21–15

Their positive directions are shown in Fig. 21-16. It is seen that these vectors are not all perpendicular to one another; however, $\boldsymbol{\omega}$ of the top can still be expressed in terms of these three components.

Since the body (top) is symmetric with respect to the z or spin axis, there is no need to attach the x, y, z axes to the top since the inertial properties of the top will remain constant with respect to this frame during the motion. Therefore $\boldsymbol{\Omega} = \boldsymbol{\omega}_p + \boldsymbol{\omega}_n$, Fig. 21-16. Hence, the angular velocity of the body is

$$\begin{aligned}\boldsymbol{\omega} &= \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k} \\ &= \dot{\theta} \mathbf{i} + (\dot{\phi} \sin \theta) \mathbf{j} + (\dot{\phi} \cos \theta + \dot{\psi}) \mathbf{k}\end{aligned}\quad (21-27)$$

And the angular velocity of the axes is

$$\begin{aligned}\boldsymbol{\Omega} &= \Omega_x \mathbf{i} + \Omega_y \mathbf{j} + \Omega_z \mathbf{k} \\ &= \dot{\theta} \mathbf{i} + (\dot{\phi} \sin \theta) \mathbf{j} + (\dot{\phi} \cos \theta) \mathbf{k}\end{aligned}\quad (21-28)$$

Have the x, y, z axes represent principal axes of inertia for the top, and so the moments of inertia will be represented as $I_{xx} = I_{yy} = I$ and $I_{zz} = I_z$. Since $\boldsymbol{\Omega} \neq \boldsymbol{\omega}$, Eqs. 21-26 are used to establish the rotational equations of motion. Substituting into these equations the respective angular-velocity components defined by Eqs. 21-27 and 21-28, their corresponding time derivatives, and the moment of inertia components, yields

$$\begin{aligned}\Sigma M_x &= I(\ddot{\theta} - \dot{\phi}^2 \sin \theta \cos \theta) + I_z \dot{\phi} \sin \theta (\dot{\phi} \cos \theta + \dot{\psi}) \\ \Sigma M_y &= I(\ddot{\phi} \sin \theta + 2\dot{\phi} \dot{\theta} \cos \theta) - I_z \dot{\theta} (\dot{\phi} \cos \theta + \dot{\psi}) \\ \Sigma M_z &= I_z(\ddot{\psi} + \dot{\phi} \cos \theta - \dot{\phi} \dot{\theta} \sin \theta)\end{aligned}\quad (21-29)$$

Each moment summation applies only at the fixed point O or the center of mass G of the body. Since the equations represent a coupled set of nonlinear second-order differential equations, in general a closed-form solution may not be obtained. Instead, the Euler angles ϕ , θ , and ψ may be obtained graphically as functions of time using numerical analysis and computer techniques.

A special case, however, does exist for which simplification of Eqs. 21-29 is possible. Commonly referred to as *steady precession*, it occurs when the nutation angle θ , precession $\dot{\phi}$, and spin $\dot{\psi}$ all remain *constant*. Equations 21-29 then reduce to the form

$$\Sigma M_x = -I \dot{\phi}^2 \sin \theta \cos \theta + I_z \dot{\phi} \sin \theta (\dot{\phi} \cos \theta + \dot{\psi}) \quad (21-30)$$

$$\Sigma M_y = 0$$

$$\Sigma M_z = 0$$

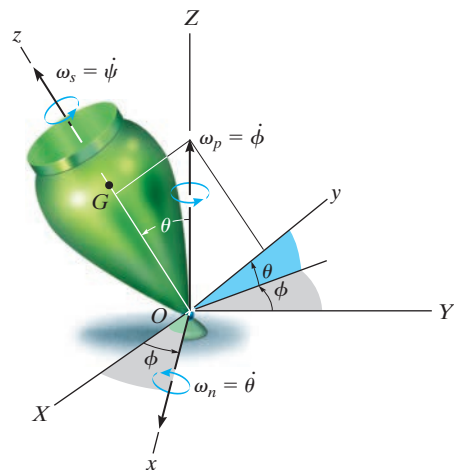


Fig. 21-16

Equation 21–30 can be further simplified by noting that, from Eq. 21–27, $\omega_z = \dot{\phi} \cos \theta + \dot{\psi}$, so that

$$\Sigma M_x = -I\dot{\phi}^2 \sin \theta \cos \theta + I_z \dot{\phi} (\sin \theta) \omega_z$$

or

$$\Sigma M_x = \dot{\phi} \sin \theta (I_z \omega_z - I \dot{\phi} \cos \theta) \quad (21-31)$$

It is interesting to note what effects the spin $\dot{\psi}$ has on the moment about the x axis. To show this, consider the spinning rotor in Fig. 21–17. Here $\theta = 90^\circ$, in which case Eq. 21–30 reduces to the form

$$\Sigma M_x = I_z \dot{\phi} \dot{\psi}$$

or

$$\Sigma M_x = I_z \Omega_y \omega_z \quad (21-32)$$

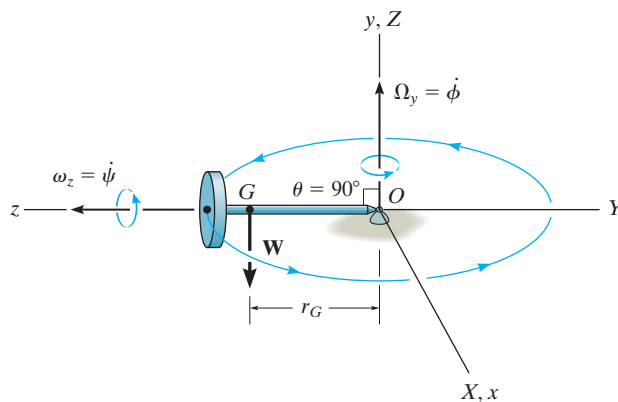


Fig. 21–17

From the figure it can be seen that Ω_y and ω_z act along their respective *positive axes* and therefore are mutually perpendicular. Instinctively, one would expect the rotor to fall down under the influence of gravity! However, this is not the case at all, provided the product $I_z \Omega_y \omega_z$ is correctly chosen to counterbalance the moment $\Sigma M_x = W r_G$ of the rotor's weight about O . This unusual phenomenon of rigid-body motion is often referred to as the *gyroscopic effect*.

Perhaps a more intriguing demonstration of the gyroscopic effect comes from studying the action of a *gyroscope*, frequently referred to as a *gyro*. A gyro is a rotor which spins at a very high rate about its axis of symmetry. This rate of spin is considerably greater than its precessional rate of rotation about the vertical axis. Hence, for all practical purposes, the angular momentum of the gyro can be assumed directed along its axis of spin. Thus, for the gyro rotor shown in Fig. 21-18, $\omega_z \gg \Omega_y$, and the magnitude of the angular momentum about point O , as determined from Eqs. 21-11, reduces to the form $H_O = I_z \omega_z$. Since both the magnitude and direction of $\mathbf{H}_O$ are constant as observed from x, y, z , direct application of Eq. 21-22 yields

$$\Sigma \mathbf{M}_x = \Omega_y \times \mathbf{H}_O$$

(21-33)

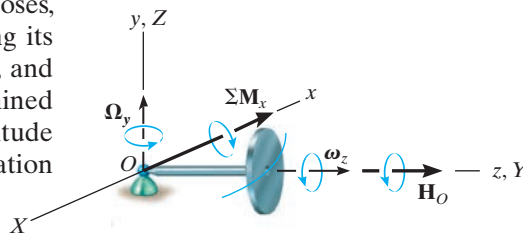


Fig. 21-18

Using the right-hand rule applied to the cross product, it can be seen that Ω_y always swings $\mathbf{H}_O$ (or ω_z) toward the sense of $\Sigma \mathbf{M}_x$. In effect, the *change in direction* of the gyro's angular momentum, $d\mathbf{H}_O$, is equivalent to the angular impulse caused by the gyro's weight about O , i.e., $d\mathbf{H}_O = \Sigma \mathbf{M}_x dt$, Eq. 21-20. Also, since $H_O = I_z \omega_z$ and $\Sigma \mathbf{M}_x$, Ω_y , and $\mathbf{H}_O$ are mutually perpendicular, Eq. 21-33 reduces to Eq. 21-32.

When a gyro is mounted in gimbal rings, Fig. 21-19, it becomes *free* of external moments applied to its base. Thus, in theory, its angular momentum $\mathbf{H}$ will never precess but, instead, maintain its same fixed orientation along the axis of spin when the base is rotated. This type of gyroscope is called a *free gyro* and is useful as a gyrocompass when the spin axis of the gyro is directed north. In reality, the gimbal mechanism is never completely free of friction, so such a device is useful only for the local navigation of ships and aircraft. The gyroscopic effect is also useful as a means of stabilizing both the rolling motion of ships at sea and the trajectories of missiles and projectiles. Furthermore, this effect is of significant importance in the design of shafts and bearings for rotors which are subjected to forced precessions.

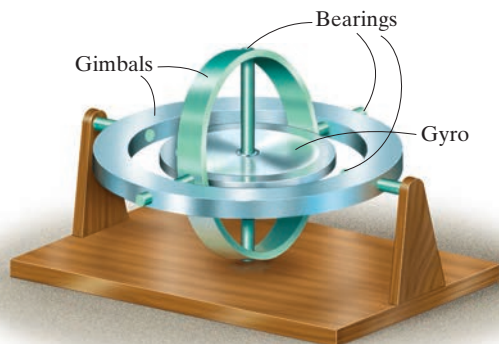
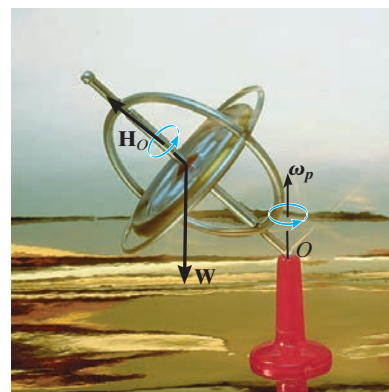


Fig. 21-19



The spinning of the gyro within the frame of this toy gyroscope produces angular momentum $\mathbf{H}_O$, which is changing direction as the frame precesses ω_p about the vertical axis. The gyroscope will not fall down since the moment of its weight $\mathbf{W}$ about the support is balanced by the change in the direction of $\mathbf{H}_O$.

EXAMPLE 21.7

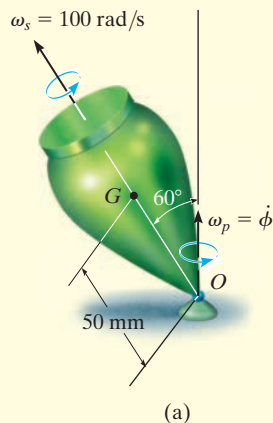
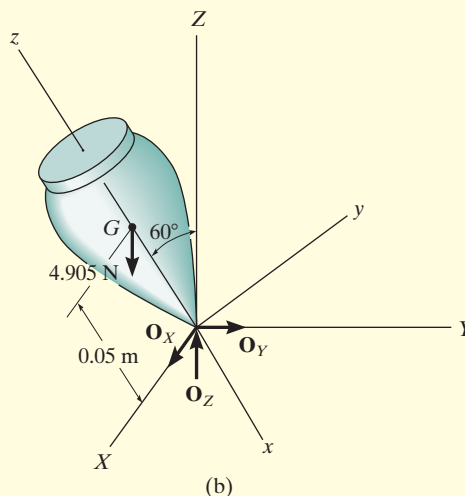


Fig. 21-20

The top shown in Fig. 21-20a has a mass of 0.5 kg and is precessing about the vertical axis at a constant angle of $\theta = 60^\circ$. If it spins with an angular velocity $\omega_s = 100 \text{ rad/s}$, determine the precession ω_p . Assume that the axial and transverse moments of inertia of the top are $0.45(10^{-3}) \text{ kg} \cdot \text{m}^2$ and $1.20(10^{-3}) \text{ kg} \cdot \text{m}^2$, respectively, measured with respect to the fixed point O .



SOLUTION

Equation 21-30 will be used for the solution since the motion is *steady precession*. As shown on the free-body diagram, Fig. 21-20b, the coordinate axes are established in the usual manner, that is, with the positive z axis in the direction of spin, the positive Z axis in the direction of precession, and the positive x axis in the direction of the moment ΣM_x (refer to Fig. 21-16). Thus,

$$\Sigma M_x = -I\dot{\phi}^2 \sin \theta \cos \theta + I_z \dot{\phi} \sin \theta (\dot{\phi} \cos \theta + \dot{\psi})$$

$$4.905 \text{ N}(0.05 \text{ m}) \sin 60^\circ = -[1.20(10^{-3}) \text{ kg} \cdot \text{m}^2 \dot{\phi}^2] \sin 60^\circ \cos 60^\circ + [0.45(10^{-3}) \text{ kg} \cdot \text{m}^2 \dot{\phi} \sin 60^\circ (\dot{\phi} \cos 60^\circ + 100 \text{ rad/s})]$$

or

$$\dot{\phi}^2 - 120.0\dot{\phi} + 654.0 = 0 \quad (1)$$

Solving this quadratic equation for the precession gives

$$\dot{\phi} = 114 \text{ rad/s} \quad (\text{high precession}) \quad \text{Ans.}$$

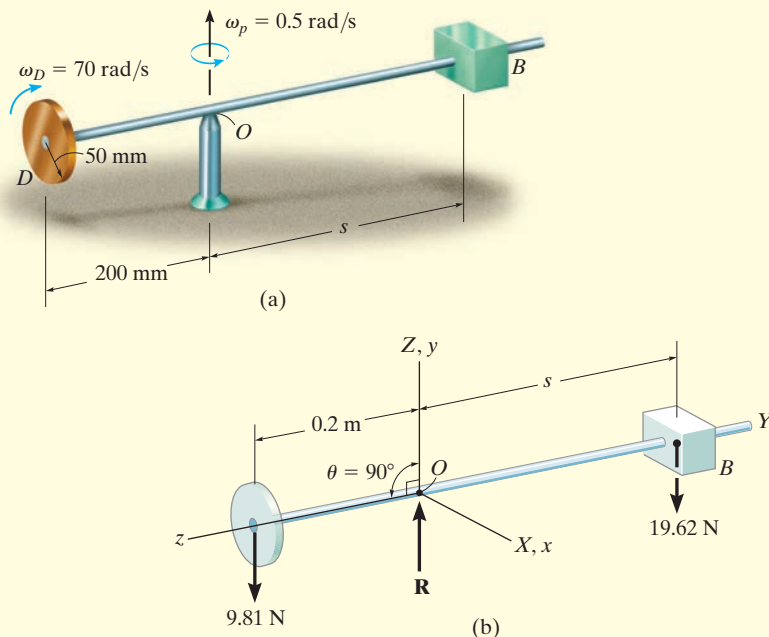
and

$$\dot{\phi} = 5.72 \text{ rad/s} \quad (\text{low precession}) \quad \text{Ans.}$$

NOTE: In reality, low precession of the top would generally be observed, since high precession would require a larger kinetic energy.

EXAMPLE 21.8

The 1-kg disk shown in Fig. 21–21*a* spins about its axis with a constant angular velocity $\omega_D = 70 \text{ rad/s}$. The block at *B* has a mass of 2 kg, and by adjusting its position *s* one can change the precession of the disk about its supporting pivot at *O* while the shaft remains horizontal. Determine the position *s* that will enable the disk to have a constant precession $\omega_p = 0.5 \text{ rad/s}$ about the pivot. Neglect the weight of the shaft.

**Fig. 21–21****SOLUTION**

The free-body diagram of the assembly is shown in Fig. 21–21*b*. The origin for both the *x, y, z* and *X, Y, Z* coordinate systems is located at the fixed point *O*. In the conventional sense, the *Z* axis is chosen along the axis of precession, and the *z* axis is along the axis of spin, so that $\theta = 90^\circ$. Since the precession is *steady*, Eq. 21–32 can be used for the solution.

$$\Sigma M_x = I_z \Omega_y \omega_z$$

Substituting the required data gives

$$(9.81 \text{ N})(0.2 \text{ m}) - (19.62 \text{ N})s = \left[\frac{1}{2}(1 \text{ kg})(0.05 \text{ m})^2 \right] 0.5 \text{ rad/s}(-70 \text{ rad/s})$$

$$s = 0.102 \text{ m} = 102 \text{ mm} \quad \text{Ans.}$$

21.6 Torque-Free Motion

When the only external force acting on a body is caused by gravity, the general motion of the body is referred to as *torque-free motion*. This type of motion is characteristic of planets, artificial satellites, and projectiles—provided air friction is neglected.

In order to describe the characteristics of this motion, the distribution of the body's mass will be assumed *axisymmetric*. The satellite shown in Fig. 21–22 is an example of such a body, where the z axis represents an axis of symmetry. The origin of the x, y, z coordinates is located at the mass center G , such that $I_{zz} = I_z$ and $I_{xx} = I_{yy} = I$. Since gravity is the only external force present, the summation of moments about the mass center is zero. From Eq. 21–21, this requires the angular momentum of the body to be constant, i.e.,

$$\mathbf{H}_G = \text{constant}$$

At the instant considered, it will be assumed that the inertial frame of reference is oriented so that the positive Z axis is directed along $\mathbf{H}_G$ and the y axis lies in the plane formed by the z and Z axes, Fig. 21–22. The Euler angle formed between Z and z is θ , and therefore, with this choice of axes the angular momentum can be expressed as

$$\mathbf{H}_G = H_G \sin \theta \mathbf{j} + H_G \cos \theta \mathbf{k}$$

Furthermore, using Eqs. 21–11, we have

$$\mathbf{H}_G = I\omega_x \mathbf{i} + I\omega_y \mathbf{j} + I_z\omega_z \mathbf{k}$$

Equating the respective $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ components of the above two equations yields

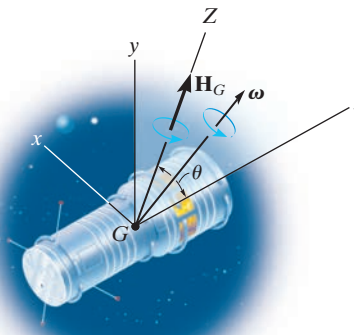


Fig. 21–22

$$\omega_x = 0 \quad \omega_y = \frac{H_G \sin \theta}{I} \quad \omega_z = \frac{H_G \cos \theta}{I_z} \quad (21-34)$$

or

$$\boldsymbol{\omega} = \frac{H_G \sin \theta}{I} \mathbf{j} + \frac{H_G \cos \theta}{I_z} \mathbf{k} \quad (21-35)$$

In a similar manner, equating the respective $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ components of Eq. 21-27 to those of Eq. 21-34, we obtain

$$\begin{aligned} \dot{\theta} &= 0 \\ \dot{\phi} \sin \theta &= \frac{H_G \sin \theta}{I} \\ \dot{\phi} \cos \theta + \dot{\psi} &= \frac{H_G \cos \theta}{I_z} \end{aligned}$$

Solving, we get

$$\begin{aligned} \theta &= \text{constant} \\ \dot{\phi} &= \frac{H_G}{I} \\ \dot{\psi} &= \frac{I - I_z}{I I_z} H_G \cos \theta \end{aligned} \quad (21-36)$$

Thus, for torque-free motion of an axisymmetrical body, the angle θ formed between the angular-momentum vector and the spin of the body remains constant. Furthermore, the angular momentum $\mathbf{H}_G$, precession $\dot{\phi}$, and spin $\dot{\psi}$ for the body remain constant at all times during the motion.

Eliminating H_G from the second and third of Eqs. 21-36 yields the following relation between the spin and precession:

$$\dot{\psi} = \frac{I - I_z}{I_z} \dot{\phi} \cos \theta \quad (21-37)$$

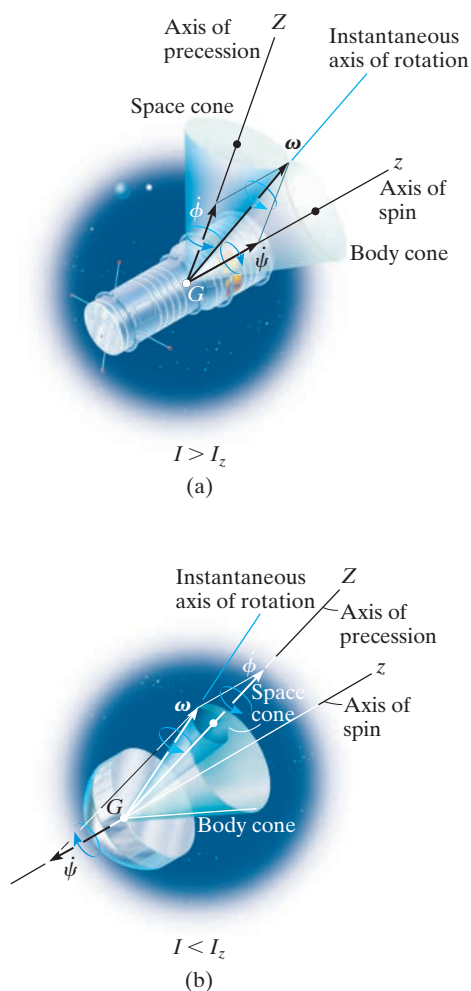
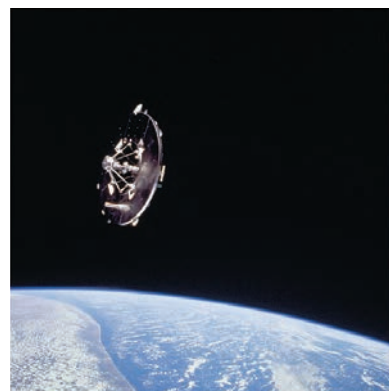
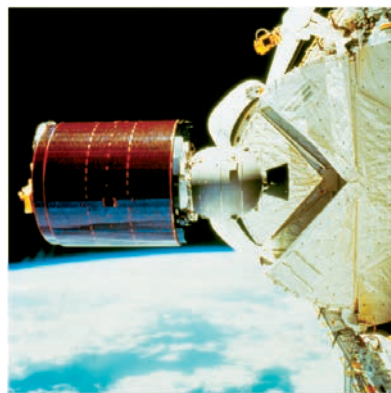


Fig. 21-23

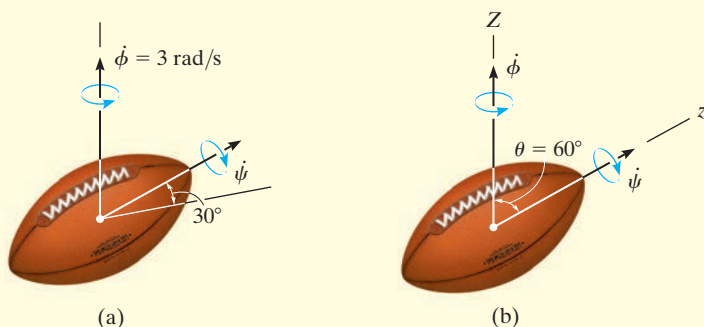
These two components of angular motion can be studied by using the body and space cone models introduced in Sec. 20.1. The *space cone* defining the precession is fixed from rotating, since the precession has a fixed direction, while the outer surface of the *body cone* rolls on the space cone's outer surface. Try to imagine this motion in Fig. 21-23a. The interior angle of each cone is chosen such that the resultant angular velocity of the body is directed along the line of contact of the two cones. This line of contact represents the instantaneous axis of rotation for the body cone, and hence the angular velocity of both the body cone and the body must be directed along this line. Since the spin is a function of the moments of inertia I and I_z of the body, Eq. 21-36, the cone model in Fig. 21-23a is satisfactory for describing the motion, provided $I > I_z$. Torque-free motion which meets these requirements is called *regular precession*. If $I < I_z$, the spin is negative and the precession positive. This motion is represented by the satellite motion shown in Fig. 21-23b ($I < I_z$). The cone model can again be used to represent the motion; however, to preserve the correct vector addition of spin and precession to obtain the angular velocity ω , the inside surface of the body cone must roll on the outside surface of the (fixed) space cone. This motion is referred to as *retrograde precession*.

Satellites are often given a spin before they are launched. If their angular momentum is not collinear with the axis of spin, they will exhibit precession. In the photo on the left, regular precession will occur since $I > I_z$, and in the photo on the right, retrograde precession will occur since $I < I_z$.



EXAMPLE 21.9

The motion of a football is observed using a slow-motion projector. From the film, the spin of the football is seen to be directed 30° from the horizontal, as shown in Fig. 21–24*a*. Also, the football is precessing about the vertical axis at a rate $\dot{\phi} = 3 \text{ rad/s}$. If the ratio of the axial to transverse moments of inertia of the football is $\frac{1}{3}$, measured with respect to the center of mass, determine the magnitude of the football's spin and its angular velocity. Neglect the effect of air resistance.

**Fig. 21–24****SOLUTION**

Since the weight of the football is the only force acting, the motion is torque-free. In the conventional sense, if the z axis is established along the axis of spin and the Z axis along the precession axis, as shown in Fig. 21–24*b*, then the angle $\theta = 60^\circ$. Applying Eq. 21–37, the spin is

$$\begin{aligned}\dot{\psi} &= \frac{I - I_z}{I_z} \dot{\phi} \cos \theta = \frac{I - \frac{1}{3}I}{\frac{1}{3}I} (3) \cos 60^\circ \\ &= 3 \text{ rad/s} \quad \text{Ans.}\end{aligned}$$

Using Eqs. 21–34, where $H_G = \dot{\phi} I$ (Eq. 21–36), we have

$$\begin{aligned}\omega_x &= 0 \\ \omega_y &= \frac{H_G \sin \theta}{I} = \frac{3I \sin 60^\circ}{I} = 2.60 \text{ rad/s} \\ \omega_z &= \frac{H_G \cos \theta}{I_z} = \frac{3I \cos 60^\circ}{\frac{1}{3}I} = 4.50 \text{ rad/s}\end{aligned}$$

Thus,

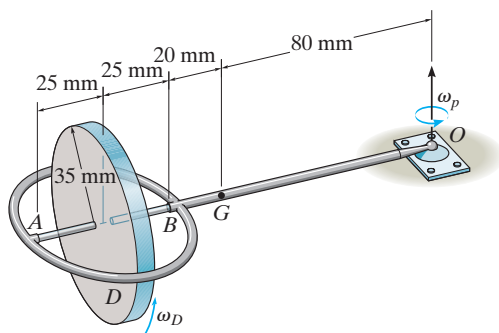
$$\begin{aligned}\omega &= \sqrt{(\omega_x)^2 + (\omega_y)^2 + (\omega_z)^2} \\ &= \sqrt{(0)^2 + (2.60)^2 + (4.50)^2} \\ &= 5.20 \text{ rad/s} \quad \text{Ans.}\end{aligned}$$

PROBLEMS

***21–60.** Show that the angular velocity of a body, in terms of Euler angles ϕ , θ , and ψ , can be expressed as $\omega = (\dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi)\mathbf{i} + (\dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi)\mathbf{j} + (\dot{\phi} \cos \theta + \dot{\psi})\mathbf{k}$, where $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are directed along the x , y , z axes as shown in Fig. 21–15*d*.

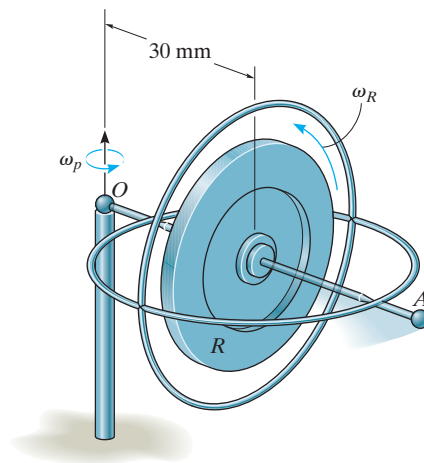
21–61. A thin rod is initially coincident with the Z axis when it is given three rotations defined by the Euler angles $\phi = 30^\circ$, $\theta = 45^\circ$, and $\psi = 60^\circ$. If these rotations are given in the order stated, determine the coordinate direction angles α , β , γ of the axis of the rod with respect to the X , Y , and Z axes. Are these directions the same for any order of the rotations? Why?

21–62. The gyroscope consists of a uniform 450-g disk D which is attached to the axle AB of negligible mass. The supporting frame has a mass of 180 g and a center of mass at G . If the disk is rotating about the axle at $\omega_D = 90 \text{ rad/s}$, determine the constant angular velocity ω_p at which the frame precesses about the pivot point O . The frame moves in the horizontal plane.



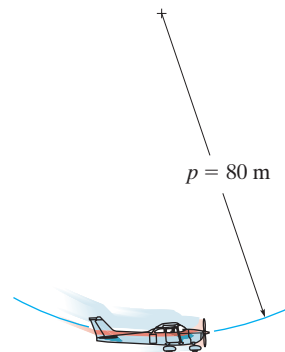
Prob. 21–62

21–63. The toy gyroscope consists of a rotor R which is attached to the frame of negligible mass. If it is observed that the frame is precessing about the pivot point O at $\omega_p = 2 \text{ rad/s}$, determine the angular velocity ω_R of the rotor. The stem OA moves in the horizontal plane. The rotor has a mass of 200 g and a radius of gyration $k_{OA} = 20 \text{ mm}$ about OA .



Prob. 21–63

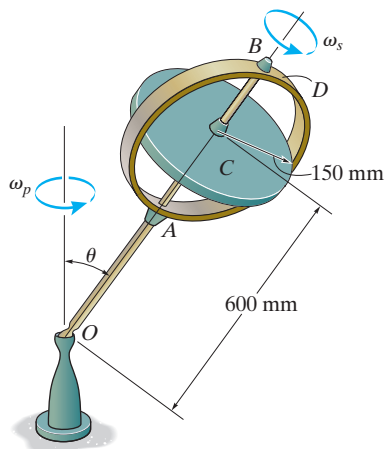
***21–64.** The propeller on a single-engine airplane has a mass of 15 kg and a centroidal radius of gyration of 0.3 m computed about the axis of spin. When viewed from the front of the airplane, the propeller is turning clockwise at 350 rad/s about the spin axis. If the airplane enters a vertical curve having a radius of 80 m and is traveling at 200 km/h, determine the gyroscopic bending moment which the propeller exerts on the bearings of the engine when the airplane is in its lowest position.



Prob. 21–64

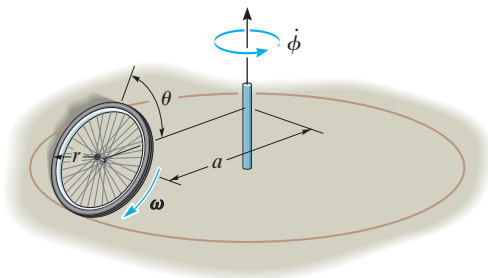
21–65. The 10-kg disk spins about axle AB at a constant rate of $\omega_s = 250$ rad/s, and $\theta = 30^\circ$. Determine the rate of precession of arm OA . Neglect the mass of arm OA , axle AB , and the circular ring D .

21–66. When OA precesses at a constant rate of $\omega_p = 5$ rad/s, when $\theta = 90^\circ$, determine the required spin of the 10-kg disk C . Neglect the mass of arm OA , axle AB , and the circular ring D .



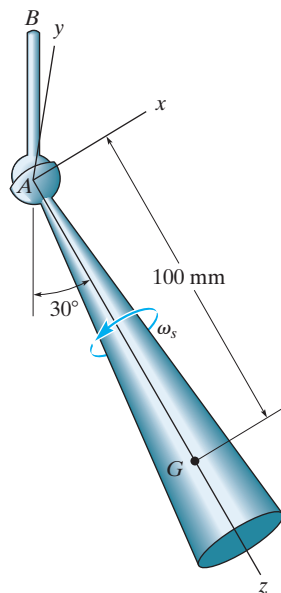
Probs. 21–65/66

21–67. A wheel of mass m and radius r rolls with constant spin ω about a circular path having a radius a . If the angle of inclination is θ , determine the rate of precession. Treat the wheel as a thin ring. No slipping occurs.



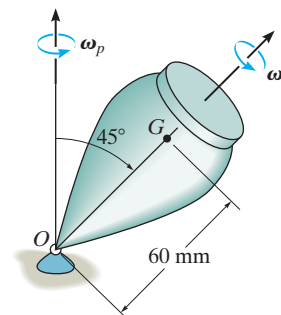
Prob. 21–67

***21–68.** The conical top has a mass of 0.8 kg, and the moments of inertia are $I_x = I_y = 3.5(10^{-3})$ kg·m² and $I_z = 0.8(10^{-3})$ kg·m². If it spins freely in the ball-and-socket joint at A with an angular velocity $\omega_s = 750$ rad/s, compute the precession of the top about the axis of the shaft AB .



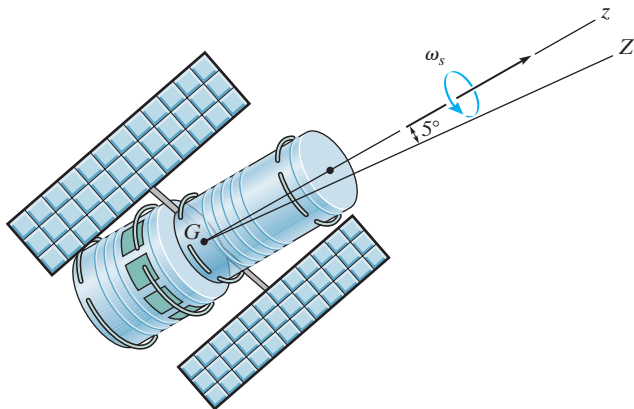
Prob. 21–68

21–69. The top has a mass of 90 g, a center of mass at G , and a radius of gyration $k = 18$ mm about its axis of symmetry. About any transverse axis acting through point O the radius of gyration is $k_t = 35$ mm. If the top is connected to a ball-and-socket joint at O and the precession is $\omega_p = 0.5$ rad/s, determine the spin ω_s .



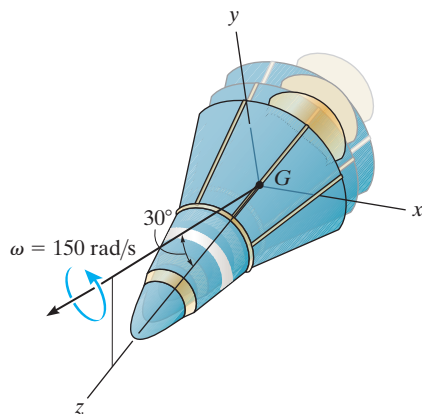
Prob. 21–69

21-70. The satellite has a mass of 1.8 Mg. The axial and transverse radii of gyration about axes passing through the mass center G are $k_z = 0.8$ m and $k_t = 1.2$ m, respectively. If it is spinning at $\omega_s = 6$ rad/s when it is launched, determine its angular momentum. Precession occurs about the Z axis.



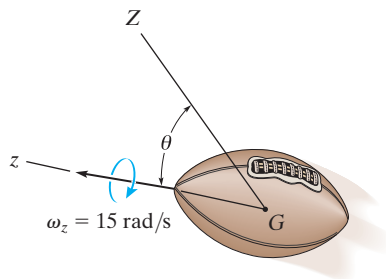
Prob. 21-70

21-71. The space capsule has a mass of 2 Mg, center of mass at G , and radii of gyration about its axis of symmetry (z axis) and its transverse axes (x or y axis) of $k_z = 2.75$ m and $k_x = k_y = 5.5$ m, respectively. If the capsule has the angular velocity shown, determine its precession ϕ and spin ψ . Indicate whether the precession is regular or retrograde. Also, draw the space cone and body cone for the motion.



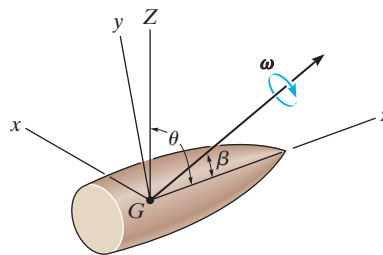
Prob. 21-71

***21-72.** The 0.25 kg football is spinning at $\omega_z = 15$ rad/s as shown. If $\theta = 40^\circ$, determine the precession about the z axis. The radius of gyration about the spin axis is $k_z = 0.042$ m, and about a transverse axis is $k_y = 0.13$ m.



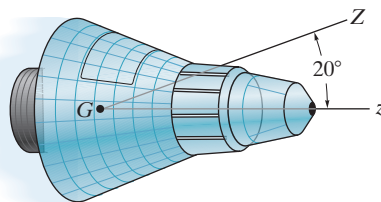
Prob. 21-72

21-73. The projectile shown is subjected to torque-free motion. The transverse and axial moments of inertia are I and I_z , respectively. If θ represents the angle between the precessional axis Z and the axis of symmetry z , and β is the angle between the angular velocity ω and the z axis, show that β and θ are related by the equation $\tan \theta = (I/I_z) \tan \beta$.



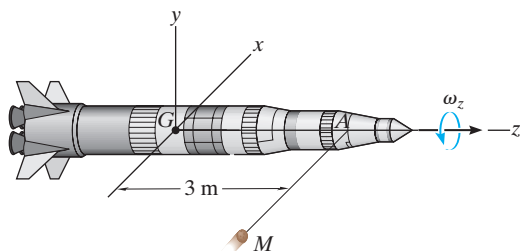
Prob. 21-73

21-74. The radius of gyration about an axis passing through the axis of symmetry of the 1.6-Mg space capsule is $k_z = 1.2$ m and about any transverse axis passing through the center of mass G , $k_t = 1.8$ m. If the capsule has a known steady-state precession of two revolutions per hour about the Z axis, determine the rate of spin about the z axis.



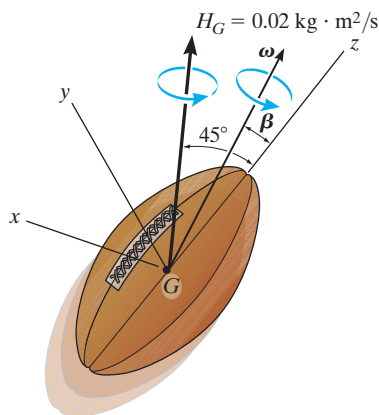
Prob. 21-74

21-75. The rocket has a mass of 4 Mg and radii of gyration $k_z = 0.85$ m and $k_x = k_y = 2.3$ m. It is initially spinning about the z axis at $\omega_z = 0.05$ rad/s when a meteoroid M strikes it at A and creates an impulse $\mathbf{I} = \{300\mathbf{i}\}$ N·s. Determine the axis of precession after the impact.



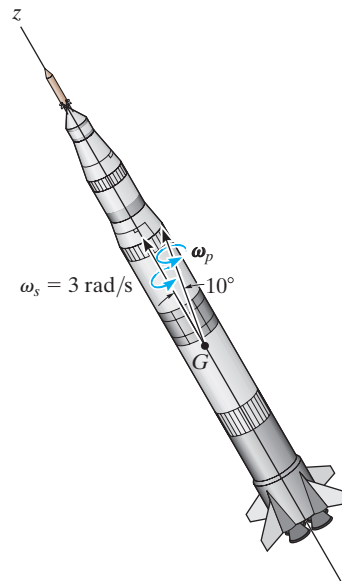
Prob. 21-75

***21-76.** The football has a mass of 450 g and radii of gyration about its axis of symmetry (z axis) and its transverse axes (x or y axis) of $k_z = 30$ mm and $k_x = k_y = 50$ mm, respectively. If the football has an angular momentum of $H_G = 0.02$ kg·m²/s, determine its precession $\dot{\phi}$ and spin $\dot{\psi}$. Also, find the angle β that the angular velocity vector makes with the z axis.



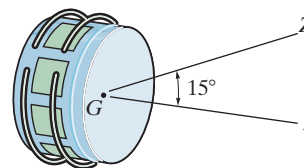
Prob. 21-76

21-77. While the rocket is in free flight, it has a spin of 3 rad/s and precesses about an axis measured 10° from the axis of spin. If the ratio of the axial to transverse moments of inertia of the rocket is $1/15$, computed about axes which pass through the mass center G , determine the angle which the resultant angular velocity makes with the spin axis. Construct the body and space cones used to describe the motion. Is the precession regular or retrograde?



Prob. 21-77

21-78. The radius of gyration about an axis passing through the axis of symmetry of the 1.2-Mg satellite is $k_z = 1.4$ m, and about any transverse axis passing through the center of mass G , $k_t = 2.20$ m. If the satellite has a known spin of 2700 rev/h about the z axis, determine the steady-state precession about the z axis.



Prob. 21-78

CHAPTER REVIEW

Moments and Products of Inertia

A body has six components of inertia for any specified x, y, z axes. Three of these are moments of inertia about each of the axes, I_{xx} , I_{yy} , I_{zz} , and three are products of inertia, each defined from two orthogonal planes, I_{xy} , I_{yz} , I_{xz} . If either one or both of these planes are planes of symmetry, then the product of inertia with respect to these planes will be zero.

The moments and products of inertia can be determined by direct integration or by using tabulated values. If these quantities are to be determined with respect to axes or planes that do not pass through the mass center, then parallel-axis and parallel-plane theorems must be used.

Provided the six components of inertia are known, then the moment of inertia about any axis can be determined using the inertia transformation equation.

$$\begin{aligned} I_{xx} &= \int_m r_x^2 dm = \int_m (y^2 + z^2) dm & I_{xy} &= I_{yx} = \int_m xy dm \\ I_{yy} &= \int_m r_y^2 dm = \int_m (x^2 + z^2) dm & I_{yz} &= I_{zy} = \int_m yz dm \\ I_{zz} &= \int_m r_z^2 dm = \int_m (x^2 + y^2) dm & I_{xz} &= I_{zx} = \int_m xz dm \end{aligned}$$

$$I_{Oa} = I_{xx}u_x^2 + I_{yy}u_y^2 + I_{zz}u_z^2 - 2I_{xy}u_xu_y - 2I_{yz}u_yu_z - 2I_{zx}u_zu_x$$

Principal Moments of Inertia

At any point on or off the body, the x, y, z axes can be oriented so that the products of inertia will be zero. The resulting moments of inertia are called the principal moments of inertia. In general, one will be a maximum and the other a minimum.

$$\begin{pmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{pmatrix}$$

Principle of Impulse and Momentum

The angular momentum for a body can be determined about any arbitrary point A .

Once the linear and angular momentum for the body have been formulated, then the principle of impulse and momentum can be used to solve problems that involve force, velocity, and time.

$$\begin{aligned} m(\mathbf{v}_G)_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt &= m(\mathbf{v}_G)_2 & (\mathbf{H}_O)_1 + \sum \int_{t_1}^{t_2} \mathbf{M}_O dt &= (\mathbf{H}_O)_2 \\ \mathbf{H}_O &= \int_m \boldsymbol{\rho}_O \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_O) dm & \text{where} \\ &\text{Fixed Point } O & H_x &= I_{xx}\omega_x - I_{xy}\omega_y - I_{xz}\omega_z \\ & & H_y &= -I_{yx}\omega_x + I_{yy}\omega_y - I_{yz}\omega_z \\ \mathbf{H}_G &= \int_m \boldsymbol{\rho}_G \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_G) dm & H_z &= -I_{zx}\omega_x - I_{zy}\omega_y + I_{zz}\omega_z \\ &\text{Center of Mass} & & \\ \mathbf{H}_A &= \boldsymbol{\rho}_{G/A} \times m\mathbf{v}_G + \mathbf{H}_G & & \\ &\text{Arbitrary Point} & & \end{aligned}$$

Principle of Work and Energy

The kinetic energy for a body is usually determined relative to a fixed point or the body's mass center.

$$\begin{aligned} T &= \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2 + \frac{1}{2}I_z\omega_z^2 & T &= \frac{1}{2}mv_G^2 + \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2 + \frac{1}{2}I_z\omega_z^2 \\ &\text{Fixed Point} & & \text{Center of Mass} \end{aligned}$$

These formulations can be used with the principle of work and energy to solve problems that involve force, velocity, and displacement.

$$T_1 + \Sigma U_{1-2} = T_2$$

Equations of Motion

There are three scalar equations of translational motion for a rigid body that moves in three dimensions.

The three scalar equations of rotational motion depend upon the motion of the x , y , z reference. Most often, these axes are oriented so that they are principal axes of inertia. If the axes are fixed in and move with the body so that $\mathbf{\Omega} = \mathbf{\omega}$, then the equations are referred to as the Euler equations of motion.

A free-body diagram should always accompany the application of the equations of motion.

$$\Sigma F_x = m(a_G)_x$$

$$\Sigma F_y = m(a_G)_y$$

$$\Sigma F_z = m(a_G)_z$$

$$\Sigma M_x = I_x \dot{\omega}_x - (I_y - I_z) \omega_y \omega_z$$

$$\Sigma M_y = I_y \dot{\omega}_y - (I_z - I_x) \omega_z \omega_x$$

$$\Sigma M_z = I_z \dot{\omega}_z - (I_x - I_y) \omega_x \omega_y$$

$$\mathbf{\Omega} = \mathbf{\omega}$$

$$\Sigma M_x = I_x \dot{\omega}_x - I_y \Omega_z \omega_y + I_z \Omega_y \omega_z$$

$$\Sigma M_y = I_y \dot{\omega}_y - I_z \Omega_x \omega_z + I_x \Omega_z \omega_x$$

$$\Sigma M_z = I_z \dot{\omega}_z - I_x \Omega_y \omega_x + I_y \Omega_x \omega_y$$

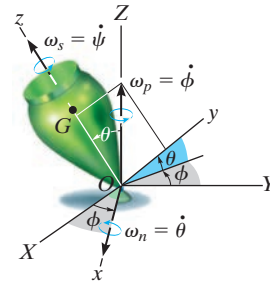
$$\mathbf{\Omega} \neq \mathbf{\omega}$$

Gyroscopic Motion

The angular motion of a gyroscope is best described using the three Euler angles ϕ , θ , and ψ . The angular velocity components are called the precession $\dot{\phi}$, the nutation $\dot{\theta}$, and the spin $\dot{\psi}$.

If $\dot{\theta} = 0$ and $\dot{\phi}$ and $\dot{\psi}$ are constant, then the motion is referred to as steady precession.

It is the spin of a gyro rotor that is responsible for holding a rotor from falling downward, and instead causing it to precess about a vertical axis. This phenomenon is called the gyroscopic effect.



$$\Sigma M_x = -I \dot{\phi}^2 \sin \theta \cos \theta + I_z \dot{\phi} \sin \theta (\dot{\phi} \cos \theta + \dot{\psi})$$

$$\Sigma M_y = 0, \Sigma M_z = 0$$

Torque-Free Motion

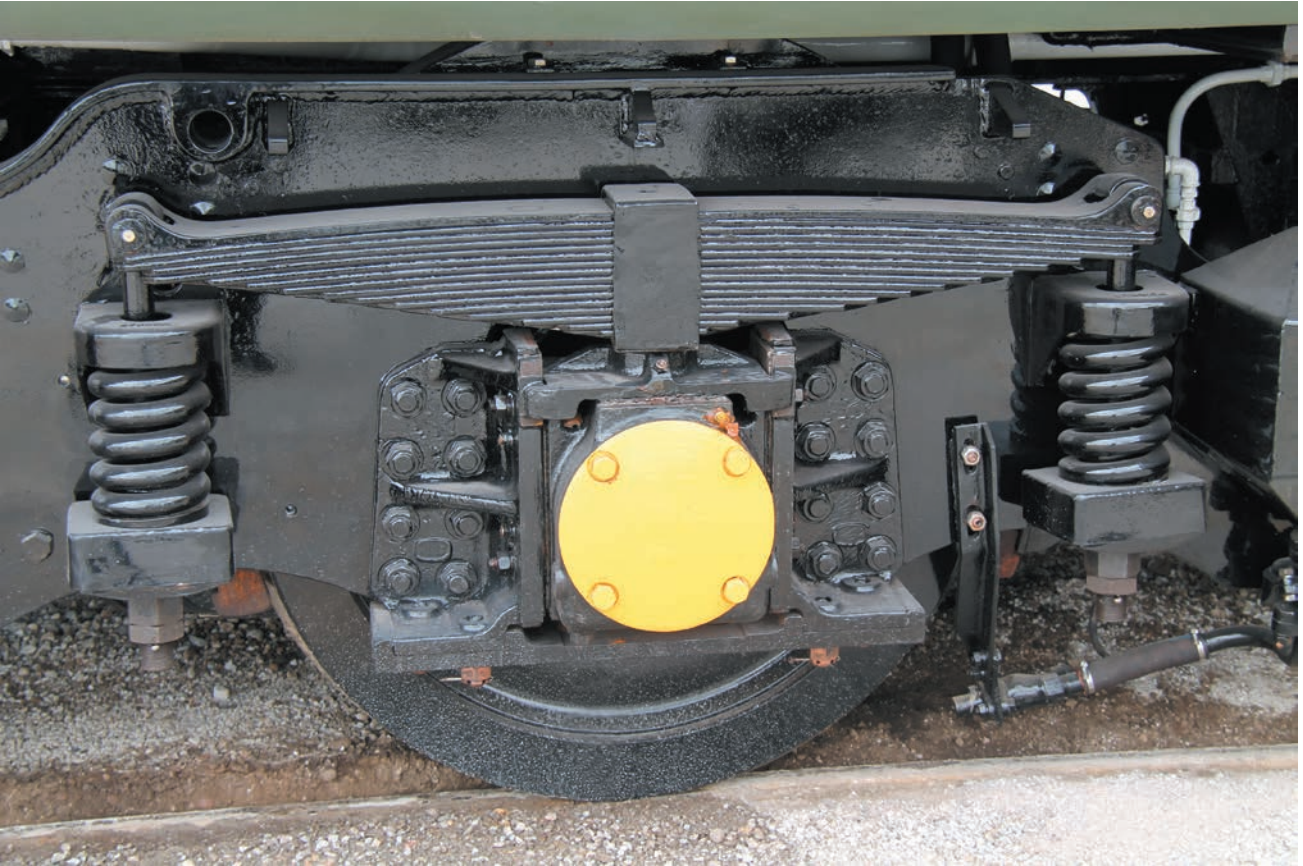
A body that is only subjected to a gravitational force will have no moments on it about its mass center, and so the motion is described as torque-free motion. The angular momentum for the body about its mass center will remain constant. This causes the body to have both a spin and a precession. The motion depends upon the magnitude of the moment of inertia of a symmetric body about the spin axis, I_z , versus that about a perpendicular axis, I .

$$\theta = \text{constant}$$

$$\dot{\phi} = \frac{H_G}{I}$$

$$\dot{\psi} = \frac{I - I_z}{I I_z} H_G \cos \theta$$

Chapter 22



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The analysis of vibrations plays an important role in the study of the behavior of structures subjected to earthquakes.

Vibrations

CHAPTER OBJECTIVES

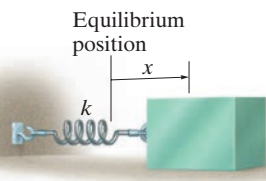
- To discuss undamped one-degree-of-freedom vibration of a rigid body using the equation of motion and energy methods.
- To study the analysis of undamped forced vibration and viscous damped forced vibration.



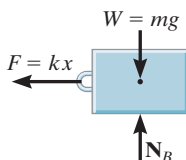
Video Solutions are available for selected questions in this chapter.

*22.1 Undamped Free Vibration

A *vibration* is the oscillating motion of a body or system of connected bodies displaced from a position of equilibrium. In general, there are two types of vibration, free and forced. *Free vibration* occurs when the motion is maintained by gravitational or elastic restoring forces, such as the swinging motion of a pendulum or the vibration of an elastic rod. *Forced vibration* is caused by an external periodic or intermittent force applied to the system. Both of these types of vibration can either be damped or undamped. *Undamped* vibrations exclude frictional effects in the analysis. Since in reality both internal and external frictional forces are present, the motion of all vibrating bodies is actually *damped*.



(a)



(b)

Fig. 22-1

The simplest type of vibrating motion is undamped free vibration, represented by the block and spring model shown in Fig. 22-1a. Vibrating motion occurs when the block is released from a displaced position x so that the spring pulls on the block. The block will attain a velocity such that it will proceed to move out of equilibrium when $x = 0$, and provided the supporting surface is smooth, the block will oscillate back and forth.

The time-dependent path of motion of the block can be determined by applying the equation of motion to the block when it is in the displaced position x . The free-body diagram is shown in Fig. 22-1b. The elastic restoring force $F = kx$ is always directed toward the equilibrium position, whereas the acceleration $\mathbf{a}$ is assumed to act in the direction of *positive displacement*. Since $a = d^2x/dt^2 = \ddot{x}$, we have

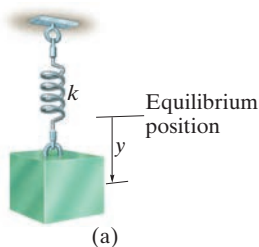
$$\rightarrow \Sigma F_x = ma_x; \quad -kx = m\ddot{x}$$

Note that the acceleration is proportional to the block's displacement. Motion described in this manner is called *simple harmonic motion*. Rearranging the terms into a "standard form" gives

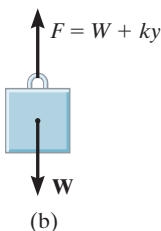
$$\ddot{x} + \omega_n^2 x = 0 \quad (22-1)$$

The constant ω_n , generally reported in rad/s, is called the *natural frequency*, and in this case

$$\omega_n = \sqrt{\frac{k}{m}} \quad (22-2)$$



(a)



(b)

Fig. 22-2

Equation 22-1 can also be obtained by considering the block to be suspended so that the displacement y is measured from the block's *equilibrium position*, Fig. 22-2a. When the block is in equilibrium, the spring exerts an upward force of $F = W = mg$ on the block. Hence, when the block is displaced a distance y downward from this position, the magnitude of the spring force is $F = W + ky$, Fig. 22-2b. Applying the equation of motion gives

$$+\downarrow \Sigma F_y = ma_y; \quad -W - ky + W = m\ddot{y}$$

or

$$\ddot{y} + \omega_n^2 y = 0$$

which is the same form as Eq. 22-1 and ω_n is defined by Eq. 22-2.

Equation 22-1 is a homogeneous, second-order, linear, differential equation with constant coefficients. It can be shown, using the methods of differential equations, that the general solution is

$$x = A \sin \omega_n t + B \cos \omega_n t \quad (22-3)$$

Here A and B represent two constants of integration. The block's velocity and acceleration are determined by taking successive time derivatives, which yields

$$v = \dot{x} = A\omega_n \cos \omega_n t - B\omega_n \sin \omega_n t \quad (22-4)$$

$$a = \ddot{x} = -A\omega_n^2 \sin \omega_n t - B\omega_n^2 \cos \omega_n t \quad (22-5)$$

When Eqs. 22-3 and 22-5 are substituted into Eq. 22-1, the differential equation will be satisfied, showing that Eq. 22-3 is indeed the solution to Eq. 22-1.

The integration constants in Eq. 22-3 are generally determined from the initial conditions of the problem. For example, suppose that the block in Fig. 22-1a has been displaced a distance x_1 to the right from its equilibrium position and given an initial (positive) velocity v_1 directed to the right. Substituting $x = x_1$ when $t = 0$ into Eq. 22-3 yields $B = x_1$. And since $v = v_1$ when $t = 0$, using Eq. 22-4 we obtain $A = v_1/\omega_n$. If these values are substituted into Eq. 22-3, the equation describing the motion becomes

$$x = \frac{v_1}{\omega_n} \sin \omega_n t + x_1 \cos \omega_n t \quad (22-6)$$

Equation 22-3 may also be expressed in terms of simple sinusoidal motion. To show this, let

$$A = C \cos \phi \quad (22-7)$$

and

$$B = C \sin \phi \quad (22-8)$$

where C and ϕ are new constants to be determined in place of A and B . Substituting into Eq. 22-3 yields

$$x = C \cos \phi \sin \omega_n t + C \sin \phi \cos \omega_n t$$

And since $\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$, then

$$x = C \sin(\omega_n t + \phi) \quad (22-9)$$

If this equation is plotted on an x versus $\omega_n t$ axis, the graph shown in Fig. 22-3 is obtained. The maximum displacement of the block from its

equilibrium position is defined as the *amplitude* of vibration. From either the figure or Eq. 22-9 the amplitude is C . The angle ϕ is called the *phase angle* since it represents the amount by which the curve is displaced from the origin when $t = 0$. We can relate these two constants to A and B using Eqs. 22-7 and 22-8. Squaring and adding these two equations, the amplitude becomes

$$C = \sqrt{A^2 + B^2} \quad (22-10)$$

If Eq. 22-8 is divided by Eq. 22-7, the phase angle is then

$$\phi = \tan^{-1} \frac{B}{A} \quad (22-11)$$

Note that the sine curve, Eq. 22-9, completes one *cycle* in time $t = \tau$ (tau) when $\omega_n \tau = 2\pi$, or

$$\tau = \frac{2\pi}{\omega_n} \quad (22-12)$$

This time interval is called a *period*, Fig. 22-3. Using Eq. 22-2, the period can also be represented as

$$\tau = 2\pi \sqrt{\frac{m}{k}} \quad (22-13)$$

Finally, the *frequency* f is defined as the number of cycles completed per unit of time, which is the reciprocal of the period; that is,

$$f = \frac{1}{\tau} = \frac{\omega_n}{2\pi} \quad (22-14)$$

or

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (22-15)$$

The frequency is expressed in cycles/s. This ratio of units is called a *hertz* (Hz), where $1 \text{ Hz} = 1 \text{ cycle/s} = 2\pi \text{ rad/s}$.

When a body or system of connected bodies is given an initial displacement from its equilibrium position and released, it will vibrate with the *natural frequency*, ω_n . Provided the system has a single degree of freedom, that is, it requires only one coordinate to specify completely the position of the system at any time, then the vibrating motion will have the same characteristics as the simple harmonic motion of the block and spring just presented. Consequently, the motion is described by a differential equation of the same “standard form” as Eq. 22-1, i.e.,

$$\ddot{x} + \omega_n^2 x = 0 \quad (22-16)$$

Hence, if the natural frequency ω_n is known, the period of vibration τ , frequency f , and other vibrating characteristics can be established using Eqs. 22-3 through 22-15.

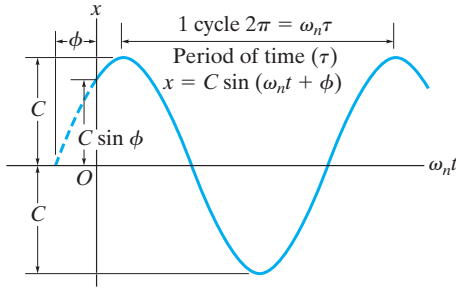


Fig. 22-3

Important Points

- Free vibration occurs when the motion is maintained by gravitational or elastic restoring forces.
- The amplitude is the maximum displacement of the body.
- The period is the time required to complete one cycle.
- The frequency is the number of cycles completed per unit of time, where $1 \text{ Hz} = 1 \text{ cycle/s}$.
- Only one position coordinate is needed to describe the location of a one-degree-of-freedom system.

Procedure for Analysis

As in the case of the block and spring, the natural frequency ω_n of a body or system of connected bodies having a single degree of freedom can be determined using the following procedure:

Free-Body Diagram.

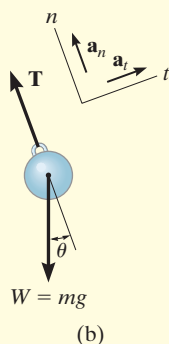
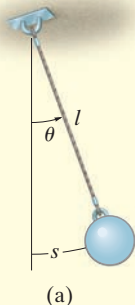
- Draw the free-body diagram of the body when the body is displaced a *small amount* from its equilibrium position.
- Locate the body with respect to its equilibrium position by using an appropriate *inertial coordinate* q . The acceleration of the body's mass center $\mathbf{a}_G$ or the body's angular acceleration $\boldsymbol{\alpha}$ should have an assumed sense of direction which is in the *positive direction* of the position coordinate.
- If the rotational equation of motion $\Sigma M_P = \Sigma (\mathcal{M}_k)_P$ is to be used, then it may be beneficial to also draw the kinetic diagram since it graphically accounts for the components $m(\mathbf{a}_G)_x$, $m(\mathbf{a}_G)_y$, and $I_G \boldsymbol{\alpha}$, and thereby makes it convenient for visualizing the terms needed in the moment sum $\Sigma (\mathcal{M}_k)_P$.

Equation of Motion.

- Apply the equation of motion to relate the elastic or gravitational *restoring* forces and couple moments acting on the body to the body's accelerated motion.

Kinematics.

- Using kinematics, express the body's accelerated motion in terms of the second time derivative of the position coordinate, $\ddot{q}$.
- Substitute the result into the equation of motion and determine ω_n by rearranging the terms so that the resulting equation is in the "standard form," $\ddot{q} + \omega_n^2 q = 0$.

EXAMPLE 22.1**Fig. 22-4**

Determine the period of oscillation for the simple pendulum shown in Fig. 22-4a. The bob has a mass m and is attached to a cord of length l . Neglect the size of the bob.

SOLUTION

Free-Body Diagram. Motion of the system will be related to the position coordinate ($q = \theta$), Fig. 22-4b. When the bob is displaced by a small angle θ , the *restoring force* acting on the bob is created by the tangential component of its weight, $mg \sin \theta$. Furthermore, $\mathbf{a}_t$ acts in the direction of *increasing* s (or θ).

Equation of Motion. Applying the equation of motion in the *tangential direction*, since it involves the restoring force, yields

$$+\nearrow \Sigma F_t = ma_t; \quad -mg \sin \theta = ma_t \quad (1)$$

Kinematics. $a_t = d^2s/dt^2 = \ddot{s}$. Furthermore, s can be related to θ by the equation $s = l\theta$, so that $a_t = l\ddot{\theta}$. Hence, Eq. 1 reduces to

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0 \quad (2)$$

The solution of this equation involves the use of an elliptic integral. For *small displacements*, however, $\sin \theta \approx \theta$, in which case

$$\ddot{\theta} + \frac{g}{l} \theta = 0 \quad (3)$$

Comparing this equation with Eq. 22-16 ($\ddot{x} + \omega_n^2 x = 0$), it is seen that $\omega_n = \sqrt{g/l}$. From Eq. 22-12, the period of time required for the bob to make one complete swing is therefore

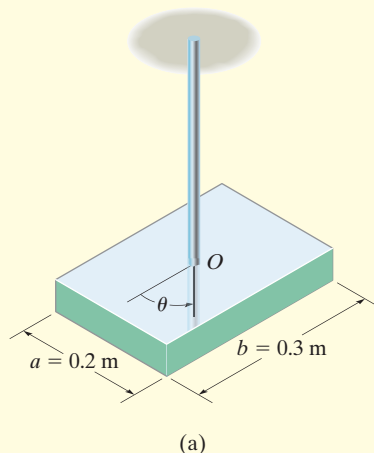
$$\tau = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{l}{g}} \quad \text{Ans.}$$

This interesting result, originally discovered by Galileo Galilei through experiment, indicates that the period depends only on the length of the cord and not on the mass of the pendulum bob or the angle θ .

NOTE: The solution of Eq. 3 is given by Eq. 22-3, where $\omega_n = \sqrt{g/l}$ and θ is substituted for x . Like the block and spring, the constants A and B in this problem can be determined if, for example, one knows the displacement and velocity of the bob at a given instant.

EXAMPLE 22.2

The 10-kg rectangular plate shown in Fig. 22-5a is suspended at its center from a rod having a torsional stiffness $k = 1.5 \text{ N} \cdot \text{m}/\text{rad}$. Determine the natural period of vibration of the plate when it is given a small angular displacement θ in the plane of the plate.

**SOLUTION**

Free-Body Diagram. Fig. 22-5b. Since the plate is displaced in its own plane, the torsional *restoring* moment created by the rod is $M = k\theta$. This moment acts in the direction opposite to the angular displacement θ . The angular acceleration $\ddot{\theta}$ acts in the direction of *positive* θ .

Equation of Motion.

$$\Sigma M_O = I_O \alpha; \quad -k\theta = I_O \ddot{\theta}$$

or

$$\ddot{\theta} + \frac{k}{I_O} \theta = 0$$

Since this equation is in the “standard form,” the natural frequency is $\omega_n = \sqrt{k/I_O}$.

From the table on the inside back cover, the moment of inertia of the plate about an axis coincident with the rod is $I_O = \frac{1}{12} m(a^2 + b^2)$. Hence,

$$I_O = \frac{1}{12} (10 \text{ kg}) [(0.2 \text{ m})^2 + (0.3 \text{ m})^2] = 0.1083 \text{ kg} \cdot \text{m}^2$$

The natural period of vibration is therefore,

$$\tau = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{I_O}{k}} = 2\pi \sqrt{\frac{0.1083}{1.5}} = 1.69 \text{ s} \quad \text{Ans.}$$

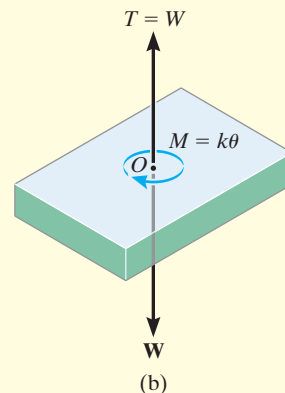


Fig. 22-5

EXAMPLE 22.3

The bent rod shown in Fig. 22–6a has a negligible mass and supports a 5-kg collar at its end. If the rod is in the equilibrium position shown, determine the natural period of vibration for the system.

SOLUTION

Free-Body and Kinetic Diagrams. Fig. 22–6b. Here the rod is displaced by a small angle θ from the equilibrium position. Since the spring is subjected to an initial compression of x_{st} for equilibrium, then when the displacement $x > x_{st}$ the spring exerts a force of $F_s = kx - kx_{st}$ on the rod. To obtain the “standard form,” Eq. 22–16, $5a_y$ must act *upward*, which is in accordance with positive θ displacement.

Equation of Motion. Moments will be summed about point B to eliminate the unknown reaction at this point. Since θ is small,

$$\zeta + \Sigma M_B = \Sigma (\mathcal{M}_k)_B;$$

$$kx(0.1 \text{ m}) - kx_{st}(0.1 \text{ m}) + 49.05 \text{ N}(0.2 \text{ m}) = -(5 \text{ kg})a_y(0.2 \text{ m})$$

The second term on the left side, $-kx_{st}(0.1 \text{ m})$, represents the moment created by the spring force which is necessary to hold the collar in *equilibrium*, i.e., at $x = 0$. Since this moment is equal and opposite to the moment $49.05 \text{ N}(0.2 \text{ m})$ created by the weight of the collar, these two terms cancel in the above equation, so that

$$kx(0.1) = -5a_y(0.2) \quad (1)$$

Kinematics. The deformation of the spring and the position of the collar can be related to the angle θ , Fig. 22–6c. Since θ is small, $x = (0.1 \text{ m})\theta$ and $y = (0.2 \text{ m})\theta$. Therefore, $a_y = \ddot{y} = 0.2\ddot{\theta}$. Substituting into Eq. 1 yields

$$400(0.1\theta) 0.1 = -5(0.2\ddot{\theta})0.2$$

Rewriting this equation in the “standard form” gives

$$\ddot{\theta} + 20\theta = 0$$

Compared with $\ddot{x} + \omega_n^2 x = 0$ (Eq. 22–16), we have

$$\omega_n^2 = 20 \quad \omega_n = 4.47 \text{ rad/s}$$

The natural period of vibration is therefore

$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi}{4.47} = 1.40 \text{ s}$$

Ans.

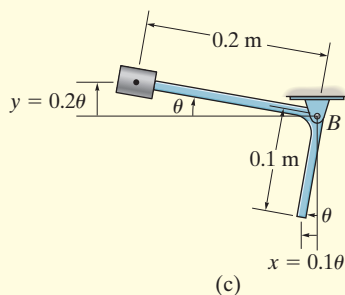
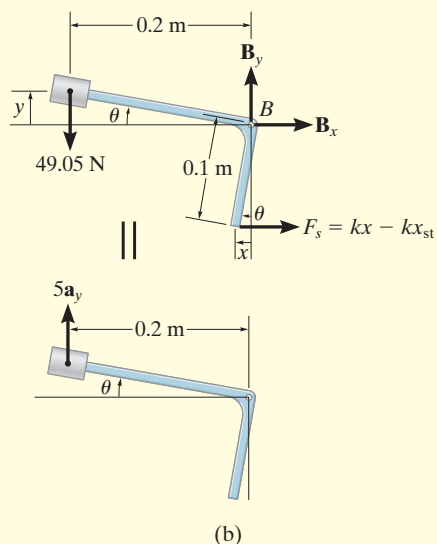
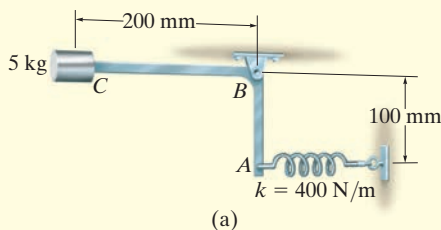
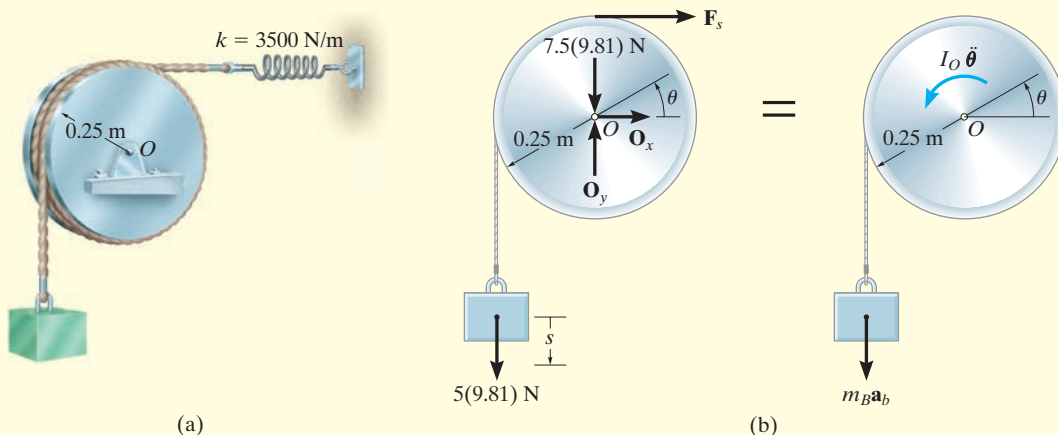


Fig. 22–6

EXAMPLE 22.4

A 5-kg block is suspended from a cord that passes over a 7.5-kg disk, as shown in Fig. 22-7a. The spring has a stiffness $k = 3500 \text{ N/m}$. Determine the natural period of vibration for the system.

**SOLUTION**

Free-Body and Kinetic Diagrams. Fig. 22-7b. The system consists of the disk, which undergoes a rotation defined by the angle θ , and the block, which translates by an amount s . The vector $I_O \ddot{\theta}$ acts in the direction of positive θ , and consequently $m_B a_b$ acts downward in the direction of positive s .

Equation of Motion. Summing moments about point O to eliminate the reactions $\mathbf{O}_x$ and $\mathbf{O}_y$, realizing that $I_O = \frac{1}{2}mr^2$, yields

$$\zeta + \Sigma M_O = \Sigma (\mathcal{M}_k)_O;$$

$$\begin{aligned} 5(9.81)(0.25 \text{ m}) - F_s(0.25 \text{ m}) \\ = \frac{1}{2}(7.5 \text{ kg})(0.25 \text{ m})^2 \ddot{\theta} + (5 \text{ kg})a_b(0.25 \text{ m}) \end{aligned} \quad (1)$$

Kinematics. As shown on the kinematic diagram in Fig. 22-7c, a small positive displacement θ of the disk causes the block to lower by an amount $s = 0.25\theta$; hence, $a_b = \ddot{s} = 0.25\ddot{\theta}$. When $\theta = 0^\circ$, the spring force required for equilibrium of the disk is 5(9.81) N, acting to the right. For position θ , the spring force is $F_s = (3500 \text{ N/m})(0.25\theta \text{ m}) + 5(9.81) \text{ N}$. Substituting these results into Eq. 1 and simplifying yields

$$\ddot{\theta} + 400\theta = 0$$

Hence,

$$\omega_n^2 = 400 \quad \omega_n = 20 \text{ rad/s}$$

Therefore, the natural period of vibration is

$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi}{20} = 0.314 \text{ s}$$

Ans.

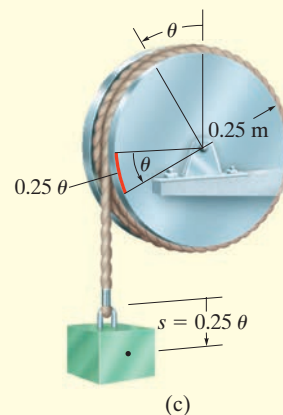


Fig. 22-7

PROBLEMS

22-1. A spring is stretched 175 mm by an 8-kg block. If the block is displaced 100 mm downward from its equilibrium position and given a downward velocity of 1.50 m/s, determine the differential equation which describes the motion. Assume that positive displacement is downward. Also, determine the position of the block when $t = 0.22$ s.

22-2. A spring has a stiffness of 800 N/m. If a 2-kg block is attached to the spring, pushed 50 mm above its equilibrium position, and released from rest, determine the equation that describes the block's motion. Assume that positive displacement is downward.

22-3. A spring is stretched 200 mm by a 15-kg block. If the block is displaced 100 mm downward from its equilibrium position and given a downward velocity of 0.75 m/s, determine the equation which describes the motion. What is the phase angle? Assume that positive displacement is downward.

***22-4.** When a 2-kg block is suspended from a spring, the spring is stretched a distance of 40 mm. Determine the frequency and the period of vibration for a 0.5-kg block attached to the same spring.

22-5. When a 3-kg block is suspended from a spring, the spring is stretched a distance of 60 mm. Determine the natural frequency and the period of vibration for a 0.2-kg block attached to the same spring.

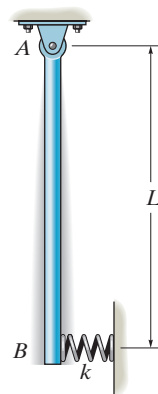
22-6. An 8-kg block is suspended from a spring having a stiffness $k = 80$ N/m. If the block is given an upward velocity of 0.4 m/s when it is 90 mm above its equilibrium position, determine the equation which describes the motion and the maximum upward displacement of the block measured from the equilibrium position. Assume that positive displacement is measured downward.

22-7. A pendulum has a 0.4-m-long cord and is given a tangential velocity of 0.2 m/s toward the vertical from a position $\theta = 0.3$ rad. Determine the equation which describes the angular motion.

***22-8.** A 2-kg block is suspended from a spring having a stiffness of 800 N/m. If the block is given an upward velocity of 2 m/s when it is displaced downward a distance of 150 mm from its equilibrium position, determine the equation which describes the motion. What is the amplitude of the motion? Assume that positive displacement is downward.

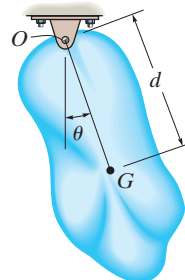
22-9. A 3-kg block is suspended from a spring having a stiffness of $k = 200$ N/m. If the block is pushed 50 mm upward from its equilibrium position and then released from rest, determine the equation that describes the motion. What are the amplitude and the natural frequency of the vibration? Assume that positive displacement is downward.

22-10. The uniform rod of mass m is supported by a pin at A and a spring at B . If B is given a small sideward displacement and released, determine the natural period of vibration.



Prob. 22-10

22-11. The body of arbitrary shape has a mass m , mass center at G , and a radius of gyration about G of k_G . If it is displaced a slight amount θ from its equilibrium position and released, determine the natural period of vibration.



Prob. 22-11